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OECD ECONOMIC SURVEYS

1998

SPECIAL FEATURES

Enhancing business dynamism

Banking sector problems

JAPAN

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF JAPAN

THE LAND

Area (1 000 sq. km), 1994	377.8
Cultivated agricultural land (1 000 sq. km, 1994)	51.7
Forest (1 000 sq. km, 1994)	251.6
Densely inhabited districts[1] (1 000 sq. km, 1990)	11.7

Major cities, October 1996 estimate (10 000 inhabitants):	
Tokyo (23 wards)	796
Yokohama	332
Osaka	260
Nagoya	215
Sapporo	177
Kobe	142
Kyoto	146

THE PEOPLE

Population, October 1997 estimate (1 000)	126 170
Number of persons per sq. km in 1997	334
Percentage of population living in densely inhabited districts in 1990[1]	63.2
Net annual rate of population increase (1986-97)	0.4

Labour force in per cent of total population, October 1997	54.3
Percentage distribution of employed persons, 1997:	
Agriculture and forestry	4.9
Manufacturing	22.0
Service	25.1
Other	48.0

PRODUCTION

Gross domestic product in 1997 (billion yen)	506 977
Growth of real GDP, 1981-97 average (annual rate, per cent)	3.1
Gross fixed investment in 1997 (per cent of GDP)	28.3

Growth of real gross fixed investment, 1981-97 average (annual rate, per cent)	3.4
Net domestic product of agriculture, forestry and fishery, at market prices, in 1996 (billion yen)	7 462
Growth of production in manufacturing 1981-97 (annual rate, per cent)	2.5

THE GOVERNMENT

Public consumption in 1997 (in per cent of GDP)	9.6
Current public revenue in 1996 (in per cent of GDP)	31.7
Government employees in per cent of total employment, 1997	8.3
Outstanding long-term national bonds in per cent of GDP (FY 1997)	54.3

	House of Representatives	House of Councillors
Composition of Parliament, September 1998:		
Liberal Democratic Party	264	103
Democratic Party	93	55
Peace and Reform (*Komei*)	47	24
Liberal Party	40	12
Communist Party	26	23
Others	30	35
Total	500	252
Last elections	Nov. 1995	July 1998

FOREIGN TRADE AND PAYMENTS

(1997, million yen)

Commodity exports (fob)	495 190
Commodity imports (fob)	372 087
Services	–65 423
Investment income	67 396
Current balance	114 363
Exports of goods and services in per cent of GDP	11.1
Imports of goods and services in per cent of GDP	9.9

	Exports	Imports
Percentage distribution:		
OECD countries	56.0	50.0
of which: North America	30.5	25.8
Far East	35.8	32.7
Other	8.2	17.3
Total	100.0	100.0
Crude material and fuels (SITC 2, 3, 4)	1.2	27.1
Semi-manufactured goods (5, 6)	18.1	18.2
Machinery and transport equipment (7)	69.0	25.4
Other (0, 1, 8, 9)	11.7	29.3
Total	100.0	100.0

THE CURRENCY

Monetary unit: Yen

Currency units per US$ average of daily figures:	
Year 1997	121.0
September 1998	134.5

1. Areas whose population density exceeds 5 000 persons per sq. km.

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Japan by the Economic and Development Review Committee on 30 September 1998.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 22 October 1998.

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The previous Survey of Japan was issued in November 1997.

Assessment and recommendations

The recession is deepening, and the outlook is uncertain

The economic situation in Japan is severe, with a deepening recession manifest in renewed deflationary pressures in goods, labour and asset markets. Policy makers have thus far not been able to prevent private-sector confidence from plunging and spending from being deferred. These factors, combined with weakness in export markets and a continuing credit crunch, have led to production cutbacks, spreading bankruptcies and a further downward adjustment in medium-term expectations. Although recovery in export markets and the appearance of the expansionary effects of the earlier stimulus package may stabilise economic activity by next year, any rebound looks likely to be limited, and the degree of economic slack may increase further. Indeed, there are a number of risks, which, if realised, would entail a more protracted downturn. To restore confidence and bring about a vigorous and sustained recovery, there is an urgent need for coherent and credible policy measures that directly address the structural and balance-sheet weaknesses of the financial sector, provide continued support to economic activity through macroeconomic policy and step up the pace of structural reforms, focusing on those which would increase economic dynamism and reduce households' uncertainty about their future income prospects. Decisive action on all three fronts is crucial to rebuilding confidence and restoring stability in Japan, as well as to Asia and the global economy.

The common view in early 1997 was one of optimism...

Looking back, it was only a year and a half ago that many observers hoped that the Japanese economy was emerging from the problems which had beset it since early in the decade and that it had probably begun a self-sustaining recovery based on domestic demand. Real GDP had turned in an impressive performance in 1996 – up nearly 4 per cent, reminiscent of the Japan of yore – led by double-digit private investment growth; inflation was virtually zero; the economy was operating at close to full capacity; the unemployment rate was at the comparatively low level of 3¼ per cent; and the perennial political issue of the current account surplus had receded into the background, having fallen to below 1½ per cent of GDP, its lowest level since the early 1980s. Aside from continuing problems in the banking sector, there was only one major macroeconomic concern: the government finances, which had worsened significantly since the onset of the period of stagnation and were set to deteriorate rapidly in the future under the weight of the increasing social burden of an ageing society. But fiscal consolidation seemed set to take care of that. Projections made in the spring of 1997, including those contained in the OECD *Economic Outlook 61,* generally called for output growth to ease a bit in the short term, while the fiscal tightening was being absorbed, and then to re-accelerate to nearly 3 per cent in 1998.

... but much of the strength in activity was ephemeral, and a series of shocks then hit the economy, the first of which was a significant fiscal tightening

Unfortunately, it did not work out that way. Upon closer examination and with the benefit of hindsight, it is now clear that, although much of the growth in 1996 represented an autonomous recovery based on private demand, the expansion was weakening and was not strong enough to absorb the significant fiscal tightening in 1997. In view of the substantial size of the budget deficit, one of the key policy questions at the end of 1996 was, quite appropriately, when and how fast to change the stance of fiscal policy. The decision had already been taken to increase the

consumption tax in April 1997 – a desirable structural change in itself, but one that had a much larger contractionary influence on domestic spending than generally expected. When combined with other increases in effective tax rates and social charges as well as an enormous cutback in government investment that was already under way, the total amount of spending withdrawn was about 11 trillion yen or just over 2 per cent of GDP. While some consolidation was clearly called for and could have been borne, this drastic tightening led to a significant weakening of activity and rendered the economy vulnerable to further deflationary influences, both foreign and domestic.

It was followed by the crisis in Asia

Few would have predicted that Japan's Asian neighbours would experience the crisis that has enveloped them over the past year or so and that their markets would shrink as precipitously as they have. The slowdown in Japan itself did not help the situation in the rest of Asia. But, superimposed on the weakening in domestic demand growth that had begun in the course of 1996 and had been hidden by the surprising amount of anticipatory spending prior to the consumption tax increase, this external shock further deflated the economy as from the summer of 1997 and raised the recessionary risks another notch.

The final blow came when two large financial institutions failed late in the year

The collapse of several financial institutions in the autumn – most prominently the failures of the Hokkaido Takushoku Bank, the nation's tenth largest, and Yamaichi, one of the big four securities firms – was decisive in pushing the economy into recession. That the financial sector in general and banks in particular would be the first to buckle was not surprising. The banks were clearly the most vulnerable to a deterioration in business conditions. Regulatory forbearance and the lack of exit in the context of the ''convoy system'' that prevailed heretofore resulted in overcapacity and therefore persistently narrow margins.

Banks' poor returns were further worsened by an increase in competition for deposits following interest-rate deregulation and a decline in the value of their franchise resulting from progressive liberalisation and the development of securities markets. But, most importantly, the lingering effects of the still ongoing asset price deflation left their balance sheets in a deteriorated state. The bursting of the asset price ''bubble'' has by now led to cumulative capital losses that on paper amount to around one *quadrillion* yen, the equivalent of about $7 trillion or two years' worth of Japan's GDP. While the financial sector has directly borne less than one-fifth of these losses, borrowers, mainly non-financial businesses in the construction and real estate sector, have been heavy losers, and lenders were accordingly left holding vast amounts of non-performing loans. This problem has worsened with the deterioration in overall economic conditions. Despite the publication of a series of industry and government estimates, the magnitude of these loans is nonetheless the subject of strenuous public debate, with some analysts claiming that they still amount to as much as 150 trillion yen (30 per cent of GDP). Partly because of the uncertainty surrounding the viability of individual banks, as well as the sudden recognition that even venerable financial institutions could henceforth close their doors, with their employees out of a job, both consumer and business confidence plunged, and risk premiums attached to Japanese borrowers, especially the banks, surged. Banks themselves feared that the strengthened supervisory framework would lead to action against those of them that had inadequate capital in relation to their assets and gradually recognised the implications of the new competitive environment for their preferences towards risk and returns. As a result lending dropped, and talk of a ''credit crunch'' spread. Spending fell sharply, bankruptcies surged and a deep recession – the worst in the post-war era – was under way.

The authorities have had to play catch-up to the deteriorating situation by reversing course on fiscal policy

It was late last year, when the seriousness of the situation became apparent, that the authorities began to make major attempts to catch up to the rapidly changing circumstances. The initial budget for FY 1998 complied with the ill-fated Fiscal Structural Reform Act (which had just passed the Diet at the end of November, setting out the means and period over which the fiscal consolidation would be achieved), and was therefore inappropriate to the conjuncture. But all the while they were also preparing a supplementary budget for FY 1997 which, it was hoped, would offset some of the downward drag from the earlier tightening. Just two months later, when the gravity of the situation became more evident, a major revision to the Act that would allow some needed flexibility and a separate supplementary budget for 1998 were also proposed and ultimately approved. The result of this process, which proved cumbersome in the circumstances, was a gradual change in fiscal policy from highly restrictive to moderately expansionary, but not before well into 1998, by which time the economy's downward momentum was substantial.

But the manner in which budgetary support was provided was inefficient

But perhaps because the government was unprepared for the possibility of having to reverse course so soon after engaging in what had been expected to be a lengthy period of consolidation, the measures it took to support economic activity were inefficient. Most particularly, the personal income tax cuts were explicitly described as being temporary, thereby reducing their potential effectiveness in stimulating demand, and much of the additional public spending took the form of still further traditional infrastructure projects, some of which have meagre social returns and few favourable spin-offs. While some valuable permanent changes were made in corporate income taxation in 1998, they amount to a rather modest reform of the overall tax system, which is in need of a substantial overhaul. The opportunity should also have been taken to remove the

distortions which result from high marginal personal income tax rates that currently reach up to 65 per cent and nonetheless yield less revenue than in any other major OECD country. Such distortions include disincentives to labour force participation, a shift towards self-employment and a handicap to entrepreneurship. The budgetary costs of bold cuts in rates could have been offset to some extent by shrinking the number and size of deductions that combine to make the tax base excessively narrow. A fundamental review of the personal taxation system is under way.

Additional public investment with meagre returns will only delay the shrinkage of the bloated construction sector

Possibly even less advisable as a form of stimulus at this point in Japan was an increase in certain kinds of traditional public investment. Public investment as a share of GDP was already twice the OECD average and has supported a construction sector which can be seen as bloated. Despite the absence of any demographic need for new housing, an already high level of business capital intensity and a surfeit of some kinds of public infrastructure, this industry is saddled with some 560 000 firms and more than one in ten employees in the country. Only partly for reasons of higher relative prices and a greater risk of natural disasters, its output accounts for 9 percentage points of GDP more than its US counterpart, for example. Delaying the inevitable downsizing of this sector can hardly be a worthwhile use of scarce public resources.

For the next fiscal year budgetary focus will be on providing further stimulus mainly through tax cuts

In planning its fiscal policy for 1999, the government has recognised that the economy's condition is still clearly too fragile to risk any kind of tightening at this point. It therefore intends to propose an unusual 15 month budget, which combines a second supplementary budget for FY 1998 and the regular FY 1999 budget, as well as to suspend the application of the Fiscal Structural Reform Law. This latter allows the authorities to avoid setting an unrealistic initial budget, only to have to follow it up immediately with one

or more supplementary budgets. The government has stated that from 1999 the top marginal income tax rate for individuals and the corporate income tax rate will be reduced substantially. This is to be financed by debt issuance but must in time be paid for by broadening the respective tax bases and by embarking on a multi-year programme to cut back on some kinds of public infrastructure. The government also appropriately intends to increase the emphasis on urban public infrastructure and raise the share of new types of public investment. In that regard, since all government spending should ideally have the same social value at the margin, stepped-up cost-benefit analysis should be carried out. Consideration should then be given to reviewing the resource allocation effects of the distinction between so-called deficit-financing and construction bonds.

In the not-too-distant future there will no longer be any leeway to use counter-cyclical fiscal policy, and official medium-term targets are not sufficiently ambitious

Even though fiscal stimulus should not be withdrawn until a self-sustaining recovery is under way, looking beyond the short term, there is not much more time available before longer-term fiscal realities will have to be recognised, no matter what the prevailing cyclical conditions. That point has not yet been reached, and insofar as fiscal stimulus itself promotes recovery, room for subsequent consolidation is created. The sustainability of even an unchanging debt level is not guaranteed in a situation where the economy's medium-term potential growth rate may slip further from today's modest pace, prices are as likely to fall as to rise and interest rates cannot be reduced much further. However, attaining a manageable fiscal position would be facilitated by the favourable impact on growth from effective structural reform. Even so, Japan's gross general government debt is now nearing the level of GDP. While there are substantial offsetting financial assets which bring net debt to nearly 20 per cent of GDP (OECD definition) at the end of 1997, assessing the true value of those assets and the state of the balance sheet overall is obscured by

uncertainty not least about contingent liabilities, most importantly in the pensions area but also possibly hidden in the accounts of various public agencies and corporations. The elderly dependency ratio is set to rise by a percentage point per year for the next twenty or more years. In these circumstances the pressures on public spending will be enormous, with pension and health care for the elderly likely to swallow up an additional 10 percentage points of GDP over the next generation. The targets in the revised Fiscal Structural Reform Law – a deficit on central and local government account of 3 per cent of GDP and the elimination of central government deficit bond issuance by FY 2005 – require only a modest average annual reduction of the deficit. These targets are not sufficient to stabilise the debt ratio at that point which could, in gross terms, easily be the highest in the OECD. In order to ensure that the fiscal position is ultimately sustainable, to limit the enormous generational inequities that are in prospect and the deleterious effects of even optimally designed tax and contribution increases, the government will therefore have to base its decisions on transparent analysis of the budgetary situation and prospects, consider carefully the value of all public spending and services and strive for the highest possible efficiency in their delivery.

Monetary policy has been used almost to its limits

The monetary authorities have also been striving to provide as much support to the economy as possible. The Bank of Japan has played a useful role in maintaining the financial system stability through the crisis period, most recently, by bringing down the call rate for the first time in three years. Previously, it allowed its balance sheet to swell as a result of enormous liquidity injections which have at least offset some of the deflationary impact of the loss of confidence resulting from the various financial institution failures, thereby stabilising market conditions and allowing longer-term interest rates to fall to record lows. But the economy's

plight has been so severe that monetary expansion has not led to increased spending. Much of the additional liquidity has gone to cautious individuals who have boosted their cash holdings at the expense of deposits in a banking system no longer seen as trustworthy. Some has been absorbed by the domestic banking system, which has not succeeded in turning it into new loans, either due to a supply-side ''credit crunch'' (the evidence of which is not entirely persuasive) or because of a cyclical fall in demand. And the rest has been reflected in declining foreign exchange reserves, as the authorities have on several occasions intervened in exchange markets to resist substantial intermittent downward pressure on the value of the yen against the dollar. But such pressures were strong at the time, given such a large interest-rate differential over the entire term structure and the massive increase in the yen's value against the currencies of its Asian trading partners in crisis as well as the domestic economy's conjunctural difficulties.

The central bank should focus on ensuring adequate growth of the money and credit aggregates

There would seem to be little choice facing monetary policy makers at this juncture. Some have suggested *raising* interest rates in order to put more spending power into the hands of the economy's creditors, but that would risk a further deterioration of overall economic conditions and a collapse of the banking system. On the other hand, it is not clear how the monetary authorities could substantially increase their support for spending and activity in current conditions without the central bank taking radical steps such as directly taking on credit risks *vis-à-vis* the non-bank private sector. In the present circumstances, it should continue to ensure that the monetary and credit aggregates grow at least as rapidly as heretofore and, if the need arises, show willingness to help finance extra public spending by purchasing ample amounts of government securities and make any limited interest-rate cuts that remain at its disposal.

The banking system problems have finally drawn public funds but with seemingly insufficient restructuring in return

The balance sheet problem in the banking system is a structural issue that has festered for so long, despite the lengthy period of low interest rates, that it has assumed macroeconomic significance. The official position since the asset price deflation first laid open a serious problem of non-performing loans had been to maintain steadfastly that it should be dealt with by the institutions themselves through gradual provisioning and write-offs which draw on current earnings and accumulated reserves. Supplying public funds was seen as a last resort because of the political fallout which resulted from a public bailout of limited size arranged for some housing loan providers a few years ago. But once the situation became clearly manifest and assumed crisis proportions last autumn, the provision of public funds was inevitable, despite continuing public resentment. The government quickly put together a plan to make available up to 30 trillion yen (about $240 billion). The objective was to bolster bank balance sheets and allow for speedy write-offs offset by public capital replenishment and to prop up the deposit insurance system which had lacked the resources to handle any further major bank failures, despite the seven-fold increase in premiums imposed last year. But the banks were reportedly so fearful of the possible conditions that could have been attached to the capital injection and by the associated stigma in accepting such funding that in the end only a small fraction of it was used – 1.8 trillion yen or about 5 per cent of banks' capital – and that in what appeared to be a rather indiscriminate fashion. While the injection successfully boosted capital adequacy ratios despite the major banks' write-off of another 10 trillion yen in non-performing loans, it was not ultimately needed by any of the 21 recipient institutions to meet the required regulatory level at the end of the financial year in March, and the amount of promised restructuring seems unlikely to have been sufficient to meet longer-term requirements.

There followed a ''total plan'' to force restructuring, but it will not succeed unless there is substantial exit from the banking system

The authorities then developed a ''total plan'' to deal with the banking sector comprehensively through improved mechanisms for the disposition of bad loans and collateral, through enhancing the power of the new supervisory body initially provided under the ''prompt corrective action'' provisions of the recent financial reforms to order restructuring, and through the establishment of a system of temporary public ''bridge banks'' to take over the operations of any institution deemed to be insolvent. Assuming that power is used appropriately, the plan is welcome, as it would limit any further negative spill-over from measures to solve the banking system problems onto the macroeconomy by maintaining lending to sound borrowers. But it may need to be supplemented by a number of other actions in order for it to have the maximum favourable impact: clear rules regarding the disbursement of public funds need to be set with a view to minimising moral hazard problems; the Financial Supervisory Authority must be adequately staffed; and the Authority must not hesitate to use its new power to close or merge even major banks (who hold two-thirds of all problem loans), if they are found to be insolvent. Such institutions cannot continue to struggle on, as they have done now for the past six years or so. Thoroughgoing rationalisation and substantial exit are crucial to bringing lending margins to levels which will allow healthy banks to resume fully their credit creation function.

Lack of transparency and disclosure is an increasingly costly problem

Another reason for unease is persistent suspicion of a lack of disclosure and transparency, a problem that permeates Japan's financial and public sectors. Financial statements are still greeted with suspicion because on too many occasions they have concealed transactions and obligations which ultimately result in insolvency. The Yamaichi case of so-called *tobashi* losses of nearly 300 billion yen is only one example. Markets are apparently now demanding a

separate risk premium attached specifically to the inadequate disclosure of Japanese securities issuers. The authorities are right to insist on at least world-class disclosure, since the markets will be slow to believe that they are being told the whole story. In this regard, the decisions to allow banks to revalue their real estate assets at market prices, to avoid capital losses on equities by backtracking on marking to market and to net out some client time deposits from their lending aggregates, even if some of these practices are common abroad, were inevitably greeted by scepticism because of their timing.

Implementation of the structural reform programme is finally under way, but the process should be accelerated

While the world's attention has been focused on the deteriorating conjunctural developments and on the distress in the financial system, structural reform efforts have been reactivated since the 1996 decision to pursue wide-ranging reform grouped together under six so-called ''pillars''. A promising start has been made, and the implementation phase is now under way in a variety of domains, most notably the financial ''Big Bang'', the permission to establish financial holding companies, a substantial revision of the law limiting store size, a new long-term care insurance programme and the abolition of the rule constraining pension fund portfolio allocation. However, even though the reform process is often cumbersome, once broad principles are agreed upon it would be desirable to see concrete changes put into effect more rapidly than has been the custom thus far. Promising to liberalise economic activity too far in the future will only serve to make many observers sceptical that such reform will ever be undertaken, even if careful planning prior to implementation is advantageous. In addition, delay may undermine the rebuilding of confidence and weaken future growth prospects, thereby making needed future fiscal adjustments more difficult to accomplish. In its own bailiwick, the government is right to shift its emphasis from discretionary to regulatory administration

with *ex post* monitoring of compliance and to separate policy planning from implementation. In this context, it deserves support in its attempt to review all regulation with a view to setting only minimal rules; this should ease the regulatory burden.

Many recent decisions, when implemented, will lead to improved productivity

There have been a number of noteworthy decisions since mid-1997. Price controls on oil products and a variety of domestic transportation sectors will be abolished. The rice price-support framework will be dismantled. Access to the accounting profession by foreigners might be reopened. Greater working-time flexibility will be encouraged, and moral support is being given to merit-based pay systems. The nation's commitment to reduce greenhouse gases will be met in part by issuing tradeable emission rights. A variety of household appliances will be subject to future recycling requirements. Postal savings and public pension funds will no longer be automatically deposited with the Trust Fund Bureau, a public wholesale bank which finances government agencies. Finally, administrative reform will reduce the number of government departments and, it is hoped, the size of the civil service. Most of these items should contribute to increasing productivity with favourable effects on supply.

But there remains a long list of areas where reform still needs to be vigorously pursued

But there remain a number of areas where change is needed and where decisions are still awaited. Interconnection fees for new common carriers in the telecommunications sector need to be set so as to encourage competitive supply. Electricity from producers other than the current set of regional monopolies should be available to a wide-range of customers. Laws concerning tenant rights and land use planning need to be re-examined to ensure that they are still providing net benefits. The proposal to expand remedies under the Antimonopoly Law so that victims of unfair trade practices may go directly to court for orders to stop them should be

implemented. The postal savings and life insurance operations are to be transferred to a new type of corporation, but the Committee believes such operations should ultimately be further distanced from the government, commensurate with its increasing degree of discretion in portfolio investment decisions. Whatever tax reforms are approved, a taxpayer identification system should be implemented without delay to minimise tax evasion. The current review of the health care system needs to result in a reform of the methods of drug pricing and setting doctors' fees as well as a widening of the base for contributions. In view of the higher rates of current and projected unemployment as well as of job turnover, it would be prudent to re-examine the adequacy of the social safety net in general and the unemployment insurance system in particular. And, perhaps most importantly, the pension systems, both public and private, must be overhauled once and for all so as to put them on a sound footing in light of their future funding difficulties.

A range of non-manufacturing industries is now feeling the impact of deregulation, with immediate benefits in some cases

A new wave of deregulation has started to shake up some of the non-manufacturing sectors such as finance, telecommunications, transport, electricity, petroleum product distribution, labour market intermediation and agriculture. Companies operating in these sectors are thus faced with heightened competitive pressures from new entrants to the markets as well as new business opportunities. It is sometimes argued that deregulation, while beneficial in the long run, reduces output and employment in the short run because of the restructuring of the companies already operating in the sectors concerned. Even within a few years, however, intensified competition often brings about increased output not only through price declines but also *via* an expanded range and quality of services offered, as seen in the telecommunications and air transport sectors. It should also be noted that certain kinds of deregulation can have immediate positive effects on output (for example, an

easing of restrictions concerning agricultural land and building construction), and it makes sense in the current circumstances to put a high priority on these in the government's deregulation efforts.

Japan needs to reinvigorate business dynamism

While deregulation has been opening up new business opportunities and international competition has been exerting pressures on companies to remain competitive, changes in the Japanese business sector overall have not been sufficient to arrest the trend declines in profitability and the rates of new company formation. For Japan to be able to continue raising the standards of living of its citizens it is necessary to reinvigorate business dynamism. A key to this is further deregulation and the enhancement of its capacity to innovate and take calculated risks, since Japan, having one of the most developed economies, has to expand the new frontiers of production.

Initiatives to improve creativity are welcome and should be pursued further...

The government has taken initiatives to enhance the nation's innovative capacity: the hitherto compartmentalised R&D system is to be improved through increased flows of information and researchers among the government research laboratories, universities and businesses; the efficiency of public funding is to be raised by introducing competitive considerations in budgetary allocation and making third-party evaluation of projects obligatory; the protection of innovators' intellectual property rights has been strengthened; and universities are to be encouraged to have their innovative work patented and sold. So as to underpin the improvement in the R&D system in the long term – and more generally to help develop more flexible and adaptable human capital – the government has embarked on education reform. The main thrust is to promote creativity through diversity, and deregulation is to play an important role in achieving this goal. The broad orientations of these reforms, *i.e.* providing greater

autonomy to the key actors involved and encouraging interactions among them, are sound and should be pursued further, even though adequate governance mechanisms should be put in place commensurate with the degree of autonomy granted.

... and so are initiatives to enhance the willingness to take risks

In an attempt to nurture the capacity to undertake risky but promising new projects, the government has readied the legal and institutional infrastructure for a potentially huge increase in the supply of risk capital, including the easing of restrictions on investment in unlisted shares by pension funds, the creation of a special over-the-counter (OTC) market for innovative companies, and measures to facilitate the transactions of unlisted shares. However, both perseverance and further measures (such as the strengthening of the price formation function in the OTC market) would be necessary before venture capital plays a significant role in financing innovative new ventures in view of difficulties in developing it in any financial system dominated by inherently risk-averse banks. Banks act as the chief monitor in the Japanese corporate governance arrangements which have had their merits in reducing the agency costs involved and in coping with temporary financial distress. But the combination of bank-based monitoring and the predominance of directors promoted from within is more likely to result in preference for continuity over change and is not conducive to calculated risk-taking. It is high time to redress the balance by strengthening the monitoring role of outsiders. In this regard, the recent initiatives by some leading companies to reduce the size of their boards of directors and increase the participation of outside directors are welcome and should be encouraged by the stock exchange as desirable practices. More frequent shareholders' derivative suits against corporate directors also represent a welcome counterbalancing move. It would, nevertheless, be advantageous to introduce the so-called

‘‘business judgement rule’’ so as to prevent an excessive degree of risk aversion on the part of corporate directors in making business decisions. Finally, the recent revision of the Commercial Code and the tax treatment of stock options should help their wide adoption, thereby aligning managers’ interests more closely with those of shareholders.

The outlook for the next 1½ years is for a minimal recovery from the current deep recession, with unemployment continuing to move up...

Bearing in mind the serious conditions in which the economy finds itself in the summer of 1998, it is clear that it will take some time for it to shake off all contractionary influences. Even with the huge set of tax and spending measures currently being implemented and proposed, real GDP is set to fall over 2½ per cent this calendar year (nearly 2 per cent for the fiscal year ending next March) and rise only modestly in 1999. Despite lower tax payments, consumers are likely to be constrained by negligible income growth resulting from weak labour market conditions. Housing construction may remain limited by sluggish household formation and the perception that housing is a poor investment. Business fixed investment should continue to respond to the economic cycle and poor profitability as well as, for smaller and medium-sized firms, the persistence of less accommodating lenders. And inventories will still be a drag on demand for several quarters to come until the current overhang is worked off. Domestic demand may, therefore, be supported only by the public works in the recent package, whose stimulus will wear off over the course of 1999, and by the permanent tax cuts starting next year. Imports should register only a limited rebound, but the uncertainty surrounding export market growth is substantial, as the risks surrounding the future course of activity in the Asian crisis countries, to say nothing of the question of Japan’s other important markets in North America, are broad. And exporters will suffer from increasing competition from their rivals based elsewhere in Asia. In such a

weak environment and with a substantial number of job losses in prospect from expected restructuring, it is highly likely that the unemployment rate will remain on a rising course for some time to come. This implies that prices could edge down. Even with respect to this weak picture, however, a number of important downside risks should be kept in mind. They relate to resolving the problems in the banking system, the pace of asset price deflation and the recovery in export market growth.

... and medium-term growth prospects are unexciting

Looking further into the future, the Japanese economy's growth performance is likely to remain mediocre for some time to come, absent major structural reform, reflecting, in part, its relative maturity and high income level. With the deceleration and ultimate decline in the working-age population, reductions in working time and a slowdown in the growth of the capital stock as well as of total factor productivity, the growth rate of potential output has already fallen significantly and is estimated by the Secretariat to be below 1½ per cent per year. To be sure, the economy can grow faster than this rate while the current slack in productive capacity is absorbed. But there is much uncertainty about the extent of such slack during a period of rapid structural change which makes some of the excess capacity obsolete. Moreover, the resulting corporate downsizing to reduce obsolete capacity works to restrain aggregate demand expansion during this period. Spending will also be held down by the inevitable fiscal consolidation, given the projected general government deficit of 5 per cent of potential GDP in cyclically adjusted terms in 1999.

The authorities should move urgently to end the recession and improve the prospects for medium-term growth by adopting a comprehensive strategy

The diagnosis is, therefore, that the economy is suffering from a combination of maladies of cyclical and longer-term nature that requires a quick and forceful response. It is still subject to chronic asset price deflation from which it has been unable to recover. This has left it with weakened business and household balance sheets and a banking system still saddled with an enormous non-performing loans problem and therefore functioning poorly. It has a massive inventory overhang which needs to be worked off. Both employees and managers are uncertain about their futures and depressed by the prospects of economic restructuring. The prescription is to move urgently in order to end the recession. A comprehensive, three-pronged approach involving measures to overcome banking system problems, insufficient aggregate demand and structural weaknesses is required. Such measures would function in a mutually reinforcing way to restore confidence both domestically and, in the current context of heightened fragility, in the rest of Asia and across the globe. Both the banks themselves and the government need to make effective and expeditious use of the newly authorised mechanisms to restructure and revitalise the sector. In this period of macroeconomic adjustment, which may well be prolonged, fiscal and monetary support is welcomed, required and should not be withdrawn prematurely. But in order to avoid unsustainable increases in public debt, fiscal policy should aim to achieve the maximum sustained impact on activity from all expenditure programmes and the fewest distortions from all taxes. At the same time, the authorities need to move ahead rapidly on a broad front with the many reforms in other areas that promise to improve the functioning of the economy's equilibration mechanisms, enhance productivity in the medium term and lay the groundwork for the future, a future which will involve a smaller role for government and its bureaucracy and a bigger part for individuals and their personal creativity, responsibility and risk-taking.

I. Origins of the current difficulties

That the current recession was caused by a combination of fiscal tightening, the effects of Asian economic turbulence and an enfeebled banking system is indisputable. An optimistic view as to the economy's underlying strength was widely held when the government decided to embark on a fiscal consolidation which had to occur sooner or later. The size of the fiscal tightening, in hindsight, was too big for the Japanese economy to absorb in the face of a series of unanticipated events including the economic crisis in some Asian countries and the failure of some major financial institutions in the latter half of the year. In order to provide a basis for an informed assessment of the current situation and policy requirements, this chapter first reviews briefly key features of economic trends in the 1990s, with an emphasis on structural changes that underlie investment and consumption behaviour. The sequence of measures that led to the significant fiscal tightening in 1997 and the economy's reaction to it are next discussed. The chapter ends with a discussion of the legacy of the asset price bubble, which remains largely in the form of problem loans which have brought the banking system near to collapse.

A brief review of developments in the 1990s

Growth with constant government support

Japan's growth performance in the 1990s has been disappointing. After peaking in February 1991, following a 51-month expansion, the Japanese economy went through an unusually long recession that lasted 32 months and then a very modest expansion, which ended in March 1997. The 1991-93 downturn was the second longest in the post-war period, the average length of recessions being about 15 months. Moreover, the subsequent 31-month upturn was clearly lacking

in vigour (Table 1, Panel A). The output gap swung from an estimated negative 3.1 per cent in 1991, (*i.e.* over-utilisation of capacity), to a positive 0.7 per cent in 1993 and then continued to widen to 2.4 per cent in 1995 (Figure 1). Until the latest downturn, however, growth outcomes had been only slightly worse than those of the OECD as a whole and similar to those in Europe (Table 1, Panel B). Such international comparisons show that it was only in housing investment and, to a lesser extent, in exports that Japan's pattern of spending growth was notably sub-par.

Table 1. **Long-term performance of the Japanese economy**

Average annual real growth rate by period in per cent[1]

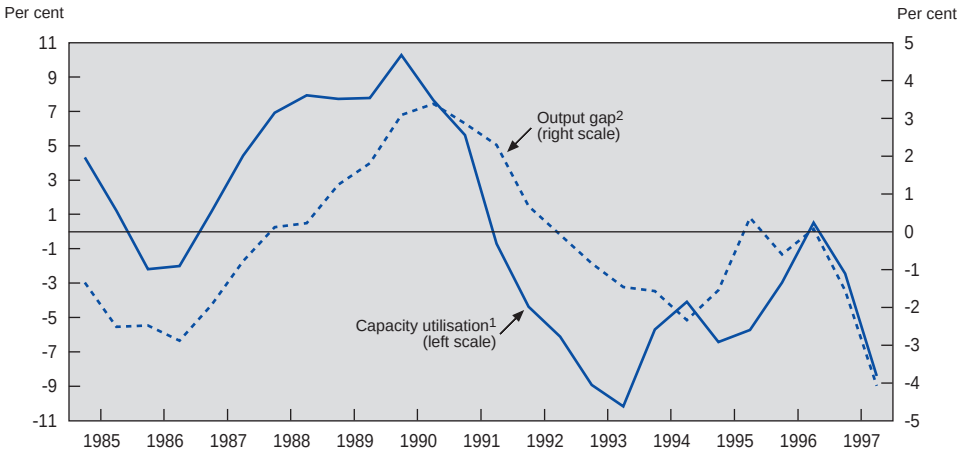
	1961-70	1971-80	1981-90	1991-97
A. Comparison with other decades				
GDP	10.2	4.5	4.0	1.7
Private consumption	9.0	4.7	3.7	2.0
Government consumption	4.8	4.8	2.5	1.9
Residential investment	16.8	3.2	3.9	–1.8
Business investment	16.6	2.8	8.1	0.6
Government investment	14.4	5.9	0.8	4.9
Exports	16.1	9.7	5.4	5.1
Imports	14.7	5.9	6.3	4.3
Compensation of employees	11.1	5.8	3.7	2.1
Disposable income	9.5	4.8	3.0	2.2[2]

	Average annual real growth rate in 1991-97				
	Japan	United States	EU	Europe	OECD
B. Comparison with other countries, 1991-97					
GDP	1.7	2.3	1.7	1.8	2.2
Private consumption	2.0	2.4	1.6	1.8	2.2
Government consumption	1.9	0.3	1.3	1.4	1.2
Residential investment	–1.8	3.8	0.7	0.9	1.7
Business investment	0.6	5.6	0.6	1.0	3.1
Government investment	4.9	–0.1	–	–	–
Exports	5.1	8.0	6.0	6.2	7.2
Imports	4.3	8.6	4.5	5.1	6.8
Compensation of employees	2.1	2.1	–	–	–
Disposable income	2.2[2]	2.1	–	–	–

1. The figures in the table are calculated by taking a simple mean of annual growth rates in each period.
2. OECD estimate.

Source: OECD.

Figure 1. **OUTPUT GAP AND CAPACITY UTILISATION**



1. Difference between the current level of the MITI capacity utilisation index for manufacturing and the average for the past 15 years.
2. Output gap as estimated by the OECD.

Source: Bank of Japan, Ministry of International Trade and Industry and OECD.

Not surprisingly, the government made vigorous efforts to reactivate the economy. It implemented seven fiscal stimulus packages in the four years from 1992 to 1995, and the Bank of Japan cut the official discount rate to 0.5 per cent in 1995 and has since kept it at this historically low level (Table 2).[1] The government also undertook a major tax reform in 1994 in order to make the tax system more suitable to a rapidly ageing society (Figure 2). Its major thrust was to shift the tax base from income toward consumption: personal income taxes were reduced by changes in the tax schedule and increases in standard personal and employment-income deductions, and the consumption tax rate was to be raised from 3 to 5 per cent in 1997. Although the reform was intended to be revenue-neutral, it had some positive effects on the economy from 1994 to 1997 because the income tax reduction preceded the rise in the consumption tax rate.

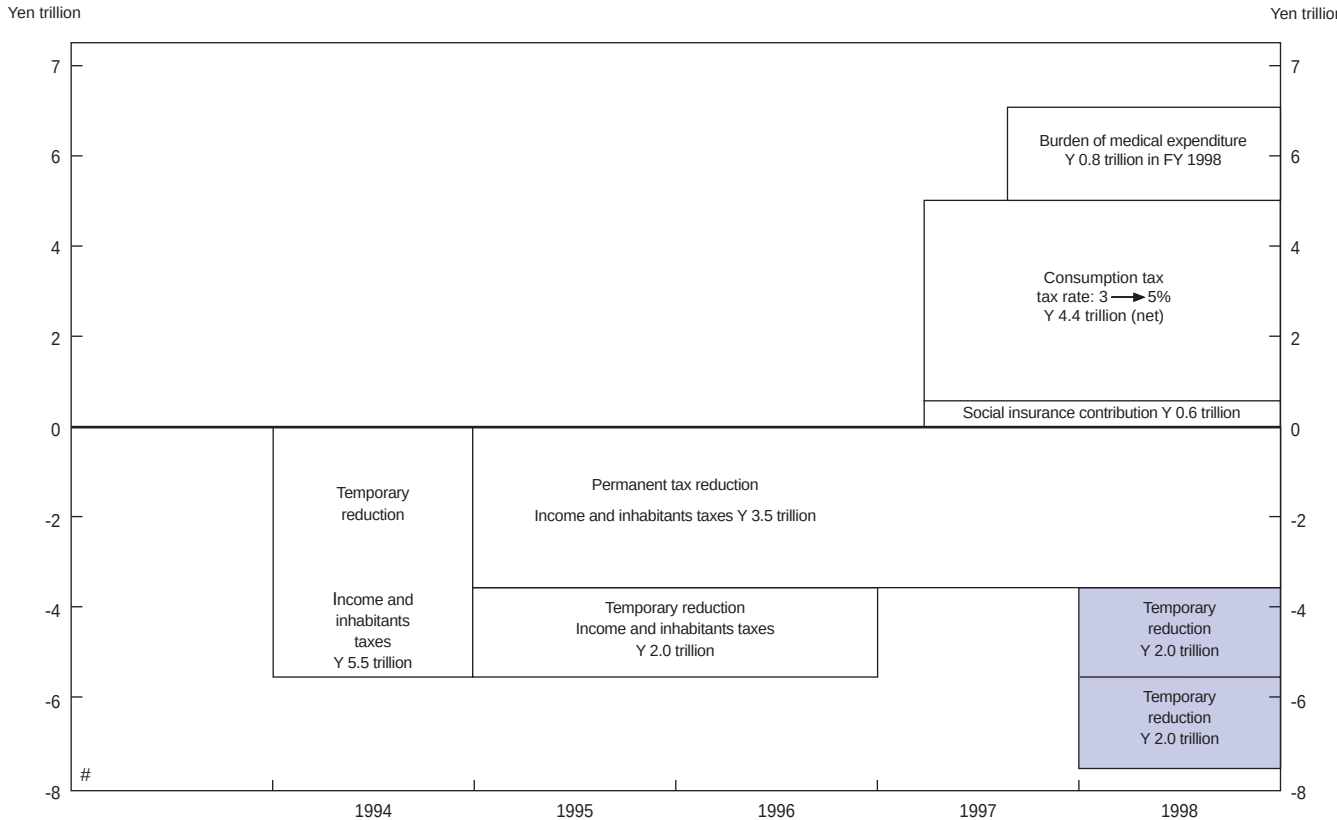
As a result of these various moves, the general government structural deficit increased by about 6 percentage points of potential GDP from 1991 to 1996 (Table 2). This implies that increasingly expansionary fiscal policy can, assuming a multiplier of one on changes in spending and taxes, account for most of the

Table 2. **Government policies to reflate the economy**

Calendar year	Real GDP growth Per cent	Cyclically adjusted net lending ratio to potential GDP Per cent	Fiscal package Date and announced amount of expenditures and tax cuts	Official discount rate Date of change and per cent
1991	3.8	1.8		August 1990: 6.0 July: 5.5
1992	1.0	0.8	March (without budgetary measures) August: Y 10.7 trillion	November: 5.0; December: 4.5 April: 3.75; July: 3.25
1993	0.3	–1.4	April: Y 13 trillion September: Y 6 trillion	February: 2.5; September: 1.75
1994	0.6	–1.7	February: Y 15.3 trillion	
1995	1.5	–2.8	April: Y 4.6 trillion	April: 1.0; September: 0.5
1996	3.9	–4.2	September: Y 14.2 trillion	
1997	0.8	–3.0	December temporary tax cut of Y 2 trillion	
1998	–2.6	–3.8	April: Y 16.6 trillion	

Source: Economic Planning Agency, Bank of Japan and OECD.

Figure 2. **TAX REFORM AND SUBSEQUENT MEASURES TAKEN**



Note: The shaded areas are not parts of the tax reform, but measures taken in order to boost the economy.
Source: Ministry of Finance.

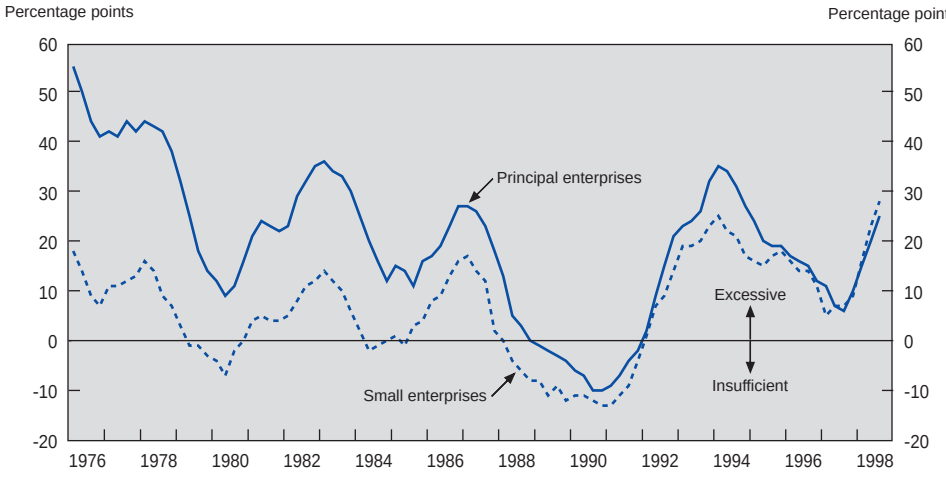
accumulated real GDP growth of 7 per cent during the same period, although Mundell-Fleming effects (in which net exports are squeezed by an induced currency appreciation) may well have restrained the foreign balance contribution. In other words, it appears that private-sector agents were unresponsive to the stimulus.[2] Discussed next are possible reasons for this sluggishness of private spending behaviour, other than the large negative wealth effects due to the asset price declines which are examined in detail further below.

Lack of dynanism in business fixed investment

One of the reasons for the sluggish business investment is the build up of a physical capital stock that was excessive in relation to requirements for the 1990s. This would have depressed the need for business fixed investment, and this spending did fall in the period 1992-94. The amount of excess capacity reported by *Tankan* respondents peaked in 1994 and fell in 1995, when the capital stock adjustment was considered to be almost at an end, judging by average historical outcomes (Figure 3). In fact, manufacturing industry increased

Figure 3. **JUDGEMENT ON PRODUCTION AND BUSINESS FACILITIES**

Manufacturing diffusion index



Note: Diffusion index = “excessive” minus “insufficient”.
Source: Bank of Japan.

investment in 1995 and 1996 (Table 3, Panel A). And this investment recovery provided a basis for judging that the economy was back on the path of self-sustaining expansion. But, the investment boom, largely confined to a few machinery sectors which responded to an increase in information-technology-related demand, was short-lived. At the same time, investment spending by firms in the construction and real-estate sectors continued to stagnate reflecting their deteriorated balance sheets, especially among small and medium-sized enterprises (SMEs) (Table 4).[3] These SMEs had increased land-related investment in the late 1980s,[4] financed by loans from banks which were in search of new clients as major companies became progressively less dependent on their loans.[5] The delayed adjustment of SMEs' balance sheets in non-manufacturing sectors partly explains the lack of dynamism in their investment in the 1990s more generally (Table 3, Panel B). On the other hand, it is the increased relocation of production to factories overseas which accounted for the atony of investment by small manufacturing firms.

Foreign direct investment to increase overseas production got an extra boost from the yen's appreciation until 1995 (Table 5, Panels A and B). While overseas production in North America and Europe has mostly been intended to substitute for exports from Japan, partly to reduce trade frictions, investments in Asian countries have usually been designed to exploit low production costs. As overseas production increased, the geographic structure of trade began to change (Table 5, Panel C): Japanese trade with other Asian countries has intensified, creating a new ''division of labour''. Japan has tended to shift low-technology production to other countries and concentrate on higher-technology production at home, exporting intermediate goods and importing finished products. At an aggregate level, the net impact of increased overseas production on the trade balance had been positive and rising until 1992, as it had led to greater exports of investment goods and component parts to Japanese subsidiaries abroad, which had more than offset the sum of the declines in exports from Japan due to substitution by overseas production and the increases in imports from foreign subsidiaries. These opposing forces were roughly offsetting between 1993 and 1995, but the net impact became distinctly negative in 1996 (Figure 4). Similar movements were observed for the net impact of increased overseas production on domestic production and employment, although it became clearly negative already in 1994 and substantial in 1996. More generally, a significant

Table 3. **Business investment by industry and firm size**

Growth rate, per cent

A. Business investment by industry (SNA base)

	All industry	*of which:* Construction	Manufacturing	Wholesale and retail	Real estate	Transport and telecommunication	Services
1990	11.1	19.7	11.6	10.3	21.0	17.9	10.3
1991	6.4	11.2	11.4	–0.9	–10.2	1.3	4.8
1992	–5.8	–1.7	–15.6	–6.8	4.3	4.3	0.1
1993	–10.4	–6.2	–18.3	–7.6	–6.0	–6.0	–13.5
1994	–5.5	–4.0	–9.2	–11.6	–33.3	9.0	–8.4
1995	4.7	–1.1	12.2	1.0	–6.3	5.8	7.2
1996	9.4	–12.2	9.1	3.2	16.5	9.1	25.0
1997	4.7	–10.7	8.5	3.6	–16.2	14.7	14.2

B. Business investment by firm size

	Manufacturing companies			Non-manufacturing companies		
	Large	Medium	Small	Large	Medium	Small
FY 1990	18.4	27.4	15.1	18.4	10.7	6.9
1991	3.8	9.3	21.2	14.1	12.9	8.4
1992	–20.4	–19.6	–21.2	–0.7	–4.5	–13.6
1993	–20.7	–24.0	–24.4	–9.0	–22.0	–14.5
1994	–9.9	–5.1	–6.0	–10.3	0.2	–2.9
1995	10.6	10.6	–4.1	–2.1	–7.5	–8.3
1996	7.3	8.7	2.2	4.1	–0.3	4.1
1997	5.7	–0.8	3.7	–6.0	–6.2	–7.9

Source: Economic Planning Agency and Bank of Japan.

Table 4. **Leverage[1] by industry and firm size**

Per cent

Fiscal year	Manufacturing		Construction		Real estate		Other non-manufacturing excluding construction and real estate	
	Large enterprises	Small and medium-sized enterprises	Large enterprises	Small and medium-sized enterprises	Large enterprises	Small and medium-sized enterprises	Large enterprises	Small and medium-sized enterprises
1985	29.2	56.8	20.3	54.5	62.6	77.4	85.2	56.0
1990	31.1	67.2	36.0	65.8	104.5	90.3	83.0	57.3
Peak[2]	(93) 38.2	–	(94) 55.2	–	(95) 277.7	–	(95) 108.1	–
1996	31.4	105.6	41.0	136.3	215.2	365.1	103.5	158.5

1. The sum of long-term loans and corporate bonds as a share of own capital.
2. The figures in parentheses show the year when the ratio peaked.

Source: Economic Planning Agency.

Table 5. **Indicators of globalisation**

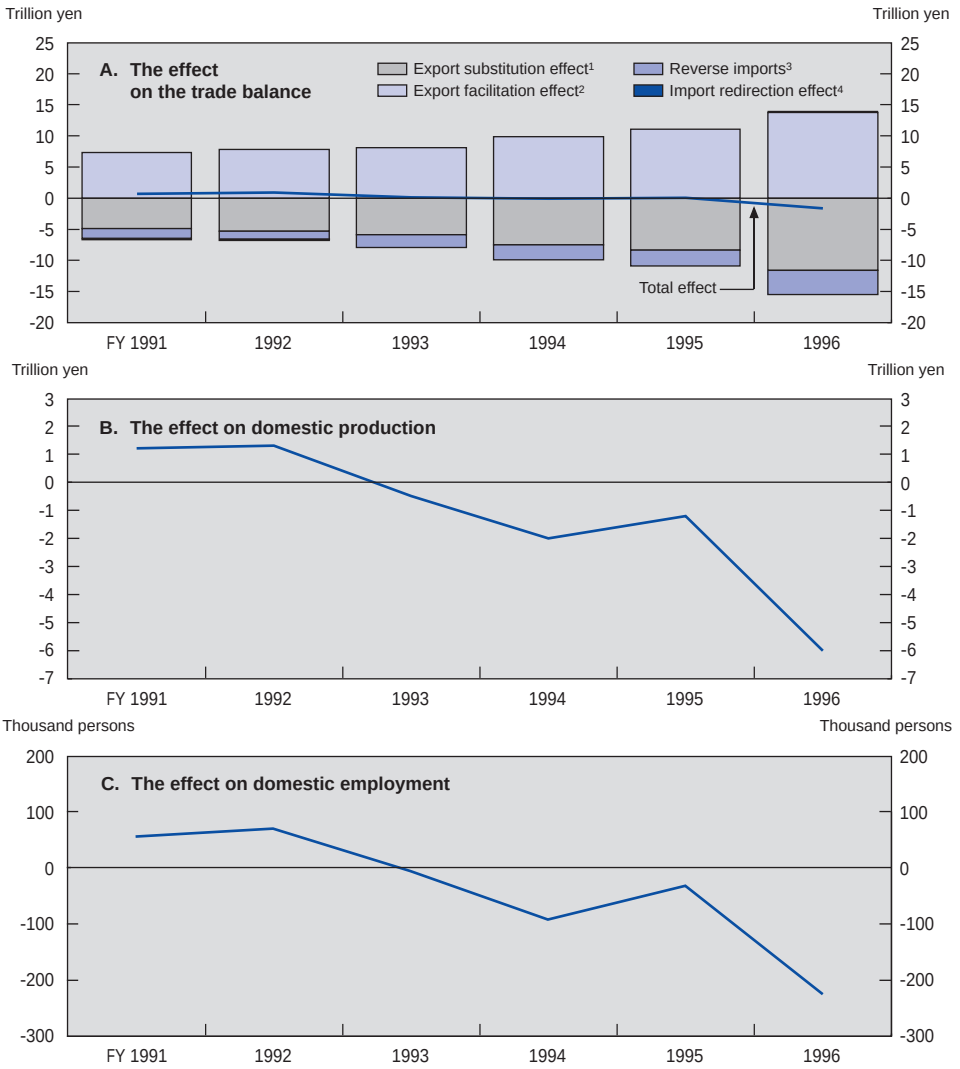
	1992	1993	1994	1995	1996	1997
	Yen billion, fiscal years					
A. Japanese direct investment abroad						
Total flows	4 431	4 151	4 281	4 957	5 409	6 623
Manufacturing	1 304	1 277	1 443	1 824	2 282	2 373
Electrical machinery	236	313	273	519	734	821
Transport machinery	156	110	214	194	436	357
Chemicals	258	204	272	208	232	370
Non-manufacturing	3 081	2 845	2 798	3 039	3 012	4 179
Finance and real estate	1 264	1 432	1 226	1 109	1 576	2 148
Services and commerce	1 332	1 004	1 176	1 550	995	1 332
Asia						
Manufacturing	400	418	540	781	747	898
Non-manufacturing	410	336	439	357	475	564
Other areas						
Manufacturing	904	859	903	1 042	1 535	1 475
	Percentage of domestic production					
B. Japanese manufacturers' overseas production[1]						
All manufacturers	6.2	7.4	8.6	9.0	11.6	13.0
Non-ferrous metals	7.8	6.5	8.8	6.7	11.1	–
General machinery	4.1	5.8	8.1	8.1	11.7	–
Electrical machinery	10.8	12.6	15.0	16.8	19.7	–
Transport machinery	17.5	17.3	20.3	20.6	24.9	–
Multinational manufacturers	17.3	18.3	22.0	24.5	27.8	31.1
Memorandum item:						
Employment in overseas plants						
Absolute number, millions	1.1	1.5	1.8	1.9	2.2	–
Per cent of domestic employment	7.2	10.8	12.0	12.8	15.4	–

	Exports Share, per cent		Imports Share, per cent	
	1991	1997	1991	1997
C. Changing Japanese trade structure				
United States	29.1	27.8	22.5	22.3
EU	18.1	15.6	13.4	13.3
Asia	37.0	44.5	43.5	48.4
China	2.7	5.2	6.0	12.4
ASEAN4	8.1	11.4	11.3	12.0
Korea	6.4	6.2	5.2	4.3
Reference total	Y 42.4 trillion	Y 50.9 trillion	Y 31.9 trillion	Y 41.0 trillion

1. 1997 figures are estimates.

Source: Ministry of Finance and Ministry of International Trade and Industry, *Basic Survey of Business Activities Abroad.*

Figure 4. **NET IMPACT OF RELOCATION OF PRODUCTION ABROAD**

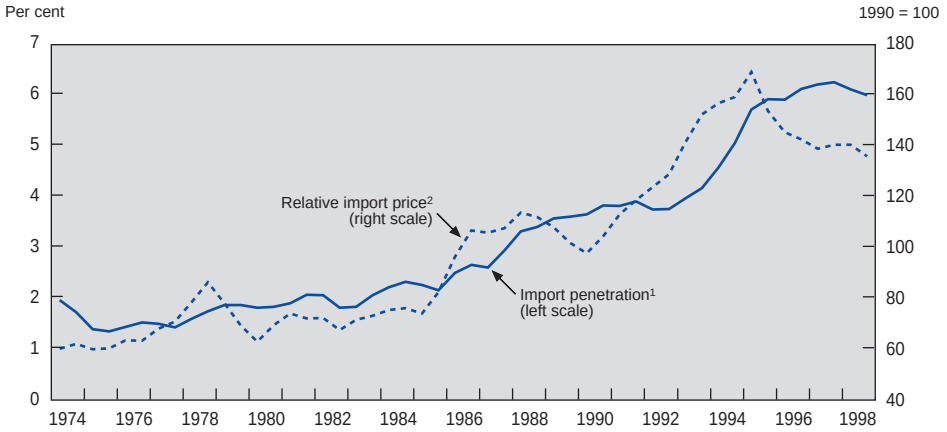


1. Substitution of exports from Japan by overseas production.
2. Increased exports of capital and intermediate goods from Japan to factories overseas.
3. Imports of products made in Japanese overseas factories.
4. Reduction of imports of materials to Japan due to overseas production (too small to be observed in the graph).

Source: Ministry of International Trade and Industry, *Basic Survey of Business Activities Abroad*.

increase in import penetration observed from 1993 to 1995 (Figure 5) was a drag on domestic production, although this influence has since been attenuated by a rise in relative import prices due to the yen's depreciation.

Figure 5. **IMPORT PENETRATION AND RELATIVE IMPORT PRICES**



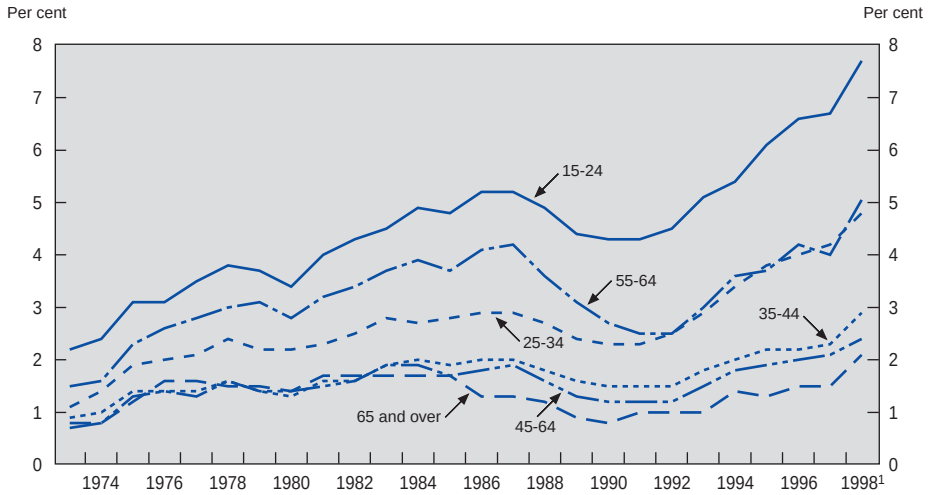
1. Import penetration is measured as the ratio of manufactured imports to total domestic demand.
2. The relative price of imports is measured by the ratio of the domestic demand deflator to the import price of manufactured products, 1990 = 100.

Source: OECD.

Sluggish household spending

The sluggishness in household consumption growth in the 1990s reflected not only the weak state of the economy overall, but also adverse wealth effects (see below) and factors such as the effects of the changes in labour markets and the ageing society.[6] Perhaps the most relevant aspect of the changes in labour markets in this regard is the significant rise in unemployment[7] which raised concerns about job security and reduced the prospensity to consume. The unemployment rate increased from 2.1 per cent in 1990 to 3.4 per cent in 1997. The evolution of the unemployment rate by age group shows a remarkable increase in joblessness among young and elderly workers (Figure 6), the former caused mainly by an increase in voluntary unemployment and the latter by corporate

Figure 6. **UNEMPLOYMENT BY AGE**



1. Unemployment rates for 1998 are through August.
Source: Management and Coordination Agency, *Labour Force Survey.*

restructuring. A study by the Economic Planning Agency (1997) shows that the propensity to consume is affected by the share of private-sector employees who have experienced either an employer-ordered job change, paid suspension of employment or encouragement to take voluntary retirement and acceptance thereof, in addition to wealth and other variables. Concerns about post-retirement incomes might also have discouraged spending, as people were beginning to understand the future financial difficulties of the pension system. According to a recent survey, more and more people, not only the elderly but younger people as well, now worry about their old-age incomes. Therefore, households might cut spending to provide for their retirement, regardless of their current age.[8]

Consumption data by household type may provide some further insight[9] (Table 6). First, non-employee households (those headed by the self-employed and retired) cut consumption significantly.[10] The decrease in consumption of the self-employed (‘‘individual proprietors’’) was due to more competitive pressures as a result of the yen’s appreciation and progress in deregulation. As for elderly households (‘‘non-occupation’’), the decrease was caused by adverse wealth

Table 6. **Consumption growth by household type**

Real consumption growth rate in per cent

	All households	Employee households						Non-employee households		
		Yearly income quintile groups[2]						Total	Individual proprietors	No occupations
		Total	I	II	III	IV	V			
1989-91[1]	1.0	1.1	1.0	0.8	1.9	1.2	0.7	1.2	0.8	2.6
1992-94[1]	–0.4	–0.4	0.5	–0.1	–0.6	–0.2	–0.8	–0.7	0.3	–0.8
1995-97[1]	–0.5	0.0	–1.1	–0.5	0.2	0.6	0.2	–1.1	–1.1	0.0

1. Average of three yearly growth rates.
2. Group I is the lowest income group and V is the highest.

Source: Management and Co-ordination Agency, *The Family Income and Expenditure Survey*.

effects and falling interest incomes which are a side-effect of the low-interest-rate policy. Second, employee households also decreased their spending, with higher income households cutting back first in 1992-94 reflecting negative wealth effects and then their lower income counterparts in 1995-97 due to stagnant labour-market conditions.

Similar factors also restrained residential investment. Eroded confidence seems to have had a significant impact, since the capital cost of housing investment fell as a result of lower land prices, construction costs and interest rates. In addition, the stock-adjustment mechanism depressed rental housing construction in particular, which continued to decline except for two years affected by temporary factors[11] (Table 7). Because rental housing is a kind of ''inferior good'' (relative to owner-occupied housing) in Japan, most rental units are occupied by younger people.[12] Expectations of a shrinking population of young people might have made the adjustment more severe. Owner-occupied housing construction might also have been affected by demographic factors: the population of house-buying age (taken here to be 34 to 43 years) fell by 19.1 per cent between 1990 and 1996. The weak development in housing starts caused the second-hand housing market to be much thinner, with an increasing concentration of all transactions on cheaper structures.

Table 7. **Housing starts by house type**

Growth rate, per cent

	Total	Owner occupied	Rental	Prefabricated	Employer-provided
1988	1.2	–5.1	0.0	18.8	7.1
1989	–1.9	–2.8	–4.8	6.4	21.3
1990	2.7	–3.5	–1.4	21.7	19.9
1991	–19.7	–9.6	–27.6	–19.8	19.5
1992	2.4	8.5	15.1	–28.7	–13.9
1993	5.9	11.2	–1.2	19.5	–11.7
1994	5.7	7.9	–10.2	44.0	–12.7
1995	–6.4	–6.2	–7.0	–5.6	–5.7
1996	11.8	19.7	12.4	–0.8	3.6
1997	–15.6	–25.6	–14.7	1.0	–12.5
1998[1]	–13.7	–13.5	–13.6	–13.6	–23.2

1. January to August.
Source: Ministry of Construction.

Front-loaded fiscal tightening in 1997 and the reaction of the economy

The measures implemented to set the course for fiscal structural reform

The expiration in December 1996 of the 2 trillion yen special tax cut implemented earlier and the consumption tax rate increase from 3 to 5 per cent in April 1997[13] implied substantial fiscal tightening in 1997, well before passage of the Fiscal Structural Reform Act in November of that year. In addition, there were two types of increase in social security charges. First, the contribution rates of various social insurance schemes were raised with implementation dates depending on the scheme. Second, the portion of medical treatment costs borne by patients was raised in September 1997. The effects of all these measures on households in FY 1997 were estimated to be 7.8 trillion yen in total (Box 1). Moreover, public investment fell in 1997 by 4.8 trillion yen in 1990 prices. This is because expenditure in 1996 was raised to some extent by implementation of the fiscal package in September 1995, and the initial FY 1997 budget cut such spending. Hence, once fiscal year figures are converted into calendar year figures, the total amount of fiscal tightening in 1997 was estimated to be about 10.8 trillion yen (just over 2 percentage points of GDP). The government kept this tight policy in place until December 1997: the Fiscal Structural Reform Act capped the increase in each spending category so as to set the central and local government budget deficit (on a national accounts basis) on a downward course toward the Act's goal of 3 per cent of GDP by the year 2003 (see Table 15 in Chapter II).

This fiscal tightening was motivated by the thinking that it was necessary to undertake budgetary consolidation straightaway because social-security-related expenditures were expected to increase further as rapid ageing proceeded. The ratio of people aged 65 years and over to those 20 to 64 years old was projected to rise from 0.23 in 1995 to 0.65 in 2050, according to the official medium projection, published in January 1997, which may turn out to be rather optimistic, considering the latest fertility outcomes (see below). The tax reform described above was an attempt to prepare for the ageing society. Poor economic performance and fiscal activism to boost the economy had worsened the fiscal situation: the gross debt of general government was about 96 per cent of nominal GDP in

Box 1. **Alternative fiscal policies in 1997**

The following measures were taken on the revenue side in 1997. *First*, as a part of tax reform starting in 1994, a special income tax cut was allowed to expire in December 1996. *Second*, the consumption tax rate was raised by two percentage points in April 1997. *Third*, contribution rates of various social insurance schemes were raised. *Fourth*, the portion of medical treatment costs paid by patients was increased in September 1997. The effects of these measures on households' disposable income are summarised below.

	Trillion yen
Rise in consumption tax rate from 3 to 5 per cent	4.4
Abolition of special tax cut	2.0
Rises in various social insurance contribution rates	0.6
An increase in medical expenditures paid by patients	0.8
Total	**7.8**

Source: Ministry of Finance and Economic Planning Agency.

In addition, public investment decreased in 1997 by 4.8 trillion yen in 1990 prices. Therefore, the total amount of fiscal tightening was estimated to be about 10.8 trillion yen in calendar year 1997. The OECD Secretariat's INTERLINK model has been used to examine what would have happened without any fiscal tightening in 1997.

Simulation results

Deviation from baseline in per cent

	1997	1998	1999
GDP level	1.8	2.2	2.3
Inflation (consumption deflator)	–0.8	0.2	0.4
Saving ratio	1.1	1.4	1.5
Current balance (% of GDP)	–0.4	–0.4	–0.4
Government financial balance (% GDP)	–1.4	–2.0	–1.9
Government debt (% of GDP)	0.8	2.4	4.2
Nominal short-term interest rate	0.0	0.0	0.0
Effective exchange rate	0.5	0.3	0.1

(continued on next page)

(continued)

The results suggest that under the alternative scenario the level of real GDP would have been higher by about 1¾ per cent in 1997 and 2¼ per cent in 1998 and 1999. The growth rate in 1997 would then have been 2.6 per cent rather than the actual 0.8 per cent and the fall in output this year – attributable to the crises in Asia and in the domestic financial sector – would have been somewhat smaller. It should be noted that a new version of the INTERLINK model was used for this simulation. The major difference from the previous model is the private consumption equation. The new consumption equation has been estimated as part of a pooled estimation covering 19 OECD countries. The ''standard'' country equation includes a new term in the change in the unemployment rate, as well as the traditional terms in real disposable income, real short-term interest rates and inflation. Japan's consumption is found to be far less sensitive to income than in the other countries and than in the previous model. The real interest rate and inflation effects are also smaller, although no different than those of the other countries in the estimation. As a result primarily of the higher income elasticity of consumption spending, if the old equation had been used, the results would have shown consumers spending more and saving less, thereby boosting real GDP by about 1 percentage point more in each year.

March 1997, just before implementing the increase in the consumption tax rate, according to official figures (see Chapter II). This encouraged the government to pursue fiscal consolidation as soon as the economy picked up.

The Secretariat has prepared an alternative scenario of neutral fiscal policy assuming there was no fiscal tightening, either on the revenue or expenditure side. Box 1 provides a summary of the simulation. Neutral fiscal policy in 1997 would have increased the level of real GDP by about 2 per cent in 1997 and 1998: the year 1997 would have been another good year, even if a recession had still got under way by year-end.

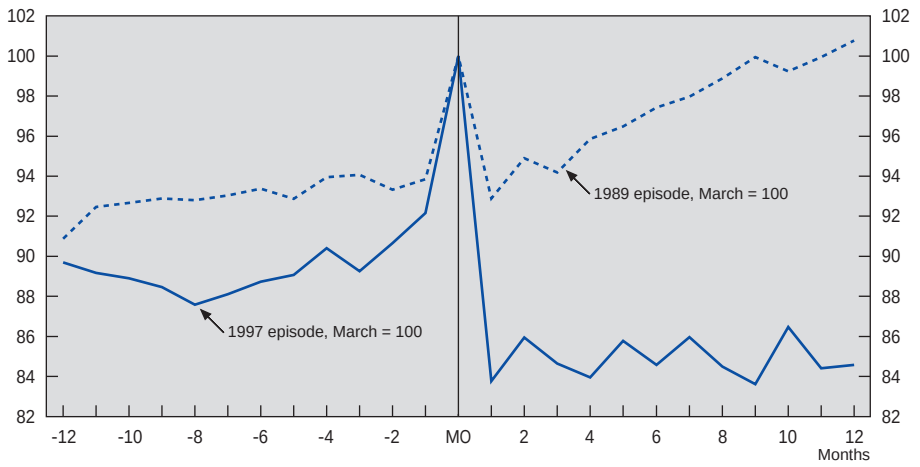
The economy's sudden turnaround

Although this fiscal tightening appeared to be justified by the relatively good growth performance in 1996, with real GDP expanding 3.9 per cent, the highest in the 1990s and the best outcome of any of the seven major OECD countries, policy makers and most other observers seem to have overestimated the underlying strength of the economic upturn, which had been affected by a shift forward in demand due to the consumption tax hike. The size of that shift was much

larger than what could have been anticipated based on experience with the original imposition of the tax in 1989 which involved an even larger tax change (Figure 7): retail sales, for example, fell about 16 per cent immediately following the change, to a level some 5 per cent below year-earlier levels, and never recovered, whereas in 1989 the decline had been less than half as large, and a steady rebound had ensued. Thus, for the year 1997 as a whole consumption stagnated, with year-on-year declines for most of the remainder of 1997 following the winter surge. Expenditures on durable goods were cut severely, although service expenditures increased slightly.[14] Despite gently declining lending rates, housing starts continued to decline throughout the year, with the year-on-year drop reaching more than 20 per cent in the fourth quarter[15] (Figure 8).

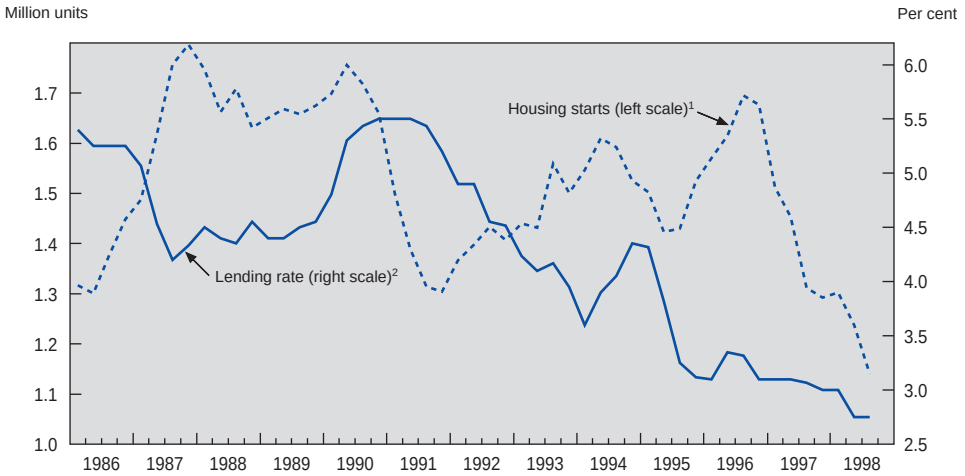
These weak household expenditures caused unexpected stockbuilding. Although business enterprises tried to reduce inventories, the manufacturers' inventories to shipments ratio rose steadily as from the second quarter of 1997 (Figure 9).[16] The transportation equipment industry suffered the most, with inventories up 11.6 per cent in 1997, hit by a sharp drop in domestic auto sales.

Figure 7. **RETAIL SALES AROUND THE TIME OF CONSUMPTION TAX INCREASES**[1]



1. Retail sales deflated by the consumer price index.
Source: OECD.

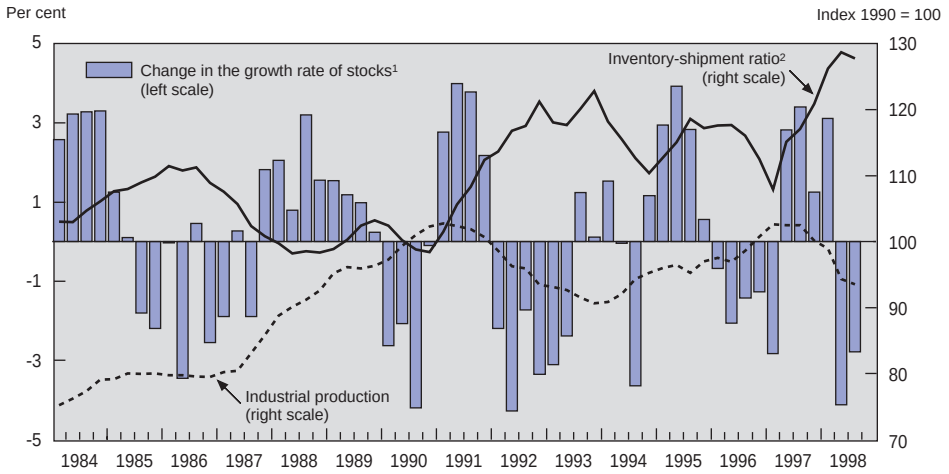
Figure 8. **HOUSING STARTS AND INTEREST RATES**



1. Seasonally adjusted at annual rate.
2. Housing Loan Corporation loans.

Source: Economic Planning Agency and Bank of Japan.

Figure 9. **STOCKS IN MANUFACTURING INDUSTRY**

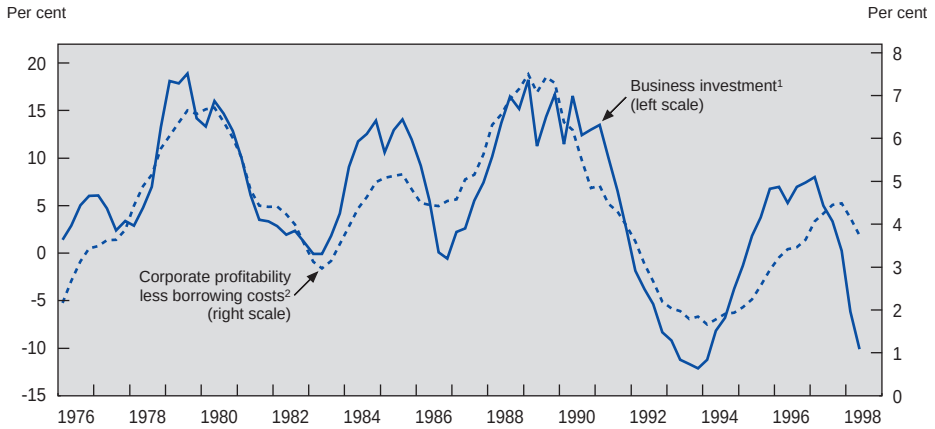


1. Change in the year-on-year growth rate of stocks of finished products in manufacturing industry.
2. The inventory-shipment ratio covers finished products of manufacturing industry.

Source: Ministry of International Trade and Industry.

To reduce the undesired inventories, industrial production was cut back. The adverse effects on corporate profits were felt gradually, especially in the second half of the year, although manufacturing still enjoyed buoyant overseas demand and benefited from the yen's depreciation compared to the previous year.[17] Decreased profits weakened investment growth, which had slowed down compared to the previous year (Figure 10).

Figure 10. **BUSINESS INVESTMENT AND CORPORATE PROFITABILITY**



1. Nominal investment, year-on-year increase.
2. Rate of return on physical assets minus long-term interest rates, 4-quarter moving average.

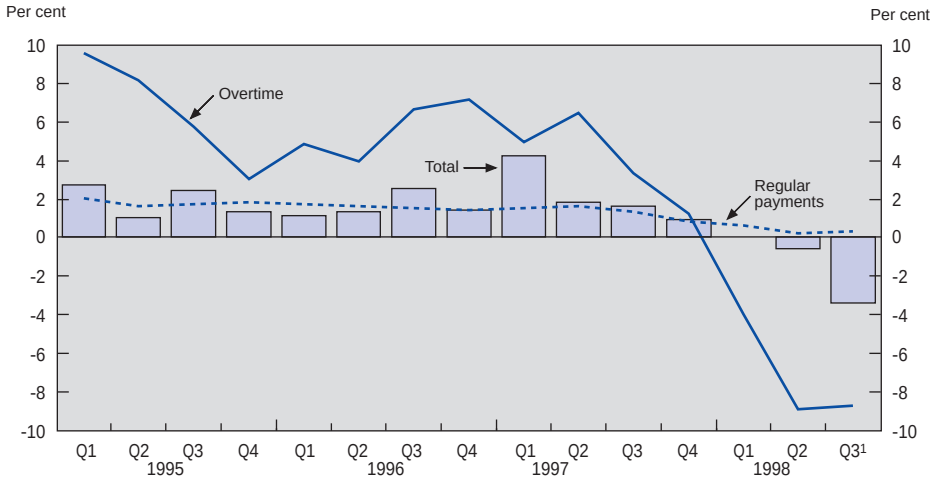
Source: Economic Planning Agency, *Report on National Accounts* and Ministry of Finance, *Financial Statements of Incorporated Businesses.*

Weak demand for goods and services began to affect employment in the third quarter, when employment in manufacturing industry started to fall year on year. The construction industry, which had played the role of absorber of employment earlier in the 1990s, followed suit in the next quarter, hit by decreases in residential as well as public investment. Involuntary unemployment started to increase in the fourth quarter (year-on-year base), with the unemployment rate reaching 3.5 per cent in the fourth quarter.

For a number of reasons, including corporate restructuring, the male labour force participation rate declined in the third quarter of 1997. However, resulting negative income effects encouraged female workers to enter the market, which partially offset the traditional ‘‘buffer’’ role of female workers. Wage growth slowed, but remained positive (Figure 11). In addition to the slowdown in growth of overtime and bonus payments, regular payments also stagnated, reflecting an increase in part-time workers.

Figure 11. **WAGE AND SALARY DEVELOPMENTS**

Per cent change over four quarters



1. The third quarter of 1998 is the July-August average.

Source: Ministry of Labour, *Monthly Labour Survey.*

A rare source of strength in 1997 was exports, which increased by more than 10 per cent, supported by strong growth in the US and EU economies, while weak domestic demand depressed imports. As a result, the current account surplus reached 2.3 per cent of GDP (Table 8).[18] The effects of the ‘‘Asian crisis’’ have nonetheless been felt since the third quarter. Growth of exports to the ASEAN4 (Thailand, Indonesia, Malaysia and the Philippines) and Korea

Table 8. **External trade**

	1994	1995	1996	1997	Seasonally adjusted annual rates 1997 1st half	1997 2nd half	1998 1st half
	Trillion yen						
A. Balance of payments							
Trade	14.7	12.3	9.1	12.3	11.0	13.6	16.1
Exports	39.3	40.3	43.6	49.5	49.0	50.0	49.7
Imports	24.6	27.9	34.5	37.2	38.0	36.4	33.6
Services	–4.9	–5.4	–6.8	–6.5	–6.7	–6.4	–6.2
Investment income	4.1	4.2	5.8	6.7	6.5	6.9	6.3
Transfers	–0.6	–0.7	–1.0	–1.1	–1.2	–0.9	–1.1
Current account	13.3	10.4	7.2	11.4	9.6	13.2	15.0
Per cent of GDP	2.8	2.2	1.4	2.3	1.9	2.6	3.0
	Percentage change from previous year						
B. Trade in goods (customs basis), 1995 = 100							
Volume							
Exports	1.5	3.2	1.2	11.8	12.2	11.4	2.1
United States	2.2	–6.2	–4.0	13.2	14.3	12.2	7.7
EU	–6.0	3.2	–1.0	18.1	14.2	22.0	18.4
Asia	10.2	9.9	1.7	6.4	9.5	3.5	–12.9
Imports	13.3	11.7	5.6	1.7	2.6	0.9	–4.7
Manufactures	19.9	23.7	8.3	3.1	5.2	1.1	–5.2
Unit value							
Exports	–0.8	–0.7	6.4	1.9	2.6	1.2	0.2
Imports	–7.6	0.5	14.0	6.0	10.2	2.0	–5.4

	1996	1997	1997 Q1	1997 Q2	1997 Q3	1997 Q4	1998 Q1	1998 Q2
	Year-on-year growth rate							
C. Effects of Asian crisis								
Export value								
World	7.7	13.9	12.2	18.1	13.3	12.2	3.9	0.7
Asia	8.9	8.6	8.0	14.5	9.1	3.5	–11.8	–18.0
ASEAN4	10.6	4.4	10.5	13.1	3.6	–8.2	–30.1	–35.9
Korea	9.0	–1.2	7.6	2.8	–7.8	–6.5	–38.5	–38.8
Import value								
World	20.4	7.8	18.8	7.6	5.1	0.8	–9.2	–10.6
Asia	23.0	6.6	16.5	9.0	4.2	–1.9	–9.0	–12.3
ASEAN4	25.9	8.2	23.0	13.3	3.7	–4.9	–15.7	–16.7
Korea	7.0	1.6	–0.9	4.9	4.4	–1.8	–4.5	–13.4

Source: Bank of Japan, *Balance of Payments Monthly*; Japan Tariff Association, *Summary Report of Trade of Japan* and OECD.

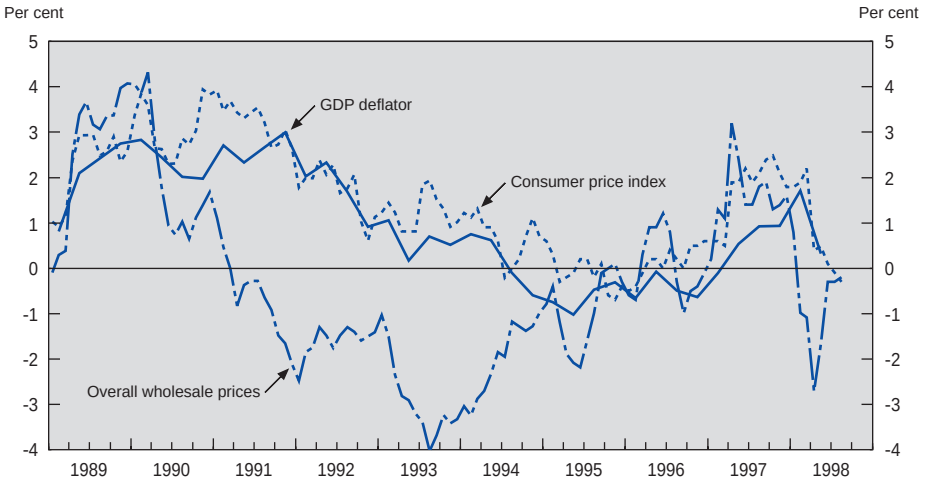
slowed in the third quarter and fell absolutely in the fourth quarter from year-earlier levels. Weak demand in these countries helped drive international commodity prices down and brought substantial terms-of-trade gains to Japan. However, rising import penetration, which could have been expected from the depreciation of Asian currencies against the yen, has not taken place.[19] In *Economic Outlook 62* of last December, the OECD Secretariat initially estimated that the ''Asian crisis'' would lower the real GDP growth rate of the total OECD by 0.3 per cent in 1997. Japan would be the non-crisis country most affected, with its real GDP growth rate reduced by 0.6 per cent and net exports as a percentage of GDP by 0.2 per cent. However, the overall effects on the economy ultimately seem to have been smaller in 1997, perhaps 0.4 to 0.5 per cent. The size of the negative shock on exports of goods and services is now estimated to have been about 0.15 percentage point of GDP based on the assumption that the growth rate of exports to ASEAN4 and Korea registered in the first semester of 1997 would have continued in the second half in the absence of the crisis. The effects on the economy will be larger in 1998: in *Economic Outlook 63* (June 1998), the Secretariat estimated the effect on Japanese real GDP growth rate to be –1.3 percentage points in 1998 (see Chapter II).

Reflecting excess supply in local markets, domestic prices remained weak, once adjusted for the effects of the April increase in the consumption tax rate and the September change in the costs of medical treatment (Figure 12),[20] although the deceleration was limited, indicating that economic slack was not yet substantial (see below).

The negative shock from the household sector resulting from the fiscal tightening thus propagated to the whole economy, beginning in the third quarter, when the ''Asian crisis'' started to affect the region. This development in turn eroded consumer and business confidence: the implied propensity to consume from the Household Survey fell continuously as from August.[21] The already fragile situation was then severely compromised by the collapse of several financial institutions in the autumn and by subsequent instability in the financial markets (see below). Consumer and business sentiment was damaged: the propensity to consume dropped below 70 per cent in December; investment plans were postponed; and safe assets such as cash and deposits provided by sound banks and the postal savings system were sought out by households and firms alike. In response to these adverse developments, the estimated output gap

Figure 12. **INFLATION INDICATORS**

Percentage changes from previous year



Source: OECD.

widened by 1.5 percentage points to 1.9 per cent in the second half of 1997. The government was eventually forced to reverse the fiscal tightening. In hindsight, it is obvious that fiscal tightening was too much for the moderate recovery to bear and left the economy open to adverse shocks, first from Asia and then from the domestic financial system.

The legacy of the asset price bubble of the 1980s

The magnitude of capital losses

The speculative build-up in asset prices which occurred toward the end of the 1980s was arguably the greatest such ‘‘bubble’’ in any OECD Member country in the post-war era. As such, it is by no means surprising that reversing it has caused so much distress and dislocation to the economy. Overall, the nation has had to confront cumulative capital losses of around one quadrillion yen (about $7 trillion) (Table 9) which represents some two full years’ worth of

Table 9. **Impact of bursting the asset price bubble**

Capital changes in Y trillion, fiscal years

	1990	1991	1992	1993	1994	1995	1996	Cumulative since 1989-90 peak
Total nation								
Gross assets	–37.7	–203.4	–446.7	–86.5	–21.8	–101.3	–69.9	–967.3
Of which:								
Equities	–306.7	–4.8	–178.1	4.1	48.4	0.0	–24.5	–461.6
Land	–	–195.5	–232.6	–83.1	–44.2	–49.5	–37.9	–642.8
Net worth	–58.9	–203.3	–435.5	–99.5	–28.2	–89.9	–61.5	–859.0
Non-financial corporations								
Gross assets	–20.4	–69.0	–155.9	–35.9	6.7	–29.9	–30.3	–334.7
Of which:								
Equities	–106.9	–3.9	–79.4	–0.0	21.0	0.9	–11.7	–180.0
Land	–	–62.6	–60.7	–30.1	–1.6	–9.8	–3.3	–168.1
Net worth	–22.8	–76.3	–152.9	–50.4	1.8	–32.3	–25.1	–312.6
Financial institutions								
Gross assets	–95.8	–13.8	–69.6	10.0	6.2	–12.5	–5.5	–181.0
Of which:								
Equities	–123.1	–2.9	–44.5	6.6	11.2	–1.0	–9.2	–144.5
Land	–	–5.8	–12.1	3.2	–1.9	–4.8	–2.8	–24.2
Net worth	–112.5	–7.8	–65.8	12.0	12.0	3.3	–10.4	–169.2
General government								
Gross assets	–	–11.5	–19.4	–15.1	–14.4	–28.1	–10.0	–98.5
Of which:								
Equities	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Land	–	–11.1	–13.9	–7.9	–6.7	–7.1	–7.1	–53.8
Net worth	–	–11.3	–19.4	–14.8	–15.0	–28.2	–10.1	98.8
Private non-profit institutions serving households								
Gross assets	–	–1.2	–2.3	–0.6	–0.2	–0.4	0.8	–3.9
Of which:								
Equities	–	0.0	0.0	–0.0	–0.0	–0.1	0.0	–0.1
Land	–	–2.4	–2.9	–1.2	–0.7	–0.8	–0.3	–8.3
Net worth	–	–1.2	–2.3	–0.6	–0.2	–0.4	0.8	–3.9
Households (including private unincorporated non-financial enterprises)								
Gross assets	–	–107.9	–199.6	–45.0	–20.1	–30.5	–24.8	–427.9
Of which:								
Equities	–76.8	2.0	–54.2	–2.5	16.1	0.2	–3.7	–118.9
Land	–	–110.2	–139.5	–43.6	–29.9	–26.6	–21.1	–370.9
Net worth	–	–106.7	–195.1	–45.7	–26.7	–32.4	–16.7	–423.3

Source: Economic Planning Agency, *Annual Report on National Accounts*, 1998 and OECD.

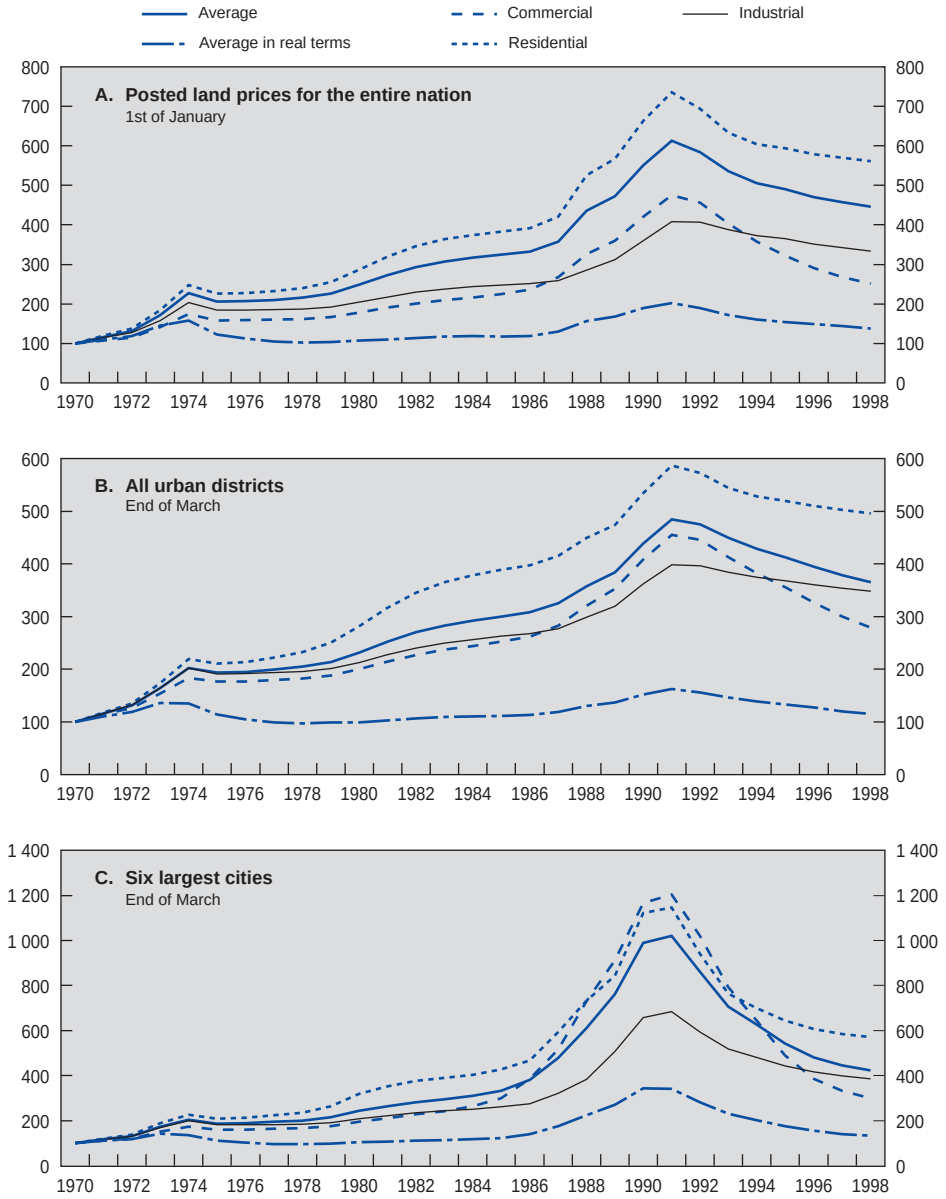
Japanese GDP and over 14 per cent of the value of the nation's total assets at the end of 1989. Around two-thirds of the total loss was seen in the value of real estate, whose value had fallen by around 27 per cent from the peak by March 1997. More than all the remainder occurred in equities markets, which plunged by just over half, while other assets in aggregate have risen in value. Little of the decline in asset values has been matched by corresponding falls in liabilities; these have dropped by only a little more than 100 trillion yen. Thus, net worth has been reduced in a largely corresponding way. Most of the correction was experienced earlier in the decade, thereby helping to bring about Japan's ninth post-war recession (by its own reckoning) in 1991-93. What is remarkable is that the economic recovery in 1994-97 was not sufficient to stabilise asset prices and bring to a close the persistent asset price deflation.

Despite some earlier hints of actual or imminent stabilisation, land and equities prices have continued to decline. After peaking in 1991 at around six times their 1970 levels, land prices have subsequently fallen by an average of around 4½ per cent per year (Figure 13, Panel A). The sharpest declines were experienced in the commercial sector, with industrial land prices falling only gently, albeit to lower levels in relation to the 1970 base year than for other land categories. For their part, residential real estate prices fell no more than those of other kinds of land but have cumulatively appreciated the most over the past few decades. A different source provides figures for all urban real estate prices which tell a similar story (Panel B). However, narrowing the sample to the six largest cities (Panel C) shows a much more pronounced bubble in these conurbations (with about double the amount of appreciation over 1970 levels). Commercial land prices, which had experienced the biggest bubble, have since fallen the furthest. While average prices earlier this year were approaching levels seen in 1986, commercial real estate values had fallen to 1985 levels, with cumulative declines of over 75 per cent since the 1991 peak. Other kinds of land have maintained much more of their value, with cumulative price reductions of 50 per cent for residential and 44 per cent for industrial land.

Equities prices have been more volatile. Prices plummeted in 1990 and again in 1992, but since then there has been no unique trend established for the shares of listed firms (Figure 14, Panel A). Trading volumes have remained a fraction of what was recorded in the bubble era, even during the most recent period of turbulence (Panel C). At their best in the intervening period, in the

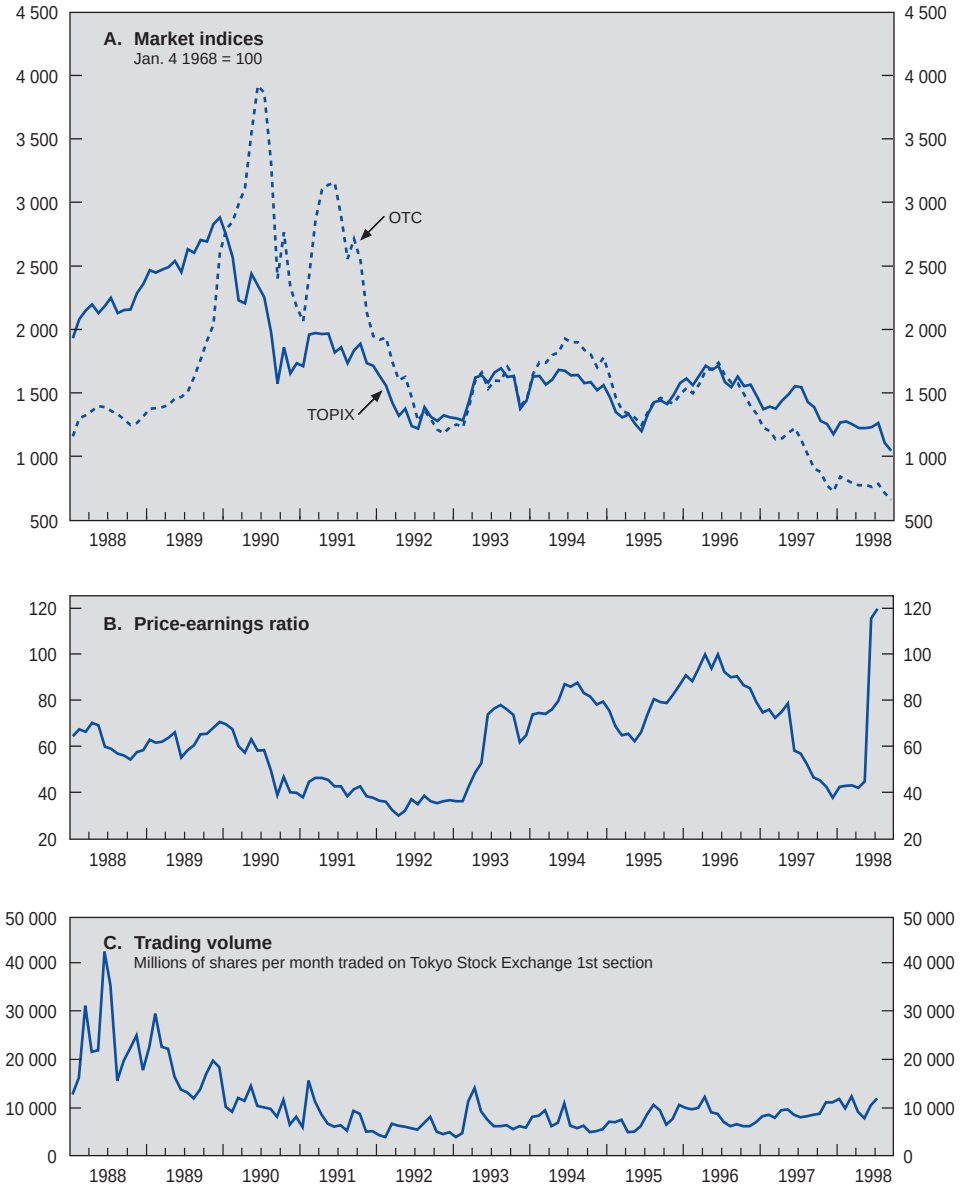
Figure 13. **LAND PRICES**

1970 = 100



Source: National Land Agency and Japan Real Estate Institute.

Figure 14. **STOCK MARKET INDICATORS**



Source: Bank of Japan and Nihon Keizai Shimbun.

spring of 1996 when the economy was showing signs of entering a self-sustaining recovery, equities prices had recovered more than a quarter of their earlier peak-to-trough decline. But since then the trend has been negative, especially in the second half of 1997 when prices dropped by around a quarter. Nevertheless, price-earnings ratios remain far higher than in almost any other nation's share markets (Panel B). The fall in share prices for smaller firms since the rebounds in 1994 and 1996 has been even more severe (Panel A). Unofficial indices of over-the-counter (OTC) equities prices show that the decline from the July 1990 peak to the December 1997 trough was 82 per cent, with the bulk of the fall (about seven-eighths) incurred by the end of 1992. Nonetheless, the decline steepened again in 1997, with a loss in value of over 40 per cent. Figures for the distribution of holdings of corporate equity by agents show that banks (15 per cent) and financial institutions other than insurance companies, pension funds and investment funds (15 per cent), as well as non-financial enterprises (27 per cent) are more significant shareholders than in other major OECD countries (Thompson, 1998).

Impact on households

The capital losses have had a number of obvious economic effects depending on the sector. Households have been the biggest losers, with reductions in net worth resulting from price declines of possibly one-sixth or some 450 trillion yen by 1998.[22] Most of this has been on their holdings of land. This works out to an average of about 20 per cent of annual disposable income. The resulting amount of negative equity for the owner of a typical 65 square metre apartment in Central Tokyo purchased at the peak would reach 12 to 15 million yen ($100 000), and a steadily rising number are at least six months in arrears on their mortgages. Their reactions have been several. First, they have probably boosted saving to try to restore some of this wealth, even if actual saving rates have edged down in order to buffer the simultaneous slowdown in disposable income growth. Using a wealth effect on consumption of some ¾ per cent[23] would imply that consumer spending has been restricted on average by around ½ trillion yen per year (around 0.2 percentage point). Second, they have become reluctant to invest in new housing, despite substantially improved affordability resulting from declining prices and interest rates, perhaps due to extrapolative price expectations (see above).

Third, they have become extremely risk averse in their portfolio allocation behaviour. By international standards, Japanese households have long held a relatively high proportion of their financial assets in the form of interest-bearing deposits (Figure 15, Panel A). Moreover, while elsewhere this proportion has declined over the past decade, in Japan it has risen to a share of about three and a half times the corresponding US one, for example. Similarly, the portfolio share held in the form of securities, which had been in line with the other major OECD countries in the bubble years, is now easily the lowest, at a mere 7.3 per cent by the end of 1997 (Panel B). The share held in the form of equities has plunged from as much as 13.9 per cent at the peak of the bubble to only 4.8 per cent at the end of 1997, less than one-quarter of the corresponding US figure. Instead, households have shifted their portfolios, as mentioned, towards bank deposits, currency and, most importantly, to insurance and pension funds, which are themselves merely an alternative form of deposit in a different kind of financial institution. These rose from 18.6 per cent of personal financial assets at end-1988 to 28.2 per cent by end-1997. It is likely that this portfolio behaviour has restrained the development of securities markets and prevented the realisation of efficiencies allowed by disintermediation in general.

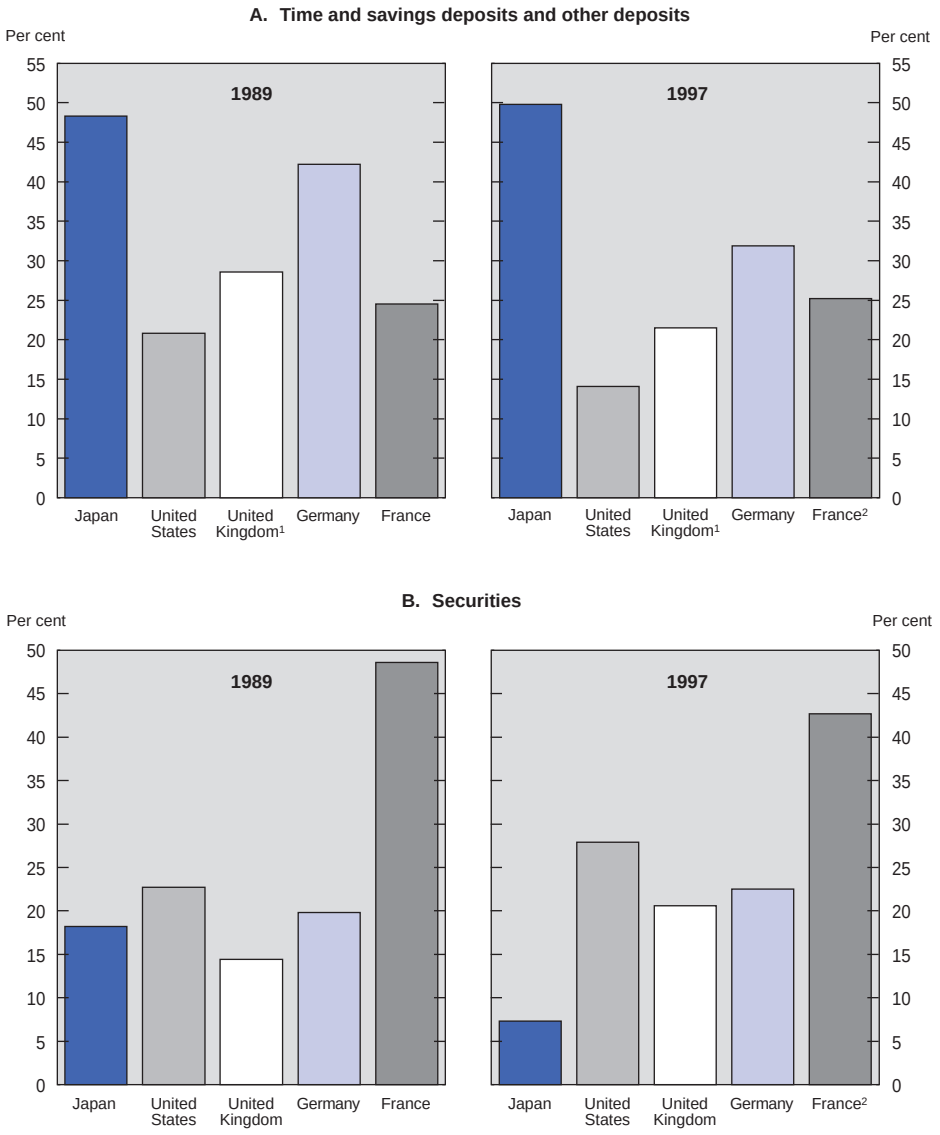
Finally, households' portfolio capital outflows, and those of other Japanese agents, may well have been influenced by the bursting of the bubble. Given the loss of wealth, especially early in the decade, less capital would likely flow abroad, raising the value of the yen in the short run, thereby cutting the external surplus and eventually lowering its value. This matches the pattern of developments in the 1990s.

Impact on non-financial corporations

Non-financial corporations have also suffered enormous losses on their asset holdings, about half each in land and equities.[24] The overall decline in their net worth is equivalent to 16 per cent of their total balance sheets at the outset. It is also nearly twice the total amount of current profits earned by all incorporated businesses with capital of at least 10 million yen over the intervening period (an average of about 30 trillion yen per year according to figures compiled by the Ministry of Finance). This deflationary pressure on business sector balance sheets has proved too much for many to bear, especially in an environment of slow growth (''stagnant sales''), and the number of bankruptcies has surged

Figure 15. **PERSONAL SECTOR FINANCIAL ASSETS**

Per cent share in total



1. Includes currency and demand deposits.
2. Data is for 1996.

Source: Bank of Japan.

(Figure 16, Panel A), although only back to levels which were commonplace in the years from the mid-1970s to the mid-1980s before the boom got under way. What is more novel about the firms going out of business in the 1990s is their average size: the total amount of liabilities of such firms in 1997 – about 14 trillion yen (2¾ per cent of GDP, compared with about 1¾ per cent in the United States in 1991, the peak value during the last recession) – was about six times the average seen in that earlier decade and double the amount incurred in the previous recession of the early 1990s. It has also left the quality of balance sheets impaired for a number of firms, even among the largest (Table 10): 20 per cent of firms followed by a reputable local ratings agency are termed to be of ''speculative standard'', up from 17 per cent at the end of the 1980s. Even exchange-listed firms – not only in the beleaguered financial sector – have been going out of business in the past year, a quite unusual development by Japanese standards and something some have alleged to be a result of a ''credit crunch'' (see below).[25]

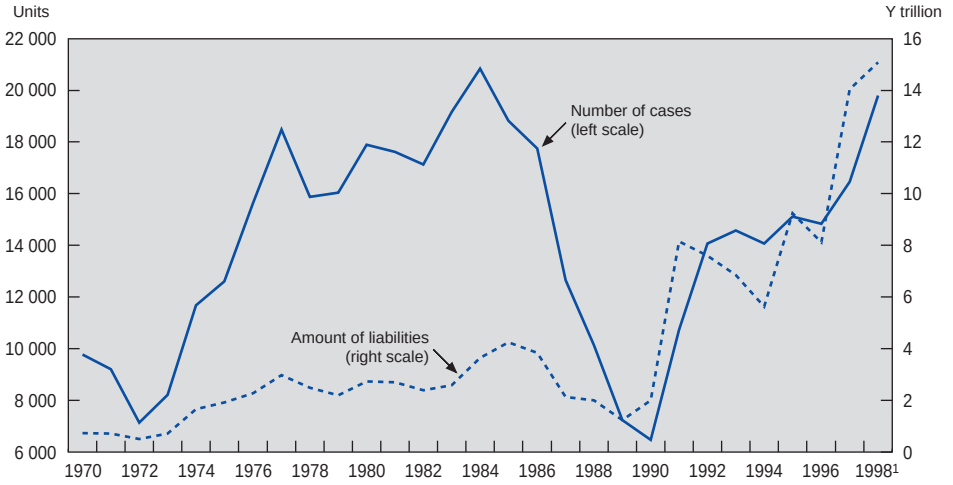
Corporations have also, like households, changed their financial behaviour, but, as was pointed out above, the process of cost-cutting disintermediation has been slow. Although intermediated borrowings declined in importance, they were still nearly half of all liabilities in 1997, about four times the comparable shares in the United States and United Kingdom, albeit still less than in Germany (Figure 17). The other side of the same coin shows that securities' share nearly doubled to 27 per cent, a proportion still less than half of the other five largest Member countries except Germany. A different data set for Japan alone and for a slightly different time period shows that what disintermediation there has been in the 1990s has been limited to the issuance of bonds, while stocks' and even commercial paper's shares have fallen markedly (Figure 18, Panel A). In flow terms, borrowing from private financial institutions has fallen steadily over the 1990s and has even become negative in recent years (see below) (Panel B). Foreign credit would seem to have replaced domestic institutional lending. Most recently, thus far in 1998, public financial institutions have compensated for some of the decline in lending by banks and other private financial institutions (see Chapter II).[26]

Impact on the general government

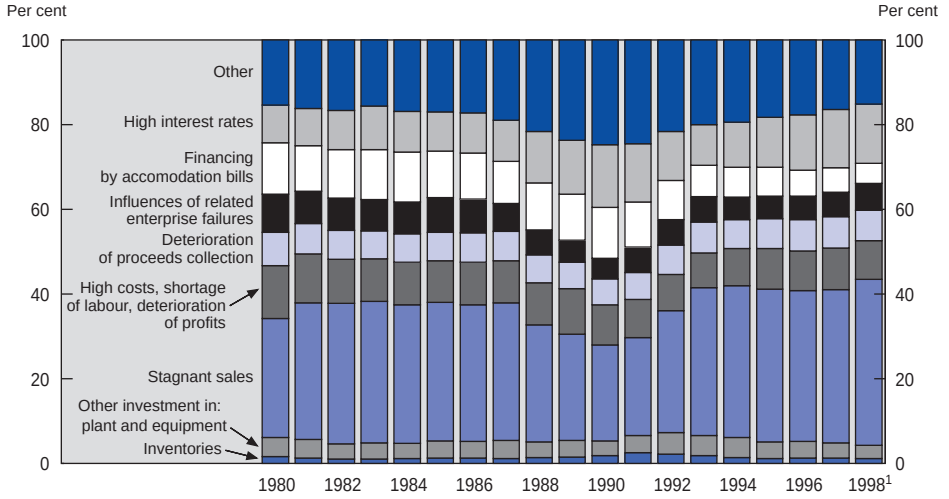
The general government balance sheet has also been harmed, but in this case the damage has been to a large extent indirect, as direct asset holdings are fairly

Figure 16. **CORPORATE BANKRUPTCIES**

A. Corporate bankruptcies



B. Suspension of business transactions with banks, by cause
Corporations capitalized at Y 1 million or more



1. Figures for 1998 are based on January to August outcomes.
Source: Tokyo Shoko Research Ltd. and Federation of Bankers Associations of Japan.

Table 10. **Senior debt ratings of Japanese issuers**

	March 1998							
	I	II	III	IV	V	VI	VII	Total
A. Banks								
Main banks	2	2	4	3	7	1	0	19
Regional banks	0	13	23	48	10	2	0	96
Total	2	15	27	51	17	3	0	115
B. Other issuers								
Exchange listed	6	36	87	176	433	149	38	925
OTC	0	0	0	16	115	27	1	159
Unlisted	0	0	1	1	3	2	2	9
Total	6	36	88	193	551	178	41	1 093
	December 1989							
	I	II	III	IV	V	VI	VII	Total
A. Banks								
Main banks	3	4	14	0	0	0	0	21
Regional banks	0	9	18	21	0	0	0	48
Total	3	13	32	21	0	0	0	69
B. Other issuers	6	46	103	186	561	170	16	1 088

Notes:

I. Highest standard of investment safety. Capacity, as evidenced by earning power, to pay interest and principal is extremely strong and is not likely to be affected by adverse economic conditions.

II. High standard. Capacity to pay interest and principal is strong but is subject to slight changes under adverse economic conditions.

III. Good standard. Capacity to pay interest and principal is strong but is susceptible to adverse economic conditions.

IV. Medium standard. Capacity to pay interest and principal is adeqate. It is, however, more susceptible than ''III'' to adverse economic conditions.

V. Lower medium standard. Capacity to pay interest and principal lacks basic stability, particularly under adverse economic conditions.

VI. Speculative standard. Capacity to pay interest and principal is considerably uncertain even under normal economic conditions.

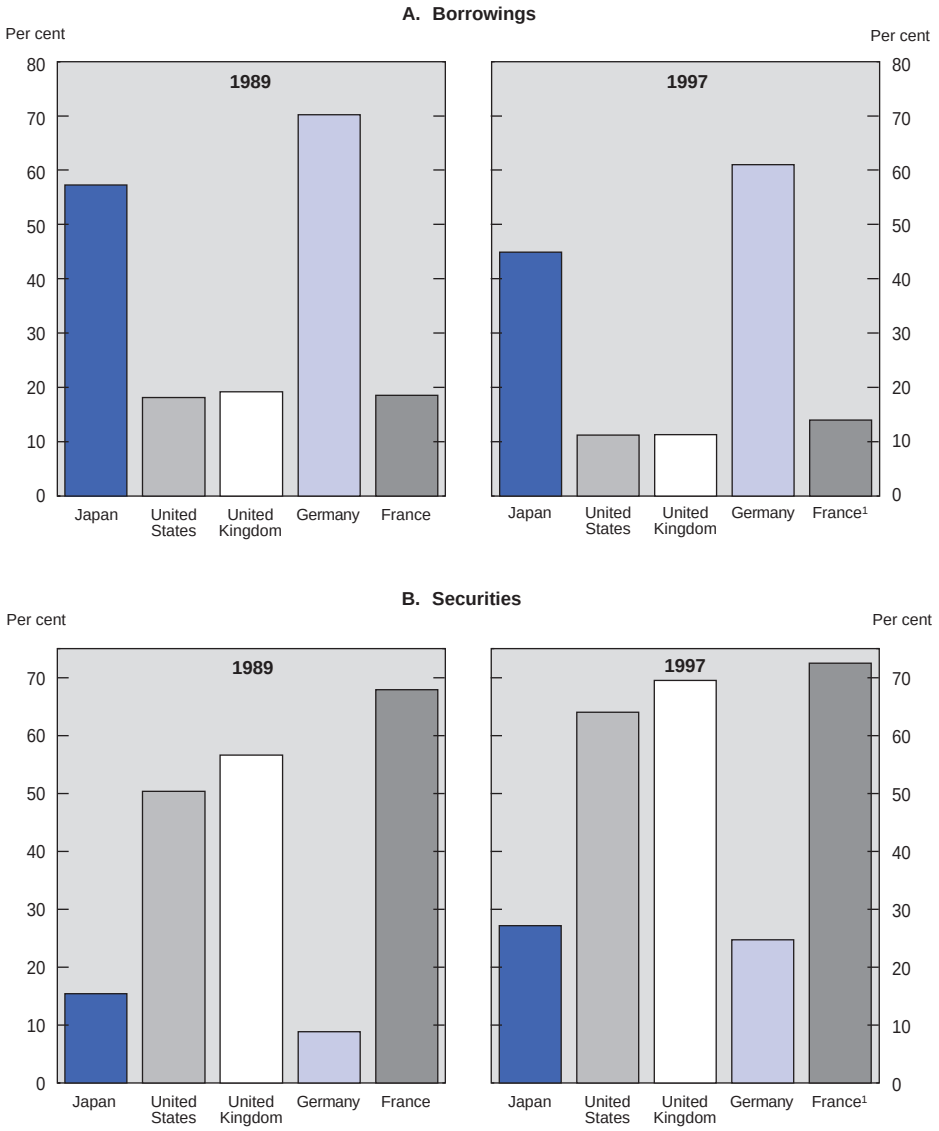
VII. Highly speculative standard. Capacity to pay interest and principal is inadequate.

Source: Mikuni and Company, Ltd.

limited.[27] Capital losses on land have cumulated to around 100 trillion yen, more than 3 per cent of GDP per year. But there is little doubt that the bursting of the bubble has been responsible for the need to provide the economy with the series of budgetary stimulus packages which have boosted the structural deficit over the last seven years by a cumulative 130 trillion yen (26 per cent of GDP) and left the nation with a bloated construction sector. (See Chapter IV).

Figure 17. **NON-FINANCIAL ENTERPRISES' FINANCIAL LIABILITIES**

Per cent share in total

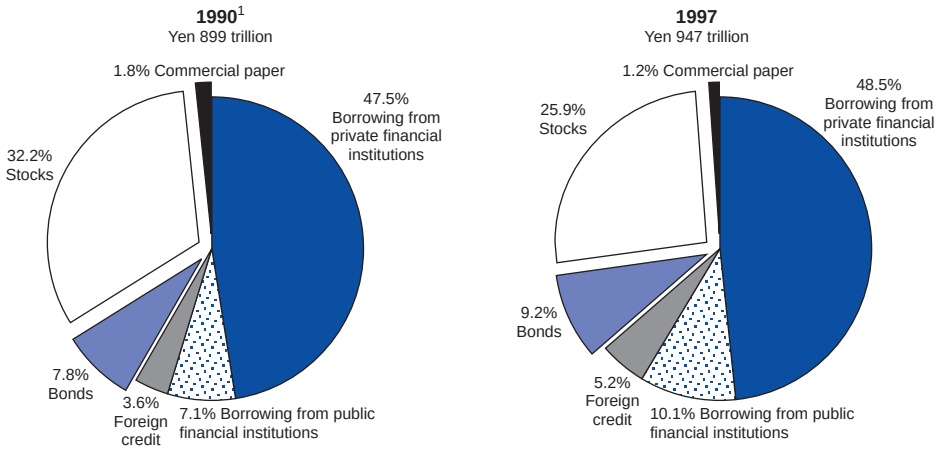


1. Data is for 1996.

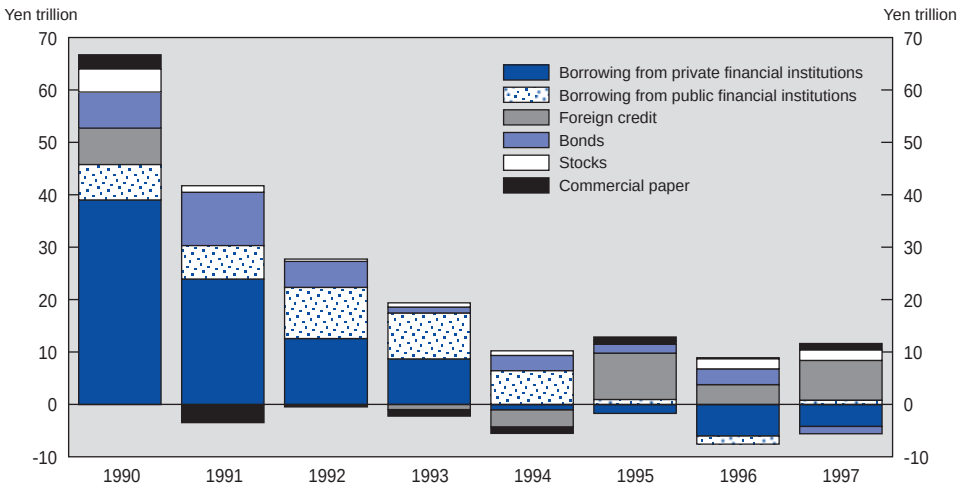
Source: Bank of Japan.

Figure 18. **SHIFTING SOURCES OF FINANCING FOR THE CORPORATE BUSINESS SECTOR**

A. Stocks of outstanding liabilities



B. Flows of funds raised



1. The stocks figure has been adjusted to make it consistent with the corresponding figure in 1997, since official data as from 1995 are on a market value (rather than book value) basis. To make the adjustment the ratio of market to book value at end-1990 was assumed to be equal to that ratio at end-95, modified by the percentage change in the Nikkei 225 index.

Source: Bank of Japan.

These and other problems in the financial system

But perhaps the most deleterious effects of the asset price deflation have been on the financial institutions, even though the direct impact on their balance sheets has only been about 20 per cent of the total, primarily through their holdings of equities. These institutions have traditionally held a combination of securities issued by their borrowing corporate customers. Substantial capital gains were earned on these holdings in the 1970s and 1980s, but the ''hidden reserves'' thereby established have since been largely wiped out, thereby squeezing their capital base:[28] *i)* by the steady erosion of market values; *ii)* by their rising book value resulting from the banks' willingness to realise some of the capital gains in order to meet capital adequacy requirements; and, *iii)* by the unwinding of these often interlocking shareholdings[29] and the diminishing role of the main bank in Japanese corporate governance more generally.[30] They have also suffered to the extent that the value of the collateral – mostly land – that they hold against their loans has dropped precipitously,[31] which, when combined with the failure to realise sales expectations formed in the bubble era, has helped to generate a bad-loans problem whose uncertain size has been plaguing analysts and policy-makers alike for much of the decade. This will be discussed at greater length in the following chapter.

Banks' financial positions in the 1990s have been adversely influenced by a number of factors other than the direct effects of domestic asset price deflation. First, the increasing level of competition brought about by the slow but steady process of financial liberalisation, when combined with market shrinkage and minimal exit, has brought about extremely narrow lending margins (see below). Second, even if the unanticipated decline in underlying inflation has provided a capital gain to all net creditors, including banks, it has harmed banks' customers, their real net worth in particular. This, along with their capital losses on their asset holdings, has thereby worsened the problems of adverse selection and moral hazard which are innate in banking markets (Mishkin, 1998). Further, these problems emanating from asymmetric information have also been amplified by the heightened uncertainty resulting from the asset price deflation, the 1991-93 and 1997-98 recessions and the financial failures which have multiplied in the recent past. The textbook reaction to this situation is that existing institutions should sort out the insolvent from the fundamentally healthy financial and non-financial firms through exit, *i.e.* bankruptcy, and the stage should then be set for

recovery. But in Japan's case the informational problems have not been overcome, as the regulatory authorities have, up until quite recently, generally maintained an attitude of regulatory forbearance in the extreme (often labelled the ''convoy'' system),[32] whereby insolvent institutions have either been kept afloat by a lack of disclosure (with obvious further implications for information flows, moral hazard and risk premia – see Chapter II), or have been merged forcibly with a healthier rival. In any case, banks had no incentive to develop any expertise in the areas of risk analysis and management with the official strategy of complacency. The results have been chronically poor returns, as well as a failure to restrain excessive risk-taking since the onset of the crisis (the much-discussed problem of moral hazard), and therefore an amplification of its duration and attitude. This institutional failure is attributable to a combination of the ''socialisation of risk'' (Mikuni, 1996) inherent in the convoy system, insufficient resources and expertise provided to the banking supervisory system (which has remained a problem to this day), as well as its discretionary and non-transparent nature, and, some argue, the practice of retiring public officials taking executive positions in the banks over which they have exercised their supervisory responsibilities.[33] With the recent regulatory and institutional reforms, described in Chapter III, which address these issues, the number of such appointments is expected to continue to decline.

The 1997-98 crisis in east Asia constituted another shock which has further harmed banks' financial positions.[34] The channels of influence are two-fold: first, the deflationary impact *via* the appreciation of the effective rate of the yen, the reduction in export market growth and the heightened uncertainty have all further accentuated the informational problems just discussed; second, the banks have been major lenders in the region (Table 11). While their ongoing problems have forced them to retrench their overall international lending positions in recent years,[35] little of that retrenchment has been in Asia (excluding Hong Kong, China and Singapore), whose share of Japanese banks' foreign lending rose from 26 per cent at end-1995 to 30 per cent at both end-1996 and end-1997. The aggregate amount of their loans to the crisis countries was some $100 billion or about 13 trillion yen. This represents about 2.6 per cent of all banks' total loan portfolio, but about 4.3 per cent of the loans outstanding by the 19 major banks and 96 per cent of their total capital. While it is not possible to know how many of these loans will ultimately have to be written off, especially since a substantial

Table 11. **International lending position of Japanese banks by geographic zone**

	End-1995			End-1996			End-1997		
	$ billions	Per cent of total bank lending	Per cent of Japanese bank international lending	$ billions	Per cent of total bank lending	Per cent of Japanese bank international lending	$ billions	Per cent of total bank lending	Per cent of Japanese bank international lending
Total	429.3	27.9	100.0	389.4	23.5	100.0	379.8	20.5	100.0
of which:									
Countries within BIS reporting area	165.6	19.0	38.6	169.7	17.1	43.6	163.4	14.6	43.0
Developing countries	132.7	21.5	30.9	140.3	19.8	36.0	135.6	17.5	35.7
of which:									
Asia	112.9	36.8	26.3	118.6	32.3	30.5	114.7	30.1	30.2
of which:									
China	17.6	36.5	4.1	17.8	32.3	4.6	19.6	31.0	5.2
Indonesia	21.0	47.1	4.9	22.0	39.7	5.7	22.0	37.7	5.8
Korea	21.5	27.7	5.0	24.3	24.3	6.2	20.3	21.5	5.3
Malaysia	7.3	43.6	1.7	8.2	36.9	2.1	8.6	31.1	2.3
Thailand	36.8	58.7	8.6	37.5	53.5	9.6	33.2	56.4	8.7
Other	8.7	15.3	2.0	8.7	7.3	2.2	11.0	13.9	2.9
Latin America	14.3	6.8	3.3	15.4	6.4	4.0	14.7	5.2	3.9
Middle East	3.6	7.0	0.8	2.8	5.8	0.7	3.4	6.5	0.9
Africa	1.8	5.7	0.4	3.5	6.8	0.9	2.8	4.8	0.7
Eastern Europe	4.7	5.2	1.1	4.0	3.9	1.0	4.2	3.4	1.1
Developed countries	28.2	17.8	6.6	25.5	14.9	6.5	23.6	12.1	6.2
Offshore banking centres	263.7	39.6	61.4	219.7	33.1	56.4	216.4	29.6	57.0
of which:									
Cayman Islands	26.6	21.7	6.2	43.4	30.1	11.1	52.2	28.1	13.7
Hong Kong, China	133.1	55.3	31.0	87.5	42.2	22.5	76.3	36.0	20.1
Singapore	76.9	40.0	17.9	58.8	31.1	15.1	58.6	30.1	15.4
Panama	13.6	48.9	3.2	14.8	52.0	4.0	15.7	51.0	4.1

Source: Bank for International Settlements and OECD.

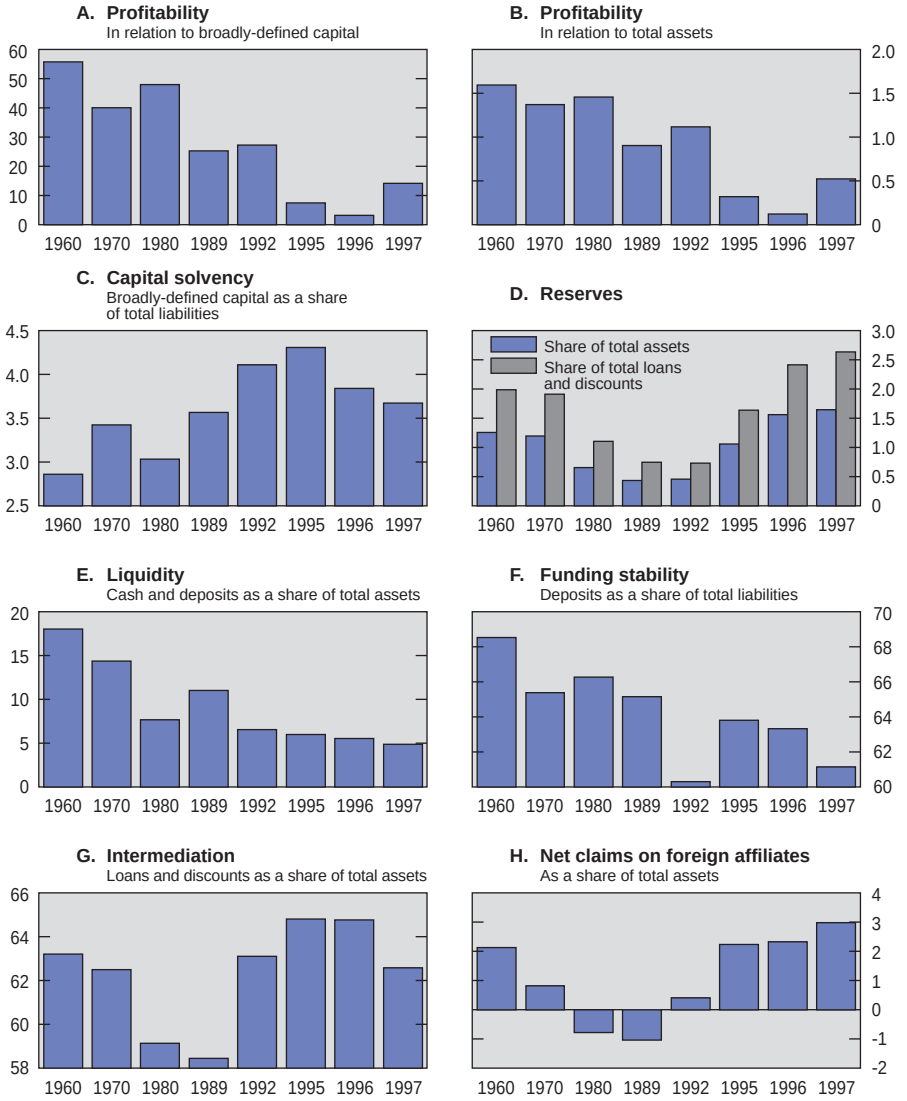
proportion are said to be to foreign affiliates of Japanese firms, it seems fair to say that the share is likely to be considerable and that it may accentuate the need to boost loan loss reserves in the near future.

The outcome of these various influences on the Japanese banking system has been generalised weakness, a situation where even the strongest banks are saddled with substantial amounts of non-performing loans and with a chronic shortage of capital resulting from an equally chronic low return on assets which renders them largely incapable of taking over even the healthy parts of the weakest banks. Core business profitability in relation to both broadly-defined capital and assets has been trending down, with the latter figure no better than 0.5 per cent in recent years (Figure 19, Panels A and B). This has contributed to a shift in position with regard to foreign affiliates: whereas they were a source of funds in the 1980s, they have increasingly been the recipients of funds in the 1990s (Panel H), especially during November and December 1997 when capital outflows of this nature reached 8.4 trillion yen ($65 billion). The earlier trend to improved capital solvency has been reversed (Panel C), even if BIS capital-adequacy ratios have been maintained. Balance sheet liquidity has been falling (Panel E), as has funding stability, as proxied by the share of deposits in total liabilities (Panel F). Massive provisioning has taken place. By end-1997, reserves of over 2½ per cent of total loans and discounts had been set up; by March 1998, the major banks' loan loss reserves reached 3.9 per cent of total loans. After such provisioning and other extraordinary items, pre-tax income of the major banks turned slightly negative already in FY 1996, before plunging to –4.6 trillion yen in the most recent financial year (Table 12).

This weakness has not escaped the market's nor ultimately the public's notice. Whereas traditionally the ''convoy'' system implied that bank liabilities had no more risk attached to them than government debt and depositors had little reason to monitor the condition of the deposit-taking institution, periodic and increasingly frequent failures of both banks and other major financial institutions (Table 13) have laid to rest the idea that credit risk in Japan is non-existent. In October 1997 a Kyoto-based regional bank closed, with bad loans exceeding capital by 15 billion yen. But the highest profile cases all came in November 1997: first that of a moderate-sized brokerage house (Sanyo Securities), then a major bank (Hokkaido Takushoku) and finally one of the nation's ''Big Four'' securities firms (Yamaichi). The effect on global market perceptions

Figure 19. **SOME BANKING SECTOR INDICATORS**[1]

Per cent



Note: Care should be taken in interpreting trends in some measures as definitions are not constant.

1. Domestically licensed banks. Prior to 1989 excludes regional banks which are members of the Second Association of Regional Banks.

Source: OECD calculations based on Bank of Japan, *Economic Statistics Annual,* Table 13.

Table 12. **Recent financial outcomes of the banks**

Y billion, except as noted
Fiscal years

	1995		1996		1997		Sum 1992-97[2] All banks
	Major banks[1]	All banks	Major banks[1]	All banks	Major banks[1]	All banks	
Operating profits	4 771	8 410	4 564	8 220	3 568		29 345
Current profits							
Before tax	3 283		–162		–4 554		n.a.
After tax	3 565		–146		–3 878		n.a.
Net unrealised capital gains on equities	16 523	24 165	8 753	14 028	2 246		n.m.
Gains from sales of equities	3 420		3 489		4 773		n.m.
Loss on disposal of non-performing loans	11 087	14 077	6 218	9 024	10 740		45 021
Non-performing loans – new standards	–	–	–	–	21 981		n.m.
As a percentage of total loans	–	–	–	–	6.11		n.m.
Old standard	21 868	34 799	16 441	27 900	15 830		
As a percentage of total loans	5.58	4.48	4.16	3.91	4.40		
Loan loss reserves	10 345	15 053	9 388	14 989	14 116		n.m.
As a percentage of non-performing loans	(47.3)	(43.2)	(57.1)	(53.7)	64.2		n.m.
Capital-asset ratio (per cent)	9.10		8.95		10.24		n.m.

1. City banks, long-term credit banks and trust banks. Results of the Hokkaido Takushoku Bank excluded as from 1997.
2. Figures for 1997 are the cumulative sum for 1996 plus the 1997 outcome for the major banks.

Source: Ministry of Finance and Federation of Bankers Associations of Japan.

Table 13. **Recent collapses of major financial institutions in Japan**

Name of company	Date of collapse	Proceedings of liquidation	Amount of capital just before collapse (in yen)	Excess liability covered by deposit insurance (in yen)
Yamaichi Securities	1 December 1997 (cessation of business)	Full winding-up of operations. Some employees and branches taken over by Merrill Lynch	n.a.	(22.5 billion and rising)
Tokuyo City Bank	25 November 1997	Operations to be transferred to Sendai Bank; part of assets and deposits to 77 Bank and other banks	n.a.	Not disclosed
Hokkaido Takushoku Bank	17 November 1997 (release of liquidation plan)	Operations will be transferred to North Pacific (Hokuyo) Bank and Chuo Trust Bank. Branches in Hanshu will be sold to other banks. Bad loans will be transferred to Resolution and Collection Bank. Closure 16 November 1998	6.78 trillion	Around 0.84 trillion in October 1997 and 1.17 trillion in March 1998
Sanyo Securities	2 November 1997	Company to be liquidated	n.a.	(106 billion)
Kyoto Kyoei Bank	14 October 1997 (release of liquidation plan)	Operations will be transferred to Kofuku Bank and bad loans will be transferred to Deposit Insurance Corp.	341 billion	15 billion estimate
Nissan Mutual Life Insurance	April 1997	Ordered to stop operations. Operations taken over by newly formed Aoba Life Insurance	n.a.	n.a.
Kizu Credit Union	24 February 1997	Operations were transferred to Resolution and Collection Bank	1.17 trillion	1.034 trillion
Osaka Credit Union	20 January 1997	Operations were transferred to Tokai Bank. Bad loans were transferred to Resolution and Collection Bank	333.2 billion	169.7 billion
Hanwa Bank	21 November 1996 (ordered to stop operations)	Operations will be transferred to newly formed Kii Yokin Kanri Bank; assets will be bought by Deposit Insurance Corp.	507.4 billion	77 billion estimate
Taiheiyo Bank	17 September 1996	Operations will be transferred to newly formed Wakashio Bank	636.6 billion	117 billion

Table 13. **Recent collapses of major financial institutions in Japan** *(cont.)*

Name of company	Date of collapse	Proceedings of liquidation	Amount of capital just before collapse (in yen)	Excess liability covered by deposit insurance (in yen)
Cosmo Credit Co-operative	25 March 1996	Operations were transferred to Tokyo Kyodou Bank (now Resolution and Collection Bank). Bad loans were transferred to Tokyo Society of Credit Cooperatives	439.3 billion	125 billion
Hyogo Bank	29 January 1996	Operations were transferred to newly formed Midori Bank	2.53 trillion	473 billion
Tokyo Kyowa Credit Association and Anzen Credit Bank	20 March 1995	Operations were transferred to Tokyo Kyodou Bank (now Resolution and Collection Bank). Bad loans were transferred to Tokyo Society of Credit Cooperatives	121.4 billion (Tokyo Kyowa) 122.7 billion (Anzen)	40 billion

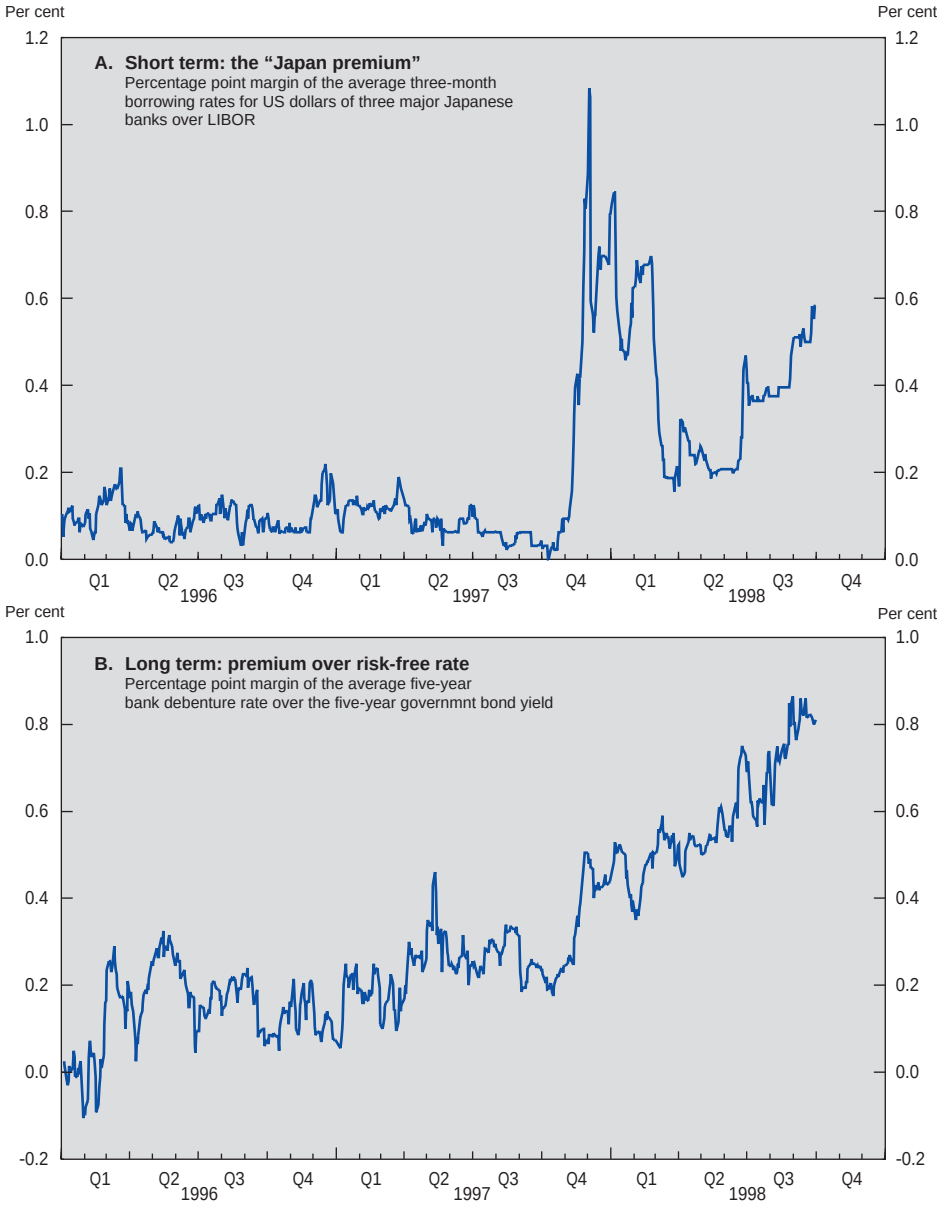
Source: *The Banker*, January 1998, p. 44, and OECD.

of risk was immediate and dramatic, with the ''Japan premium'' (the extra amount that Japanese borrowers – here the banks – have to pay over the market rate for short-term loans) moving up from the 10 basis points that it had been averaging over the previous year to as much as 100 basis points in the wake of these failures (Figure 20, Panel A). A similar effect has been visible in the market for long-term funds. The amount Japan's banks have to pay over government borrowing costs experienced a rising trend to around 25 basis points before the crisis, but it then doubled to over 50 basis points thereafter and reached around 75 basis points in the summer of 1998 when the outlook for banks' finances deteriorated further (Panel B). As disturbing as this temporary cost disadvantage has been the fact that although the short-term premium has sometimes fallen back, it remains persistently at least 20 basis points (double its pre-November level), and the long-term premium has stubbornly refused to come down at all. Should these premia endure, the impact on profitability in the sector would be dramatic, since, as mentioned above, margins in the industry were already paper thin beforehand. Finally, depositors began to shift their financial assets away from those institutions deemed the most vulnerable to their stronger rivals – often foreign-owned, and from deposits to currency.[36]

Another pressure on banks' capital requirements over the past few years has been the impact of the yen's depreciation on the size of their balance sheets. This has been estimated to be worth 1 trillion yen in extra assets for each additional yen rise in the exchange rate against the dollar. Thus, the decline in the yen from 80 to 130 in the three years to FY 1997 would have caused the sector's balance sheet to balloon by 50 trillion yen. The authorities have recently estimated that a ten-yen rise in the value of the dollar cuts the industry's capital adequacy ratio by 0.2 percentage points, or about the same as 1.6 trillion yen in Tier II capital.

Faced with the need to evaluate credit risks more strictly and the prospect of Prompt Corrective Action being imposed as from 1 April 1998 if capital adequacy ratios were not satisfied, the banks had strong incentives to try to boost their capital and reduce their assets, their loans in particular. Accordingly, several issued substantial amounts of subordinated securities (see Chapter II).[37] And there has indeed been a reduction in loans outstanding as from late in 1996, a development that has become known as a ''credit crunch''. The evidence of a shift in credit *supply* is, however, mixed. That such a shift has been perceived by actual

Figure 20. **INCREASED CREDIT RISK PREMIA IN THE BANKING SYSTEM**



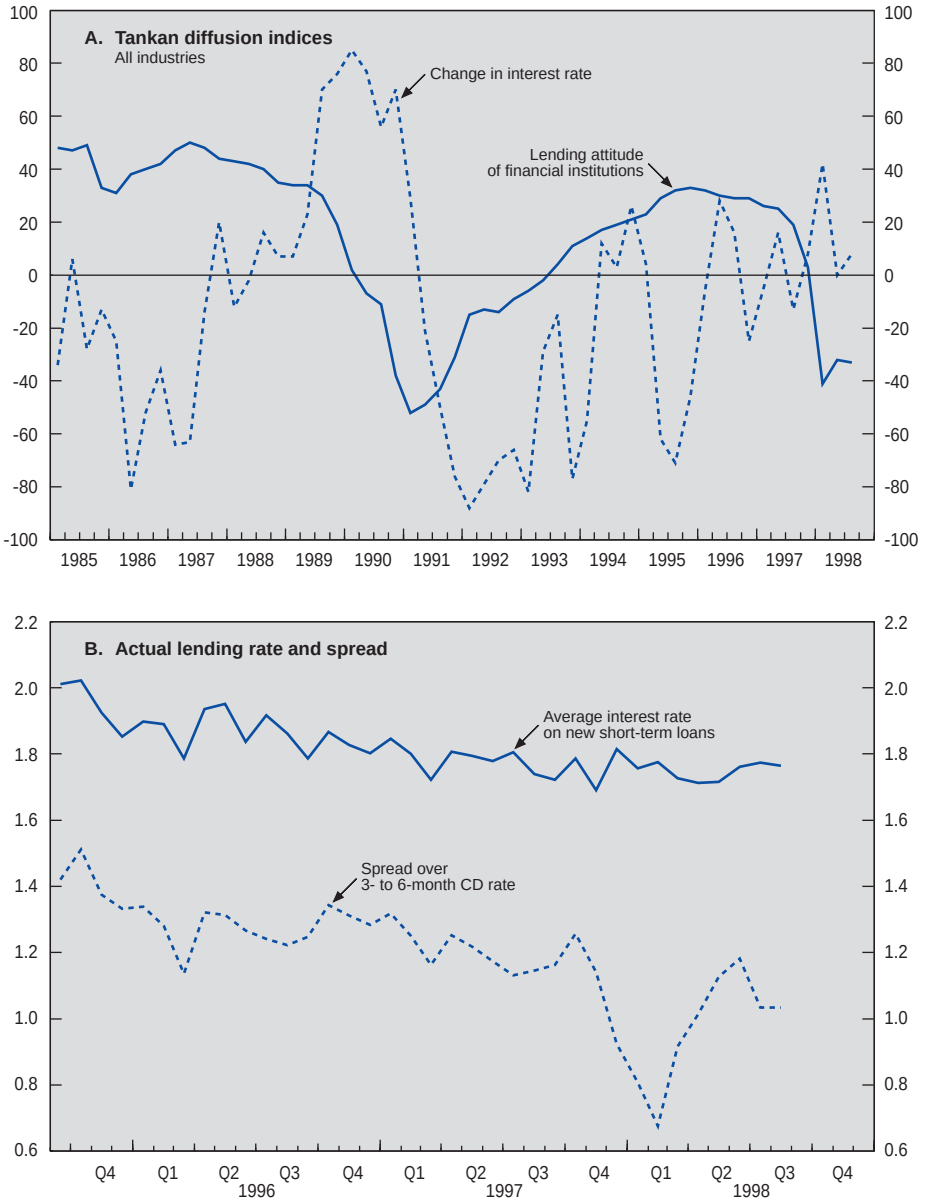
Source: Bloomberg, Bank of Japan and OECD.

and potential borrowers cannot be denied: respondents to the quarterly *Tankan* survey reported severe changes in the lending attitudes of financial institutions in December 1997 and March 1998[38] (Figure 21, Panel A). But many analysts argue that desired borrowing has been shrinking even faster than the supply of loanable funds, and indeed, even the *Tankan* figures show an even more severe shift in perceived lending attitudes in the period immediately after the bubble burst in 1990-91 as the economy was heading into its previous recession. While it is true that loans and discounts were gently falling on a year-over year basis already in late-1997 and that the rate of decline picked up to over 2 per cent in the spring of this year, the importance of supply and demand factors cannot be separately identified.[39] Furthermore, if a shortage of capital was really restraining the supply of loans by domestically licensed banks, then a shift towards foreign banks (who suffer from no such capital shortage) should be perceptible. However, while foreign banks did increase their market share until around the middle of 1997, the trend began to reverse in November, just when the supposed crunch became severe.

The extent to which banks are truly increasing their sensitivity to credit risk may be gauged by looking at the sectoral distribution of loans and discounts outstanding. The share of loans[40] going to real estate firms (surely the riskiest of sectors) has risen from around 11½ per cent in the early 1990s to 12½ per cent at end-1997 and nearly 13 per cent in March 1998. And loans to individuals have climbed from 15¼ per cent at the peak of the boom to 18½ per cent most recently. On the other hand, small firms – also likely to be among the riskiest borrowers – have seen their share decline from over 54 per cent when the bubble burst to 53 per cent at end-1996 and below 52 per cent most recently. The evidence is, therefore, mixed as to the role of enhanced attention being paid to credit risk. This might be attributable to continuing regulatory forbearance (Ogawa *et al.*, 1998). Another possibility related to the existence of a credit crunch is that banks have ‘‘pushed’’ otherwise sound firms, even large firms, into bankruptcy. Although the number of bankruptcies has indeed surged (as was documented above), most large-scale victims had been facing financial difficulties for a considerable time and benefited from a substantial degree of patience by bankers before collapse became inevitable (Ogawa *et al.*, 1998).

Another piece of evidence contrary to the existence of a true credit crunch is that the price of new loans shows no sign of having risen, even though the cost of funds to the banking system, as proxied by the 3 to 6-month CD new issue rate

Figure 21. **CREDIT CRUNCH?**



Source: Bank of Japan.

trended up over the year ending last February, thereby squeezing lending margins on short-term loans to less than a percentage point, 37 per cent lower than the CY 1996 average (Panel B).[41] And even before this latest decline, net interest margins were a fraction of those being earned in other OECD countries.[42] Thus, while individual institutions may face a shortage of financial capital on their balance sheets, it is fairly clear that there is an excess of economic capital in the sector[43] and that until this excess is corrected and loan pricing improves there will continue to be chronic banking sector problems resulting from inadequate returns.

Ultimately, what is of interest here is the impact of banking sector problems on the macroeconomy. Other than the possibility of a credit crunch and dislocations resulting from forced bankruptcies, however, this sector has come to be perceived as a test case for the success of regulatory reform in general and therefore of Japan's ability to adapt to the globalisation pressures which have enveloped it of late. Thus, even if banking sector problems do not play much of a direct role in restricting business investment or other forms of spending (Gibson, 1997), the ''animal spirits'' of agents both at home and abroad may have been and may continue to be adversely affected by the sector's difficulties.

II. Policy responses and economic prospects

The macroeconomic context

The situation which policymakers have confronted over the past year has been complex and frequently changing. Analysis points in a straightforward fashion to a deficiency of aggregate demand, but its extent was not widely appreciated for quite some time, and as a result the authorities were forced to engage in steadily increasing economic support through 1998. A good deal of the deficiency resulted originally from the withdrawal of government spending and the higher taxes imposed on the private sector over the course of 1997. A second contractionary shock came with the financial crises in neighbouring Asian countries which began in the summer of 1997, before worsening in the autumn and again in the spring of 1998. Further deflationary effects may have resulted from an apparent deterioration in credit supply conditions, at least for smaller firms, in part attributable to the imminence of the new financial era which was to begin in April 1998 and banks' need to prepare their balance sheets to avoid falling foul of capital adequacy requirements and the associated ''prompt corrective action'' (PCA) power of the authorities. But the impact of the problems in the banking system was amplified by the collapse of several major financial institutions in the autumn of 1997, although the related loss of liquidity in the system was to a large extent offset by central bank action. Nonetheless, the resulting uncertainty and loss of confidence, combined with the unprecedented rise in unemployment and the increasingly discussed problems in both the public and private pension systems, boosted saving rates, cut desired investment, left an inventory overhang and magnified the effect of the limited liberalisation of capital flows which took place in April, putting enormous downward pressure on the yen.

The fiscal policy reversal

''Gradualism'' in changes in fiscal policy

The fiscal tightening in 1997 turned out to be too much for the economy to bear: the economy is officially recognised to have fallen into recession in April 1997. Although the slow recovery of consumption expenditures attracted attention, the government retained its original contractionary stance until December 1997. The government's first reaction was to promote further deregulation: in November, a deregulation package was announced. However, as the initial estimate of GDP figures for the third quarter showed only a weak rebound from the sharp decline in the second quarter and the effects of the ''Asian crisis'' began to be felt, the government decided that there was a need to shift directions just after passing the Fiscal Structural Reform Act at the end of November: yet another temporary tax cut was implemented in the context of the FY 1997 supplementary budget, which passed the Diet in early February 1998 with initial effect later that month (Table 14). The supplementary budget also contained measures to support the financial system. The Fiscal Investment and Loan Programme (FILP) was revised as well in order to raise lending by government financial institutions by about 1.6 trillion yen (17.4 per cent),[44] in order to deal with the ''credit crunch'', which was expected to worsen toward the end of March, the end of the banks' accounting year (see Chapter I). Although the revisions to the initial budget and FILP can be considered as a first step toward expansionary policy, they turned out to be insufficient to boost the economy. For example, the March 1998 *Tankan* survey by the Bank of Japan showed a sharp deterioration in business sentiment, despite the February income tax cut. First-quarter national accounts portrayed an economy in deep recession. This sharp deterioration of the economic situation decreased tax revenues in FY 1997 more than previously expected (Table 14). The shortfall amounted to about 2¼ trillion yen, but was partly offset by expenditure savings of some ¾ trillion yen. The difference will be temporarily financed by drawing down accumulated reserves, which will increase the issuance of deficit-financing bonds in FY 1998 by that amount.[45]

The government also maintained its tight policy stance in the initial FY 1998 budget, the first subject to the Fiscal Structural Reform Act. For example, public investment was cut by 7.8 per cent compared to the initial

Table 14. **Central government budget**

	Trillion yen								Percentage increase		
	FY 1996	FY 1997				FY 1998			FY 1997	FY 1998	
	Settled (a)	Initial budget (b)	First supplementary budget	Total budget	Settled (c)	Initial budget (d)	First supplementary budget	Total budget (e)	(c)/(a)	(d)/(b)	(e)/(c)
Total expenditure	78.8	77.4	1.1	78.5	77.8	77.7	4.6	82.3	–1.3	0.4	5.8
General spending	48.1	45.1	1.7	46.8	46.3	44.5	4.6	49.1	–3.7	–1.3	6.0
of which:											
Public works	12.3	9.7	0.8	10.5	10.5	9.0	2.8	11.8	–14.9	–7.8	12.4
Other	35.8	35.4	0.9	36.3	35.8	35.6	1.7	37.3	0.0	0.6	4.2
Debt servicing[1]	16.1	16.8	–0.5	16.3	16.0	17.3	0.1	17.3	–0.6	3.0	8.1
Transfers to local government	13.9	15.5	0.0	15.5	15.5	15.9	0.0	15.9	11.5	2.6	2.6
Other spending	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	–100.0	–	–
Total revenue	60.1	60.7	–0.6	60.1	57.7	62.1	–1.5	60.6	–4.0	2.3	5.0
Taxes	52.1	57.8	–1.6	56.2	53.9	58.5	–1.5	57.0	3.5	1.2	5.8
Non-tax revenues	8.0	2.9	1.0	3.8	3.8	3.6	0.0	3.6	–52.5	24.1	–5.3
Deficit[2]	21.7	16.7	1.8	18.5	18.5	15.6	6.1	21.7	–15.1	–6.6	17.3
Per cent of GDP	4.3	–	–	–	18.5[4]	–	–	n.a.	–	–	–
Financing:											
Construction bonds	10.7	9.2	0.7	9.9	9.9	8.4	4.1	12.5	–7.5	–8.7	26.3
Deficit bonds	11.0	7.5	1.0	8.5	8.5[4]	7.1	2.0	9.1	–22.7	–5.3	7.1
Memorandum item:											
Net bond issuance[3]	19.5	–	–	–	26.4	–	–	n.a.	35.4	–	–
Per cent of GDP	3.9	–	–	–	5.2	–	–	n.a.	–	–	–

1. Including amortisation payments on existing debt.
2. The deficit is equal to the gross issuance of government debt.
3. Increase in outstanding stock of ordinary government bonds.
4. Although the deficit was 20 trillion yen (the difference between total expenditure of 77.8 and total revenue of 57.8 trillion yen), 1.5 trillion yen will be drawn from reserves and, therefore, the deficit will be only 18.5 trillion yen. The decline in reserves will be financed by the issuance of 1.5 trillion yen of deficit bonds in FY 1998.

Source: Ministry of Finance and Bank of Japan.

FY 1997 budget and by 14.6 per cent compared to the revised FY 1997 budget (Table 14). But a total reversal took place in April when the principle of revising the Fiscal Structural Reform Act was agreed and a fiscal package with additional expenditures and tax cuts worth over 16 trillion yen (about 125 billion dollars) was announced. The revisions to the Act and the supplementary budget finally passed the Diet in late May, and the measures incorporated in the package started to be implemented in June 1998. The Fiscal Structural Reform Act was revised along the three lines described in Table 15. The newly introduced cyclical flexibility with respect to reducing the issuance of deficit-financing bonds[46] allowed an additional tax cut in FY 1998 (which was incorporated in the fiscal package); as a result, such issues were then projected to rise from 8.5 trillion yen in FY 1997 to 9.1 trillion yen in FY 1998 (Table 14). The April package included a tax cut – yet again of a temporary nature – amounting to 4 trillion yen (2 trillion yen more in 1998 and 2 trillion yen in 1999) and social infrastructure investment of 7.7 trillion yen (6.2 trillion yen for central government, both with and without local government participation, and 1.5 trillion yen for local governments on their own) (Table 16). The package was larger than any of its predecessors in the 1990s (Table 17), especially in the amount of expenditures devoted to public works and tax cuts, which increase aggregate demand[47] more readily than, say, additional loans provided by government financial institutions. An additional boost to demand was also intended by front-loading the public investment in the initial budget, with a target of over 81 per cent of all contracts being finalised by the end of September. This will be followed by the extra public investment resulting from the supplementary budget. However, the effects of the measures may have been limited in two ways. First, the initially temporary nature of the tax cut may have meant that it did not stimulate much consumption. Second, there is uncertainty as to how much local governments will be able to increase their public works on their own because their financial situations have already substantially deteriorated. Despite these uncertainties, the measures are expected to have a sizeable impact on the economy in the latter half of the fiscal year (see below).

The responses of the government until April could be characterised as ‘‘gradualism’’. This approach in what amounts to be a ‘‘crisis’’ situation could be accounted for by the otherwise laudable objective of maintaining consistency with past policies. However, the cost of the gradualism was negative reactions from the markets in the form of falling stock prices and the yen’s depreciation

Table 15. **Summary of the Fiscal Structural Reform Act**

	Initial	Revised
	Legislated in November 1997	Legislated in May 1998
1. Size of fiscal deficit	The sum of the central and local government fiscal deficit as a percentage of GDP, around 6 per cent in FY 1997, should be brought down to 3 per cent or less by FY 2003 (Section 1, Article 4, Chapter 1).	The target year for the mid-term fiscal consolidation of both 1 and 2 is postponed from FY 2003 to FY 2005.
2. Special deficit-financing bond issues	The amount of such bonds should be reduced every fiscal year, and issurance should be terminated by FY 2003 (Section 2, Article 4, Chapter 1).	More flexibility is introduced in issuing deficit-financing bonds: the amount of such issues is allowed to exceed that of the preceding year under such circumstances as catastrophic disasters and noticeable stagnation of economic activities.[1]
3. Spending limits on major programmes	From FY 1998 to 2000 (defined as the intensive reform period), the magnitude of expenditures in major programmes is subject to the numerical upper limit set in each programme (Chapter 2).	Social security expenditure will not be subject to the ''cap'' stipulated in the Act, although its growth will be strictly contained to the extent possible.
4. Guidelines on fiscal policy	Fiscal policy will be conducted in such a manner that the sum of taxes, payroll contributions and fiscal deficit as a percentage of national income will not exceed 50 per cent (Section 1, Article 6, Chapter 1).	

1. Under the government order, ''noticeable stagnation of economic activities'' may mean any of the following:
   a) Real GDP growth rates (s.a.a.r.) are less than 1 per cent in the two most recent quarters.
   b) Monthly economic indicators of consumption, investment and employment show clear stagnation, in addition to a real GDP growth rate (s.a.a.r.) of less than 1 per cent in the most recent available quarter.
   c) The economy falls into a situation of stagnation as in *a)* or *b)*, due to unexpected shocks.

Source: Ministry of Finance.

Table 16. **Fiscal measures in the April 1998 package**

Yen trillion

	Fiscal measures	April package	FY 1998 supplementary budget allocation
Tax reductions, etc.	Temporary income tax reductions[1]	4.0	1.4
	Special taxation measures[2]	0.3	0.1
	Welfare benefits, etc.	0.3	0.3
Social infrastructure investment	Public investment[3]	6.0	3.4
	Environment and new energy	(1.6)	(0.8)
	Telecommunications network, and science and technology	(1.0)	(0.8)
	Social welfare, medical care and education	(1.0)	(0.5)
	Improvement of distribution and transportation system	(0.8)	(0.4)
	Emergency disaster prevention	(0.8)	(0.4)
	Inducement of private investment through the redevelopment of downtown districts	(0.8)	(0.4)
	Disaster relief	0.2	0.2
	Public investment by local governments	1.5	–
Subtotal		12.3	5.4
Others	Land mobilisation	2.3	0.4
	Integration of lands and urban redevelopment	(1.2)	(0.2)
	Prior land acquisition for future public works		
	Central government	(0.3)	(0.2)
	Local government	(0.8)	–
	Measures for small and medium-sized enterprises	2.0	0.3
	Measures for employment[4]	0.05	0.001
Total		16.6	6.1

1. April package contains Y 0.6 trillion by local governments in FY 1998 and Y 2.0 trillion scheduled for FY 1999.
2. Y 0.1 trillion is for FY 1998, while Y 0.3 trillion is the full-year effect as from FY 1999.
3. The rest of the expenditure is expected to come from local governments.
4. The rest of the expenditure is expected to come from special accounts.

Source: Ministry of Finance.

Table 17. **Comparison with previous fiscal packages**

Yen billion, project cost base

	Total amount	Social infrastructure investment					Tax cuts	Others
		Public works		Building and equipment	Public works by local government	Sum of 2-5		
		General public works	Disaster relief					
	1	2	3	4	5			
28 August 1992	10 700	3 400	500	550	1 800	6 250	0	4 450
13 April 1993	13 200	3 640	530	1 150	2 300	7 620	150	5 430
16 September 1993	6 150	1 000	450	0	500	1 950	0	4 200
8 February 1994	15 250	3 590	0	610	300	4 500	5 850	4 900
14 April 1995	4 620 [1]	205 [2]	718 [2]	154 [2]	0	1 077 [2]	0	3 543 [1]
20 September 1995	12 810	3 930	700	910	1 000	6 540	0	6 270
24 April 1998	16 650	4 500	200	1 500	1 500	7 700	4 600	4 350
Cumulative	79 380	20 265	3 098	4 874	7 400	35 637	10 600	33 143

1. OECD estimate.
2. Central government budget figures, rather than project cost figures.

Source: Economic Planning Agency and Ministry of Finance.

until August, thereby leading to further worsening of consumer and business sentiment. The market appeared to be puzzled by the government's reaction to the ''crisis'' situation: the April package was announced about two weeks after the initial budget passed the Diet. A technical reason for the gradualism is that the Fiscal Structural Reform Act did not, at the time, allow for any flexibility in dealing with business cycles in the initial budget, whereas it had nothing to say about what could be implemented in the form of supplementary budgets. These are only indirectly constrained by the end-point targets for fiscal deficits and deficit-financing bond issuance (Table 15).

As stated above, there is a difference in the government's treatment of initial and supplementary budgets. Although the government sets an upper limit on general expenditures, it is not applied to supplementary budgets. Given the difficulty of issuing deficit-financing bonds, the government depends on the supplementary budget for an increase in government spending through issuing construction bonds. While the distinction between the two is sometimes believed to be used to good measure to fend off low priority spending proposals, the effect is probably merely to deflect such pressures to items financeable by contruction bonds. Such expenditures are legally required to be allocated to public works, loans and investment. The range of public works is also required to be determined in the Diet every fiscal year. Because the use of revenues raised by construction bonds is thus narrowly specified, it is difficult to allocate the money efficiently. The efficiency of public investment allocation would gain from a more extensive use of cost-benefit analysis in the budgetary process. Such analysis has just started to be used in some ministries and will be extended more generally in the near future.

Following the government's significant loss in the upper house elections in July, a newly formed cabinet adopted a fiscal stance guaranteeing the continuation of the current expansionary posture. Guidelines for the second FY 1998 supplementary budget and the initial FY 1999 Budget, a so-called 15-month budget, call for permanent personal income and corporate tax cuts of a total of more than 6 trillion yen per year and for higher spending following suspension of the constraints imposed by the revised Fiscal Structural Reform Act due to the current recession. In order to ensure continued stimulus in FY 1999 initial budget public investment levels will be maintained and a special allocation will ensure at least steady overall capital outlays. Most recently, in October 1998, the

government was considering expanding and advancing the supplementary budgetary measures, including gift vouchers (time-limited coupons distributed to every adult resident).

Sustainability of government debt

As stated in Chapter I, the fiscal tightening in 1997 was driven by the recognition that the government's financial position had deteriorated and was therefore deserving of prompt attention. But the seriousness of the situation has been the subject of substantial debate, in part because available measures of government debt provide quite different indications. Because of the importance of the debt situation for assessing fiscal sustainability, the following paragraphs provide a detailed analysis of the data.

In December 1996, the ratio of debt to GDP was 82.7 per cent on a gross basis but only 15.4 per cent on net basis, according to the OECD definition, for example. But other versions are also cited; they differ along the following dimensions: gross or net base, coverage of general government and degree of consolidation.

The gross debt figure was some 96 per cent of GDP on the official, incompletely consolidated basis in March 1997 (Table 18), up from 12 per cent of GDP in March 1969 (Figure 22, Line A). In its calculations, the OECD Secretariat excludes the agencies of the central government which are labelled ''non-profit institutions controlled by government'' (*jigyo-dan*);[48] the most important among these institutions have functions similar to those of public financial corporations. The effect of this exclusion is to reduce the gross debt ratio (Figure 22, Line B); by FY 1996, this downward adjustment was 11.4 percentage points of GDP.

There is room for some further consolidation. In FY 1996, the central government held 6.8 trillion yen of short-term government securities,[49] while its agencies held 2.7 trillion yen and the social security funds 5.2 trillion yen in long-term government bonds (figures bolded in Table 18). If this amount (14.7 trillion yen in total) were consolidated, both official and OECD gross debt figures would have been lowered by 2.9 percentage points of GDP in 1996 (Figure 22, Line C). Other approaches to consolidation would make even larger reductions.[50]

Table 18. **Detailed balance sheet for general government, FY 1996**

Official presentation, yen trillions

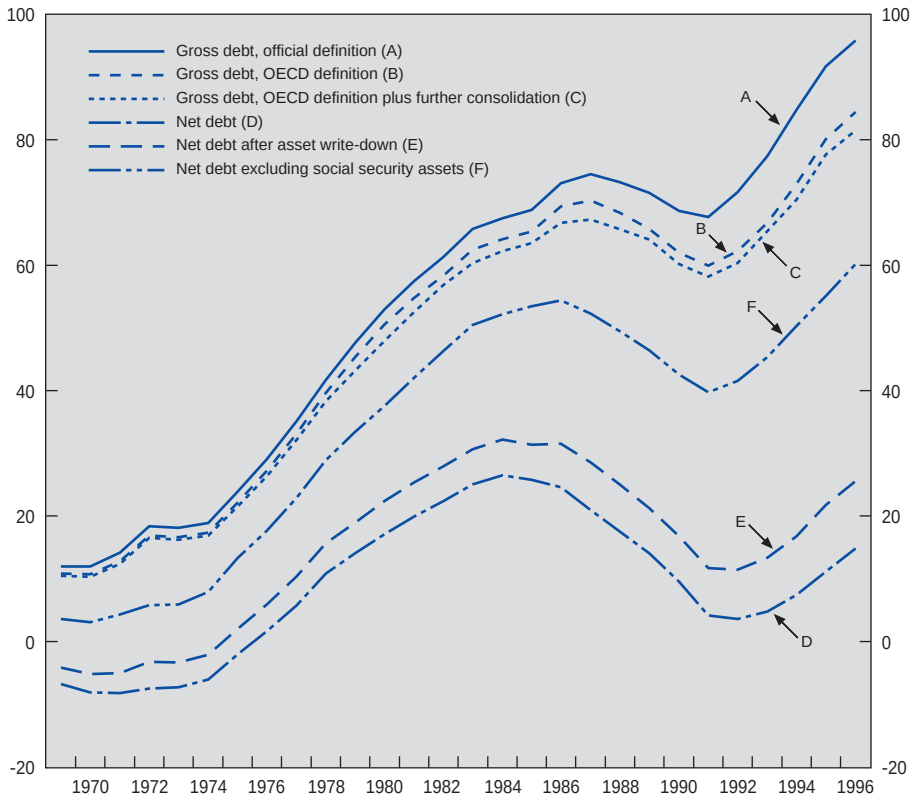
	Central	Agencies	Local	Social security	General[1]
Gross assets					
Total	84.4	63.9	43.3	228.5	408.1
As per cent of GDP	16.8	12.7	8.6	45.4	81.0
Of which:					
Cash and deposits	0.1	37.7	29.4	36.1	103.1
Short-term government securities	**6.8**	0.0	0.0	0.0	6.8
Long-term bonds	0.0	7.6	0.0	16.1	23.7
Government	0.0	**2.7**	0.0	**5.2**	7.9
Public corporations	0.0	0.9	0.0	6.0	7.0
Other	0.0	4.0	0.0	4.9	8.9
Corporate equities	0.5	0.0	0.1	0.2	0.8
Loans	9.0	13.3	9.0	7.2	***36.5***
Life insurance	0.0	4.2	0.0	19.5	***23.7***
Transfers from general government	4.0	0.0	0.0	0.0	4.0
Other	63.9	1.0	4.9	149.5	209.3[1]
Deposits with Trust Fund Bureau	9.2	0.6	0.0	142.5	152.3
Investments	24.7	0.0	4.7	3.6	***23.0***
Non-life insurance	0.0	0.0	0.2	0.0	0.2
Foreign trade credits	0.2	0.0	0.0	0.0	0.2
Foreign claims	0.4	0.2	0.0	2.0	2.7
Other	29.4	0.3	0.0	1.3	***30.9***
Gross liabilities					
Total	310.7	57.6	124.9	1.5	482.7
As per cent of GDP	61.7	11.4	24.8	0.3	95.8
Of which:					
Short-term government securities	29.8	0.0	0.0	0.0	29.8
Long-term bonds	245.3	0.6	39.6	0.0	285.5
Loans	30.1	46.4	85.1	1.5	161.1
From public sector	30.1	46.4	75.4	1.5	151.4
From private sector	0.0	0.0	9.7	0.0	9.7
Trade credit	0.0	0.0	0.2	0.0	0.2
Other	5.5	10.5	0.0	0.0	6.0
Investments	0.0	10.5	0.0	0.0	0.5
Foreign debts	1.2	0.0	0.0	0.0	1.2
Other	4.3	0.0	0.0	0.0	4.3
Net liabilities	226.4	–6.3	81.6	–227.1	74.6
As per cent of GDP	45.0	–1.3	16.2	45.1	14.8

1. General government is not the sum of its components in all cases because of consolidation.
Source: OECD calculations based on Economic Planning Agency submission.

Debt figures for the major OECD economies, before any consolidation of government-held financial assets takes place (Table 19, Line 1), show Japan to be one of the more heavily indebted major countries. Differences among countries

Figure 22. **VARIOUS DEFINITIONS OF JAPAN'S GOVERNMENT DEBT**

FY 1969-96, per cent of GDP



Source: Economic Planning Agency, Bank of Japan and OECD.

emerge when consolidation takes place (Line 2), and these have to do with the proportion of assets (including social security assets) which are also liabilities of other parts of the general government sector. For example, in the United States, social security funds are invested in government bonds which are netted out in consolidation. In Japan, however, a large portion of social security funds is held in accounts outside of the general government, in particular in the Trust Fund Bureau (TFB), a branch of the Ministry of Finance; most (42 per cent worth 28 per cent of GDP) are invested in a wide range of public agencies through the

Table 19. **Major seven OECD countries' consolidated debt and assets, CY 1996**

Per cent of GDP

	United States	**Japan**	Germany	France	Italy	United Kingdom	Canada
1. Unconsolidated gross debt	84.6	85.7	64.9	76.6	133.0	65.4	109.6
Minus consolidation	21.5	3.0	4.7	13.6	9.3	5.7	12.1
Of which:							
Consolidation in social security funds (SSF)[1]	12.5	1.0	2.7	4.1	1.3	0.0	4.1
2. Consolidated gross debt	63.1	82.7	60.2	63.0	123.7	59.7	97.5
3. Consolidated government assets	14.7	67.3	12.1	22.2	13.3	15.7	29.1
Breakdown of assets							
SSF holdings of non-government assets	0	44.0	3.4	5.5	1.3	0	2.2
Assets held by other parts of government	14.7	23.3	8.7	16.7	12.0	15.7	26.9
Cash, deposits, etc.	4.4	7.3	6.8	2.9	3.8	6.6	2.7
Trade credits	0.3	0.0	0.0	3.8	0.0	0.7	0.6
Bonds	4.2	0.0	0.0	0.0	0.3	0.3	0.9
Loans	1.9	3.2	0.0	2.9	0.1	4.9	2.5
Shares	1.1	0.1	1.9	7.1	7.7	0.0	14.4
Miscellaneous	2.8	12.7	0.0	0.0	0.1	3.2	5.8
4. Net debt (*line 2 minus line 3*)	48.4	15.4	48.1	40.8	110.4	44.0	68.4
Memorandum item:							
Total SSF	12.5	45.0	6.1	9.6	2.6	0.0	6.3
Maastricht definition of gross debt	–	–	60.4	55.4	124.0	54.5	–

Notes: Figures for consolidated gross and net debt correspond to those presented in *Economic Outlook* 63 except for Germany (unconsolidated in EO63) and the United Kingdom (which have been revised). Japan: Government non-profit institutions not included. Germany: Trade credits included in other assets. Social security funds are estimated from 1989 figures. Consolidated assets split proportionally between sectors based on unconsolidated assets for both Germany and France. Italy: Trade credits not included.

1. The term social security funds generally reflects all types of assets connected to social insurance programmes, including health services.

Source: OECD.

TFB *via* the Fiscal Investment and Loan Programme. Therefore, those assets are, by definition, not consolidated in the figures reported by either the Japanese authorities or the OECD. Bringing TFB into the general government sector would raise gross debt by around 5 percentage points of GDP near the beginning of the historical sample and over 10 percentage points by FY 1996. However, net debt is insensitive to this modification, because the TFB holds no net assets itself.

Compared with other major economies, the Japanese government holds large amounts of financial assets (Table 19, Line 3). These appear mainly as social security funds and in the miscellaneous category. Assuming the official value placed on these assets is accurate, Japan's net debt position is the lowest among the countries examined here (Table 19, Line 4), although, because of the recent large fiscal deficits, it has been rising rapidly (Figure 22, Line D). However, the true value of many of these assets is difficult to estimate, in part due to their lack of liquidity. In fact, the solvency of many of the entities which have benefited from government loans and investments is heavily dependent on government operating subsidies. There is no procedure for writing down the value of loans whose repayment may be doubtful; liabilities accumulate until they are seen as clearly not payable (such as in the current cases of the JNR Settlement Corporation and the National Forest Service), at which point the government obligation to stand behind the debt is called upon, and it is taken over (see Box 2). Therefore, the net debt figure is heavily dependent on the quality of assets on the balance sheet.

For illustrative purposes, the net debt figures may be adjusted for assumptions regarding the assets which *are* on the balance sheet and whose value is likely to be the most fragile; these are rather arbitrarily assumed to be ''loans'', ''life insurance'', ''investments'' and the residual ''other'' category under the heading ''other financial assets and liabilities'' (Table 18, last column, figures are bolded and in italics). In FY 1996, such assets amounted to 114 trillion yen after official consolidation or nearly 23 per cent of GDP. Assuming arbitrarily that their recorded value were double their true value, OECD net debt figures would be raised by 11 percentage points of GDP, to more than 25 per cent of GDP by FY 1996 (Figure 22, Line E).

The issue of whether to consider social security funds in the same fashion as other components of general government has been much discussed in the literature. Since social security assets are earmarked to meet future pension liabilities,

Box 2. **JNR Settlement Corporation's debt**

Japan National Railways (JNR) was a state-run company in the railroad business with a monopoly on the movement of freight and a heavy presence in the movement of passengers. In terms of total transportation (air, sea and land) it had over half the market in both categories in the mid-1950s. However, its performance was poor because it was subject to political interventions, inefficient management and unstable industrial relations. JNR incurred steady losses from FY 1964 until FY 1986 when it was privatised.

In April 1987, JNR's railway business was taken over by seven newly established private companies (six for passengers and one for freight). At the same time, JNR Settlement Corporation (JNRSC) was set up to assume JNR's debt worth 25.5 trillion yen. Whatever debt was left after disposing of any JNR assets would be borne by the general public. By FY 1997, JNRSC's debt had increased to 27.8 trillion yen. This is because cumulative expenditures, which were mainly interest and pension payments and amounted to 15.4 trillion yen, exceeded cumulative revenues, which were earned from sales of assets such as land[51] and stocks of privatised companies and amounted to only 14.3 trillion yen. The following scheme to dispose of JNRSC's debt was devised by the government in preparing the initial FY 1998 budget. All JNRSC's financial debt worth 24.3 trillion yen would be assumed by the general account of the government. The remaining pension liability of 3.5 trillion yen would be assumed by a new company to be financed by subsidies from the general account, asset sales and contributions from the privatised JNR companies. A new law to implement this scheme was submitted to the Diet, but it failed to pass in the last ordinary session partly because the privatised companies opposed the plan. This opposition resulted from the fact that the previous agreement was that their contribution was to be 360 billion yen, but the just-approved legislation requires them to raise it to 540 billion yen.

This case provides a lesson. When a public agency – inside or outside the general government sector – faces financial difficulty, the government's implicit guarantee may be called upon and its general account will take it over. However, because the JNRSC was outside general government on an SNA basis, the balance sheet of general government will not reflect its financial situation, no matter how serious, until that take-over occurs. Therefore, the quality of ''off-balance sheet'' assets and implicit liabilities matters in judging the government's financial health.

which are not included in the gross liability figures, it might be argued that they are different from other financial assets of government, and some might go so far as to favour their exclusion from the debt calculations altogether. Excluding the assets of the social security funds from the (net) debt calculations on the grounds that they are not available in the medium term for meeting general budgetary

obligations might better represent the government's financial position. The resulting series – identical for both the official and OECD definitions – can be seen (Figure 22, Line F) to have reached over 60 per cent of GDP at the end of FY 1996, after remaining in the 40 to 55 per cent range for some 15 years. Even if it is a bit radical to eliminate all the pension funds from the general government, in Japan's case moving in that direction has some justification. Many companies have taken over the responsibility of paying public as well as private pensions to their employees and have accordingly been building up assets to fund this future liability. The official figures of social security funds include not only ''privatised'' assets but the ''pure'' private part of assets (because their asset management is subject to strict government regulations), estimated to be about 18.7 trillion yen (about 3.8 per cent of GDP in FY 1995) (Murakami, 1997). To the extent that public pension funds are included, various institutional differences in pension systems affect the amount of assets that are counted. The most transparent way to handle these factors is to calculate the present discounted value of both sides of the balance sheet. It is widely accepted that Japan suffers from underfunding: Roseveare *et al.* (1996) calculate a deficiency of up to 237 per cent of 1994 GDP, depending on the productivity growth and discount rate assumptions. For this reason, a reform of the system is under discussion (see Chapter III).

Summing up, Japan's public debt has been on a fast rising trend in relation to GDP since early in the decade. Only part of that has been offset by increases in financial assets, mostly held in the social security system. Excluding such assets general government debt was already 60 per cent of GDP in FY 1996. After absorbing some additional liabilities in the near future this net debt measure looks set to rise to around 85 per cent of GDP at the turn of the century. While that may still be sustainable, the rate of increase must soon be brought down for the dynamics of the debt not to become explosive, even if the pensions problem is separately resolved.

Toward tax reform

In the April fiscal package documentation, it was made clear that reform of the income and corporate tax would be examined. Permanent income tax cuts combined with broadening the tax base are clearly worth considering not only as a means to boost the economy but more importantly to improve incentives and to

avoid restraining the economy's long term potential growth rate (see below). Compared to an increase in public investment, permanent income tax cuts may do less to boost demand because of greater leakages into saving, but they are more neutral in resource allocation (see the discussion of the unsustainable situation in the construction industry in Chapter IV) and can be expected to have more durable effects through enhanced confidence.

Most proponents of reform are in agreement that the worst feature of the current system is the high marginal personal tax rates at the upper end of the distribution: with top national rates of 50 per cent and local residents' tax of an additional 15 points, the resulting burden is sure to have some adverse incentive effects on both labour force participation, entrepreneurship (see Chapter IV) and avoidance/evasion. In most years, however, typical deductions for a single-earner family of four are said to be over 3.6 million yen (over $25 000), exempting around 20 per cent of all salaried workers from paying any income tax; this year, with the various tax cuts, the threshold for such a family unit is nearly 5 million yen (around $35 000), and fully 30 per cent of such workers will not pay any tax. Thus, the problem confronting reformers is that the condition of the public finances will not permit a tax cut with substantial revenue losses,[52] so that the proposed reduction in marginal rates will have to be eventually paid for by a combination of the endogenous expansion of the base and explicit base broadening. While the cited figures for the threshold seem very high by OECD standards, cutting it at the same time as cutting top marginal rates would make the system less progressive, which might be especially difficult to do in an egalitarian country like Japan. It may be desirable to lower all tax rates, not just a single statutory marginal tax rate, to benefit a wider range of people, especially when it may be necessary to confront vested interests when expanding the tax base.

The need to expand the tax base is further exemplified by the following facts. First, Japan has uniformly higher personal income tax rates (Table 20) but lower revenues (Table 21) than, for example, France. Second, it is generally believed that the self-employed bear less of a tax burden than they ought to if their income were known to the tax authorities. On the other hand, employees are subject to tax withholding from their salaries.[53] Generous tax deductions available to employees are designed to achieve a better balance in the tax burden in relation to the self-employed. As a result, the minimum taxable income is rather high in Japan, compared to that of other countries (Table 21). Third, as ageing

Table 20. **Effective tax rate under the individual income tax**

Single earner couple with two children, per cent, 1998

Gross employee income Million yen[1]	Japan		United States[2]	United Kingdom	Germany	France
	Before special tax cut	After special tax cut				
3	0.0	0.0	7.2	16.3	0.0	0.0
5	3.5	2.1	12.3	19.4	6.0	3.0
10	11.8	11.2	19.7	27.4	21.6	10.0
20	25.9	25.6	28.1	33.7	35.3	21.3
30	33.7	33.5	33.3	33.7	42.2	31.2
50	43.9	43.7	38.0	38.0	47.7	39.1
Memorandum item:						
Minimum taxable income	3.6 (3.0)[3]	4.9 (3.0)[3]	2.5 (1.8)[3]	1.1	3.7	3.2

1. The following exchange rates are used: 1 dollar = 119 yen, 1 pound = 195 yen, 1 DM = 67 yen, 1 FF = 20 yen.
2. The state income tax rates in New York are taken to represent the local inhabitant tax rate in the United States.
3. Minimum taxable income for local income taxes.

Source: Ministry of Finance, *Japanese Tax System 1998*.

proceeds, the tax burden borne by younger people is expected to increase while elderly people are enjoying several tax advantages. Generous tax deductions are available to the elderly. A lower tax rate is also applied to interest income earned by the elderly. The deduction for pension benefit payments allowed to the elderly also seems difficult to justify. The earnings test for pension recipients also discourages those over 60 from working (see Table 30 below). Fourth, non-working spouses also benefit from generous spousal tax deductions which tend to reduce women's working hours.[54] Eliminating all disincentives for the labour supply of the elderly and women would help to maintain the financial health of the pension system.

Tax collection could also be strengthened by an increase in the number of staff working for the tax authorities,[55] the introduction of a taxpayer identification number system[56] and/or effective use of withholding taxes in order to minimise tax evasion: unless a taxpayer has a significant amount of income other than from earnings, (s)he is not obligated to file a tax return and merely pays *via* withholding by his employer. Thus, paying tax on interest and dividend income and capital gains is based on self-assessment and what often amount to final withholding taxes. A recent legal change allowed for the creation of a nation-wide

Table 21. **International comparison of tax revenues**

Ratio of tax revenue to national income, 1995

	United States	United Kingdom	Germany	France	Canada	Italy	**Japan**	**Japan** (adjusted)
Corporate income taxes	3.0	3.8	1.3	1.9	3.6	4.2	4.3	3.5
Individual income taxes	11.6	10.9	12.4	7.1	16.8	12.5	6.1	6.0
Consumption-based taxes	5.7	13.8	12.6	14.0	11.5	13.1	4.3	4.3
Property-based taxes, etc.	3.6	4.2	1.3	4.8	5.3	4.1	3.4	4.3
Total taxes	23.9	32.6	27.5	27.8	37.2	33.9	18.1	18.1
Including social security contributions	31.9	39.7	45.3	51.3	45.1	47.9	28.5	28.5

Note: Japan's adjusted figures are calculated according to the Japanese authorities' view that enterprise tax and mineral product tax should be classified in other taxes which belongs to ''Property-based taxes, etc.'' here.

Source: OECD *Revenue Statistics 1965-96*.

10-digit computer identification system, but it explicitly prohibits its use for this purpose. The efficiency of the income tax system will increase in importance as deregulation proceeds. Income redistribution will then be more dependent on the income tax system than various government interventions such as official price setting. ''Losers'' from increased competition could be provided with a fuller social safety net *via* the social security and income tax systems, thereby enabling the government to pursue further deregulation. The income tax system should also be assessed as to its generational equity. The uniformly beneficial treatment of all the elderly described above is not desirable because there are significant income and wealth differences among them. Taxes on wealthy elderly people should be levied more heavily, which could in turn allow reduced taxes on younger people.

As for corporate tax, the national corporate tax rate was lowered by 3 percentage points and the local corporate enterprise tax rate by 1 percentage point this year, a move toward comparable levels in other developed countries, and the tax base was expanded (Table 22). Nonetheless, the total tax rate on corporations remains high by international standards, and the government has accordingly agreed to bring it down by a further 6 percentage points in the FY 1999 budget. The question which remains is how to implement this cut. As for local tax, a reform of the corporate enterprise tax is under consideration. The following three facts are worth considering in the discussion. First, the base of this tax is corporate profits,[57] the same as for the national corporate tax. Therefore, tax revenues vary pro-cyclically, and even in boom periods only about half of all

Table 22. **Effective tax rate under the corporation tax**

Per cent, 1998

	Japan	United States	United Kingdom	Germany[2]	France[3]
National tax	31.1	31.9	31.0	36.1	41.7
Local tax	15.3	8.8[1]	0.0	15.6	0.0
Total	46.4	40.8	31.0	51.7	41.7

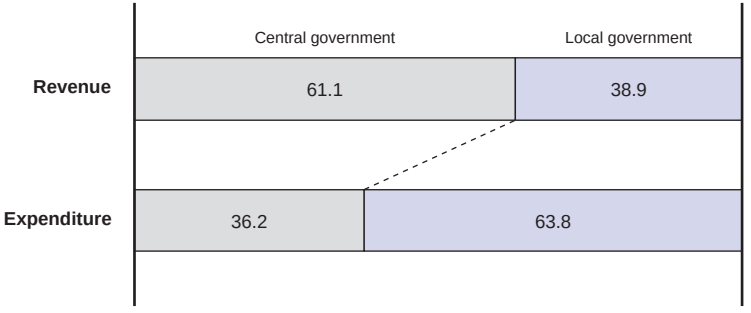
1. Corporation tax in California is taken to represent the local corporation tax in the United States.
2. Basic tax rate in Germany if calculated under the assumption that the dividend payout ratio is 30 per cent. Surtax is levied at 5.5 per cent of corporation tax. National corporation tax in Germany is jointly levied by the Federal Government and the states with the proceeds split evenly.
3. Surtax in France is levied at 25 per cent of corporation tax (only on companies whose annual sales are greater than 50 million francs).

Source: Ministry of Finance, *Japanese Tax System 1998.*

corporations pay the tax, because the rest register losses.[58] Second, revenues and expenditures at the local government level are quite unbalanced (Figure 23). This structure is sustained by large transfers from central to local government. Third, the revenues generated by this tax differ significantly among prefectures, even more than aggregate local tax revenues (Figure 24). In sum, local governments depend on volatile and geographically biased tax revenues, necessitating heavy interventions by central government. Greater income stability would be desirable, taking account of the services provided by the local governments, the higher risk-pooling ability of central government, and the promotion of decentralisation, a central government objective. Stable revenues would allow local governments to play a more important role in the ageing society, by providing more welfare services. It is argued that a shift in the tax base to some measure of firms' benefits from the local community, such as their property used, employees and sales, would make it possible to lower the tax rate and still maintain tax revenue. It would also prevent firms from free riding on local public goods and secure a more stable revenue source for local governments. This last goal might be better achieved, administrative and compliance burdens minimised and distortions resulting from taxing based on size avoided,[59] if central government collected all corporate tax and distributed a greater share to local authorities by some sort of equalisation mechanism.

Figure 23. **REVENUE AND EXPENDITURE STRUCTURE**

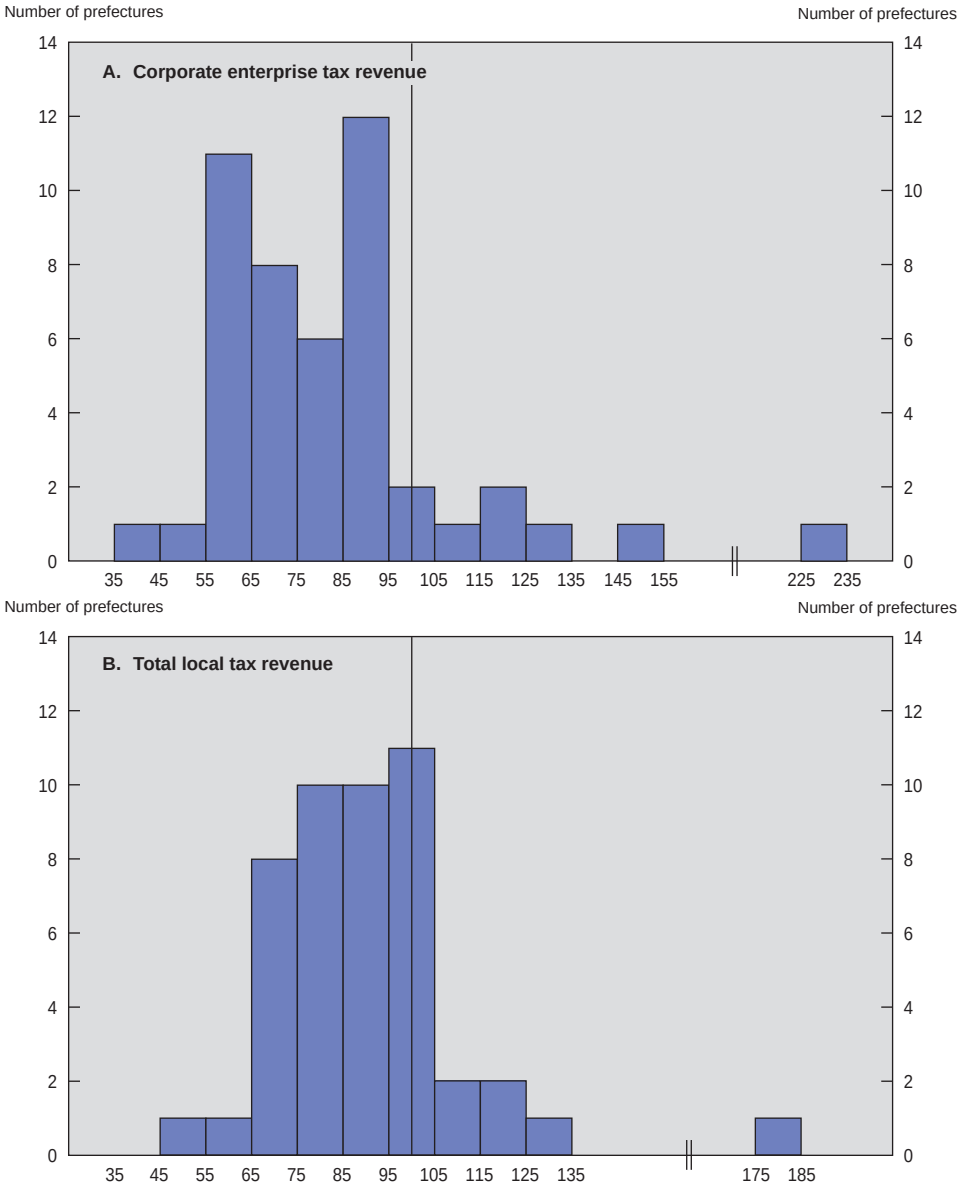
Per cent, FY 1996



Source: *Ministry of Finance Statistics Monthly,* April 1998.

Figure 24. **PER CAPITA DISTRIBUTION OF LOCAL TAX REVENUE**

National weighted average = 100, FY 1997, 47 prefectures



Source: *Ministry of Finance Statistics Monthly*, April 1998.

The monetary policy dilemma

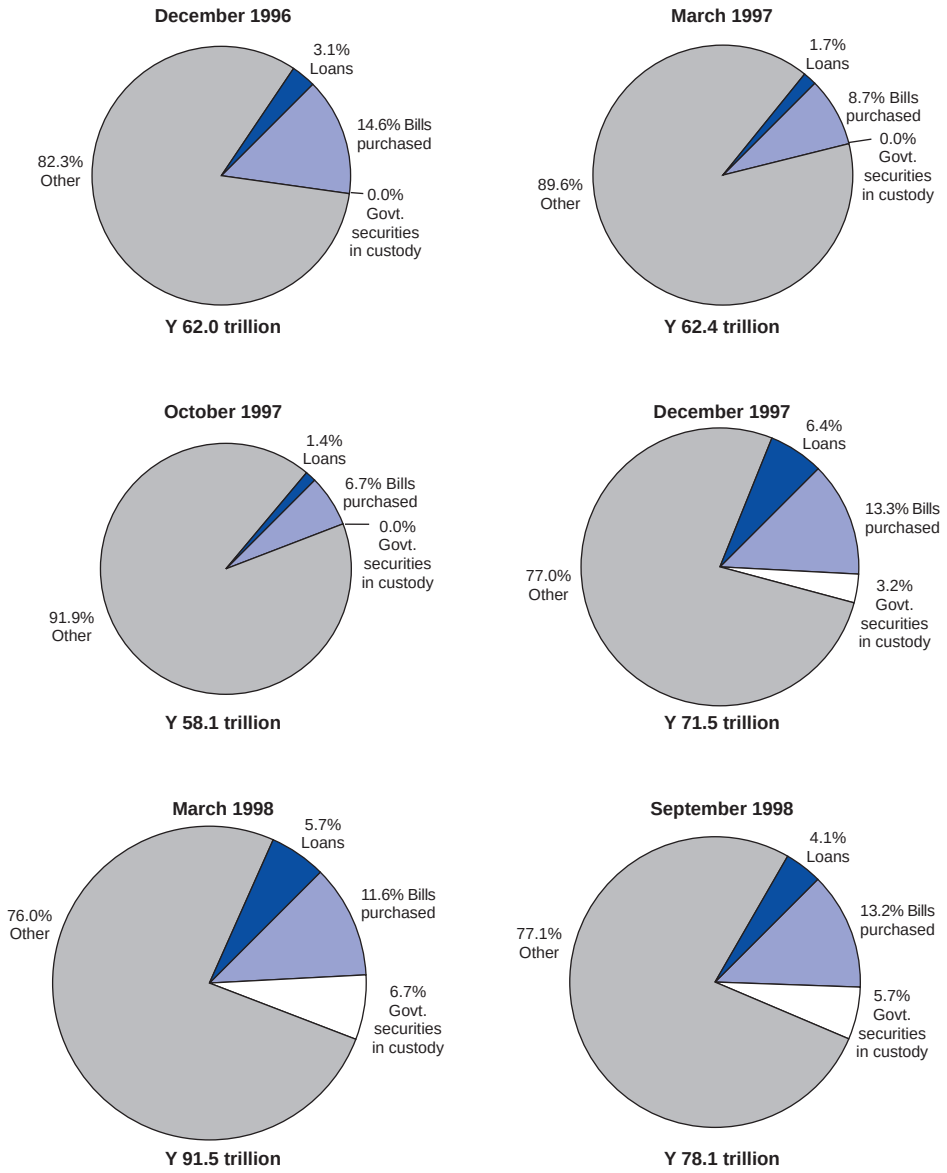
How to give additional support

The consensus view in the autumn of 1997 was that the economy would perform in a satisfactory fashion with a steady rise in employment (albeit some degree of excess capacity) and persistently low underlying inflation, despite the restraint resulting from the marked tightening in fiscal policy. Monetary conditions were termed ''exceptionally accommodating'' by the Committee in last year's *Survey*, but this was regarded as appropriate until the recovery became firmly established. In the event, of course, things turned out rather differently, and the debate quickly turned to whether the monetary authorities could do even more to support the weakening economy. With the official discount rate (ODR) at the low level of 0.5 per cent now for the past three years and the target short-term rate – the rate on uncollateralised overnight call loans – on average slightly below that level, there would seem to be little that could be done on the interest-rate front.[60] But the Bank of Japan has indeed been very active in ensuring that the spillover from the banking sector crisis onto the real economy was minimised. First, it made massive new loans to the institutions in question (up to a peak of 3.8 trillion yen in November 1997) to allow them to continue to function while winding down their operations.[61] Second, it engaged in substantial injections of liquidity through various means, including repurchase operations, a new tool in its arsenal. It also bought a great deal of commercial paper – up to around 6 trillion yen in February 1998, nearly half the outstanding stock. The result of this can be measured by pointing to the Bank's balance sheet which surged from 58.1 trillion yen at the end of October 1997, just prior to the onset of the crisis, to a peak of 91.5 trillion yen at the end of the fiscal year in March, an annualised growth rate of nearly 200 per cent (Figure 25). Some of that can be attributed to double counting,[62] but even after correcting for that, the rise was remarkable, representing some $210 billion or nearly 5½ per cent of annual GDP. But despite its best efforts, there was a noticeable increase in short-term market rates over the early winter, as exemplified by the rate on three-month certificates of deposit, initially as a result of a flight to quality: rates on government bills actually fell during the early part of the crisis period (Figure 26, Panel A).[63]

As mentioned in the previous chapter, one of the effects of the multiple financial failures in late 1997 was to make the market much more sensitive to credit risk. To counter this as well as the usual financial year-end liquidity risk

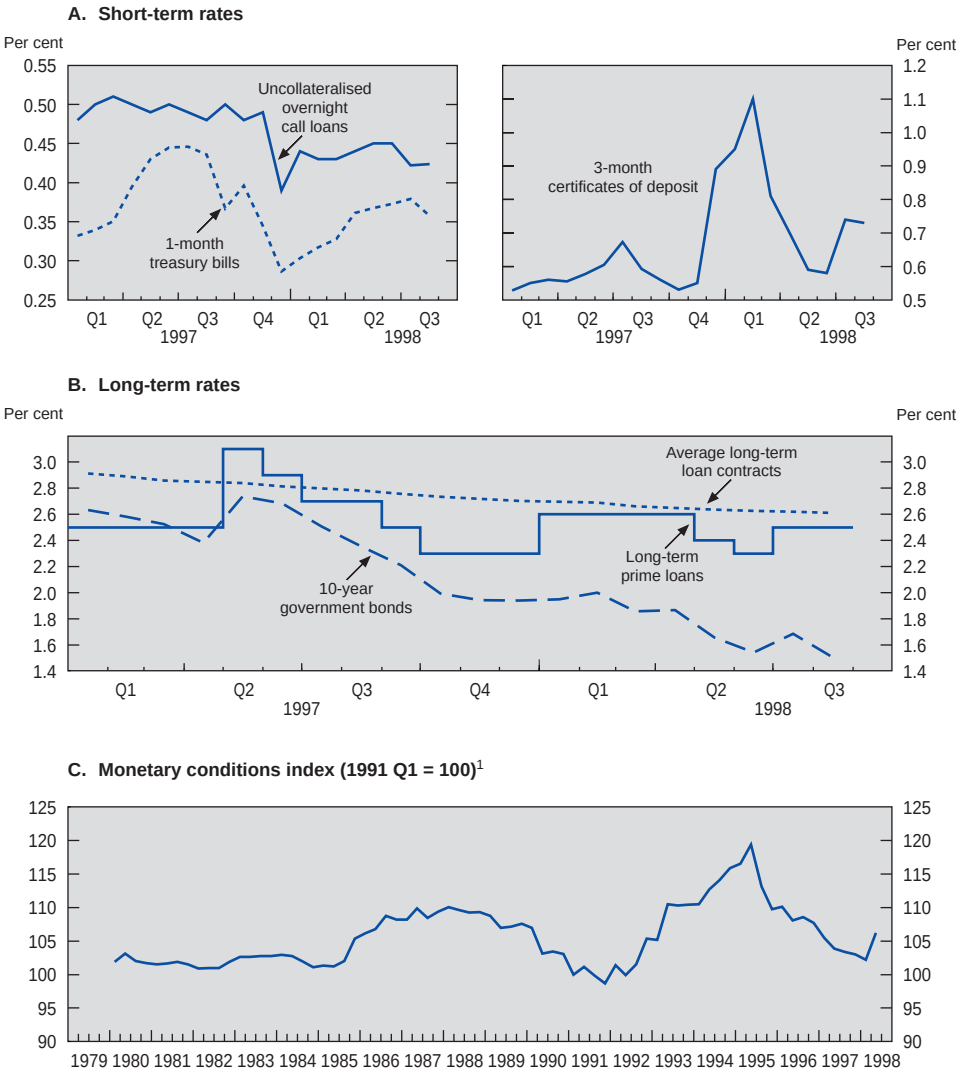
Figure 25. **BANK OF JAPAN ASSETS**

End of period



Source: Bank of Japan.

Figure 26. **INTEREST RATES AND MONETARY CONDITIONS**



1. Defined as the level change in the real three-month deposit rate adjusted by the four-quarter percentage change in the GDP deflator (adjusted for the 1989 and 1997 tax hikes) plus one-quarter of the percentage change in the real effective exchange rate of the yen (unit labour cost definition) against 29 OECD and 12 non-OECD trading partners. The weight (one quarter) was derived from INTERLINK model simulation properties.

Source: Bank of Japan and OECD.

– which, in 1998, was enhanced by the uncertainty and fear surrounding the advent of Prompt Corrective Action (PCA) – in the winter the Bank also provided the market with a great deal of one- to three-month funds by buying bills and corporate paper with maturity beyond the end of March.[64] Offsetting that, it drained shorter maturity funds by selling Bank bills. In the Spring, there were reports that banks had raised effective lending rates for corporate borrowers by 25 to 50 basis points since the turn of the fiscal year in order to start charging for credit risk. However, official figures for domestically licensed banks show only a slight uptick in rates charged for new long-term loans in April, which was, in any case, largely reversed in May, when new loan rates over all maturities averaged only 1.85 per cent. In the summer the Bank announced that it would undertake similar injections to overcome upward pressure on market rates before the end of the first half of the fiscal year[65] and that it might review and update its methods of market operations to reflect ongoing changes. The Policy Board also began to deliberate on the possibility of buying corporate bonds to provide added liquidity.

Contractionary influences have also been offset to some extent by movements in longer-term interest rates, which have continued to trend down (Figure 26, Panel B), with only a fairly brief interruption during the November-January crisis period (when liquidity preference rose strongly and rates moved up across the board) and another reversal in June, when action was taken on the banking front and talk of possible permanent tax cuts first surfaced. The fundamental reasons for lower long-term government bond rates have been:

- lower inflation expectations (indeed, the risk of deflation has again come to the fore, as it did in the strong yen period of the mid-1990s);
- a reduced demand for capital by the private sector in light of initially stagnating and eventually collapsing business investment activity;
- the term-structure effects (related to the first two factors) resulting from the spreading belief that short-term interest rates would remain low for the indefinite future;
- and, to a lesser extent, the aforementioned flight to quality because of the bankruptcy risk associated with private-sector borrowers.[66]

The ten-year bond rate reached record low rates of less than 1.5 per cent in early June, and the rate on the benchmark government bond (maturing in September 2005) touched 1.115 per cent on 2 June, a level said to have surpassed

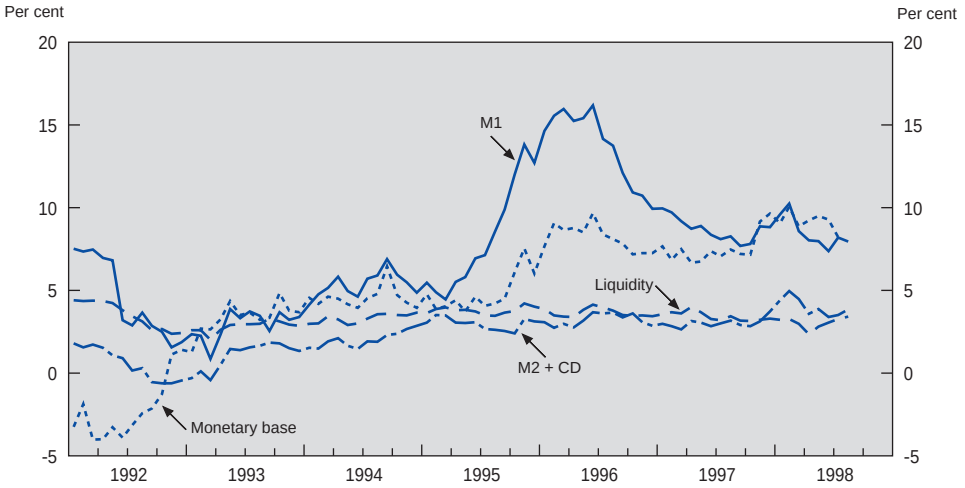
the all-time global low of 1.125 per cent set on four- to five-year loans in Genoa in the year 1619;[67] more recently it has even dipped to as low as 0.7 per cent. The resulting flattening of the yield curve might be thought of as a sign of probable further deterioration in economic conditions, but recent evidence has shown that the term structure contains far less information in the Japanese case than it does for other major industrialised countries (Bernard and Gerlach, 1998), and it may be the case that its slope is less meaningful when interest rates are near zero.

Up against a liquidity trap

But the impact of keeping the financial system flush with liquidity has not been entirely as hoped for. Rather than showing up as faster growth in money and real activity, or at least in a generalised rise in asset prices, much of the effect seems to be initially difficult to account for, thereby earning the situation the label ''liquidity trap''.[68] There are several possible explanations. Some of the high-powered money has been absorbed by the banking system, most of which may be being used to deal with cash-flow shortages resulting from the expansion in non-performing loans (NPLs). In any case, banks' usual credit creation function has probably been impaired by their balance sheet problems. A larger part of the ''missing liquidity'' has been accounted for by unusually large increases in cash in circulation, especially in the form of high-denomination (10 000 yen) bills, whose value jumped by 4.2 trillion yen (10.3 per cent) in the year to end-April due to the rising fear of bank failures and the continued low level of interest rates.[69] The year-long downtrend in narrow money growth was brought to a halt and even temporarily reversed around year-end, with M1 rising in a range of 8 to 10 per cent year on year (Figure 27). But given the apparent reluctance of the banking system to make loans and discounts and the weak demand for them (see Chapter I), the money multiplier (the ratio of broad money to the monetary base) has continued to trend down, although no sign of a quickening in the rate of decline is apparent. Thus, growth in broad money (M2 plus CDs) briefly edged up from a stable year-on-year rate of around 3 per cent through most of 1997 to a peak of 5 per cent in February 1998, before heading back down.[70] Broadly-defined liquidity and credit did not even manifest that temporary pickup, however, with their annual growth rates holding near 3 per cent. Another source of leakage of the additional liquidity has been a decrease in foreign exchange reserves (see below).

Figure 27. **MONEY SUPPLY**

Per cent change over 12 months



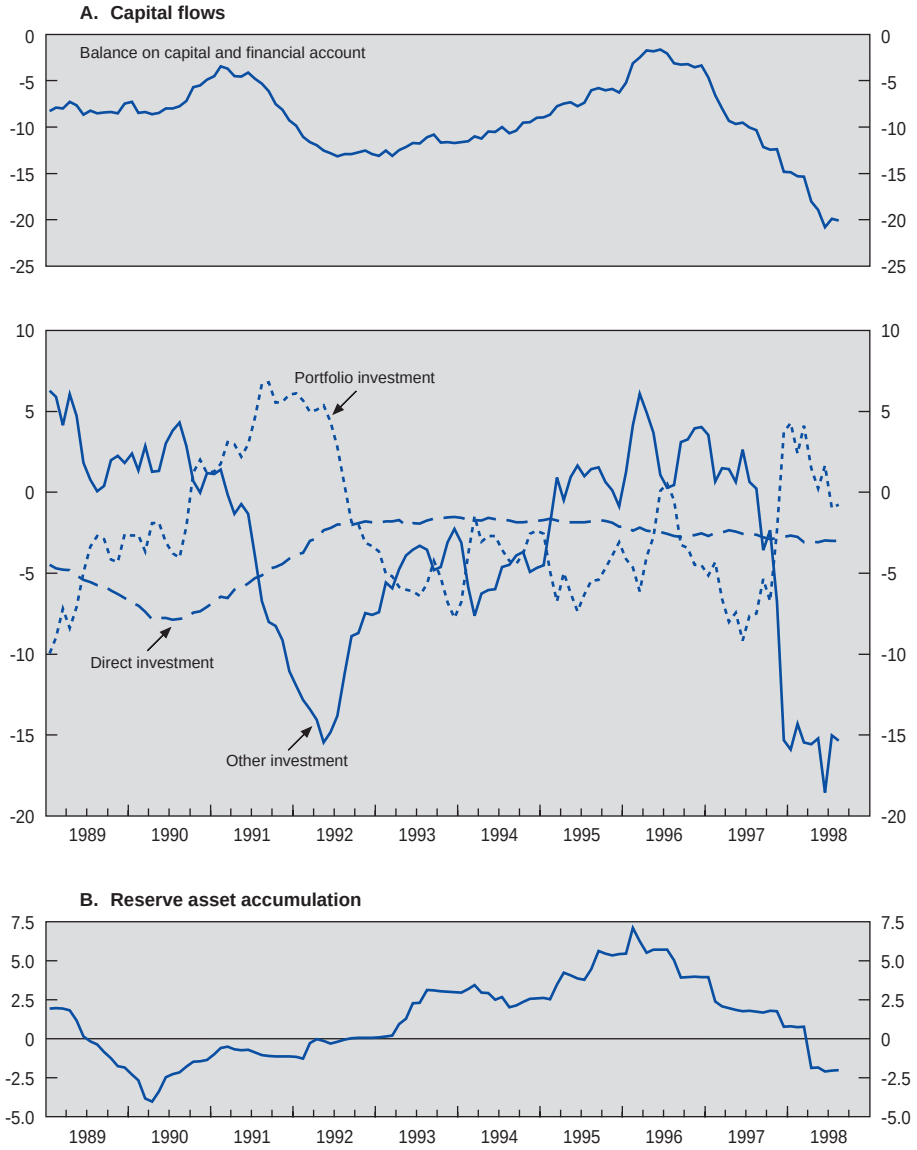
Source: Bank of Japan.

Capital outflows and downward pressure on the yen

Capital outflows have been increasing steadily in significance since the middle of 1996. At that point, after a four-year-long contraction, the 12-month moving sum of outflows reached a trough of less than 2 trillion yen (0.3 per cent of GDP), but by April 1998 it had jumped to 18.0 trillion yen (3.6 per cent of GDP), with peaks in December 1997 and April 1998 of around 40 trillion yen at annual rates (Figure 28, Panel A). A small proportion of that is ascribable to increased net outflows on direct investment account, which have continued in the last few years in spite of the falling yen. Portfolio investment flows contributed to the widening capital account deficit for a time in 1996 and early 1997, but since then have moved sharply into surplus: banks and other financial institutions have been selling their holdings of foreign debt securities, perhaps in order to shrink their balance sheets to meet capital-adequacy requirements, and non-residents have been buying increasing amounts of Japanese debt securities, at least until the end of 1997. The biggest shift in flows towards deficit has therefore been in

Figure 28. **CAPITAL FLOWS AND RESERVE ASSET ACCUMULATION**

Yen trillion, moving 12-month sum



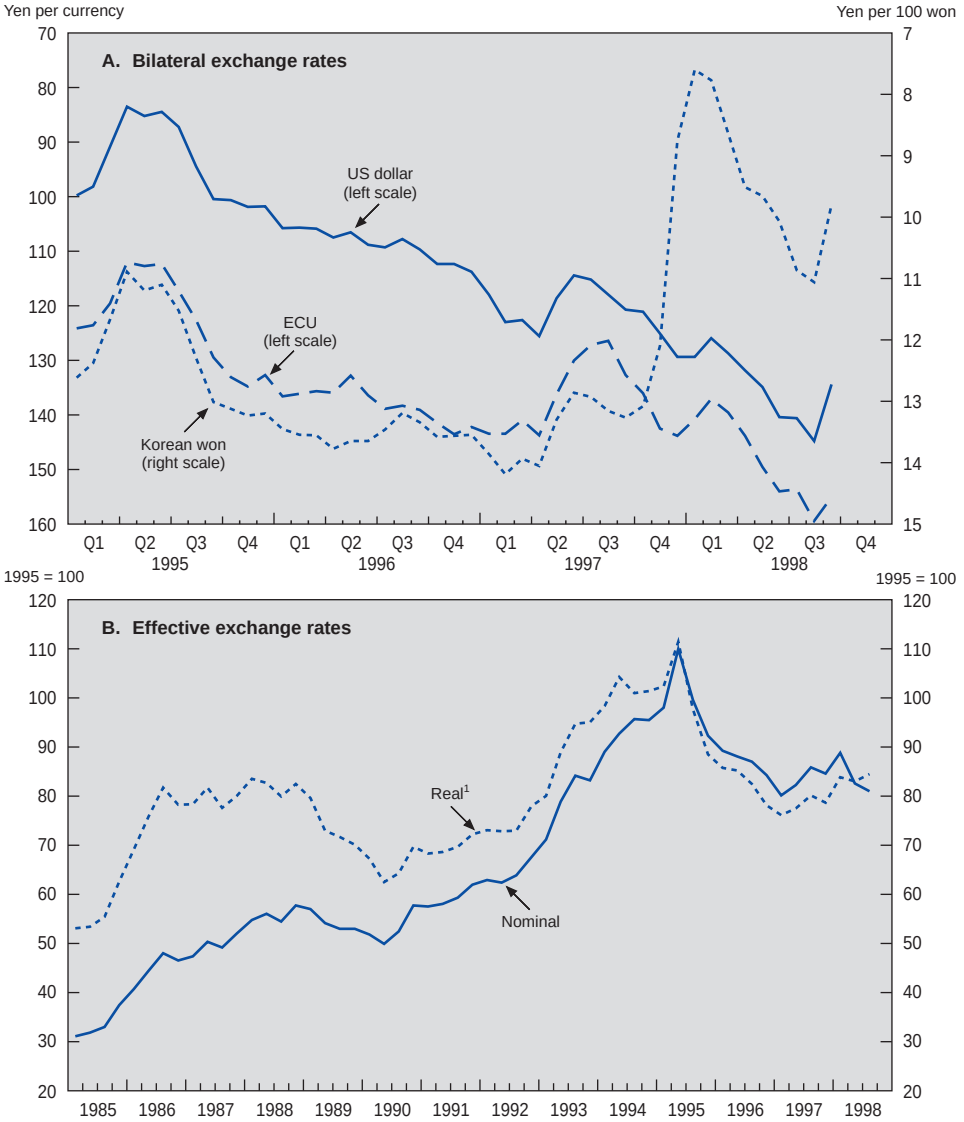
Source: Bank of Japan.

the residual ‘‘other investment’’ category, whose 12-month moving sum was near zero from 1995 to the spring of 1997, before sizeable outflows got under way. Banks once again were heavily responsible, as, despite the aforementioned balance-sheet requirements, Japanese institutions increased their foreign loans and deposits in 1997 and their foreign counterparts reduced their deposits in Japan. Overall, Japanese banks boosted their net foreign asset position (almost entirely in short-term form) by 13.4 trillion yen during the last three quarters of 1997, although more than half of that was unwound in the first quarter of 1998.

These capital outflows have no doubt contributed to the weakness of the yen against the dollar. They are probably in large part attributable to persistent unfavourable, interest differentials against the United States, which have stabilised in the range of 4 to 5 percentage points. Such wide spreads imply – under covered interest arbitrage – that agents expect the yen to appreciate substantially against the dollar over the life of the assets in question, no doubt because of the high and rising current account surplus (which reached nearly 3 per cent of GDP in the first half of this year) and the expected associated accumulation of net foreign assets (unlike the trend over the past 3½ years). However, in order to achieve long-term external equilibrium, if that expected appreciation is to bring the currency to a level which is continuously being revised down, then it will depreciate in the short run to a point from which the appreciation can be expected to occur. This expected appreciation may be another factor holding back domestic investment and boosting domestic saving, thereby lifting the current account further into surplus in the short run and ratifying the original capital outflows.

However, while the yen weakened in almost equal measure against most other OECD Member country currencies over the year to August, it appreciated sharply against the currencies of most other Asian countries, the Korean won for example (Figure 29). With a trade share of over 40 per cent, the appreciation of the yen against these currencies meant that in effective (trade-weighted) terms the currency actually appreciated some 15 per cent since from April 1997 to February 1998, thereby undoing more than one-third of the depreciation incurred since the 1995 peak, preventing overall monetary conditions from easing. Despite a substantial reversal of this appreciation over the following six months, conditions were in fact much tighter than they were in the last recession (Figure 26, Panel C). In real terms the yen appreciation was more limited, as Japan’s price

Figure 29. **EXCHANGE RATES**



1. Trade-weighted effective exchange rate, adjusted for relative unit labour costs in manufacturing, covering 29 OECD countries plus the emerging Asian countries (China; Chinese Taipei; Hong Kong, China; India; Indonesia; Malaysia; the Philippines; Singapore and Thailand) plus Argentina, Brazil and Russia.

Source: Bank of Japan and OECD.

stability has contrasted with persistent inflation registered by its trading partners. Nonetheless, the real effective exchange rate remained quite high over the past year – around the same level as was reached during the boom years of the mid- to late 1980s. Furthermore, it rose substantially further – and monetary conditions tightened sharply – in recent weeks when fears over economic prospects elsewhere, especially in the United States, led to a sudden change in interest rate expectations and an abrupt unwinding of the so-called ''carry trade'' in which highly leveraged investors borrowed yen at low interest rates and exchanged the proceeds for dollars which were then placed in the long-term Treasury bond market. Once the upward move in the yen got under way in late summer, it fed on itself, and the yen rose sharply from a trough of around 147 to the dollar to as high as 112 in early October.

Despite the favourable effect on activity which could have been expected from a lower value of the currency, the authorities on occasion tried to stem its depreciation over the past year: the reasoning was that a declining yen would harm trade prospects of the neighbouring crisis countries, raise the risks of a devaluation of the Chinese currency and augment the risks of protectionist retaliation by trading partners, especially the United States with whom Japan already had a large bilateral current account surplus. It might also have worsended the ''credit crunch'' by boosting banks' foreign asset position, thereby requiring additional capital or, failing that, an offsetting cut in assets, such as loans. Support has most often taken the form of statements favouring a stronger yen, but on occasion the authorities have engaged in active intervention in the exchange markets to prevent excessively rapid declines. Reserve accumulation continued to dwindle in 1997 before reversing as from December (Figure 25, Panel B). The first such intervention took place in April 1998, when markets were reacting to a warning by a foreign ratings agency that it might cut its rating of Japan's sovereign debt. Reserve assets fell by more than twice the previous largest decline ever recorded. But the first-round impact was very short-lived, as there was no follow-up and the fundamentals had not changed, though the market might have become more sensitive to the risk of yen appreciation. In June there was another episode of intervention, this time in co-ordination with the US authorities. This round seems to have been more effective, most likely because it was joined by the United States, even though the immediate gains were nearly all reversed within about a week, at which point the authorities began to

reveal their ''total plan'' for the banking system (see below) and the markets took heart. The effects of the intervention on the domestic money market were quickly sterilised in order to keep liquidity flush at a time when the ''Japan premium'' was building again (to around 50 basis points) and downward pressure on the share price of the Long Term Credit Bank was growing. Nevertheless, the apparent appetite for dollar purchases by financial institutions around mid-year was so strong[71] that they were willing to bid up three-month Euroyen rates to 0.8 to 0.9 per cent (near which it remained through the summer) in order to swap the proceeds into dollars. Some members of the governing party even began to demand the imposition of short-term capital controls.

The new Bank of Japan Law and the monetary policy debate

In the midst of these difficult times facing the monetary authorities significant changes were made to the monetary policymaking process. The new Bank of Japan Law, designed to enhance independence and transparency, took effect on 1 April 1998. Its main features were to increase membership of the Policy Board from seven to nine (of which only three are insiders), to require Diet approval for the Cabinet nominations for all Board members and to require the Bank to hold Monetary Policy meetings regularly and to publish their minutes quickly. In addition, as a result of retirements and the resignations of a number of senior officials who accepted responsibility for scandals involving bribery and early access to Bank information, only one of the Bank's top six officials from 1997 remained in office after the new law took effect.

The published minutes reveal that the debate about the direction of policy has been quite lively. The Board has actually debated whether interest rates should *rise* (as many outsiders have demanded) in order to defend the exchange rate and to support the incomes of those who rely on interest income and to ease the burden on insurance companies who are struggling to find ways of earning sufficient investment returns on their assets to honour their obligations.[72] But, at its May meeting, members shared the view that ''raising interest rates was not a subject for further discussion at present'' in view of the problems it would create for the real estate, construction and financial sectors and the adverse impact on overall employment. The Board was, in fact, closer to making a further *cut* in policy-controlled rates, especially given the ongoing decline in inflation expectations which has raised real rates. But most members of the Board preferred to

wait and see the effects of the aforementioned budgetary stimulus package over the summer before resorting to further policy easing. In the event, with the yen showing signs of stabilisation, a further decline in real output in the second quarter and a lack of any sign of improvement in the third quarter, and a deepening sense of crisis in global financial markets, in September the Bank decided to guide the call rate down to 0.25 per cent, its first cut in three years, and to state clearly that it would go further if necessary to stabilise market conditions.

Debate has also focused on whether a new monetary strategy is needed. One possibility would be to target inflation as several other OECD Member central banks have done. In the other countries, however, the idea was to bring down an unacceptably high rate of inflation and/or prevent it from rising to such a rate, whereas in Japan's case advocates suggest that the goal would be to raise the measured inflation rate from current near-zero rates to, say, 1 to 2 per cent per year. Some observers (Krugman, 1998) have called for a similar and superficially more radical approach by advocating that the Bank promise to be ''irresponsible'' and to boost *expected* inflation to something around 5 per cent in order to reduce expected real interest rates to sufficiently negative levels that agents would prefer to spend than hold assets or add to their wealth. In either case it is quite unclear how this could be achieved without the Bank directly assuming substantial credit risks *vis-à-vis* the non-bank private sector. Furthermore, whether it could be managed so deftly as to prevent the process from getting out of hand, given the risks of time inconsistency, and permanently damaging the central bank's reputation, especially in this new era of greater independence, is doubtful. Finally, targeting some money or credit aggregate would require volatile interest rates and may still run up against zero interest-rate constraints.

Dealing with the banking system's balance sheet problems

As shown in Chapter I, the long-standing concerns over the banking system's problem loans, which have largely resulted from the asset price deflation of the 1990s but also from the lack of consideration of credit risk throughout the economy, burst forth in late 1997 in the form of a series of financial institution failures. It then became clear that the authorities had first to stabilise the situation and then to confront the market's growing conviction that the

economy could not escape from the doldrums without a full and complete disclosure and purging of all bad loans from banks' balance sheets, a move which would require substantial public monies. The willingness to use public funds for this purpose had been lacking ever since the 1995-96 *jusen* episode when the government reluctantly agreed to use 680 billion yen (about $7 billion) of such funding to bail out these specialised housing loan providers. This was an extremely unpopular decision, and the authorities were not prepared to repeat it on a much larger scale without clear evidence of the inevitability of such action. The financial institution failures provided just such evidence.

Provision of public funds

In late December the government announced that the Prompt Corrective Action powers would be applied in a flexible manner for a year (until April 1999) for banks not operating internationally and that up to 30 trillion yen (about $240 billion) of public funding would be provided by the end of the 1997 fiscal year through the Deposit Insurance Corporation (DIC), ten trillion of which in the form of government bonds and the rest as borrowing with government guarantees. The magnitude of the funding made available was possibly determined by the amount of problem loans[73] which the banks had just declared in their half-year financial statements dated end-September: according to official figures these reached 28.1 trillion yen (4.0 per cent of total loans and 5.6 per cent of GDP), slightly higher than the previous estimate of 27.9 trillion yen at end-March.[74] Another way of assessing the loan books of the banks is to classify them by the probability of their being repaid. In 1997 the government had asked the banks to evaluate their loans and to allocate them to one of the four classifications that its banking inspectors use to evaluate risk. The outcome, reported in Table 23 (Panel A), showed that already early in the latest economic downturn, only 87.7 per cent of all bank loans were seen as reasonably certain to be repaid. Of the other 76.7 trillion yen worth of loans, the large majority were those that were judged to be recoverable but would have to be collected carefully (Class II or substandard), while only the residual 11.4 trillion yen in the two lowest classes were clearly bad loans. One important question that remains is what proportion of Class II loans is already[75] or will ultimately become non-performing. Some evidence (Bank of Japan, 1998*a*), drawn from a sample of 18 banks, points to a figure of around one-sixth of Class II loans in 1993-94 that had to be disposed of

Table 23. **Loan classification based on banks' self-assessment**[1]

Yen billion

	Total credit exposure	*of which:*				Total of II through IV	Per cent of total exposure
		Class I	Class II	Class III	Class IV		
A. March-September 1997							
Major banks	431 682	377 347	45 302	6 931	2 103	54 336	12.6
Regional banks	139 153	124 692	13 060	998	403	14 461	10.4
Regional banks II	54 028	46 118	6 927	794	189	7 910	14.6
Regional banks total	193 181	170 810	19 987	1 792	592	22 371	11.6
Total all banks	624 863	548 157	65 289	8 723	2 695	76 707	12.3
B. 31 March 1998							
Major banks	452 374	402 018	45 418	4 816	122[2]	50 234	11.1
Regional banks	145 492	130 245	14 414	833	0	15 247	10.5
Regional banks II	50 640	44 285	5 931	424	0	6 355	12.5
Regional banks total	196 132	174 530	20 345	1 257	0	21 602	11.0
Total all banks	648 506	576 548	65 763	6 073	122[2]	71 836	11.1

Notes: Definitions on class:

I. Credit exposures which have not been classified as Class II, III or IV.

II. Credit exposures on which banks have judged adequate risk management on an exposure-by-exposure basis will be needed.

III. Credit exposures on which banks have serious concerns in terms of their ultimate collection and, thus, are likely to incur losses but have difficulties with rational estimation of when or how much losses will actually occur.

IV. Credit exposures which banks have judged to be non-collectable or of no value.

1. Data is aggregation of each bank's data collected by the Ministry of Finance between March and September 1997 and the Financial Supervisory Agency at end-March 1998. A number of closing and merging banks are excluded.

2. Due to write-offs, there are virtually no Class IV loans at the financial year end (31 March). However, where banks' trust accounts have a different year-end, some Class IV loans remain.

Source: Ministry of Finance and Financial Supervisory Authority.

(provisioned or written off) within three years. Another trenchant question is how many bad loans are held by affiliated companies and, therefore, not included in the banks' total.[76] These questions should be answered next year when full consolidation for certain affiliates where there is less than a 5 per cent equity holding, will be required.

The 30 trillion yen of public funds was comprised of 13 trillion yen to bolster bank balance sheets and 17 trillion yen to prop up the deposit insurance system. This dual approach amounted to a carrot-and-stick strategy: the smaller amount provided an incentive for banks to speed their write-offs and replenish their capital, while the larger amount was an implicit threat that the authorities could henceforth allow any institution to fail without having trouble finding the funds to guarantee deposits.[77]

The bank aid legislation was approved by the Diet in February, and a seven-person Examination Board of Financial Crisis Management (EBFCM) quickly set about fixing conditions for banks to apply for a share of the 13 trillion yen. Initially, it appeared to the banks that the conditions might be draconian, with a substantial risk of closure for several of the major institutions, and, as a result there were few applicants. In the end the authorities asked the industry not to undermine the credibility of the whole endeavour and to accept the funding on the promise of fair treatment. On the basis of a seven-page set of eligibility criteria,[78] 21 institutions (including all but one of the major banks) made formal applications in March. Nearly all asked for 100 billion yen, mostly in the form of subordinated debentures or loans with a call option at the discretion of the banks.[79] The final allocation is summarised in Table 24. The total injection was only 1.8 trillion yen (or 4.7 per cent of total Tier I plus Tier II capital) out of the 13 trillion yen made available, and only two applicants were not given what they applied for.[80] The average cost of the capital injection to the borrowing institutions appears to have been lower than what they would have had to pay in the market: for example, Sumitomo Bank had just borrowed $1.8 billion (about 234 billion yen) at LIBOR plus 380 basis points and the Industrial Bank of Japan $1 billion in the form of perpetual preferred securities[81] with a coupon of 8.79 per cent (325 basis points over LIBOR).[82] To judge whether the structure of rates charged across the different institutions was equitable it is possible to compare them with their financial ratings, as given in the Table. Differences in rates reflected the instrument mix more than the credit risk.[83]

Table 24. **Summary of public capital injection, March 1998**

	Total injection (billion yen)	*of which:* Preferred shares (billion yen)	Subordinated debentures (billion yen)[1]	Subordinated loans (billion yen)	Dividend rate %	Interest spread over Euroyen (%) First 5 years	Thereafter	Credit ratings March 1998 Standard and Poors	Moodys[2]
City banks									
Tokyo Mitsubishi	100		100			0.9	2.4	A+	B(–)
Dai Ichi Kangyo	99	99			0.75			BBB+	C(–)
Sakura	100		100			1.2	2.7	BBB	D+
Sumitomo	100		100			0.9	2.4	A–	C
Fuji	100		100			1.1	2.6	BBB+	D+
Sanwa	100		100			0.55	1.25	A–	C
Tokai	100			100		0.9	2.4	A–	D+
Daiwa	100			100		2.7	3.95[3]	BBB	E
Asahi	100			100		1	2.5	A–	D+
Long term credit banks									
Industrial Bank of Japan	100		100			0.55	1.25	A–	C(–)
Japan LTCB[4]	176.6	130		46.6	1	2.45	3.95	BBB	E
Nippon Credit Bank[4]	60	60			3			n.a.	E
Trust banks									
Mitsubishi Trust Bank	50		50			1.1	2.65	A–	D
Sumitomo Trust Bank	100		100			1.1	2.6	A–	D
Mitsui Trust Bank	100		100			1.45	2.95	BBB+	E
Yasuda Trust Bank	150		150			2.45	3.95	BB+	E
Toyo Trust Bank	50		50			1.1	2.6	n.a.	D
Chuo Trust Bank	60	32		28	2.5	2.45	3.95	n.a.	E(+)
Regional banks									
Yokohama Bank	20			20		1.1	2.6	BBB	D+
Hokuriku Bank	20			20		2.45	3.95	n.a.	E
Ashikaga Bank	30		30			2.95	4.45	n.a.	E
Total	1 815.6	321	1 080	414.6					

1. These debentures are generally of a consol nature and are therefore considered upper Tier II capital. The only exceptions are those issued by Sanwa Bank and the Industrial Bank of Japan whose debentures are of fixed (10-year) duration and are therefore lower Tier II capital, which is limited to no more than half of Tier I capital.
2. Possible change is shown in parentheses.
3. Applies only after ten years and three months.
4. These banks were granted only part of the injection for which they applied.

Source: Deposit Insurance Corporation and ratings companies.

What was achieved by the injection? According to the EBFCM it prevented a small decline in the capital adequacy ratios of applicant banks and allowed a small rise instead (Table 25), with the biggest favourable impact on the ratios of the long-term credit and trust banks. But there was no real consolidation that resulted, possibly intentionally because of the fear of intensifying the credit crunch by ending traditional banking relationships (see below). Rather, each bank set its own targets in a number of dimensions for the next three years. The banks' goal is to cut the number of directors by 23 per cent, the number of employees by 13 per cent and the number of outlets by 10 per cent in Japan and by 26 per cent abroad. However, no penalties will be imposed if these objectives are not realised and, even if they are, their impact on profitability will be limited, as labour costs are tiny in relation to loans outstanding.[84] And moral hazard might well be worse in the future because of the blanket nature of the programme. But, most importantly, without any reduction in the supply of banking services, healthier banks

Table 25. **Impact of public capital injection on applicant banks' capital adequacy ratios**

Trillion yen, except as otherwise noted

	Year[1]	Total	City banks	Long-term credit banks	Trust banks[2]	Regional banks[3]
Own capital	1997	41.7	29.6	5.3	5.3	1.6
Before injection	1998	38.8	27.2	5.3	4.9	1.4
After injection	1998	40.8	28.1	5.8	5.4	1.5
	2001	43.2	30.0	6.0	5.6	1.5
Of which:						
Tier I capital	1997	21.5	14.9	2.7	3.0	1.0
Before injection	1998	20.8	14.3	2.8	3.0	0.9
After injection	1998	21.2	14.4	3.0	3.0	0.9
	2001	23.7	16.2	3.2	3.4	0.9
Risk assets	1997	422.9	284.0	65.3	55.3	18.3
Before injection	1998	401.2	273.4	59.3	51.1	17.4
After injection	1998	406.3	277.2	59.8	51.9	17.4
	2001	395.9	266.1	61.4	51.6	16.9
Capital adequacy ratio (%)	1997	8.9	9.0	8.1	9.5	8.9
Before injection	1998	8.8	8.6	9.0	9.6	8.8
After injection	1998	9.1	8.8	9.7	10.5	9.1
	2001	9.9	9.8	9.7	10.9	9.9

1. 31 March. Figures for 2001 are banks' own projections.
2. Excluding Nippon Trust Bank which did not apply.
3. Includes only Yokohama Bank, Hokuriku Bank and Ashikaga Bank.

Source: Financial Crisis Management Committee.

were implicitly penalised, as they will face continuing downward pressure on loan margins from their weaker rivals who might otherwise have had to exit or to seek a merger partner as the Long Term Credit Bank did only three months later.[85]

Persistent disclosure concerns

The lack of adequate disclosure and transparency in financial statements has been of long-standing concern to analysts of the problems of Japan's financial institutions.[86] Thirty-five client firms helped Yamaichi, for example, hide enormous so-called *tobashi* losses off its books. In that case the surprise was compounded when in June it was revealed that such unreported liabilities were in fact 291 billion yen rather than the originally announced 265 billion, thereby moving the firm to a position of insolvency and imposing losses on the mainly insurance-company holders of its subordinated debt.[87] Bad loan disclosure has been a particular problem, with far higher amounts revealed when the institutions which collapsed have opened their accounts up to scrutiny.[88] The result is twofold: first, investors and analysts alike are sceptical of any official and unofficial figures for the size of the non-performing loan problem. Second, an added risk premium directly attributable to a lack of disclosure has become evident: a new form of security called ''collateralised loan obligations'' (CLOs), backed up by portfolios of loans to similar US blue-chip borrowers, yields 7 to 10 basis points more when issued by a Japanese bank than a US bank.[89] It is therefore crucial that, at a minimum, the authorities follow through with the promise that problem loans will be reported on a consolidated basis with world class accounting and disclosure standards by the end of the current financial year.[90]

The authorities made a number of important decisions regarding banks' accounts for the 1997 fiscal year. On the one hand they required them to move to US standards for reporting their problem loans. In particular, loans would be considered to be non-performing if they were as little as three (rather than six) months in arrears and also if they were restructured at all (rather than only if the interest rate was cut below the discount rate, currently 0.5 per cent). However, in order to mitigate banks' reluctance to lend, they also announced three changes which all had the effect of improving the capital adequacy ratio[91] (used to determine permission to maintain international operations) without changing underlying balance sheet conditions, and which may have further harmed the

credibility of the accounting system (Ito and Szamosszegi, 1998). First, following authorisation by the Basle Committee on Banking Supervision and the practice in some OECD countries such as Germany and the United Kingdom, they allowed banks to write up the value of their land assets from acquisition cost (the norm in most OECD countries) to current market price, which even if much depreciated in recent years, provided an even bigger boost to own capital than the public capital injection.[92] Second, somewhat inconsistently, they let banks choose whether or not to shift the valuation of their stock portfolios back to acquisition cost in order to avoid the sensitivity of their balance sheets to the drop in equities prices. All but three of the major banks took advantage of this opportunity.[93] Last, following international practice, they allowed banks to deduct from loan balances the amount of time deposits held by the client at the bank to the extent that they mature after the loan comes due; this is estimated to have raised the BIS ratio by an average of around ¼ percentage point at the nine city banks.

In the event the major banks were nearly all able to meet the ratio with a comfortable margin to spare at the end of FY 1997: as shown in Table 11 (Chapter I), the average outcome was 10¼ per cent, 1.3 percentage points higher than the previous year. Operating profits shrank 22 per cent,[94] and after disposing of around 10¾ trillion yen in bad loans,[95] even larger gains on the sale of equities than last year were not enough to prevent a huge current loss. Self-assessment of loan classification (Table 23, Panel B) showed that write-offs had resulted in a near-stabilisation of Class II loans and progress in reducing exposures subject to even greater risk as a result of write-offs. Non-performing loans were provided on the basis of both the old standards and the new (which include loans in arrears by three, rather than six, months or more and restructured loans with any changes in terms or conditions, rather than just those whose revised interest rate is below the official discount rate). The former showed a slight reduction over the year in absolute terms (from 27.9 trillion yen to 25.0 trillion yen for all deposit-taking institutions) but a slight rise as a share of total loans. Shifting to the broader definition boosted the aggregate for major banks by 7.5 trillion yen (51 per cent) to 22 trillion yen (6.0 per cent of total loans) and by 10.2 trillion yen (41 per cent) to 35.2 trillion yen (5.1 per cent of total loans) for all deposit-taking institutions. Finally, loan loss reserves at all banks represented 59.9 per cent of estimated non-performing loans under the new standards, ranging from a low of 40 per cent to as much as 86 per cent. However, those estimates may quickly

prove too optimistic. For example, reserves against Indonesian exposure ranged from less than 6 per cent up to 19 per cent, whereas non-Japanese banks are said to have reserved around 30 per cent.

Moving towards a ''total plan''

But the pressure coming from the financial markets did not abate after the turn of the fiscal year nor after the major banks reported their financial results in late May. The chorus of voices grew louder in demanding further government action to deal with the banks' balance sheet problems, the ''Japan premium'' began to widen again, and, as banks' ratings were being downgraded, their shares fell, in some cases, sharply. The authorities had already twice announced packages of changes which they believed would ease the situation by encouraging the banks to dispose of their foreclosed real estate collateral.[96] In October 1997 they had promised to suspend the 0.15 per cent national land holding tax,[97] to review the regional tax on the sales of fixed assets and land and to ease the rules on creating firms to securitise real estate loans; these changes were implemented in FY 1998. In April 1998 they had agreed to set up an *ad hoc* committee to try to quickly settle disputed claims on real estate collateral[98] and to ease the rules pertaining to the auctioning of such collateral.[99] They had also announced that private debt collectors (''servicers'') would be allowed to be established and that the tax law would be changed so that selling problem loans at a loss would no longer usually be treated as a taxable gift. The package also contained a series of moves to promote securitisation and the establishment of an asset-backed securities (ABS) market. Finally, they had indicated that the private Co-operative Credit Purchasing Corporation (CCPC),[100] whose mandate to buy loans had expired at end-March, would see its powers renewed and strengthened.

But in June the government began to promise a ''total plan'' to rehabilitate the banking sector. The plan was worked out by July, and after the summer elections, it was presented to a special session of the Diet. Heavily revised, it was passed in October. Part of the scheme, loosely based on US practice,[101] would work as follows. The new Financial Supervisory Agency (FSA) and the Bank of Japan have been between them undertaking audits of all major banks in recent months. The authorities would use already existing PCA powers in any cases where capital adequacy is found to be substandard. But the new legislation allows it to dismiss existing management and dispatch administrators to any bank determined to be insolvent.[102] These trustees would continue to lend to sound

borrowers so that a chain reaction of bankruptcies does not ensue. If no private institution can be immediately found to absorb the bank, then its Class III and IV loans (as determined by the Examination and Judgment Committee of the EBFCM) would be sold at market prices to a new institution patterned after the US Resolution Trust Corporation formed out of a merger of the Resolution and Collection Bank (RCB)[103] and the Housing Loan Administration Corporation, while remaining assets and operations would be transferred to a new, specific ''bridge bank''. That bank would formally be owned by the DIC. It would then engage in selling the transferred assets. Such bridge banks would normally have a one-year lease of life. If, at the end of that time, no buyer or merger partner had yet been found, then any remaining operations would have to be liquidated.

The legislation approved by the Diet authorises the imminent establishment of a new ''financial revitalisation committee'', composed of a chairperson (Minister of State) and four prominent private individuals. This committee will oversee the FSA in its inspection and supervision activities and will also assume responsibilities for financial crisis management, dealing with failures and designing systems for resolving failed institutions. While the bridge bank scheme was retained, the menu of choices available to regulators was broadened to include temporary nationalisation through March 2001, a procedure almost certainly to be used for the Long Term Credit Bank. The original funding mechanism decided last winter was abolished. In its place a second supplementary budget for FY 1998 earmarked a much larger amount for dealing with the banking system problems. In total, 60 trillion yen (over 12 per cent of GDP) will be available; of these 17 trillion yen from the original plan remain on hand to compensate depositors in failed banks, while an additional 43 trillion yen (rather than the original 13 trillion yen) will be at regulators' disposal,[104] consisting of 25 trillion yen for recapitalising weak but viable banks and 18 trillion yen for actions such as the establishment of bridge banks and temporary nationalisation of weak banks. The legislation makes public capital injections possible at any capital-adequacy ratio, but only at the bank's request. Closure is not obligatory at any ratio; for those which are ''critically undercapitalised'' (whose ratios are below 2 per cent for internationally active banks or 1 per cent of those without any international operations), the committee can force banks to choose between recapitalisation, a merger, significant reduction of their operations or closure. Management is to be replaced only at significantly or critically undercapitalised banks (where capital falls below half of targeted levels).

The plan provides the authorities with the means to carry out sectoral restructuring and mitigate the fall-out on credit supply, a primary concern after the closure of Hokkaido Takushoku Bank led to an especially severe downturn in its regional base of Hokkaido. But whether it will succeed is unclear because of a number of uncertainties.[105] First, the authorities will have to be willing to resort to the powers they have allotted themselves if there is ultimately to be any further reduction in the number and lending capacity of banks. One major worry is the limited size, independence and experience of the FSA. It has only 151 permanent inspectors at headquarters plus 420 at regional bureaus under its authority and acknowledges a need for more.[106] Another question is whether or not any *major* banks (other than LTCB) that might truly be insolvent could be so declared in view of multiple recent official pronouncements that such is not the case; major banks hold about two-thirds of all non-performing loans and whether capacity can be sufficiently shrunk without further withdrawal by some of them remains to be seen.[107] In addition, the issues of whether banks under special public administration may be subject to bureaucratic or political intervention and whether the plan entails an implicit guarantee to borrowers – solvent or otherwise – have been raised. Finally, well-capitalised banks which do not wish to request public capital injections cannot be forced to accept such funds. They will be faced only with the Prompt Corrective Action powers, which have yet to be used, and exhorted to disclose and write off their bad loans, much as they have been in the past. However, whether or not to seek the power to force such injections is now being considered.

The plan should suffice to avoid any near-term crises, even if it is less clear whether it will alleviate the credit crunch, especially if major banks choose not to apply for funds, preferring to try and meet capital adequacy requirements through balance-sheet shrinkage and to avoid the possibility of management replacement and sanctions and shareholders' equity mark-downs. As with the early-1998 plan, it is uncertain whether the injection of public funds will be conditional on substantial restructuring and what role moral hazard effects will play. Finally, while the ultimate fiscal costs are unknown, they are likely to be large if the plan is successful, but the macroeconomic costs could well be lessened, depending on how the funding is financed and how implementation is perceived by Japanese households and resident and non-resident investors alike.

Recent developments and the short-term outlook

The national accounts figures for the first half of 1998 showed the economy was in critical condition, with real GDP falling by 3.8 per cent (s.a.a.r.).[108] Negative growth outcomes have now been recorded in three consecutive quarters following the brief slight rebound during the summer of 1997. Household spending decreased in the first semester, with falls of 1.3 per cent in private consumption and 2.0 per cent in residential investment. Monthly indicators show both aggregates are still sluggish and have probably declined once again in the third quarter; only household appliance sales have been rising in recent months. Rising inventories of unsold houses have led to further reductions in housing starts, portending persistent cuts in residential construction over the rest of the year. On a year-on-year basis private consumption expenditure has declined for ten consecutive months (Table 26), despite the tax cuts conducted in February and over the summer. The propensity to consume out of disposable income picked up in the second quarter, but fell back below 70 per cent again in July and August. These weak developments have been affected by the steadily worsening labour market situation. The unemployment rate rose rapidly throughout the first half of the year, reaching a peak of 4.3 per cent, the highest on record, in June and August. Increases in joblessness have been especially large for youths and older workers, reflecting both severe corporate restructuring and a lack of hiring. The job offers to job seekers ratio declined to 0.50 in July and August, the lowest ever. Total wage and salary payments began to decline year on year in January and have continued to deteriorate since then.

The business sector was a drag on the economy in the first half. Attempted inventory adjustment began cautiously, despite the significant decline in industrial production that began in the fourth quarter of 1997. With demand falling even faster, stocks continued to build and by April reached their highest level in relation to shipments since January 1975. That ratio started to decline at last only in May, but progress has thus far been meagre, especially for production goods, as demand has fallen nearly as fast as production. Further cuts in output were made from the second quarter on in several industries such as automobiles, steel and chemicals. In addition to fundamentals, the Asian and financial sector shocks eroded business confidence and brought about a fall of 14.9 per cent (s.a.a.r.) in business investment in the first half. The June and September 1998 *Tankan* surveys by the Bank of Japan showed substantial further deterioration in

Table 26. **Recent monthly indicators**

Year-on-year growth rates

	1997		1998		1998		
	Q3	Q4	Q1	Q2	June	July	August
Household consumption expenditure (all households)	1.9	–2.3	–4.8	–1.3	–1.0	–4.0	–2.4
Large retail store sales	–1.6	–3.2	–8.2	–0.7	–4.8	–4.0	–5.1
Auto sales	–10.3	–13.7	–21.5	–2.1	–2.2	–0.5	–3.1
Household appliance sales	–3.7	–4.5	–14.5	3.6	3.2	8.3	6.3
Housing starts (s.a.a.r. 10 000 units)	131	129	130	124	121	110	118
Machinery orders	1.3	–13.4	–5.8	–21.7	–18.6	–24.1	–25.0
Public works contracts	–1.7	–9.1	–2.2	–6.2	0.8	–10.7	3.5
Exports volume (s.a.)[1]	1.2	2.4	–3.4	–0.6	–2.4	2.9	–3.5
Imports volume (s.a.)[1]	2.9	–2.3	–0.4	–6.7	9.4	–0.4	–2.6
Production (s.a.)[1]	0.2	–2.5	–1.3	–5.1	1.7	–0.6	–1.3
Inventories (s.a.)[1]	1.0	1.6	0.7	–1.9	–0.4	–0.7	–1.3
Inventory ratio (s.a. 1995 = 100)	100.3	107.0	112.9	111.0	111.0	111.2	114.8
Job offers to job applicants ratio (s.a.)	0.73	0.69	0.61	0.53	0.51	0.50	0.50
Unemployment rate (s.a. %)	3.4	3.5	3.6	4.2	4.3	4.1	4.3
Overtime working hours	1.1	–2.5	–5.4	–9.3	–9.9	–8.8	–8.1
Number of employees	1.0	1.0	0.0	–0.6	–0.8	–1.0	–0.3
Total wage payments	1.5	0.7	–0.1	–0.6	–0.4	–2.5	–3.8
Domestic wholesale prices	1.4	0.9	0.3	–2.3	–2.1	–2.2	–2.1
Consumer price index	2.1	2.2	2.0	0.4	0.1	–0.1	–0.3
Money supply (M2 + CD)	3.0	3.3	4.7	3.7	3.4	3.5	3.9
Number of bankruptcies	9.5	19.0	20.9	28.9	22.8	28.4	7.9

1. Change from previous period.

Source: Economic Planning Agency and Bank of Japan.

sentiment. They also showed that business investment plans for this fiscal year are weaker than at the same time last year, especially for small and medium-sized companies. Machinery order statistics and shipments of capital goods also imply further falls in investment since the spring. This weak business-sector activity is likely to have negative effects on the labour market.

As expected from the earlier tight fiscal policy stance, public spending contributed negatively to growth in the first half of the year. Public works contracts have only just begun to show any signs of increasing as from August. The effects of the April fiscal package, including frontloading of the initial

budget outlays, are expected to be felt only slightly in the third quarter and mostly in the fourth quarter because of lags between orders and progress in construction.

As for the external side, exports of goods and services dropped by 6.2 per cent (s.a.a.r.) in the first half because of the ''Asian crisis'' and a slowdown of exports to the United States. As shown in Table 8 in Chapter I, the values of exports to the ASEAN4 and Korea declined by about 30 per cent (year-on-year) in the first and second quarters on a customs basis. Imports of goods and services also fell by 10.7 per cent in the first half because of weak demand. The latest figures show that these downtrends may be coming to an end. With imports falling faster than exports in volume terms and favourable movements in the terms-of-trade, the current account has continued to widen, reaching nearly 3 per cent of GDP on average in the first half of the year

The outlook for the second half of 1998 and 1999

The projections of the second half of 1998 and 1999 are based on the following fiscal assumptions:

- the April fiscal package will be implemented steadily, aside from local governments' own public investment worth 1.5 trillion yen. While this spending might be officially implemented, it is assumed that it is offset by other spending cuts by these governments, given their fragile fiscal position and incoming indicators;
- there will be a permanent personal income tax cut amounting to 4 trillion yen in 1999 – the top marginal rate will fall from 65 to 50 per cent;
- the corporate tax rate will be lowered from 46.36 per cent in FY 1998 to 40 per cent, as the government has promised, which the OECD Secretariat estimates will reduce revenue by about 3.2 trillion yen;
- application of the Fiscal Structural Reform Act will be suspended, and the combined second FY 1998 supplementary and initial FY 1999 budget will include 4 trillion yen for traditional and new infrastructure investment;
- in late 1999, the social security contribution rate will be raised from 17.35 to 19.5 per cent of standard earnings, and the benefits will be re-rated in line with the movement of real average earnings;[109]
- the exchange rate is fixed at a level of 131.8 yen to the dollar.

In the second half of 1998, despite a boost from expansionary fiscal policy, the economy is projected to continue to shrink, contracting by 1.3 per cent (s.a.a.r.), which would result in a fall in real GDP of 2.6 per cent for the calendar year 1998 and 1.9 per cent for the fiscal year (Table 27). Some of the effects of fiscal stimulus are expected to be carried over to the first half of 1999. Consumption is expected to increase only slightly next year. A further deterioration of labour market outcomes is likely to constrain disposable income growth. Progress in corporate restructuring is expected to raise the unemployment rate to 4¾ per cent by the second half of 1999. Businesses are expected to continue holding back on production, until inventories return to normal values in relation to shipments possibly in the second half of 1999, and to similarly restrain capital expenditure until confidence is restored that the upturn has begun. In the central projections, the contribution of net exports to real GDP is expected to shrink from 1¼ percentage points in 1997 to less than ½ percentage point in 1999 because of a loss of competitiveness, still weak export market growth and a slight recovery in imports. Overall, output is expected only to inch up next year, pushing excess capacity up further. In view of such a substantial amount of slack, prices and wages are likely to edge down and imports to grow only modestly. Assuming something of a rebound in export market growth, especially in Asia, the current account could continue to rise, reaching 3¾ per cent of GDP next year.

Major uncertainties number possibly three; either of the first two, if realised, might be sufficient to push the economy into a more intense period of recession and deflation. First, the ‘‘Asian crisis’’ may be more prolonged and severe, continuing its drag on the economy; similarly, contagion effects on other emerging markets may be more marked; and the very recent appreciation of the yen may be sustained. The effects of the significant depreciation of Asian currencies have not yet been felt, at least in bilateral trade, but may be having depressing effects on the economy through increased competition in third-country markets, especially the United States to which Japanese export growth has slowed noticeably. Second, the bad loan problem still needs careful attention. The Japanese banking system remains in fragile condition, and further closures of major banks, even with the new structures in place, could shake consumer and business confidence. Third, the details of the government’s fiscal policy in 1999 are still unclear at this stage: for example, possible changes in personal income tax rates

Table 27. **Short-term outlook**[1]

	1997	1998	1999	1998		1999	
				1st half	2nd half	1st half	2nd half
Demand and output (volumes)							
Consumption							
Private	1.1	–1.8	0.1	–1.3	–1.5	0.4	0.7
Government	–0.1	0.5	0.5	–0.4	0.9	0.3	0.5
Gross fixed investment							
Public[2]	–11.1	1.1	7.9	–5.4	14.3	8.6	0.8
Private residential	–15.7	–13.5	–4.0	–2.0	–11.9	–1.8	0.1
Private non-residential	4.3	–9.6	–3.7	–14.9	–8.4	–3.1	0.0
Stockbuilding[3]	0.0	–0.1	–0.2	–0.2	–0.2	–0.2	–0.1
Total domestic demand	–0.5	–3.4	–0.2	–4.3	–1.9	0.2	0.4
Exports of goods and services	10.8	–1.8	5.0	–6.2	0.3	6.8	6.5
Imports of goods and services	–0.2	–8.0	1.8	–10.7	–4.3	3.8	4.3
Net exports[3]	1.3	0.7	0.5	0.4	0.5	0.5	0.4
GDP	0.8	–2.6	0.2	–3.8	–1.3	0.7	0.8
Inflation and capacity utilisation							
GDP deflator	0.6	0.8	–0.2	1.1	0.1	–0.1	–0.8
Private consumption deflator	1.6	0.7	–0.4	0.5	–0.4	–0.4	–0.5
Unemployment rate	3.4	4.2	4.7	4.0	4.4	4.6	4.7
Output gap	–0.7	–4.7	–5.6	–4.1	–5.3	–5.5	–5.6
Memorandum items:							
Net government lending							
(per cent of GDP)	–3.3	–11.1[4]	–7.6				
Gross debt	87.4	99.5[4]	107.8[4]				
Net debt	18.5	29.9[4]	37.5[4]				
Short-term interest rate	0.6	0.7	0.5	0.8	0.6	0.5	0.5
Long-term interest rate	2.4	1.5	1.1	1.8	1.2	1.0	1.1
Current account (per cent of GDP)	2.3	3.2	3.8	3.0	3.3	3.7	3.9

1. Provisional projections assuming an exchange rate of 131.8 to the dollar. Definitive projections will be released in *OECD Economic Outlook 64* in November.
2. Including public corporations.
3. Contribution to GDP growth.
4. Including 5.3 percentage points of GDP for the assumption of debts of Japan National Railway Settlement Corporation and the National Forest Service.

Source: OECD.

other than the top rate have not been decided, the pension contribution rate may be frozen; and the fiscal stance of local governments remains a source of uncertainty.

Assessing the sustainable growth rate

Looking beyond the short term, it is useful to have some estimates of the scope for further expansion. As discussed in Box 3, the potential growth rate is estimated to be in the 1¼-1½ per cent range, and the output gap could be around 5 per cent in the second half of 1998. Potential growth could decline further toward 1 per cent per year in the next few years, assuming annual total factor productivity growth stays near the ½ per cent where it has been in the recent past. However, the myriad structural reforms currently being implemented or expected to be in the next few years, should eventually raise that figure. Nevertheless, strengthening downward pressure from the labour input side (through a faster shrinkage of the working-age population) will set in around the year 2000, even if it could be approximately offset by a pickup in capital input growth following the current recessionary episode. In any case, given the substantial slack in the economy at present, supply conditions will not constrain growth in the next few years, so long as that slack is not drawn down too quickly.[110]

Over the longer term, much is determined by the future course of the population. Official population projections from the Ministry of Health and Welfare were last revised in January 1997 (Table 28). As fertility continues to slide,[111] contrary to previous expectations, and the rise in life expectancy exceeds projected rates, it is now believed that Japan's peak population will only be around 1 per cent higher than current levels (rather than 3 per cent), and shrinkage will get under way around the year 2008, not after 2010. Once under way, the process would quickly accelerate, and the population would finish the next century only just above half what it had been at its start. While reduced fertility makes for lower total dependency until around 2020 in the latest projections, longer life expectancy raises elderly dependency. Both measures of dependency are projected to peak around 2050, slightly later than previously expected, and at much higher levels. On average over the period out to 2025 the shrinkage in the working-age population may be initially mild, held down by assumed increases in labour force participation, but it will pick up speed as the demographic transition accelerates. Hours worked per person may continue to fall, possibly to something around European levels, which would imply a contribution of around –¼ per cent per year to potential output growth. The contribution of changes in labour inputs might, therefore, average approximately –0.4 percentage point per year over the next 25 years, around –¼ percentage point in the years

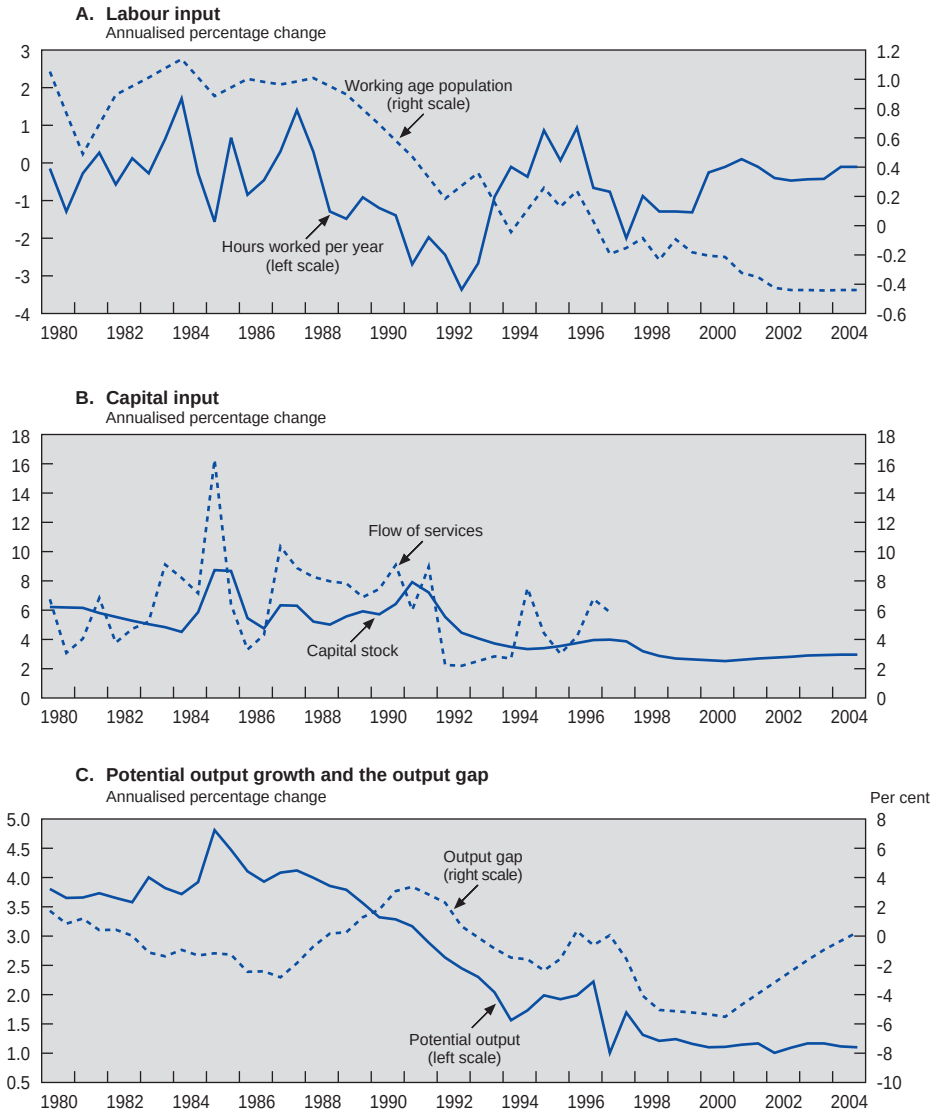
Box 3. **Potential growth rate**

Two broad approaches are possible to try to gauge how much of a shift in potential growth there has been. First, if there is a stable relationship between real GDP growth and changes in measures of labour market conditions such as the unemployment rate or the ratio of job offers to applicants, the estimated relationship can be used in an Okun-type approach to the calculation of the rate of growth which is just sufficient to stabilise these labour market outcomes. Unfortunately, the Secretariat was unable to find any such stable relationship for Japan. The second approach is based on growth accounting within a production function framework. For most of the 1980s aggregate potential labour inputs grew at an annual rate of around 1½ per cent (Matsuura *et al.*, 1998).[112] This began to slow in 1988 when the average number of hours worked per year began to fall (Figure 30). This fall was interrupted from 1994 to 1996 but resumed in 1997. The increase in the working-age (15 to 64) population has also eased, due to the well known ageing of Japan's population which is certain to continue well into the next century. Indeed, the size of the working-age population started to fall absolutely in 1996. However, the number of labour market participants above 65 continues to rise at a better than 4 per cent annual rate, and if participation rates for this group remain fairly stable (near 25 per cent) as they have over the past 15 years, this should slow only gradually toward 2¾ per cent over the next decade. Nevertheless, maximum labour inputs grew by less than ½ percentage point per year in the second half of the 1980s and shrank outright in the 1990s, despite a trend increase in labour force participation.

As mentioned above, the measured capital stock was growing at an average of 5¾ per cent in the 1980s and by more than 7 per cent in 1991, but by 1994 that growth had slowed to less than 3½ per cent and is currently around 3 per cent. If the flow of capital services is taken to be proportional to the product of the stock and its utilisation rate, at least for the manufacturing sector (with the rest assumed to be operating at full capacity), then capital input growth dropped from around 6 per cent per year from 1976 to 1991 to not much more than 4 per cent on average since then.

Using a production-function framework the residual is total factor productivity growth. This seems to have averaged around ¾ per cent per year in the decade prior to the bubble era and then, probably for cyclical reasons, jumped to around 1¾ per cent until 1991 before falling back below the pre-bubble pace at a bit more than ½ per cent per year. Thus, overall, the actual growth of around 1 per cent per year over the past seven years has been managed by a shrinkage in labour input that has been more than offset by productivity growth and capital deepening, combined with a widening in excess supply. But both productivity growth and capital deepening are contributing less than they had been previously, even prior to the bubble era. Potential growth looks to have slipped to below 2 per cent even prior to the most recent 1997-98 downturn, and the economy's resources were utilised at average historical levels late in 1996. Since then potential output growth has eased still further to the 1¼-1½ per cent range because of the demographic slowdown, the deceleration in capital stock growth and a slowing trend in total factor productivity growth.

Figure 30. **SOME DETERMINANTS OF POTENTIAL GROWTH AND THE OUTPUT GAP**[1]



1. Figures for the period 1998 to 2003 are projections based on work for the forthcoming *Economic Outlook*.
Source: OECD.

Table 28. **Changing population projections for Japan**[1]

	Projections made in September 1992						Projections made in January 1997					
	Total (thousands)	Share 0-14	Share 15-64	Share 65+	Total dependency ratio	Elderly dependency ratio	Total (thousands)	Share 0-14	Share 15-64	Share 65+	Total dependency ratio	Elderly dependency ratio
2000	127 385	15.2	67.8	17.0	47.5	25.1	126 892	14.7	68.1	17.2	46.8	25.3
2005	129 346	15.6	65.2	19.1	53.3	29.3	127 684	14.3	66.1	19.6	51.2	29.6
2010	130 397	16.4	62.4	21.3	60.4	34.1	127 623	14.3	63.6	22.0	57.2	34.6
2015	130 033	16.3	59.5	24.1	68.0	40.5	126 444	14.2	60.6	25.2	65.0	41.6
2020	128 345	15.5	59.0	25.5	69.4	43.2	124 133	13.7	59.5	26.9	68.2	45.2
2025	125 806	14.5	59.7	25.8	67.5	43.2	120 913	13.1	59.5	27.4	68.0	46.0
2030	122 972	14.2	59.8	26.0	67.2	43.5	117 149	12.7	59.3	28.0	68.6	47.1
2035	120 132	14.6	58.8	26.6	70.0	45.2	113 114	12.7	58.3	29.0	71.4	49.7
2040	117 290	15.3	56.7	28.0	76.4	49.4	108 964	12.9	56.1	31.0	78.1	55.1
2045	114 432	15.8	55.8	28.4	79.2	50.9	104 758	13.1	54.9	32.0	82.0	58.2
2050	111 510	15.7	56.1	28.2	78.3	50.2	100 496	13.1	54.6	32.3	83.0	59.1
2060	105 516	15.5	58.6	25.8	70.5	44.1	91 848	12.9	56.0	31.0	78.5	55.4
2070	101 023	16.6	58.2	25.2	71.9	43.3	83 773	13.5	56.5	30.0	76.9	53.0
2080	98 249	17.2	57.1	25.6	75.1	44.9	77 375	14.0	56.0	30.0	78.6	53.5
2090	95 732	17.2	58.4	24.5	71.3	41.9	72 068	14.2	56.3	29.5	77.5	52.3
2100	–	–	–	–	–	–	67 366	14.6	56.5	28.8	76.9	51.0

1. Middle series of estimates for future population as of 1 October of the stated year.
Source: National Institute of Population and Social Research.

to 2010 and more than –½ point per year in the following 15 years. Of course, uncertainty remains substantial over the longer term and is compounded by the possibility of increased inward migration.

However, that is not the end of the story. There is a possible further indirect downward effect coming from the life cycle theory of saving: as the population ages, saving should decline, and to the extent that this is reflected in a drop in investment (the Feldstein-Horioka effect), capital inputs may be adversely affected, possibly by an additional –¼ point per year (Matsuura *et al.*, 1998). Thus, if total factor productivity growth does not benefit significantly from structural reforms,[113] potential growth would fall to 1 per cent per year on average over the next decade or so and then further to around ½ to ¾ per cent thereafter. Clearly, in the future realised GDP growth will be increasingly determined by productivity performance, with smaller contributions from factor input trends.

III. Enhancing medium-term prospects through structural reform

All OECD Member countries agree on the importance of continuing structural reform to ensure optimal economic performance over the medium term. Structural policies must be frequently re-evaluated to verify that they are consistent with all the institutional and technological changes that are the hallmark of the ever more open world economy. In Japan's case there is possibly even greater reason to focus on these reforms: it is widely believed that its poor macroeconomic performance over the past seven years is in large part directly attributable to a failure to adapt its economic structures to the needs of a modern, wealthy nation in open competition with more flexible and dynamic rivals. This chapter will review some of the more important reforms that are being considered or have been implemented over the past year beyond the raft of changes being implemented in the banking sector (already discussed).

The overall reform strategy: careful planning but slow implementation

A renewed impetus was given to structural reforms when in 1996 the Hashimoto government designated six priority areas (administration, fiscal structure, social security, economic structure,[114] financial system and education) and announced that achieving significant progress in these areas was one of the most important tasks of the government. As noted in the previous *Survey*, the pace of reform appears to have picked up since this announcement. Three factors can account for this acceleration. First, under-performance of the economy in the 1990s made people realise that reforms must be undertaken more quickly. In fact, the good growth outcome in 1996 did not discourage reform efforts, probably because people had become convinced of the necessity of the reforms.

Second, the government's strategy of simultaneously dealing with many issues probably made it easier to persuade interest groups to support the programme. Third, Mr. Hashimoto showed strong leadership in advancing the agenda. The reform process in Japan is often said to be slow and cumbersome due to the ever-present need for consensus building prior to action and lengthy delays between agreement and implementation. Although this observation still seems valid, the pick-up in the pace in 1996 and 1997 is encouraging. For example, in the case of the administrative reform it took only about a year and half since a council had started discussion to legislate a basic law to restructure ministries and agencies. This shows that the pace of reform can be speeded up if there is strong political leadership. But there remains such a tremendous amount of administrative discretion about the timing, scope and detail of reforms that overall progress is very much hostage to continued political leadership.

The Administrative Reform Committee, which ceased to exist in December 1997, gave a major push to promoting regulatory reform in many areas by showing the merits and demerits of each regulation to the public and by recommending specific measures. Most of its recommendations were incorporated into the Deregulation Action Plan and are being implemented. A newly set-up committee within the government, though not an independent one like the Administrative Reform Committee, is expected to play a similar role. At the end of March 1998, the previous Deregulation Action Plan ended and another three-year Plan was launched. Although the basic framework is the same, the new committee is also to undertake ''horizontal'' work, which is applicable to many areas, such as a review of the professional qualification system, specifying criteria for granting regulatory approvals, an obligatory review of new regulations after a certain period, etc.

Progress in regulatory reform: potentially more than meets the eye but considerable uncertainty

Not too many years ago it was commonly believed that the Japanese economic system was overtly designed to favour production over consumption. This seems to be in the process of changing. An example of this evolution came in June of this year, when the Ministry of International Trade and Industry proclaimed that trade and industrial policies should be tailored to benefit individuals

and not business. Since much of the existing economic regulation[115] seems designed to dissuade producers from competing with one another, it follows from that revised view that the government is gradually modifying its attitude towards regulation: instead of prospectively imposing regulation, the government has recently expressed its wish to set minimal rules and to check private-sector compliance retrospectively. To this end all permitting and licensing regulations (around 12 000) are to be reviewed by March 2001. For example, the 1998 Deregulation Action Plan charges the Fair Trade Commission with conducting surveys and making proposals regarding the business fields into which entry is restricted, ''supply-and-demand'' adjustment and other regulations, a reform which could prove crucial in reducing the problem of regulatory capture.

The government continues to advance its regulatory reform agenda, although many measures are put into effect only three or more years after their approval, and the regulatory process still lacks coherence and transparency. It also remains unclear if competitive markets will indeed develop in order to generate the hoped for consumer benefits. A new three-year plan got under way in April 1998. It comprised 624 items in 15 fields. The new Deregulation Subcommittee at the Administrative Reform Headquarters will monitor progress.

Recent OECD work (OECD, 1997) pointed to substantial gains in output (up to 6 per cent)[116] which could be expected from regulatory reform. In November 1997, the Economic Planning Agency estimated that a package of 120 items in seven fields including deregulation would raise output by 60 trillion yen ($480 billion or 10 per cent of GDP) in the long run. The group of items with the largest estimated impact relate to land use: by easing the regulated floor area coverage ratio in commercial districts of major cities,[117] introducing fixed-term house leaseholds and promoting house construction in suburban areas, gains of 3.3 trillion yen per year and 46 trillion yen in the long run were projected. Another related inefficiency results from the rent controls permitted under the long-standing Rental Land and House Law: by making eviction very difficult, this Law has had the effect of shifting the distribution of rental housing toward smaller apartments, especially those with single rooms, which are most often occupied by younger people with higher mobility. Finally, property market turnover is encumbered by the land use planning law that requires the National Land Administration to be informed in advance of any transaction involving more than 2 000 square metres in order that it can verify that the price has been properly set.

Many of Japan's other key sectors are in the process of being deregulated with a view to enhancing their efficiency and lifting productivity to world class levels. One measure of progress is the price gap between Japan and the rest of the world. Official Japanese figures comparing Tokyo and New York City, for example, show that the implicit purchasing power parity for deregulated items fell from 199 yen/dollar in 1991 to 171 in 1996, a drop of 14 per cent, while the decline for other items was only 10 per cent (from 158 to 142) (Economic Planning Agency, 1997). For example, in *telecommunications* international telephone services *via* the internet were liberalised in August 1997, and several foreign and domestic firms have announced their intentions to offer new services of various kinds. The main Japanese international long-distance carrier (KDD) has just launched a domestic long-distance service. WorldCom's application to offer long-distance services has already been approved, and British Telecom hopes to begin service in January. Prices have continued to fall rapidly (Table 29): for example, calling long-distance within Japan has become 77 per cent cheaper than it was little more than a decade ago. A key current issue which will determine the degree of competition in the future is the price to be charged by NTT for interconnection as from the year 2000 when the legal basis for the determination of this price may shift from historial to long-run incremental costs. NTT's initial proposals were for much higher rates than foreign best-practice levels. But pressures are strong for those rates, when they are ultimately decided on, to be much lower. It is possible that the government may require new common carriers to contribute to a fund to help subsidise NTT's mandatory provision of unprofitable services. Like ATT in 1984, NTT will be split into several pieces between 1 April and 20 December 1999, but all under the umbrella of a single holding company. Thus, while it has been decided to allow the two regional firms to adopt different tariff schedules and enter each other's market, it remains to be seen whether effective competition between the two will result. Two important additional problems are that about 10 per cent of all phone districts in Japan still lack a local internet access point, raising costs by as much as a factor of ten, and that those wishing to have local fixed-point telephone service in Japan must still pay an installation charge and a subscription charge to the monopoly provider amounting to 72 800 yen (about $600).

In the *energy* sector, a panel has recommended that the government should abolish price controls on petroleum products and regulations on refinery mergers and construction; a bill is expected to be sent to the Diet in FY 2000 with

Table 29. **Falling prices for long-distance telephone calls**

Yen

	NTT	New common carriers/KDD[1]
A. Maximum[2] day-time domestic long-distance rates for a three-minute call		
September 1987-February 1988	400	300
February 1988-January 1989	360	300
February 1989-March 1990	330	280
March 1990-March 1991	280	240
March 1991-April 1992	240	200
April 1992-June 1992	240	180
June 1992-October 1993	200	180
October 1993-November 1993	180	180
November 1993-March 1996	180	170
March 1996-January 1998	140	130
February 1997-February 1998	110	100
February 1998-July 1998	90	90
July 1998-Now	90	69
	KDD	**New common carriers**
B. Rate for three-minute call to the United States		
Prior to September 1986	1 530	n.a.
September 1986-September 1988	1 240	n.a.
September 1988-October 1989	890	n.a.
October 1989-November 1989	890	680
November 1989-April 1990	730	680
April 1990-September 1990	680	680
September 1990-October 1993	680	670
October 1993-November 1993	670	670
November 1993-December 1994	670	660
December 1994-December 1994	600	660
December 1994-November 1995	600	580
November 1995-December 1995	480	580
December 1995-November 1996	480	470
November 1996-December 1996	450	470
December 1996-Now	450	440

1. Lower of KDD and new common carrier.
2. Prior to March 1991: over 320 km (340 km for new common carriers); from then to March 1996: over 160 km (170 km); since then over 100 km.

Source: Economic Planning Agency.

implementation the following year. The April 1996 abolition of the law limiting the importation of oil products has helped push average retail prices of petrol down from 107 yen/litre to 92 yen, with some retailers charging as little as 79 yen.[118] Deregulation of petrol stations also started in 1996 and is now finally

leading to long-overdue cuts in capacity: some 10 000 of the existing 58 000 stations (50 per cent of which are said to be unprofitable) are expected to close in coming years. Self-service stations are now permitted, but thus far there are only 20, although 50 more are expected to open in the autumn. It will take time for motorists to adjust to this new possibility. In the *electricity* sector a deregulation bill, if necessary, would be sent to the Diet in 1999, but complete free entry at the retail level has been rejected for now. It is not clear whether the UK example of splitting generation, transmission and distribution will be followed. ''Large'' users (as yet undefined) will be allowed to buy from independent power producers (IPPs),[119] who will pay a fee for using utility firms' transmission lines. These fees will have to be set at a level which would not distort competition between the IPPs and existing utility firms. But few of the benefits may go to small users as downstream prices will remain subject to controls. Average prices were officially estimated to be 15 per cent above US levels in April 1997 (at the prevailing exchange rate of 125 yen to the dollar).

In April, a new agreement to last four years in the *aviation* sector formally took effect. It will boost the number of flights and airlines serving Japan and allow code-sharing. Both Japanese and US carriers have been accorded more extensive ''beyond rights'' in a symmetric manner. On the other hand, the key domestic reform is the promised elimination of demand-supply balancing regulation by 1999. This deregulation is being proposed in order to reduce the excess costs of the system over those in the United States, for example; these are estimated to be around 20 per cent. A bill will be sent by the Ministry of Transport to the Diet in FY 1999 which will allow price discounting beyond the current 25 per cent limit and pay subsidies on unprofitable but indispensable routes. But what will happen to rules on airport landing slots for domestic flights is unclear. Fare controls in *other transport* sectors are also to be abolished next year, with ceiling prices alone remaining in place. This should have some beneficial effect for consumers, as taxi fares are still much higher than abroad,[120] but price gaps for trains and buses are no longer large, and comparisons often reflect a relative lack of subsidy in Japan. Coastal shipping costs are raised by inefficient port practices (see below) and by the existence of a ''tonnage-capping'' cartel which forces new entrants to buy and scrap existing capacity to keep overall supply unchanged. This is to be phased out, but the transition period may be as long as 15 years.

The Large Scale Retail Store Law has since 1973 regulated the floor size, shop hours and closing days of retailers and has long been seen as a barrier to entry by new, often foreign, chains. In recent years it has been liberalised several times, and the policy concerning such stores has been fundamentally changed. A new law will be implemented by June 2000; it will eliminate the use of supply/demand adjustment mechanisms and will require local governments to apply transparent criteria that are limited to environmental factors, as set out in guidelines issued by the central government. Nevertheless, US officials have expressed their fears that local governments may try to block new stores on unreasonable or arbitrary grounds. Access to Japan's professions may also be eased: in April the Ministry of Finance decided to offer the examination for foreign-trained accountants in 1999 for the first time since 1975, to adjust professional requirements to reflect international standards and to negotiate reciprocal certification with Japan's trading partners. In August, the government changed the law governing legal practise by foreign legal consultants in order to ease their access to the market, but several contentious issues remain outstanding, especially the prohibition of hiring of Japanese lawyers by foreign law firms and of partnerships between foreign and Japanese lawyers (Report by the Deregulation Subcommittee of the Administrative Reform Committee, December 1997). Finally, foreign investment is to be allowed in cable television, whose licensing procedures are to be streamlined.

Financial reform: finally coming on-stream

The number one priority of the government's reform efforts thus far has been the financial sector which has been excessively compartmentalised, costly and lacking in innovation. The symptoms of malaise are several. For example, Japanese banks, which not so many years ago occupied most of the top spots in world rankings, have now largely disappeared from the first division. Confidence in them fell so severely last autumn that depositors shifted nearly 2 trillion yen (about $15 billion) in deposits from them to foreign banks. One third of all futures contracts on the Nikkei 225 are traded in Singapore; while Tokyo had once been the world's top centre of equity investment, it has now unofficially fallen behind London as well as both New York and Boston; OTC trading in Tokyo is only 1 per cent of what is traded on the NASDAQ; and foreign firms in

increasing numbers are applying to delist their shares from the Tokyo Stock Exchange (ATT, Texas Instruments, Anheuser Busch, Chrysler).

The ''Big Bang'', announced in late 1996, is now gradually being implemented. In December 1997 the ban on financial holding companies was scrapped. In April 1998 a number of previously legislated changes took effect. As mentioned in the previous chapter the new Bank of Japan Law was one of these. Deregulation of trading commissions on the value of equities transactions in excess of 50 million yen was another. In addition, the Foreign Exchange and Foreign Trade Law has been amended so that firms no longer have to pass all their foreign currency transactions through authorised banks (netting is now permitted, thereby saving an estimated 175 billion yen per year), and other institutions can engage in such transactions on a commercial basis. The result is that other financial institutions have become exchange market participants, and even small stores are now accepting payment in foreign currency.[121] At the same time individuals are now permitted to hold accounts abroad directly (without having to invest in a mutual fund). The new Financial Supervisory Agency began operations in June. It is responsible for surveillance of all private financial institutions, including brokerages and insurance companies. The overwhelming majority of its staff were transferred from the Ministry of Finance – there were only 26 new hires – and will eventually return to it, but its top officials are said to have cut their ties to the Ministry. Nevertheless, the question of its independence remains paramount, and the administrative procedures to be used in the supervisory process need to be made transparent.

In June 1998 the Diet passed the Financial System Reform Bill. As a result, non-life insurance premia were deregulated in July, and, effective December 1998, banks will be allowed to sell mutual funds over the counter; these funds will be able to incorporate (so that income earned from them can be taken in the form of dividends); brokerages will merely have to register in order to be authorised; equities may be traded off exchanges; insider trading will be subject to stricter penalties; and in December mandatory, industry-financed[122] Investor and Policyholder Protection Funds will be set up in the brokerage and insurance sectors to provide a safety net[123] of up to 10 million yen per brokerage client and of 90 per cent of voluntary reserves paid in to insurance firms by their policy holders. Brokerage commissions are also to be fully deregulated by the end of 1999,[124] cutting an expected 20 to 30 per cent from retail fees. Controls on

banks' securities subsidiaries are to be lifted by March 2000. And complete decompartmentalisation – allowing entry by banks, insurance companies and securities firms in one another's businesses – will proceed in stages in FY 1999 and 2000.

There are other important changes under way as well. In June, the Ministry of Finance reorganised their ordinances (*Tsutasu*) and communication notes (*Jimurenraku*) through which they had shown their interpretations of laws and regulated operations of financial institutions, or which had been administrative guidelines, sometimes not based on specific legal authority. They abolished more than 500, which were considered to be irrelevant in the post ''Big Bang'' era, out of more than 600 outstanding. As a result, the following deregulations were realised: issuance of commercial paper by financial institutions was permitted; the minimum issuance of certificates of deposit was abolished; and terms for the sales of privately-placed bonds on the secondary market were eased. However, the Ministry also reinforced the regulation by upgrading about 60 ordinances and communication notes, such as submission of financial statements to the authorities, to ministry orders (*Shorei*) or notices (*Kokuji*); it also integrated the rest into about 30 ordinances or notes. These measures are intended to enhance the transparency of banking supervision. The government also submitted a law to the Diet that would enable non-banks,[125] one of the few profitable parts of the financial sector, to issue bonds. As a reaction to and in anticipation of all these liberalisations, there has been a significant amount of restructuring in the sector of late (see Box 4).

Bankruptcy procedures

Bankruptcy should play a major role in the process of structural reform as it is a mechanism to adjust interests among stakeholders and realise more efficient use of resources through liquidation or debt restructuring. Lowering exit costs facilitates new entry into markets by unbinding resources tied to poorly performing companies. The Japanese bankruptcy system consists of five legal procedures, two for liquidation and three for restructuring.[129] Because they were legislated separately and long ago, the system is lacking in coherence and is out of date. Many specific problems have been pointed out. For example, applicants have to decide which procedure should be applied. However, what is important to the financially distressed firms is to protect their assets from being reclaimed by

Box 4. **Financial sector restructuring and recent developments**

A great deal of restructuring has been announced, much of which involves either reductions in financial institution cross-holdings (especially insurance companies selling their shares in banks) or greater foreign investment and market share.[126] In the brokerage industry six firms (other than the much-publicised cases of Sanyo and Yamaichi) have closed or will do so. A variety of others have announced their intent to merge. Nikko, one of the largest, will form a joint venture with Salomon Smith Barney in order to acquire foreign know-how. And Merrill Lynch has taken over some of Yamaichi's offices and employees in order to start up a full-service retail brokerage operation in Japan. The Mitsubishi group looks set to establish a large-scale securities firm and to enter the insurance, pensions and investment trust markets. Also, one of the major trading firms (Nissho Iwai) has become the first non-financial company to enter the brokerage business. And Daiwa Securities, which is joining an alliance with Sumitomo Bank and Sumitomo Trust Bank, is radically scaling back its international operations. In the banking sector, the Long Term Credit Bank is to be temporarily nationalised, and then, once its balance sheet is cleaned up, it will be either sold back to the private sector or wound up. There are two mergers which have been announced (Hanshin and Midori and Fukutoku and Naniwa), both of which will receive public funding through the DIC; and the city banks Asahi and Tokai have just announced a widespread co-operation agreement. Some banks are forming alliances with foreign investors, such as Nippon Credit Bank with Bankers Trust and Fuji Bank with Goldman Sachs in mutual funds. In many cases banks are withdrawing from foreign markets, especially in Hong Kong, China and Paris.[127] They are also forming joint ventures in asset management (Industrial Bank of Japan with Nomura) to try to gain some scale economies.[128] Many regional banks (64 in all) will entrust their foreign exchange back-office operations to city banks. Foreign firms are also moving into the insurance sector: GE Capital is taking over all the new business of Toho Mutual Life; Meiji Life is allying itself with Dresdner Bank; and Aoba, the firm which took over Nissan Mutual Life when it collapsed last year, is to be auctioned off later this year by its owners, the Life Insurance Association of Japan.

creditors, rather than the restructuring versus liquidation decision. As for restructuring, the procedures of the Corporate Reorganisation Law are cumbersome, taking one year in practice, sometimes longer, before approval of the reorganisation plan, thereby raising the costs of the procedure.[130] As a result, many resort to out-of-court solutions. Although it is sometimes observed that ''main banks'' step in to help financially distressed firms, this intervention is considered to be rather exceptional, especially for SMEs. The Composition Law requires a firm to be already insolvent before it can apply for restructuring at which point it must

file a rehabilitation plan. Hence, the bankruptcy system is being reviewed. First, the system should be sufficiently flexible that smooth paths between restructuring and liquidation are possible. Second, simple but effective restructuring procedures should be available, without heavy involvement of the courts.

In response to these concerns the Ministry of Justice has just announced that it will present revisions to the Composition Law in 1999. These changes will allow firms to seek settlements with their creditors by petition with a moratorium period before submission of a rehabilitation plan.

Agriculture: struggling with policy reform

One of the sectors most in need of structural reform is agriculture. It has benefited from a historic Japanese custom of considering food and especially rice production to be inherently different from that of other goods and services – because of its traditional spiritual aspect – and a more recent but equally strongly held view that self-sufficiency is crucial for an island nation. However, the sector has indeed been shrinking, even if zoning restrictions slow the process:[131] in value-added terms from 2.5 per cent of GDP in 1990 to 1.9 per cent in 1995 and in employment terms from 8.8 to 7.3 per cent over the same period. Also, several years ago, as part of the Uruguay Round trade agreement, Japan agreed to open its rice market to limited quantities of imports. These began in 1995, with total imports equal to some 4 per cent of domestic consumption. The target is to boost this to 8 per cent by the year 2000. Thus far, however, the bulk of these imports has gone either to overseas food aid, processed food or inventories. Greater competition in rice retailing has also been encouraged – the number of vendors has risen sharply – by changing entry terms from Ministry permission to registration in November 1995. In November 1997 the government announced new price policies which aim to bring supply and demand into balance by reinforcing the present diversion scheme, to stabilise rice farmers' incomes by introducing a new type of payment and to further introduce market mechanisms. In the spring of 1998 the government stopped regulating the free-market price of rice. Looking forward, a debate is under way over whether corporations should be allowed to own and operate farms: in September 1998 a government advisory panel recommended that the 1961 Basic Law be amended to allow this under certain conditions. Such a change is seen as a way to boost

domestic production, which is still viewed as the basis for a steady food supply and therefore the primary goal of agricultural policy. In order to maintain supply capacity and preserve the environment, the panel is also recommending the examination of the possibility of introducing direct payments for hilly and mountainous areas where the costs of production are high because of diseconomies of small farm scale. Some of the enabling legislation is to be submitted to the Diet next year, with the remainder as soon as possible thereafter.

Environmental policy: meeting the Kyoto commitments

In December 1997 Japan hosted the Third Session of the Conference of the Parties to the United Nations Framework Convention on Climate Change to discuss the problem of global warming. Agreement was reached there, in the form of the Kyoto Protocol, for the Parties to reduce their emissions of greenhouse gases by varying amounts relative to the 1990 baseline level. Japan's binding commitment was to cut its emissions by 6 per cent for the average of the period 2008-12. The target will not be achieved easily: despite only modest economic growth since 1990, emissions in 1996 were 9 per cent greater than those in the 1990 base-year period; moreover, Japan's economy is already relatively energy efficient.

In response to the Kyoto Protocol, the government decided on a Guideline of Measures to Prevent Global Warming in June 1998. The objective of this guideline is to identify measures to be promoted urgently to meet the 6 per cent emission reduction target. On the energy demand side, the target is to further improve energy efficiency so that 56 million kilolitres oil equivalent would be saved by the year 2010. In order to achieve this target, the Law Concerning Rational Use of Energy was substantially strengthened in June. On the energy supply side, non-fossil-fuel energy such as nuclear and new and renewable energy will be vigorously promoted. In order to meet the Kyoto target, nuclear plant capacity needs to be expanded from 42. 5 GW in FY 1996 (12.3 per cent of total energy use) to 66 to 70 GW in 2010 (17.4 per cent). The share of new and renewable energy in total primary energy supply needs to be tripled from 1.1 to 3.1 per cent as well. It is also essential to review daily life styles in order to meet the Kyoto target. For this purpose, such measures as a broad discussion about the introduction of Daylight Saving Time, and education and public relations about

energy and the environment, are to be further promoted. In October, the Bill for the Promotion of Measures to Cope with Global Warming was passed by the Diet. The core objective of this bill is to urge all the relevant social actors to make their best efforts to mitigate climate change on a voluntary basis. These efforts include the formulation of action plans and public reporting of the existence and implementation of these plans.

There have also been recent developments on the recycling front. A 1995 law requiring the recycling of packaging and containers came into force in April 1997, and by the year 2000 it will be extended to include all such items made from paper and plastic. In May 1998 the Diet ratified the Appliance Recycling Law, which requires that a variety of consumer durables be returned *via* retailers to manufacturers for appropriate disposal.

Commercial policy: keeping trade tensions to a minimum

Perhaps in part because so much of the attention of Japan's trading partners has been on its economic weakness over the past year, problems in the commercial policy domain have been relatively few. A number of issues have been successfully dealt with. With respect to the United States, Federal Maritime Commission sanctions against three Japanese carriers – imposed because of US dissatisfaction regarding the prior consultation system in force at Japanese ports[132] – were indefinitely suspended in intergovernmental talks last autumn. Japan subsequently requested consultations regarding their withdrawal. The bilateral film dispute ended when the WTO upheld the Japanese position in January 1998. As mentioned above, in March the two nations came to a civil aviation accord, and bilateral deals have also been made in the areas of wood products, sound recordings, tomatoes, telecommunications equipment, drug approval and spirits.[133] They are also of like mind in wishing to keep electronic commerce free of tariffs and customs charges and to encourage the development of self-regulation in the sector. A bill was also presented to the Diet in March which would relax the rules governing inspections of foreign auto parts. And with a view to improving market access, consultations between Japan and the European Union on a Mutual Recognition Agreement (MRA) are under way, covering as many sectors as possible.

But, inevitably, some disputes have not been resolved, and others have cropped up. Problems with the United States remain over the so-called ''third sector'' insurance market (covering cancer, nursing care, personal accidents, etc.). Taking into account the high dependency of foreign insurance providers on this sector and their efforts to serve specific consumer needs, Japanese firms' access to it has in effect been restricted. In 1996 the two nations agreed that entry should be liberalised in the year 2001 so long as other life and non-life insurance sectors were fully deregulated by the end of June 1998. Japan believes that this condition has been met, but the United States is not satisfied with the pace of new product approval and with what may be only a superficial end to price-setting by ratings organisations. Complete and rapid opening of all parts of the insurance market would obviously be in the interest of Japanese consumers. Separately, in February the US Commerce Department ruled that five Japanese producers were dumping stainless steel wire rods with dumping margins of 24.4 to 31.4 per cent, and in September, US steel producers launched an antidumping suit against competitors from a number of countries including Japan.

But now that most sectoral dossiers have been dealt with, the focus of US attention has turned to deregulation more generally. Current discussions are being held under the Enhanced Initiative on Deregulation announced at Denver in 1997. The United States is of the view that Japan's 1995-98 Deregulation Action Plan was ineffective because many of its important recommendations have not been acted on; indeed, implementation lags have been long, although by 31 March 1998, 2 793 out of 2 823 items in the Plan had been at least partly implemented. But a recent Rand survey of US businessmen found that Japan's non-tariff barriers make conducting business there as difficult as in China and tougher than in Europe and Korea (Wolf, 1998).

In July 1998 Japan also decided to bring its fifth case before the WTO claiming that Canada's discriminatory measures regarding auto import tariffs[134] relating to the implementation of its 1965 auto pact with the United States are inconsistent with Canada's WTO obligations, including most favoured-nation treatment. In September, it also brought the case of Massachusetts' sanctions against firms doing business with or in Myanmar to the WTO for a ruling on its permissability.

Finally, in multilateral matters: the timing of Japan's April 1998 decision to remove Generalised System of Preferences (GSP) (zero-tariff) treatment of imports of 122 items from Korea, Taiwan, Singapore, Hong Kong, China and

New Caledonia was criticised in some quarters, given the extreme pressure some of these economies are under. And its hesitation to move forward at this time with other APEC members toward free trade in nine previously agreed sectors (''early voluntary sector liberalisation'') because of its unwillingness to open up its forestry and fisheries sectors did not set a good example.

Public sector reform: streamlining but no change in bureaucratic incentives

Reform of the public sector – often referred to in Japan as ''administrative reform'' – is one of the key objectives of the current government. Two of basic ideas are to shift from discretionary to regulatory administration and to separate policy planning from implementation.[135] Sweeping proposals were made in 1997, but the final Administrative Reform Bill, passed in June 1998, did not contain some of the more significant elements previously suggested. First, the privatisation of the postal savings and insurance schemes was dropped; instead, responsibility for the provision of services by the post office, including its saving and insurance schemes, will be transferred to a new Postal Service Agency at some point between 2001 and 2003 which will then be corporatised. Such a large institution, in competition with private-sector financial institutions in a deregulated, market-based system, should be operated without any cross-subsidisation from other services. Postal savings funds will no longer automatically be transferred to the Ministry of Finance's Trust Fund Bureau to be invested in the government's Fiscal Investment and Loan Programme, while the postal life insurance fund has had substantial discretion in portfolio investment decisions since its establishment. Second, the Bill contained no clear separation of the financial and fiscal responsibilities of the Ministry of Finance, despite the demonstrated risks of an over-concentration of power in one Ministry. While the Ministry's supervisory role was transferred to the new Financial Supervisory Agency (FSA) in June 1998, there had been calls for financial planning functions to go with it, as there had been suggestions that the National Tax Administration should be split off from the Ministry. However, in the autumn agreement was reached with the opposition parties to present a bill in the next Diet session to transfer all responsibilities for financial regulation from the Ministry to a new

‘‘financial revitalisation committee’’ (which would also oversee the FSA) by the end of 1999. In the interim the Ministry will retain joint responsibility (with the FSA) for financial crisis management.

However, the Bill did make some useful changes by providing for the streamlining of the existing 22 government departments into a single cabinet office and 12 ministries and agencies, together with a reduction in the number of bureaux from 128 to around 90 by the year 2001. It also called for a 10 per cent reduction in the civil service, whose current size is around 510 000. Finally, it gives greater power to the Prime Minister, who will be able to propose fundamental national policies at cabinet meetings, and the cabinet secretariat will be responsible for the design of basic budget, macroeconomic and security policies. This may clarify the lines of power. However, any improvement in efficiency may be limited unless regulatory incentives, bureaucratic discretion and the lack of accountability and transparency in the decision-making process are changed.

Following a series of ethics scandals in the public sector[136] an ethics law is again under discussion, as it was in 1996. Each ministry and agency is drawing up a voluntary code, but a standardised ceiling on gifts and entertainment will be set. Private-sector firms such as Nissan have also curtailed their entertainment activities. With a gradual shift in the nature of regulation away from bureaucratic discretion (see above), there has also been a decline in the number of retiring public officials taking managerial positions in business enterprises over which they have previously exercised regulatory responsibilities, known as *amakudari*. In recent years a steadily increasing share of public works has been open to tendering, and now the government’s price estimates are being disclosed. In March the government submitted the Access to Government Information Act to the Diet. It will initially not apply to 82 public corporations, but it provides that the government shall devise legislative and other measures to deal with their information disclosure practices. While public disclosure should improve, possibly heading off the most egregious of government waste[137] and dishonesty, it may nevertheless remain less complete than in many other OECD Member countries. And government arbitrary behaviour has not been eradicated. In January it announced that it wished to add 360 billion yen to the debt allocated to the seven privatised rail companies formed out of the Japan National Railways in 1987. This is being strongly opposed by the firms (see Box 2).

The government has also adopted the idea of so-called ‘‘private financing initiatives’’ (PFIs) for infrastructure investment. Japan has heretofore had a ‘‘third-sector system’’ of joint public-private ventures. Many ventures were set up, in particular in the late 1980’s when various policy supports became available. A good number have since come to face difficulties because the sharing of responsibility between the public and private sectors was unclear: unless more than 25 per cent of the equity was obtained by a local government, it has no authority to audit; private banks continued to provide loans, depending on ‘‘loss compensation’’ – a form of capital injection by the local government. Although the hope is that PFIs will allow the development of a project finance market, thereby easing demands on the public sector’s budget, the system needs to be carefully designed so that the problems of the ‘‘third-sector system’’ can be avoided. The draft law which has been submitted to the Diet seems to offer so much public policy support to PFI projects that it risks substantially weakening private incentives for efficiency gains. In this respect the Japanese approach seems very different from the UK model established in 1992. Pending passage of the legislation, one of the first planned PFI might be to transform the existing ‘‘third-sector’’ project to build the new Chubu Airport, in which Toyota would have the largest private equity share.

One area in which the public sector remains in need of reform, as pointed out in several previous OECD Surveys of Japan, is the timeliness and transparency of budget information. Monitoring is rendered extremely difficult for three reasons. First, the general government accounts are available only at an annual frequency and then only with a lag of over one and a half years, while details by major government sector are subject to an even longer lag. However, the government has begun this year to include a projection of the deficit of central and local governments each time that it submits initial and supplementary budgets to the Diet. Second, the budget itself and other government financial statements are not presented in a way which is readily identified in terms of national accounts aggregates. A lack of consolidation of the large number of accounts is an especially important problem. Third, changes in spending are often announced in the form of supplementary budgets, but later, when financing plans are provided, it is revealed that they have been partly offset by changes elsewhere, depriving them of their originally expected effects. Improved disclosure of fiscal data would serve to dispel uncertainties, thereby allowing policy makers to make better informed decisions and outside analysts to proffer higher-quality advice.

Labour market reform: increasing flexibility without adding to insecurity

During the 1950s and 1960s, the Japanese labour market paradigm was dualistic: lifetime employment for most employees of large corporations and considerable job switching for the rest. Thereafter, the lifetime employment system became increasingly pervasive among smaller firms. The system worked well in the sense that it kept unemployment at very low levels. In recent years, however, pressures on employers have brought about some reduction in the employment guarantee, and this process has probably been stepped up over the past year as the economy has weakened. One result has therefore been a substantial rise in joblessness, especially among youths and the elderly. In order to deal with this severe situation, in June 1998 a Cabinet unemployment committee agreed that public works contracting should be brought forward as much as possible, especially in regions of the highest unemployment, and confirmed that the Emergency Measures for Developing Employment, already included in the April stimulus package, should be steadily implemented. Some of these measures are: public financial support to firms hiring older workers is being extended to those over 45 rather than the current 55;[138] subsidies to firms in troubled industries to maintain employment levels are being increased;[139] and funding for a programme that pays for temporary employee transfers and leaves of absence is being expanded. However, other OECD countries' experience with such schemes has been, at best, mixed.

The shrinkage of lifetime employment has brought about an increase in the importance of the mid-career job market: for example, a recent survey showed that 28 per cent of 1994 university graduates had resigned from their first jobs within three years. At the same time the rising joblessness among school leavers has boosted the supply of workers willing to work in temporary positions, which are attractive to firms not wishing to make longer-term commitments. Until December 1996 temporary help agencies had been restricted to filling positions for 16 occupations. At that point the number was raised to 26, and in May 1998 a panel of the Ministry of Labour recommended that nearly all jobs be eligible. It also recommended that the duration of such contracts be limited to a maximum of one year. The authorising legislation will be presented shortly to the Diet, with implementation set for FY 1999 (Table 30). Similarly, although some liberalisation with respect to private job placement firms was undertaken in 1997, there remain job categories to which such services could usefully be extended.

Table 30. **Implementing the OECD Jobs Strategy – an overview of progress**

Jobs strategy proposal	Action taken	OECD assessment/recommendations
I. Increase the flexibility of employment		
– Allow temporary employment agencies to play a larger role.	– Further expansion in the number of job categories allowed to use temporary workers was recommended by a Ministry of Labour panel.	– Further reduction of restrictions on these firms would be beneficial.
– Make corporate retirement allowances portable.	– No action.	– Such a measure would help workers change companies in mid-career or find employment after employer downsizing.
II. Taxes and transfers		
– Relax the earnings test for pension recipients.	– No action.	– The earnings test discourages persons over age 60 from working.
III. Active labour market policies		
– Permit private job placement firms to play a more important role.	– No action.	– The number of job categories should be further expanded to help match firms with workers possessing needed skills.
– Improve the functioning of the Public Employment Security Office.	– Employment information is to be provided through Internet on a trial basis as from the end of FY 1998.	– The measures should be implemented on a full scale as early as possible.
– Evaluate the effectiveness of active policies.	– No action.	– Evaluation of policies in place could identify needed changes to improve their effectiveness.
IV. Enhance product market competition		
– Accelerate deregulation.	– Most recent plans appear to speed up the pace of deregulation.	– Removal of entry barriers and price controls would promote the growth of new businesses and employment.
– Further strengthen competition policy.	– The FTC has been given slightly more resources and has begun to widen the scope of its investigations.	– Effective enforcement of competition policy will help realise the benefits of deregulation. The FTC needs to upgrade its economic expertise.

Table 30. **Implementing the OECD Jobs Strategy – an overview of progress** *(cont.)*

Jobs strategy proposal	Action taken	OECD assessment/recommendations
V. Increase labour force skills		
– Improve the quality and availability of training outside of firms.	– No action.	– This is important in the context of more workers changing companies and receiving less on-the-job training.
– Establish a systematic recognition of qualifications.	– No action.	– Will help to raise labour mobility.
– Improve the education system.	– Educational Reform Programme is being implemented.	– Emphasis on providing greater autonomy to educational institutions is appropriate, but their accountability should be enhanced.
VI. Technology and innovation		
– Increase the emphasis on basic research and improve the efficiency of public spending on technology.	– Science Basic Plan is being implemented to promote open and competitive R&D system (see Chapter IV).	– This is a promising start.

Source: OECD.

Another way the government is encouraging greater flexibility is by allowing more elastic working-time arrangements through reform of the working hours system. These arrangements have contributed to a reduction in the number of hours worked per year. Revisions to the Labor Standards Law passed in September 1997 permit employers to schedule ten- rather than nine-hour days and 52- rather than 48-hour weeks within a negotiated annual total. The 40-hour workweek has been applicable to all industries since April 1997, with an overtime premium of 25 per cent (35 per cent on holidays). An increasing number of firms are shifting to annual salary systems for white-collar workers. But even in this system, a premium has to be paid for overtime work.

Another pillar of Japan's labour market has been seniority-based pay. But here too market pressures have succeeded in convincing some firms to replace that system by one which is based to a greater extent on merit. Fujitsu is bringing in such a change in October for its entire staff of 46 000. Mitsui will follow suit in April 1999. And the Bank of Tokyo Mitsubishi will allow employees the choice of dropping out of the company's pension schemes and pay them 10 per cent of the corporate revenues they generate instead. In July 1998, a government white paper gave its support to a shift away from seniority towards merit as the basis of pay. Consistent with that the September revisions to the Labor Standards Law, effective FY 2000, expand and further liberalise the so-called ''discretionary work system'' under which pay depends on notional rather than actual hours worked.

Overall, the *OECD Jobs Strategy* recommendations for Japan have been the object of some government actions in about half of all cases, but in none would it be the case that the proposal could be said to have been fully adopted. Furthermore, a new question must now be raised: whether in an era of higher unemployment, especially long-term unemployment, the current unemployment benefit programme, related social safety net system and active labour market policies are adequate to the task.

Health insurance reform: progress but some way to go

Health care spending has been rising fairly rapidly in recent years in Japan, even if it remains low in relation to GDP by OECD standards. However, this fiscal year may see the first decline since the universal care system came into

existence in 1961. There are three reasons for this: first, overall wages and prices have been stable; second, the increase in co-payments imposed last year seems to have had a marked effect on volumes; and, last, drug prices were cut last year by an average of 2.8 per cent. Despite the short-term success in holding down spending, reform aimed at boosting efficiency is on the agenda. One element of reform has been completed: a long-term care insurance programme to cover the estimated 2.8 million mostly elderly who need such care was approved by the Diet in December 1997 to take effect in April 2000. The government hopes that the reform will solve the problem of those unnecessarily in hospitals (currently believed to number around 100 000), because of a shortage of nursing homes and in-home care services and because hospitalisation is cheaper to the individual, and that this will save it some 340 billion yen per year. The overall cost of the programme is estimated at 4.2 trillion yen per year,[140] half of which will be covered by budgetary revenues (half national and half local). The remainder will be paid for by a new charge of 2 500 yen per month on all those over 40, half of which will be paid for by employers; retirees will pay the premium as a deduction from their pensions. Some of the public financial burden will fall on local governments, and given that their finances vary, so might the quality of care provided, even though the Ministry of Health and Welfare has specified minimal national standards. Even though private companies will still be prevented from owning and managing nursing homes, the introduction of long-term care insurance is expected to promote market mechanisms not only in in-home care services but also to some extent in nursing homes, since physicians may own such nursing homes. Beyond the public finances, the biggest economic impact should be the expansion in nursing home construction and in the training and employment of appropriate care providers for in-home or nursing-home care services.

The Ministry set up a Council on Health Insurance and Welfare in November 1997 to deal with the next stage of reform. Agreement has been reached that an element of capitation should be introduced into doctors' fees. Current work is focusing on a new medical insurance scheme for the elderly. The other main focus of reform efforts has been drug authorisation and pricing. The Ministry had proposed to implement a reference price system (effectively ceilings), based on the German model, with patients paying any excess, but critics said this risked lowering the amount of research efforts in the area. The Japan Medical Association (JMA) has suggested changing the current system whereby

hospitals buy drugs directly and health plans pay the higher government-set price, thereby providing an incentive for hospital doctors to overprescribe. Instead, the JMA proposes that insurers should buy the drugs directly, pay hospitals an administration fee and eliminate the margin.

The Ministry has also indicated that the health insurance contribution system is in need of rebalancing: currently a much higher rate is imposed on regular wage and salary payments than on bonuses[141] or overtime payments. Even though bonuses have fallen as a share of total employee compensation, some movement toward equalisation is likely to be proposed in the overall reform planned for FY 2000. Finally, another official concern is over a longer-term excess supply of physicians. There are currently about 250 000 in Japan, nearly one for every 500 population,[142] and the trend increase has long been 5 000 per year. It is believed that in about 20 years this trend will generate more doctors than are needed (an excess of 6 000 in 2020 and 14 000 in 2025), and discussions are under way over whether to impose a *numerus clausus* or tighten entry standards into the profession.

Pension reform: boosting financial returns to offset the effects of ageing populations

Japan's pension situation – both private and public – is serious. OECD figures point to a probable rise in public pension expenditure from 6.8 per cent of GDP in 1995 to 14.3 per cent in 2020; the increase of 7½ percentage points of GDP is probably unequalled by any other Member country.[143] The public system has an adverse impact on the inter- and intra-generational resource allocation, and it has negative effects on the labour force participation decision for women and the elderly (Horioka, 1998). Recent work in generational accounting (Table 31) has shown an enormous generational imbalance (the excess of the net tax paid by future generations over that paid by those born today) of nearly $250 000 per person; without changes, future generations would face lifetime tax payments several times as high as today's new-born Japanese and those who are 25 or 30 have a similar burden to look forward to (Takayama *et al.*, 1998). Contribution rates for Employees' Pension Insurance are scheduled to rise again in FY 1999 from 17.35 to 19.5 per cent of salary, but benefits cannot be maintained at current levels in relation to wages and salaries unless contribution rates rise to

Table 31. **Generational accounts for 17 countries, 1995**[1]

US dollars, thousands

	Education as consumption					Education as a transfer				
	Age					Age				
	0	25	65	Unborn	Generational imbalance[2]	0	25	65	Unborn	Generational imbalance[2]
United States	86	175	–96	130	44	29	173	–96	74	45
Japan	**143**	**295**	**–48**	**386**	**243**	**73**	**295**	**–48**	**319**	**246**
Germany	165	310	–207	317	152	97	303	–207	249	152
France	151	322	–200	223	72	82	319	–200	161	79
Italy	114	184	–138	265	151	68	184	–138	210	142
Canada	114	184	–85	114	0	56	193	–84	58	2
Australia	80	148	–13	105	25	49	142	–13	73	24
Belgium	94	273	–166	148	54	43	271	–166	90	47
Denmark	84	251	–172	124	40	–18	232	–172	26	44
Netherlands	110	237	–91	194	84	49	237	–91	137	88
New Zealand	57	104	–50	55	–2	18	95	–50	16	–2
Norway	106	143	–171	174	68	1	127	–171	57	56
Portugal	62	85	–49	99	37	44	85	–49	73	29
Sweden	184	295	–111	144	–40	122	284	–111	84	–38
Argentina	23	34	–43	36	13	14	32	–43	24	10
Brazil	14	31	–33	27	13	10	30	–33	22	12
Thailand	8	14	4	1	–7	6	14	4	–2	–8

1. Assuming productivity growth of 1.5 per cent per year and a discount rate of 5 per cent.
2. Excess of ‘‘unborn’’ over ‘‘0’’.

Source: Auerbach *et al.* (1998, forthcoming).

34.3 per cent by the year 2025. Public support for maintaining benefits is, however, weak.[144] Besides cutting benefits, other possibilities for dealing with the problem include a further and/or faster rise in the retirement age,[145] to broaden the base for contributions to include income from part-time employment[146] and, similarly, to introduce National Pension contributions from full-time homemakers, to try to attract more people into the labour force (especially women and the elderly) and to pay for the rising pension burden through increases in taxes.[147] In addition, as from FY 2000 the existing funds will be managed by a new organisation according to prospective returns considerations. In June, a panel of the Ministry of Health and Welfare recommended that the 130 trillion yen in public pension deposits should be invested 65 to 80 per cent in domestic bonds and 5 to 15 per cent in domestic equities; this should achieve annual returns of 4.2 to 4.8 per cent. It is possible that the new oversight organisation will be explicitly banned from instructing managers to invest in specific stocks and from engaging in what are commonly known as ''price keeping operations'' (PKOs). For example, calls were made by some politicians in March 1998 for up to a trillion yen of these public funds to be used to try to raise equity prices at the end of the fiscal year so as to eliminate the capital losses that holders, especially banks, suffered during the year.

The situation of corporate pension funds is no better.[148] Employee pension funds are believed to be underfunded by at least half, since the 25 firms whose shares are listed on US stock exchanges have to disclose their unfunded liabilities,[149] and for them underfunding amounted to over 40 per cent (and they are a relatively prosperous group of firms). Companies are reacting by trying to cut benefits and topping up their contributions. In recent years some employee funds have begun to fail, and by FY 1997 there were 14 such failures.[150] There is a rescue reserve fund operated by the Pension Fund Association, but it contains only 8 billion yen, and in April it refused to come to the aid of two funds which had earlier failed and whose beneficiaries have since been left at a loss. Besides lifting the ''5:3:3:2 rule''[151] which constrained portfolio allocation in December 1997 (in advance of previous promises),[152] the government has liberalised fund management so that it is no longer the exclusive domain of trust banks and life insurance companies.[153] ''Investment advisers'' are now allowed to manage tax-qualified pension plans, and foreign firms have gained substantial market shares in this business.

A number of further reforms are being discussed. First, because there is so much variation in corporate pension plans, the Ministry of Health and Welfare will propose a basic framework law. This may include the equalisation of the tax treatment of corporate and other private pension schemes (for employees of small firms and the self-employed), but it does not look likely that this will deal with the lack of portability in the system as it stands now, a problem which will grow ever more serious as the prevalence of lifetime employment continues to wane (Table 30). Second, it is possible that defined contribution plans may be allowed as from FY 1999. This would create another actor on the financial scene pressing for higher investment returns. Third, a ruling-party panel called in June 1998 for a drastic easing of the rules governing property-accumulation type pension plans under which interest earned on deposits of up to 5.5 million yen is exempt from tax. In particular, plan holders would be allowed to switch financial institutions based on performance. Corporate plans are already switching, at least at the margin. The recent reform allowing pension funds to be managed by investment advisers led to a 42 per cent increase in their funds under management in 1997, because insurance companies are currently offering returns of only 2½ per cent per year. Many are switching to foreign advisers[154] who now manage about 8 per cent of the overall pension market.

Consumer and competition policy: strengthening the market mechanism

Despite a new Product Liability Law that took effect in July 1995 it is still felt that consumer rights are in need of further strengthening in light of the continuing increase in the number of consumer complaints.[155] Thus, in an interim report, the Social Policy Council suggests a reform of the Consumer Contract Law which would allow consumers to void contract terms if unreasonably disadvantageous and to rescind contracts if a business failed to provide adequate information or acted in bad faith. The number of cartels exempt from the Antimonopoly Act (but subject to approval by, consultation with or notification to the Fair Trade Commission (FTC)) continues to fall in line with the 1994 decision to end such exemptions within five years: by March 1997 there remained only 12, down from a peak of 1 079 in March 1976.

A proposal to examine the desirability and feasibility of amending the Antimonopoly Act so that victims of unfair trade practices could bypass the FTC and go directly to court to seek orders to cease such practices is under study by the FTC and the Ministry of International Trade and Industry. Amendments may be presented to the Diet soon after conclusions are reached.[156] Such disputes may become much more prevalent in the coming deregulated environment, but the current judicial system makes consumer litigation lengthy and costly. In May, the Diet approved an amendment to the Act which would reduce notification and reporting requirements and alleviate compliance burdens with respect to stock-holdings, interlocking directorships, mergers and acquisitions and business transfers. In June 1998 the FTC proposed draft new merger guidelines which were interpreted in some quarters as being somewhat easier but were intended merely to attach explicit numbers to the existing guidelines. The number of mergers and acquisitions involving Japanese companies in the first half of 1998 reached 434, up 28 per cent over the same period in 1997. FTC resources also continue to rise, but by no means in similar proportions: staff has risen only from 520 in FY 1995 to 552 in FY 1998. This compares with 960 full-time equivalents at the US Federal Trade Commission,[157] where a higher proportion are said to be economists.

IV. Reinvigorating business sector dynamism

Introduction

Not many years ago, Japanese corporations were often seen as model performers. There are reasons to think that they have lost some of their dynamism of earlier decades. The manufacturing sector has come to the end of the catch-up process, and many of the non-manufacturing industries, which have long operated in a regulated environment, have accumulated inefficiencies. More generally, weak business performance has led some to question the appropriateness of the Japanese corporate system in an environment which requires rapid decision-making and calculated risk-taking to achieve high rates of return. This chapter first presents some indicators of waning business dynamism and discusses possible reasons. After describing emerging signs of renewed vigour in industrial activity as well as of changes in business practices and arrangements, it concludes with a discussion of the role of policies to promote a dynamic business sector, focusing on the creation of new business activities, improvement of innovative capacity and the evolution of the corporate system.

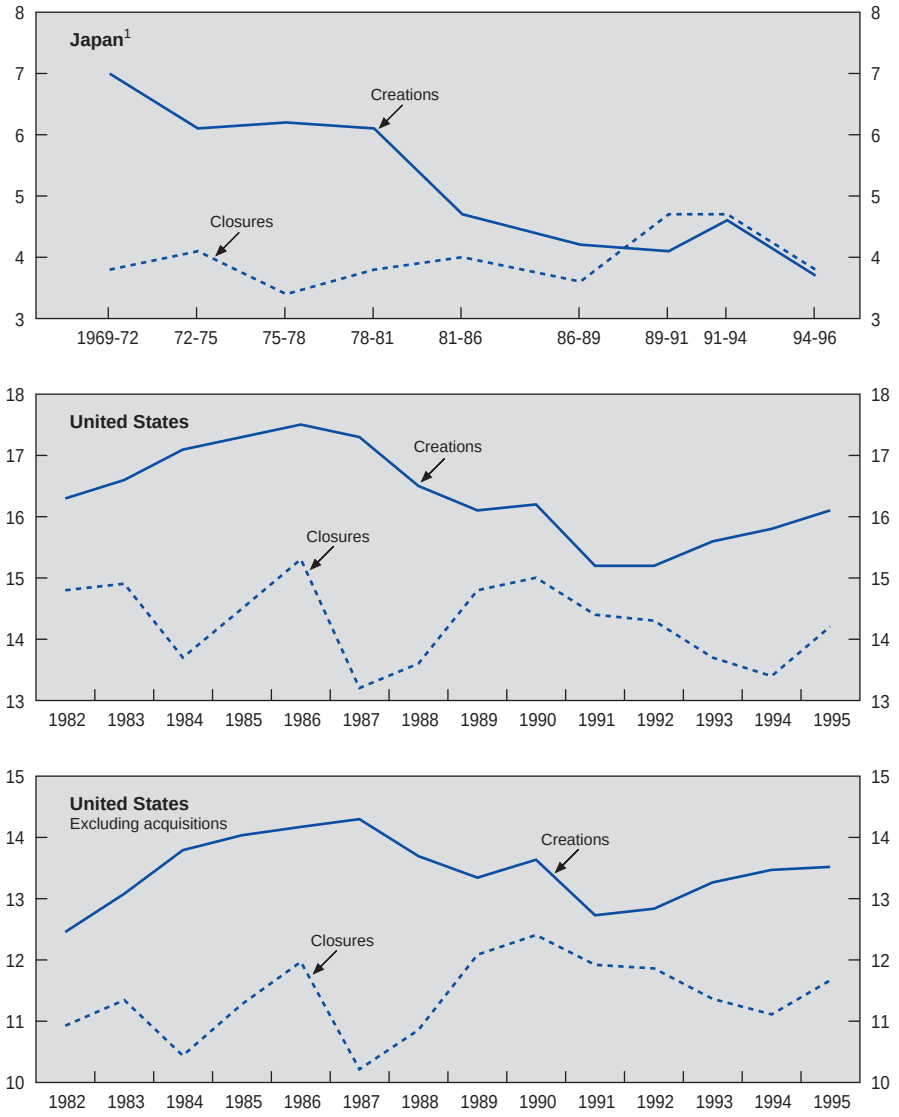
Waning business dynamism: indicators and possible reasons

Some indicators

There are indications that the Japanese business sector has lost some of its dynamism of earlier decades. One of the most worrying trends has been a declining rate of new company formation combined with a rising rate of company closure (Figure 31). To be sure, a rise in the rate of company closure as such should not be a matter for concern to the extent that closures represent an elimination of the inefficient. The firm birth rate had been over 6 per cent until the end of the 1970s but fell to about 4 per cent by the mid-1990s, whereas the

Figure 31. **NEW BUSINESS FORMATION AND CLOSURES**

Per cent



1. Annual rate.

Source: Management and Coordination Agency, *Establishment and Enterprise Census of Japan*; US Small Business Administration (1997).

death rate rose from the 3 to 4 per cent range to about 5 per cent over the same period, thus surpassing the birth rate. These trends are more marked for the manufacturing sector where the birth rate came down to about 3 per cent, compared with the situation in the service sector where the birth rate still remains at around 5 per cent. This gap in rates between the two sectors is commonly observed among the advanced industrial countries. It reflects a trend shift towards services production as well as the relatively lower capital requirements in setting up a service operation. While international comparison is difficult, both the birth and death rates are lower in Japan than in many other OECD countries, notably the United States where the rates are two to three times higher even after adjusting for acquisitions.[158] This suggests that the pace of metabolic change in business activity, a measure of the Schumpeterian process of creative destruction, is lower in Japan and has been declining.[159]

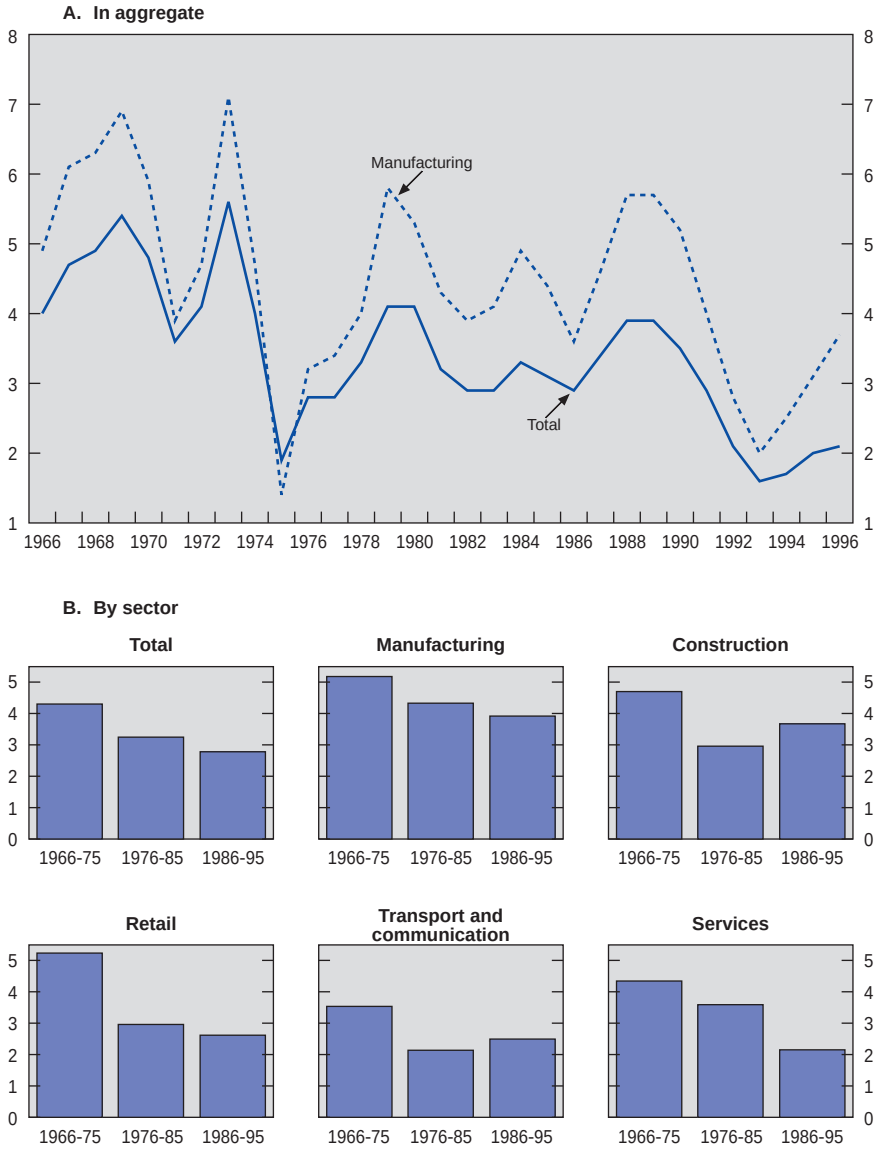
This concern should, however, be attenuated somewhat by the fact that new business activities often take place within existing companies. For example, about half of the small and medium-sized enterprises surveyed in 1996 replied that they had tried to enter new areas of business during the previous five-year period (Small- and Medium-Sized Enterprise Agency, 1996.) There are no statistics on the rate of metabolic change in existing companies more generally, but they should eventually be reflected in their profitability as stagnant companies typically lose out in a competitive environment.

Profitability in the business sector has continued to trend downwards. The ratio of current profits to total assets in non-financial corporations declined from an average of 4.3 per cent during the ten-year period 1966-75 to 3.3 per cent in the following decade, and further to 2.9 per cent in the 1986-95 period[160] (Figure 32). A similar trend has been observed for the manufacturing sector which saw this measure of profitability falling from 5.2 per cent to 4.1 per cent from the first to the last decennial period. As this is the sector which has been exposed to strong competitive pressure, often on a global scale, the declining profitability trend can be interpreted as an indication of weakening business dynamism.

Outside the manufacturing sector, on the other hand, it is less straightforward to interpret the profitability trend. Starting from a relatively high level of profitability in a regulated environment, a declining trend in profitability due to intensified competition would rather indicate enhanced business dynamism. It is

Figure 32. **BUSINESS PROFITABILITY**

Current profits as a percentage of total assets, FY



Source: Ministry of Finance Statistics Monthly.

striking that profitability rebounded in the 1986-95 period in construction and transport,[161] sectors which have typically been highly regulated, whereas it continued to decline in retail trade – which has experienced intensified competitive pressures coming from a rapid increase in supermarkets and convenience stores – as well as in services which can be characterised as being subject to atomistic competition.

Possible reasons

The end of the catching-up process

Cross-country comparison indicates that overall the manufacturing sector in Japan has broadly attained the levels of productivity of that in other high-income countries[162] (Table 32). This would generally mean that Japanese manufacturing companies cannot be expected to improve productivity any more by merely ''catching up'' to those in the more advanced countries. Within the manufacturing sector, however, the picture varies much across sub-sectors. Whereas Japan has been a leader in basic metals and transport equipment, it has lagged substantially in several including food, textiles, wood products and pulp and paper industries. The gap varies much more in Japan than in the United States or other Anglo-Saxon countries. This suggests a strong presence of country-specific characteristics in consumer preferences and entry barriers and protective measures in the Japanese market for those sectors with low relative productivity. These sectors can enhance productivity given the catch-up potential. But for most other industries incumbent companies will have to undertake greater innovation efforts, even if many of them may well remain competitive owing to their superior technical know-how pertaining to the production process, which has been the traditional strength of Japanese manufacturing companies.[163]

That Japan is reaching the end of the catch-up process can also be seen from the trends in R&D spending and royalty payments. Given the need to fill the large technology gap that existed between Japan and other developed countries at the end of World War II, R&D spending has increased rapidly and reached 3 per cent of GDP in 1995, the highest among the major OECD countries (Table 33). Such catch-up efforts have resulted in continued improvement in the balance of technology trade, which has been positive since 1993, as well as in a steady decline in the ratio of royalty payments to R&D expenditure. Most R&D spending has been allocated to applied research and experimental development.[164] Reflecting this

Table 32. **Manufacturing labour productivity levels by sector**

Value added per hour worked, leader country = 100[1]

	United States	Japan	Germany[2]	France	United Kingdom	Canada	Australia	Netherlands	Sweden
	Panel A: 1987								
Food, beverages and tobacco	**100.0**	32.3	75.3	65.3	46.1	59.6	45.9	95.4	57.3
Textiles, clothing and footwear	67.4	38.1	60.1	61.7	47.4	54.6	42.2	**100.0**	60.8
Wood products and furniture	69.5	15.6	50.2	52.4	38.1	63.8	32.7	**100.0**	64.1
Paper products and printing	97.2	47.5	61.2	65.0	64.7	81.4	53.2	62.7	**100.0**
Chemical products	80.8	52.9	60.1	58.0	59.5	68.0	44.9	**100.0**	72.4
Non-metallic mineral products	77.0	55.1	67.1	**100.0**	59.9	75.1	56.4	97.7	75.5
Basic metal products	94.4	**100.0**	80.3	77.0	74.2	89.3	57.1	80.3	93.3
Metal products	86.3	76.0	76.3	57.3	50.6	70.1	42.3	68.9	**100.0**
Machinery and equipment	99.0	85.6	73.8	**100.0**	65.4	64.2	61.1	59.1	66.5
Electrical machinery	**100.0**	82.7	67.6	90.0	51.3	66.4	35.8	93.7	75.6
Transport equipment	96.9	**100.0**	76.7	84.9	42.1	69.7	39.3	47.0	55.8
Other manufacturing	**100.0**	39.4	45.3	40.1	52.5	58.3	33.0	47.2	67.0
Total manufacturing	**100.0**	66.5	78.5	80.3	59.4	76.0	51.8	98.5	82.0
	Panel B: 1993[2]								
Food, beverages and tobacco	**100.0**	35.6	88.7	87.0	41.7	64.3	51.1	96.6	72.8
Textiles, clothing and footwear	78.3	41.9	72.1	67.1	51.5	46.3	32.3	**100.0**	66.5
Wood products and furniture	56.0	17.6	55.7	55.3	28.1	52.6	27.1	**100.0**	71.9
Paper products and printing	85.0	49.7	59.0	64.3	76.4	67.6	53.7	64.5	**100.0**
Chemical products	66.9	52.6	55.4	56.9	79.7	52.6	39.8	**100.0**	89.4
Non-metallic mineral products	81.8	62.9	79.5	99.4	70.6	78.4	77.4	**100.0**	81.0
Basic metal products	76.8	78.3	72.9	63.3	61.4	87.9	56.8	70.4	**100.0**
Metal products	68.9	67.6	64.6	46.4	42.5	54.8	35.9	54.0	**100.0**
Machinery and equipment	**100.0**	67.4	49.2	67.3	47.9	55.5	46.4	34.6	45.2
Electrical machinery	80.3	89.0	49.9	78.9	48.2	51.9	28.0	82.2	**100.0**
Transport equipment	88.4	**100.0**	68.0	85.0	47.8	71.9	45.5	41.8	49.5
Other manufacturing	**100.0**	41.4	43.1	31.4	43.5	33.5	22.1	27.0	47.4
Total manufacturing	**100.0**	76.6	79.8	84.2	64.1	71.3	52.0	95.6	91.8

1. The productivity level of the leader country in each industry is indicated in bold.
2. West Germany.

Source: Pilat (1996).

Table 33. **Selected statistics related to R&D**

	Financing			Academic performance in natural sciences						Share of graduate students by subject (%)[2]	
				Number of papers (share,[1] %) (1)		Number of citations (share,[1] %) (2)		Measure of ''quality'' = (2)/(1)			
	Total expenditure on R&D as % of GDP 1995	Share of business enterprise expenditure on R&D financed by government (%)	Share of higher education's expenditure on R&D financed by business enterprises (%)	1986	1996	1986	1996	1986	1996	Studying science	Studying engineering
Japan	3.0	1.6	2.3	7.7	9.9	6.6	7.8	0.86	0.79	10.2	42.4
France	2.3	12.7	3.3	5.6	6.6	5.1	6.5	0.91	0.98	–	–
Germany	2.3	8.8	7.9	7.8	8.5	6.6	9.9	0.85	1.16	27.8	10.0
Italy	1.0	16.7	5.6	–	–	–	–	–	–	–	–
United Kingdom	2.0	11.3	6.2	8.9	9.2	10.3	11.9	1.16	1.29	10.7	7.7
United States	2.6	17.8	5.5	37.5	34.6	54.6	51.6	1.45	1.49	5.0	10.2
Canada	1.7	7.1	10.3	–	–	–	–	–	–	–	–

1. Shares are calculated from wider samples.
2. Data for graduate students are for 1994 for Japan and the United Kingdom, 1993 for the United States, and 1991 for Germany.

Source: OECD, *Basic Science and Technology Statistics* and US Institute for Scientific Information, *Natural Science Indicators 1981-96*.

orientation toward application, the government's share in total expenditure has been relatively small – about one-quarter compared with around a half in other countries – and so has government's financial support to industry in this regard.[165]

In sectors where the catch-up process ended earlier, Japanese companies have relocated production abroad in order to benefit from cheaper labour costs, less stringent regulation and lower taxes and social insurance costs (see Chapter I). After a decade of massive investment largely concentrated in Asia, however, concerns arose about the ''hollowing out'' of Japanese manufacturing industries as it resulted not only in a closure of large factories but also a massive restructuring of clusters of related suppliers and sub-contractors[166] (Box 5). Those clusters which have been particularly hard hit are situated in central Japan, the heartland of many of Japan's technological achievements.[167] These concerns can be illustrated by the large drop in employment in ten sub-sectors of the

Box 5. **Clusters**

The phenomenon of ''clustering'' refers to the apparent tendency of firms in the same line, or related lines, of business to be geographically concentrated. Clusters are thought to arise from agglomeration economies, since concentrations of firms create larger markets for specialised labour and intermediate inputs. They can also generate informational spill-overs, enhancing the ability to stay abreast of the latest industry developments. Clusters may develop initially because of intrinsic advantages found in a region, including natural resources such as mines, natural port facilities and forests. For example, firms in the steel industry are often established close to energy supplies and good transportation networks. Or, less well understood, some regions may provide more fertile ground for enterprise development because of the presence of an enterprise culture or a more favourable institutional framework.

A manufacturing cluster can be stable over long periods of time, perhaps because its advantage increases with the size of the cluster. It is estimated that 60 per cent of Japan's clusters were created before the 20th century (OECD, 1996*b*). However, it would appear that these cumulative advantages are not altogether decisive, and the position of an entrenched cluster can be successfully challenged. As production becomes standardised over time, localisation within an industry can tend to fade away. In some cases there appears to be a kind of product cycle in which emergent new industries initially flourish in localised industrial districts, then disperse as they mature.

electronics industry where hollowing-out is thought to have been the greatest (Table 34). While hollowing out is a common phenomenon among advanced industrial countries, it nonetheless left a hole to be filled.

Table 34. **Job cuts in ten sub-sectors of the electronics industry**

Changes in the number of employees

	1991	1994	% change
Sound recording or reproducing apparatus	152 122	90 723	–40.4
Video cassette recorders	117 717	63 925	–45.7
Other applied electronic devices	51 951	36 770	–29.2
Computing machinery	164 852	149 960	–9.0
Electricity distribution and control apparatus	160 156	146 792	–8.3
Electrical equipment for vehicles	107 213	94 357	–12.0
Household appliances	136 728	123 941	–9.4
Electric motors and generators	74 739	62 535	–16.3
Electronic tubes	31 298	22 925	–26.8
Electric measuring instruments	34 570	28 102	–18.7
Total	1 982 887	1 768 753	–10.8

Source: Ministry of International Trade and Industry, *Census of Manufactures.*

Regulations in the non-manufacturing sector

It is difficult to make generalisations about the non-manufacturing sector which is rather heterogeneous. But a common characteristic across non-manufacturing industries is that they are typically subject to various regulations and often sheltered from foreign competition either by their autarkic nature or by border protection. The speed and extent of deregulation already undertaken vary from one industry to another. In some, notably distribution and telecommunications, a substantial easing of restrictions has resulted in large price cuts and expansions in both the range and amount of services provided. For example, the easing of the *Large-Scale Retail Store Law* in 1990, 1992 and 1994 allowed an opening of larger, more efficient retail outlets, which is estimated by the Economic Planning Agency to have increased annual demand by about 0.8 per cent of GDP in the first half of the 1990s. The EPA also estimates that an increase in telephony demand of 0.4 per cent of GDP over the same period is

attributable to deregulation in telecommunications. In contrast, in most other sectors deregulation has been slow and partial.[168] The result has been the persistence of inefficiencies and economic rents in these latter industries.[169]

An international comparison of productivity and efficiency in non-manufacturing sectors is even more difficult than that for the manufacturing sector. Nonetheless, some selective comparisons indicate that Japan clearly trails the United States, even though some Japanese service industries are ahead of their European counterparts.[170] Among the least efficient is the financial sector which is discussed elsewhere in the present Survey. Another is the construction sector for which a detailed comparison is made below.

Comparative national accounts data show that in recent years Japan's construction sector has been responsible for over 10 per cent of GDP, a share surpassed only by that of Korea among OECD Member countries for whom such data are available (Table 35) and greater than it has been over the entire post-war period, despite the obvious trend decline in the need for new housing, discussed above, and the progressive satisfaction of infrastructural needs. Employment figures present a similar picture, although very recent indicators point to belated substantial downsizing in the sector.[171] Looking below the surface it is obvious that the location of some of this excess is in the public investment share of GDP, which is twice as large in Japan as the OECD average. In order to ensure that this outcome is not the result of differences in relative prices across countries, it is appropriate to resort to purchasing power parities (Table 36). To some extent a higher construction share does appear to be attributable to higher absolute and relative prices: prices were 41 per cent above OECD average levels for the sector in 1996, mainly due to a widening gap in the civil engineering sub-sector which reached 67 per cent that year. But even after making the price adjustment and evaluating spending at average OECD prices, real spending per capita was nearly double the OECD mean, a difference of $2 467, most of which was attributable to business investment in buildings and to civil engineering works (*i.e.* public investment). Similar differentials – as well as a tendency for them to widen – are apparent in the GDP share figures. Thus, it would appear that higher prices and higher volumes have combined to lead Japan to devote an unusually high share of its GDP to construction, with a gap that was equal to nearly 7 percentage points over the OECD average and 9 percentage points over the US outcome in 1996.[172] However, that was a year of strong growth in the construction sector, and it is

Table 35. **The construction sector**

	Employment in construction % of total employment				Value added in construction % of total output			
	1970	1980	1990	1996[1]	1970	1980	1990	1996[1]
United States	5.3	5.7	5.7	5.5	5.1	4.8	4.5	4.1
Japan	8.1	10.1	9.7	10.5	7.7	9.4	10.1	10.3
Germany	..	..	..	8.4	..	..	..	6.1
France	9.8	8.7	7.4	6.4	..	6.9	5.2	4.3
Italy	10.2	8.0	7.0	7.2	9.5	7.4	5.9	5.0
United Kingdom	6.3[2]	6.4	6.7	..	5.7[2]	5.3	6.3	4.6
Canada	..	5.8	6.3	5.4	6.2	6.6	6.3	5.5
Australia	8.5	7.7	7.4	7.2	8.6	8.0	6.8	6.0
Austria	..	8.4	7.8	8.9	..	7.2	6.3	7.4
Belgium	8.6	8.1	6.6	7.1	7.5	7.4	5.3	4.8
Denmark	9.7	7.8	6.6	6.7	9.4	6.5	5.1	4.7
Finland	9.2	7.8	8.8	6.3	8.8	7.2	8.4	5.1
Greece	7.6[2]	..	..	..	8.0[2]	7.6	6.4	5.2
Iceland	10.7	10.1	9.1	7.9	..	6.6	6.3	5.2
Ireland	..	..	..	..	..	..	4.7	4.5
Korea	..	..	..	..	5.1	8.0	11.6	14.5
Mexico	..	..	..	..	..	..	3.6	3.7
Netherlands	..	9.8	7.5	7.2	..	6.6	5.2	4.9
New Zealand	..	..	3.6	..	6.3[2]	5.2	3.9	
Norway	..	..	..	..	..	4.5	4.1	3.6
Portugal	..	..	8.4	..	..	..	5.3	5.3
Spain	..	..	9.4	8.9	..	..	9.1	8.2
Sweden	..	7.1	6.6	5.3	..	6.7	6.9	4.8
Turkey	..	5.5	5.1	5.3	5.8	5.7	6.3	5.5
Simple average of above countries	8.5	7.8	7.2	7.1	7.2	6.7	6.2	5.8

1. Data are for 1996 or most recent year available.
2. Data are for 1971.
Source: OECD, *Annual National Accounts*.

possible that the gap may have shrunk somewhat over the intervening years. Nonetheless, for a country with a rapidly ageing population (and, therefore, less need for new housing) and an already high level of business capital intensity,[173] it is hard to avoid the conclusion that this gap is unjustified and represents a waste of resources, even allowing for the need to further develop certain kinds of social infrastructure such as sewage treatment and urban parks.

Table 36. **The size of the construction sector: an international comparison, 1985-96**

	Total				Residential buildings				Non-residential buildings				Civil engineering works			
	1985	1990	1993	1996	1985	1990	1993	1996	1985	1990	1993	1996	1985	1990	1993	1996
Real spending per head at average OECD prices (US$)																
Japan	1 635	3 216	3 808	4 966	375	1 158	1 189	1 516	485	1 062	1 465	1 870	775	996	1 155	1 580
EU[1]	1 059	1 912	2 161	2 331	453	804	925	1 126	320	620	657	617	287	488	578	587
United States	1 836	2 215	2 303	2 759	934	1 149	1 086	1 290	509	645	653	806	393	422	563	663
OECD	n.a.	2 228	2 435	2 499	n.a.	972	1 029	1 084	n.a.	696	736	737	n.a.	559	669	678
GDP shares at average OECD prices (%)																
Japan	13.9	15.6	15.8	18.2	3.2	5.6	4.9	5.6	4.1	5.2	6.1	6.9	6.6	4.8	4.8	5.8
EU[1]	10.1	10.8	11.2	10.9	4.3	4.5	4.8	5.3	3.0	3.5	3.4	2.9	2.7	2.7	3.0	2.7
United States	11.1	9.3	8.6	9.2	5.7	4.8	4.1	4.3	3.1	2.7	2.5	2.7	2.4	1.8	2.1	2.2
OECD	n.a.	11.5	11.3	11.4	n.a.	5.0	4.8	4.9	n.a.	3.6	3.4	3.4	n.a.	2.9	3.1	3.1
Comparative price level[2]																
Japan	125	130	131	141	169	133	109	134	114	112	135	124	99	145	154	167
Relative prices (GDP = 100)																
Japan	124	114	115	105	154	114	109	100	117	99	92	92	113	130	151	125

1. 12 countries in 1985, 15 countries as from 1990.
2. US = 100 in 1985, OECD = 100 as from 1990.

Source: OECD (1987), *Purchasing Power Parities and Real Expenditures, 1985;* OECD (1993), *Purchasing Power Parities and Real Expenditures, 1990,* Vol. 2; OECD (1996), *Purchasing Power Parities and Real Expenditures, 1993,* Vol. 2; for 1996: OECD Secretariat provisional estimates.

Some lacunae in the corporate system

While the legal features of Japan's corporate system are similar to those in other OECD countries, direct incentives for managers to enhance shareholder value have been lower, and this is likely to have weakened business dynamism. A majority of the members of the board of directors is promoted from inside, having made a career in the company, and sanctions through the market for corporate control are extremely limited.[174] Cross-shareholding has resulted in only about 20 to 40 per cent of shares of *keiretsu* companies being actively traded on the stock exchange. This arrangement is said to have insulated the firms from hostile takeovers and management from pressure to achieve short-term profit, thereby promoting more strategic decision-making from a long-term perspective. This has also meant, however, that the board of directors cannot function as an impartial monitor of the performance of senior corporate executives who themselves are board members and appoint other directors. As well, with directorship widely considered as a reward for dedication to a company, the board has grown in size, often counting as many as 40 directors in large companies, and ceased to be an effective body for making strategic decisions.[175]

The key institutional shareholder which exerts primary control over managers is the so-called main bank which also is a principal lender. Strong bonds between banks and companies have been formed through the dependence on bank loans rather than equity markets through most of the post-war period. Long-term relations between them have meant that bank lending has traditionally been made to a company secured by a real estate collateral, rather than in the form of project finance which requires assessment of business risks. The main bank is commonly believed to monitor managerial performance, thereby reducing the monitoring costs to other shareholders,[176] and, if necessary, to discipline managers – as a substitute for the open market for corporate control. Another often-cited advantage of main banks is lower costs of dealing with financial distress in comparison with going through a protracted formal bankruptcy procedure. It has often been claimed that these features of the corporate governance arrangements contribute to reducing the cost of capital to companies.

These corporate governance arrangements are, however, likely to have contributed to managerial aversion to large risks: directors promoted from within tend to favour continuity over change, and the reliance on banks as the chief

corporate monitor is likely to have resulted in managerial orientation towards low-risk options. The absence of a significant change in the composition of business investment towards innovation and new product development seems to be consistent with such a conjecture (Table 37). This managerial risk aversion is a plausible explanation of reduced business dynamism.

Table 37. **Investment motives**

Share of total outlays in per cent

	Capacity expansion	New product development and upgrading of existing products	Rationalisation and labour saving	R&D	Maintenance and repair	Others
1973	57.9		15.2	4.6		22.2
1974	51.7		15.8	5.0		27.5
1975	46.7		15.4	3.2	10.9	23.8
1976	51.0		14.9	3.5	13.7	16.9
1977	53.6		12.0	2.9	12.7	18.8
1978	54.4		13.2	2.8	13.3	16.2
1979	53.1		14.3	3.8	12.7	16.1
1980	53.9		13.9	4.1	11.8	16.2
1981	52.5		15.3	4.5	12.3	15.3
1982	50.8		16.0	4.8	11.7	16.6
1983	51.1		16.0	5.3	11.5	16.1
1984	48.9		14.4	6.2	16.3	14.2
1985	46.4		14.6	6.3	15.1	17.7
1986	50.0	8.2	10.5	5.8	12.4	13.1
1987	49.8	5.7	7.5	5.1	11.2	20.7
1988	53.3	7.3	9.4	5.1	12.4	12.5
1989	49.6	8.1	10.1	5.8	12.9	13.4
1990	51.0	8.8	9.9	5.4	12.3	12.7
1991	50.7	8.4	10.4	5.1	11.8	13.5
1992	50.8	7.3	9.7	4.7	13.6	14.0
1993	48.3	6.5	8.0	3.7	14.4	19.1
1994	45.1	5.7	7.3	4.1	13.7	24.1
1995	48.0	6.6	7.8	3.9	15.0	18.7
1996	48.9	7.5	8.0	4.0	14.5	17.2
1997	47.6	8.2	7.4	4.5	14.4	17.9
1998	47.3	8.4	7.5	4.5	15.0	17.3

Source: Japan Development Bank.

Some recent developments

A reviving manufacturing sector?

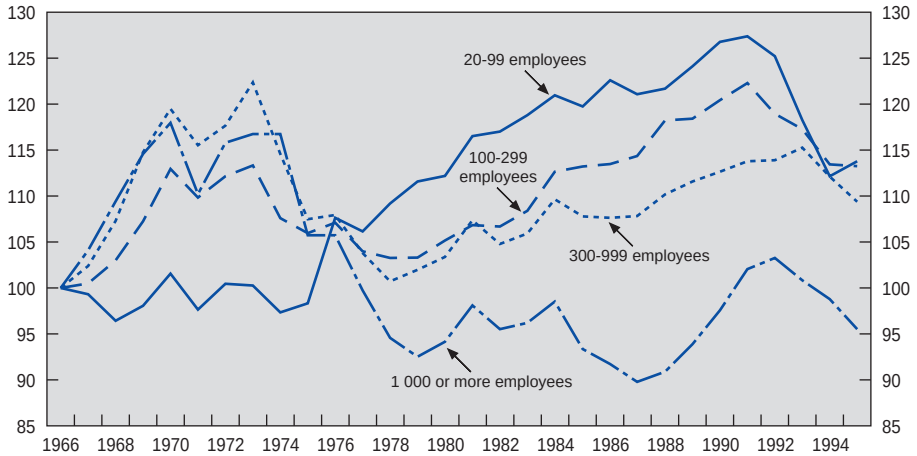
Some manufacturing clusters have successfully emerged from the ''hollowing-out'' process by reorganising their networks to more closely resemble the loose networks found in Silicon Valley or Italy's Emilia-Romagna. Vertical networks with many suppliers dealing individually with one buyer are becoming increasingly rare. More common are horizontal networks with suppliers engaging in close, but not exclusive, relations to supply a product. For example, metal parts suppliers located in the region of Nagano Prefecture have established a co-operative with other suppliers to receive orders jointly from large manufacturers. Because it groups together suppliers producing different types of products, the co-operative can receive orders more easily than a single supplier can. These changes in the traditional supplier system are not only promoting a greater independence among small and medium-sized suppliers, they also encourage the pursuit of multiple technical opportunities with many groupings and regroupings of skills, technology and capital.

The most conspicuous of the emerging clusters can be found in the so-called Tama high-tech corridor, which encompasses parts of Tokyo and of the two adjacent prefectures, Saitama and Kanagawa. Open networks of large electrical machinery makers, their research laboratories, some 20-odd universities with science and engineering departments, middle-sized companies with strong product innovation capability (many of them possessing a dominant position in niche markets) and smaller companies with specialised metalworking capabilities based on the most advanced technologies have developed there.[177] These middle-sized companies spent over 7 per cent of their sales on R&D in 1997, compared with the average for manufacturing industries of 3.4 per cent. The resilience of these companies can also be seen in their shipments which grew 6 to 8 per cent at an annual rate over the 1993-95 period, while total manufacturing shipments fell slightly.[178]

Indeed, according to a recent study of the manufacturing sector, it is those companies between 100 and 999 employees that have best survived the turbulent period characterised by the two oil shocks, the asset price bubbles as well as the large swings in the exchange rate. Figure 33 below provides the time series data on the number of manufacturing firms by employee size. The distinct reversal of

Figure 33. **THE NUMBER OF MANUFACTURING COMPANIES BY SIZE**

1966 = 100



Source: Ministry of International Trade and Industry, *Census of Manufactures.*

trend between the small firms (with 20 to 99 employees) and the large ones (employment of 1 000 persons and above) after the first oil crisis symbolises the end of the heavy-industry-led growth era. The total number of firms peaked in 1991 with the bursting of the asset price bubble, and the subsequent fall was particularly marked for small firms. The size category which recorded the smallest decline was that of middle-sized firms (employment of 300 to 999 people). It is this category of firms and the one just below (100 to 299 employees) that outperformed others in terms of the growth of value added, employment and labour productivity between 1986 and 1995.

Changes in the corporate system

Among the main features of traditional Japanese business arrangements the supplier relations (often referred to as vertical *keiretsu)* have changed the most. A study by the Small and Medium-Size Enterprise Agency (1996) reports that the percentage of companies maintaining strong ties with current suppliers has dropped to half of previous levels. In addition, purchasing companies,

pressurised by intensified international competition, have looked outside their group for lower cost suppliers, both foreign and domestic. For example, parts procurement in the auto industry has long been viewed as a model of Japanese vertical *keiretsu*, but Japanese auto makers purchased almost $20 billion in US-made parts in 1994, an eight-fold jump since 1986 (Economic Planning Agency, 1996). Another example is the household electric appliances industry which, as noted, moved their assembly operations abroad on a large scale. This resulted in an abrupt end to traditional supplier relations in that industry. For their part, suppliers have also shifted production abroad in order to cut costs and sometimes to accompany the key purchaser.

Some key aspects of the long-standing Japanese corporate governance arrangements, namely the monitoring role of the main bank and the business groups which are tied through cross-shareholding, have also been under pressure for change, though the extent of actual change seems to have been limited so far. The extent of true cross-holding within groups is difficult to estimate, but it appears to have been declining slowly but steadily. A 1994 Fair Trade Commission survey found that intra-group shareholding averaged 22.2 per cent (though it varied from 14.2 to 38.2 per cent across groups).

With alternative sources of capital widely available to Japanese firms (including from overseas sources) and a decline in investment requirements, dependence on bank loans is diminishing. As described in Chapter I, corporations relied on banks for 19 per cent of their financing in 1990-94, down sharply from 42 per cent in 1970-74 (OECD, 1996*a*). Symmetrically, because of their need to boost their capital and improve profitability in an increasingly competitive environment, banks' holding of shares in non-financial corporations has generally declined since last year.

Firms' employment and pay arrangements have remained largely unchanged until very recently. The practice of long-term employment became even more prevalent over the twenty years to the mid-1990s. The share of men in the 40 to 50 age group who have worked for a single employer increased between 1980 and 1990 before stabilising. This, together with the extension of the retirement age, raised the average job tenure for male workers from 9.8 years in 1974 to 12.8 years in 1994.[179] And the evolution of age-earning profiles does not point to any significant changes in seniority-based pay arrangements. More recently,

however, a number of large companies have announced changes in pay determination, giving more weight to individual performance than length of service (see Chapter III).

Surveys of companies show that while they plan increasingly to base their pay and promotion systems on individual performance, they continue to favour long-term employment for its well recognised merits – it contributes to maintaining employee loyalty to the company and facilitates long-term investment in training (Economic Planning Agency, 1997). It is not, however, clear to what extent merit-based pay and promotion can be compatible with long-term employment. Moreover, surveys of employees indicate a rising proportion of those wishing to change jobs in the 15 to 24 age group (it rose from 7.1 per cent in 1971 to 18.4 per cent in 1992) and 25 to 34 age group (from 5.6 to 13.7 per cent).[180] This suggests that long-term employment practices may become much less prevalent in the future, despite employers' wish to retain them.

Enhancing business dynamism

The trends discussed above point to the need to reinforce a number of changes, most of which are already in train. These include the nurturing of new business activities, improving the R&D system, reforming the education system, and facilitating the evolution of business arrangements and corporate governance practices towards those which promote flexibility and accountability.

Nurturing new business activities

The creation and expansion of new business activities are the key to raising the standards of living in advanced countries like Japan. But, as noted above, the rate of new company formation has been on a declining trend, and the emphasis on innovation and new product development in the existing companies' investment decisions has not visibly increased. The government has adopted a number of measures to promote the establishment and growth of venture companies largely by improving the provision of private risk capital, which would also be assisted by the financial Big Bang reform discussed above.[181]

The size of the venture capital stock in Japan, a little less than one trillion yen in 1997, is comparable to that of Europe but rather small compared with that in the United States. Moreover, while firm figures are difficult to obtain, so-called

‘‘angel capital’’, which in the United States is said to be twice the size of venture capital, is virtually non-existent in Japan. Long-standing problems have inhibited the supply of private risk capital in Japan, many of which have been addressed over the last several years. Three sets of measures are significant. The first, which pertains to enlarging the potential sources of risk capital, includes the extension of the loss carry-over period for tax purposes for venture companies, the favourable tax treatment of investment by business ‘‘angels’’ and the easing of restrictions on investment in unlisted shares by corporate pension funds and mutual funds. This last has opened up new investment opportunities for fund managers and enhanced the potential supply of risk capital – this would be complemented by the introduction of the limited partnership business form by which these institutional investors’ liabilities would be unambiguously limited to the amount of investment in a venture business.[182] These changes have paved the way for a large influx of funds into the venture capital industry, just as was the case following similar changes in the United States that occurred in the late 1970s.

The second set of measures is aimed at improving the marketability of the venture company shares. The practice by JASDAQ of requiring the companies wishing to be listed to show profits – one of the reasons why the average age of companies at the initial public offerings (IPOs) was 17 years in Japan as opposed to 5 years in the case of the US NASDAQ – has been circumvented by creating a special OTC market for venture companies with R&D-to-sales ratios of 3 per cent or above, but who do not have to show a profit before launching IPOs. Steps have also been taken to strengthen the market-making function of securities dealers in the OTC market, which should contribute to increasing trading volumes. Moreover, some measures have been taken to facilitate transactions in unlisted shares, which should help venture capital companies to dispose of so-called ‘‘living-dead ventures’’, that is those which are neither a big success nor a total failure, thereby raising the turnover of risk capital.

The third set of measures is directed towards helping venture firms to obtain loans. These include the provision of loan guarantees and concessionary loans to venture firms from government financial institutions. An innovative attempt has also been made to collateralise intellectual property rights but has as yet produced only a few examples. In any event, debt is not the most suitable form of financing risky ventures and can only play a supplementary role in short-term finance.

The main problem in developing venture capital is the shortage of experienced venture capitalists who can not only assess the risks of proposed projects but also closely monitor the company and intervene if necessary. With the majority of venture capital companies (165 in all as of March 1997) being subsidiaries of financial institutions, they have tended to take up late-stage ventures rather than those at the seed or the start-up phase. In addition, those working in the affiliates of financial institutions often are seconded by their parent companies for several years and hence do not accumulate much experience as venture capitalists. Even so, there are some 50 independent venture capital companies, and their cumulative experience is by now non-negligible. It is also encouraging that in 1997 the share of early-stage involvement increased despite the onset of recession.

A number of large well-established companies have tried to create in-house ventures, more as a way of shaking up the existing corporate culture and stimulating the entrepreneurial spirit of their employees, rather than identifying new business opportunities. Large companies aggressively diversified their business in the 1980s, but the results have not been satisfactory. As a consequence, there has been a return to a strategy of expansion based on core competencies, and successful new activities have tended to be related to existing lines of business. R&D activities have also tended to be decentralised to the level of business divisions, and any radically new ideas have been pursued at the firms' central research laboratories. In-house ventures too have been dealt with at the corporate headquarters. Surveys carried out by Keidanren and MITI report many difficulties with this approach. At their heart is the inevitable contradiction of asking the inherently conservative employees of large companies to take risks. Even though the success rate of new ventures is low by definition, it has been difficult for employees who volunteer for in-house ventures to obtain any positive recognition in the event of failure.

Improving the R&D system

Japan's application-oriented R&D system has acquired certain characteristics some of which are now seen as impeding creativity. R&D activities in each sector have been conducted on its own funds, and a clear ''division of labour'' has developed, with basic research undertaken in higher education institutions and applied research in industry; this may have impeded the full exploitation of

beneficial co-operation between them. The academic level of natural science studies in Japan is comparable to European countries in terms of the number of papers registered and cited in major scientific journals,[183] although there still seems to be room for improvement in research ''quality''. The application-oriented R&D activities have also led to Japan having fewer graduates in science and more in engineering than in other countries (Table 33). Once recruited by companies, these graduates have been trained and have experienced a wide range of jobs: even laboratory research staff tend to experience one or two jobs in the early stages of their careers, and some of them may be promoted to managers outside the research function.[184] The firm-specific skills obtained during job rotation, as well as close and flexible co-operation among related departments, are thought to facilitate smoother and faster development of new products in Japan.[185]

The government has taken initiatives to overcome some of the perceived weaknesses of the R&D system. The Science and Technology Basic Plan (decided in July 1996) and the Education Reform Programme (decided in January 1997, revised in August 1997) provide the framework for such an endeavour, the main features of which are summarised in Table 38. The Plan envisages a large increase in government investment in R&D (17 trillion yen in total from FY 1996 to FY 2000) which is expected to raise its ratio to GDP to a level comparable to that in other developed countries, thereby boosting basic research. Another major aim of the government is to change the current system, which is seen as highly compartmentalised and rigid, into a ''more flexible, competitive and open system''. The improvement of the system is to be achieved by increasing information flows among different sectors,[186] enhancing the efficiency of R&D activities by promoting the co-operation between academics and industry, introducing competitive factors in budgetary allocation and making third-party evaluations obligatory for publicly funded projects.

Universities are expected to become a more important source of innovation, and the transfer of the results of innovation to industry is to be facilitated. The research efforts will also be stimulated by several new incentives: patents will be counted as academic achievements like papers, and the period in which participating firms can exercise a prior executing right of outcomes of joint research is to be extended. Some universities have started to set up Technology Licensing Offices to commercialise their academic results and reinvest the profits in their research activities, and such efforts will be facilitated by legal changes.[187]

Table 38. **Government's initiatives to reform the R&D system**

Summary of measures	Major progress made
I. Summary of Science and Technology Basic Plan (July 1996): Reforms of the R&D system	
a) Measures to increase the mobility of researchers.	Introduction of a fixed-term employment contract (implemented by revised laws in 1997).
b) Investment in human resources.	Implementation of the ''support programmes of 10 000 postdoctoral fellows'' and expansion of research support staff.
c) Measures to enhance co-operation between industry, academics and government.	*a)* Easier permission to engage in side jobs by government researchers; *b)* Easier participation in joint research efforts; *c)* A law to promote technology transfer from universities to industry, which passed the Diet in May 1998.
d) Fair and strict evaluation.	Evaluation system with clear criteria, including evaluation from outside is being introduced.
Increase in government R&D investment About 17 trillion yen for government R&D expenditures cumulative from FY 1996 to FY 2000, to be achieved with a ceiling of 5 per cent growth in Fiscal Structural Reform Act.	
II. Summary of relevant parts of Education Reform Programme (revised in August 1997): Reform of universities	
a) Universities in the 21st century.	
– Reform of graduate schools.	Examine how to improve business education and strengthen research function as COE (Centre of Excellence).
– Strengthened undergraduate education.	Examine the specific ways to strengthen education, while keeping each university's merits.
– Review of management system of universities.	Ensure autonomous decision-making: establish leadership of Presidents and Deans. Increase flexibility of personnel and improve the accounting system of national universities.
– Objective evaluation mechanism of universities.	
b) Flexible higher education.	
Promotion of research activities	
a) Promotion of academic research and science and technology research.	The Academic Council will conclude around November 1999
b) Activate research activities by promoting effective co-operation between academics and industry.	
– Promote transfer of outcomes of academic studies to industry.	Implement a new law, which passed the Diet in April 1998, to promote the transfer.
– Easier utilisation of joint research facilities.	

Source: Science and Technology Agency, Ministry of Education, and Ministry of International Trade and Industry.

As for patents, two main measures have been taken. First, the patent-related laws were revised in April 1998 so as to enhance the incentive to innovate through strengthened protection of innovators' intellectual property rights.[188] Second, a more efficient exploitation of unutilised patents, officially estimated to include 360 000 out of 830 000 valid patents in total as of the end of 1997, has been promoted since FY 1997. Some of these dormant patents belong to large companies which tend to find the potential size of the new business to be generated through their application too small to act on. These patents can, however, be rather attractive to small and middle-sized companies.

Education reform

The education system was one of the six pillars of reform the Prime Minister first outlined in January 1997. The overriding objective is to boost creativity by making the system more heterogeneous. Uniformity is widely believed to be excessive, and individuality will be promoted by offering more options as from the upper secondary level and greater autonomy for individual schools and municipalities. Another goal will be to shift the emphasis slightly from knowledge acquisition to ''education of the heart''. More concretely, class time spent on fundamental subjects will be cut in order to devote more time to subjects such as the arts and philosophy/morality. It is hoped that this will help counter the recent trend toward greater juvenile delinquency, no doubt associated with a rising high school dropout rate.[189] Finally the reform aims to strengthen the system of higher education and research.

At the level of higher education, the Education Reform Programme goes far beyond the improvement in the university research function and its interaction with industry. Emphasis will be given to the expansion of graduate studies, human resource development in co-operation with the business sector (for example, through the promotion of internships, entrepreneurship and hiring businessmen as part-time lecturers) and improving the quality of education and research through the introduction of fixed-term contracts, the absence of which has impeded the mobility of faculty members and detracted from the intensity of academic competition.

The Programme also recognises the importance of adapting polytechnic institutions to the changing needs of the economy and society as well as of making their academic credits transferable to universities. Technical and

professional schools can be expected to play a significant role akin to that of community colleges in the United States in continuing education and providing training services tailored to employers' needs. More generally, improving the portability of academic credits would expand the possibilities for students to change academic orientations and allow many students who failed the entrance examinations a second chance to enter the universities of their first choice without spending an extra year (or more) at preparatory schools, which is generally seen as wasteful.

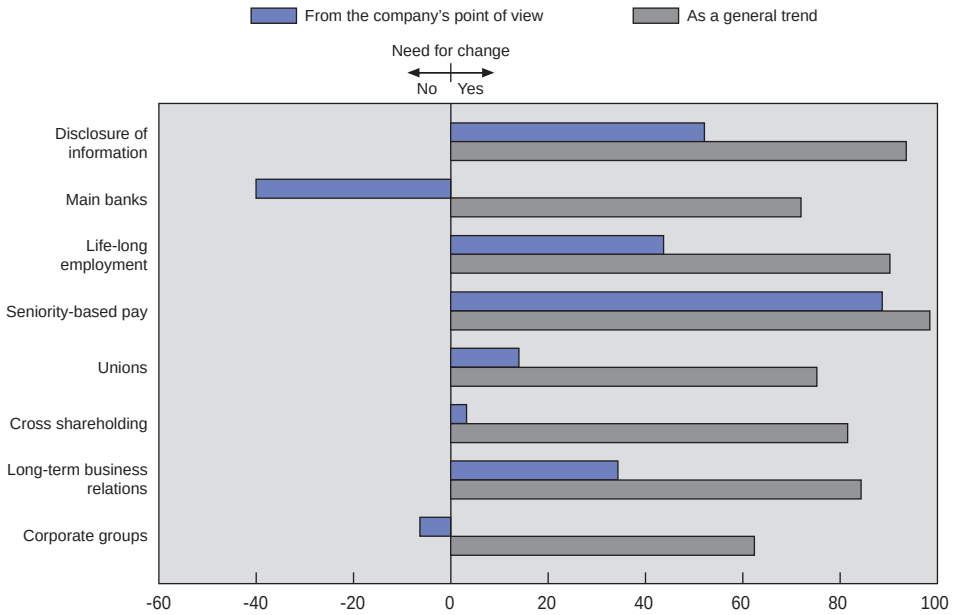
The remarkable aspect of the Programme is that linkages to other reforms are acknowledged and the importance of deregulation in education reform is clearly recognised. This is significant since, faced with the prospect of changes in demand for their services linked to the declining number of children and the rising importance of life-long learning, many (particularly private) tertiary institutions have already begun to adapt to the anticipated changes in demand. In an increasingly deregulated environment and with a greater amount of funds flowing in from industry it is important to ensure good governance mechanisms operate at academic institutions. The first step in this direction is to increase the transparency of their financial operations. Private universities have the status of special purpose corporations and are legally required to prepare balance sheets, but their disclosure is not obligatory. And the preparation of financial statements for each individual national university is not a legal requirement.

Facilitating the evolving business arrangements and corporate governance practices

A recent survey revealed a picture of companies which, while putting more importance on increased shareholder value, are struggling to ascertain what specific changes to the traditional arrangements would be necessary[190] (Figure 34). While most companies replied that as a general trend most features of the traditional business system will have to change in some form or another, they did not foresee large changes for themselves, except for seniority-based pay and to a lesser extent greater information disclosure, life-long employment and long-term business relations.

With the likely weakening of seniority-based pay and life-long employment practice the importance of external labour markets is bound to increase and so is their complexity as there will be a wider range of demands and supplies of skills

Figure 34. **THE NEED TO CHANGE THE CURRENT CORPORATE SYSTEM**[1]



1. This is measured as follows. For the need for change from the company's point of view, answers are in four categories; *a)* need to change rapidly, *b)* need to change gradually, *c)* little need for change, and *d)* no need for change. The index shown is calculated as the difference between the sum of *a)* and *b)* and the sum of *c)* and *d)*, divided by the total number of answers. For the need for change as a general trend, answers are in three categories: *a)* expect a radical change; *b)* expect a gradual change; and *c)* expect hardly any change. The index is calculated as the sum of *a)* and *b)* minus *c)*, divided by the total number of answers.

Source: Long-term Credit Bank Research Institute Consulting Inc., *Survey of opinions concerning the corporate system of the 21st century*, 1998.

and attributes to be matched. Further deregulation of placement and dispatch services will help to match the demand and supply of increasingly heterogeneous human resources. It is also desirable to reduce impediments to inter-company labour mobility. The most significant among these is the lack of portability of corporate pension funds (Table 31), which are currently provided by different independent systems so that an employee is made subject to tax as (s)he changes jobs. Pension portability can be improved through the introduction of a defined-contribution scheme akin to the 401k plan in the United States. The favourable tax treatment of lump-sum separation pay after 20 years of service needs to be reconsidered in a similar light.

Companies have been actively experimenting with different organisational forms as a way of coping with the shift in competitive advantage away from those based on economies of scale and synergies of large size towards those based on innovation and speed of decision-making. While pure holding companies have been made legal, there are a number of obstacles to employing this form of corporation, even though it has the advantage of allowing greater flexibility for individual companies to operate. Under the Commercial Code it is rather cumbersome to spin off divisions into separate companies, and it is not possible to split a company into several independent companies. Moreover, the tax law does not allow consolidation of profits, which tends to discourage the strategic use of different-sized companies.

The pure holding-company structure may contribute to streamlining the corporate governance arrangement of a group of companies in so far as their shares are not listed on the stock exchange.[191] Otherwise it would be necessary to extend the right of the holding company's shareholders to obtain a detailed reporting on each of the subsidiaries.

On a broader front, while there have been animated debates about the desirable changes in corporate governance arrangements, the majority view seems to be that it would be unproductive to consider the issue as the choice of any particular model such as an Anglo-Saxon model. The state of the art in this domain is such that there is no single magic formula, and, as the opinion survey cited above indicates, certain traditional arrangements – notably main banks and corporate groups – are strongly rooted in the Japanese corporate system. Nonetheless, there seems to be a consensus that the traditional approach has relied too heavily on insiders (directors promoted from within) and semi-insiders (the main bank and other stable shareholders) who cannot exercise an independent monitoring role, and that a better balance should be sought.

Some of the more progressive companies have recently strengthened the role of the board of directors by cutting the number of directors substantially (by two-thirds in Sony, for example) and increased the number of outside directors. Such a move should be encouraged through, for example, the stock exchange's public recognition of their desirability. Moreover, the role of an outside auditor (which was introduced in 1993 to enhance the independence of auditing from the chairman of the board of directors) needs to be strengthened further by prohibiting the appointment of any related persons.[192]

One powerful way to strengthen the incentive for corporate managers to work to enhance shareholder value is to align their mutual interests by giving stock options. This was introduced for a special category of firms (mainly venture firms) in 1995 and was generalised to all joint-stock companies in 1997. The number of companies which plan to introduce stock options stood at 141 as of June 1998 and is expected to increase significantly with the favourable tax treatment to be introduced next year.

The possibility of derivative suits by shareholders can act as a mechanism to discourage directors and auditors from acting against the interest of the shareholders. It has existed since 1950, but its use has risen since the reduction of application fees to a flat rate of 8 200 yen in 1993. This is a welcome change. However, it is necessary to introduce the so-called business judgement rule to avoid the possibility of directors' becoming excessively risk-averse in their decision-making.

Summing up

This chapter first presented some evidence of weakening dynamism of the Japanese business sector and attributed it to three main factors, *i.e.* the lag in adjustment to the end of the catch-up process in the manufacturing sector, regulation in the non-manufacturing sector and, more generally, poor adaptation of the corporate system to exploiting new business opportunities.

There are, however, encouraging signs of changes for the better, and the government has taken measures to facilitate the desirable changes. The emerging new types of industrial clusters based on open network relations centred around innovative middle-sized companies (notably in the greater Tama area) have already achieved excellent results, and so have middle-sized companies more generally. The government has readied the legal and institutional infrastructure for a leap in the supply of risk capital to finance new business ventures, including the easing of the restriction on investment in unlisted shares by pension funds, the creation of a special over-the-counter market for innovative companies, and measures to improve the transactions of unlisted shares. It has also taken initiatives to improve the hitherto compartmentalised R&D system, by increasing the flows of information and researchers among the government research laboratories, universities and businesses, and by enhancing the efficiency of R&D

funding through the introduction of competitive considerations in budgetary allocation and the obligation of strict third-party evaluation of publicly funded projects. Universities are now encouraged to have their innovative work patented and sold through a Technology Licensing Office. As well, the patent-related laws have been revised to strengthen the protection of innovators' intellectual property rights, and greater use of dormant patents, many of which are potentially valuable to middle-sized companies, has been promoted. Nonetheless, further measures would be conducive to new innovative business activities. These include the introduction of limited partnerships, which would clear up legal ambiguities which currently exist for pension funds' investment in venture capital, and the strengthening of the price formation function of securities dealers in the over-the-counter market.

Further substantial deregulation is inevitable, as it is widely accepted in Japan that many existing regulations have outlived their usefulness and are impediments to greater efficiency. As industries are deregulated, they are faced with the challenge of coping with much more intense competitive pressures. The problem is compounded in some sectors such as finance and real estate, which have been seriously hit by the bursting of the asset price bubble, and construction, which has seen a rise in employment in the 1990s due partly to public works spending (see above). On the other hand, further deregulation will open up new business opportunities. Deregulation in telecommunications and airlines has already induced new entrants to these sectors, and the prospective easing of restrictions on placement and worker dispatching services is likely to result in the expansion of business in this domain. Other reform efforts which are likely to enlarge business opportunities include allowing corporations to own and operate farms, further zoning derestriction (particularly with regard to the use of agricultural land in urban areas) and deregulation in medical and social assistance services.

More generally, to cope with these structural changes requires different types of human capital than in the past. The attributes of human capital favoured in the 21st century will include flexibility, adaptability and creativity. Fully aware of these needs, the government has embarked on education reform as the sixth pillar of its structural reform efforts. The main thrust of the education reform is to promote creativity through diversity, and deregulation is to play an important role in achieving diversity. At the primary and secondary level,

diversity will be promoted by allowing greater autonomy for individual schools and municipalities and by offering more options as from the upper secondary levels. Exposure to information technology is also to be boosted. Beyond that, emphasis is being given to expanding the graduate programmes and stimulating the mobility of professional staff. Some efforts are also being made to increase the transferability of academic credits earned in polytechnics to universities. It would be desirable to improve the portability of credits across different tertiary institutions more generally and to allow greater flexibility than hitherto to individual institutions in creating new faculties. At the same time, monitoring of the quality of education services should be strengthened through third-party evaluations, and good governance mechanisms should be put in place commensurate with the degree of autonomy granted.

The on-going structural changes are also exerting pressures on traditional business arrangements and corporate governance practices. Among these, the long-term supplier relations (sometimes referred to as vertical *keiretsu*) have been changing relatively rapidly along with the crumbling of vertically integrated industrial clusters. In most other respects, changes have been slow, and surveys indicate that companies are struggling to determine desirable models, except for seniority wage systems for which there seems to be a consensus that they will have to give way to merit-based pay. With an increased importance of external labour markets in prospect, the set of measures the government has adopted or plans to take to improve their functioning, for example, through deregulating labour placement and dispatch services, is welcome and should be reinforced by further measures to do away with obstacles to job switching, notably by improving the portability of corporate pensions. As well, there is a clear need to restore balance between different corporate stakeholders. The traditional Japanese corporate governance approach has had its merits in reducing the agency costs involved in monitoring and in coping with temporary financial distress, but it gives too much power to insiders. It is high time to redress the balance by strengthening the monitoring role of outsiders. In this regard, the recent initiatives by some leading companies to reduce the size of their boards of directors and increase the participation of outside directors are welcome and should be encouraged by the stock exchange as a desirable practice. More frequent shareholders' derivative suits against corporate directors also represent a welcome counterbalancing move. It would, nevertheless, be advantageous to introduce the

so-called ‘‘business judgement rule’’ so as to prevent an excessive degree of risk aversion on the part of corporate directors in making business decisions. Finally, the revised tax treatment of stock options should help align managers’ interests more closely with those of shareholders.

Notes

1. However, the yen rose substantially in the first half of the decade, thereby imposing effectively tightening monetary conditions (see Figure 26 below).
2. There have been debates about possible reasons for a decline in multiplier effects of fiscal policies. There is no consensus as to what factor is the most responsible for the decline. A few examples of the possible reasons discussed in the literature other than Mundell-Fleming effects are: an increase in the propensity to import, a decrease of inputs of intermediate goods in the production process and an increase in the share of the service sector in the economy. See Asako (1997) for a survey of the literature.
3. Firms' ability to borrow and invest was curtailed by the reduced value of collateral (land) on their balance sheets. Most research finds significant effects in investment functions, but only in the case of non-manufacturing SMEs (see Miyagawa and Ishihara, 1997; Ogawa *et al.*, 1996; Yoshikawa and Obara, 1997).
4. According to a survey by the National Land Agency, land acquisition peaked in 1989, more than half of which was purchased by companies with capital of 10 to 100 million yen. This share was disproportionate, since companies with capital of less than 100 million yen were responsible for about a quarter of total business investment according to another survey by the Ministry of Finance.
5. For example, the share of loans to SMEs in total loans outstanding increased from 34.5 per cent in 1975 to 71.3 per cent in 1990 and then levelled off.
6. Although many empirical studies have been conducted, the causes of the slow growth rate of consumption in the 1990s have not been conclusively identified, nor have the magnitudes of the adverse wealth effects. Taking account of lack of liquidity of real estate, which is a current policy issue (see Chapter II), changes in the values of financial assets and real estate are likely to have different impacts on consumption. Liquidity constraints may be a more important factor explaining consumption in Japan, if it is accepted that 40 per cent of all households are subject to such constraints in Japan (Ogawa, 1992, and Economic Planning Agency, 1996), while a corresponding earlier figure for the United States is only 17 per cent (Hayashi, 1987), for example. However, according to a survey by Muellbauer (1995), these estimates are not at all precisely determined.
7. Japan's Ministry of Labour (1998) estimates that 0.8 percentage points of the 1.3 percentage point rise in the unemployment rate from 1990 to 1997 was the result of deficient demand, with the remainder (0.5 percentage point) attributable to a rise in equilibrium unemployment. The OECD's corresponding figures are 1.0 and 0.3 percentage points, respectively.

8. The current pay-as-you-go pension system will continue to transfer income from the younger generation to the elderly (see Chapter III). The point is that there is uncertainty about how large the transfer will be: if younger people do not know to what extent their future benefits will match their contributions, they tend to regard the contribution as a kind of ''tax'' and save money on their own for precautionary reasons.
9. The data in Table 7 are from the monthly Household Survey which is administered to a small sample of Japanese households and may yield different outcomes from aggregate figures in the annual National Accounts.
10. This brought real consumption growth for ''total households'' down to negative rates for five consecutive years from 1993 to 1997, which is the first time this has happened since the start of the Household Survey in 1962.
11. In 1992, construction of rental housing was encouraged by an easing of zoning restrictions. In 1996 it benefited from the anticipated rise in the consumption tax rate (see below).
12. The effects of the Rental Land and House Law are discussed in Chapter III below.
13. The Liberal Democratic Party (LDP), the largest party in the Diet, increased its number of seats in the general election for the Lower House held in October 1996, campaigning on a platform of completing the tax reform. On 18 December 1996, the LDP and its coalition partners reached agreement not to continue the temporary special tax cut for another year. On 13 December 1996, the bill to freeze the consumption tax at 3 per cent, which had been submitted by an opposition party, was rejected in the Diet. It should be recalled that the consumption tax was raised in 1997 in the context of the 1994 tax reform which included income tax reductions.
14. According to the Household Survey, in 1997 real expenditure on goods decreased by 1.1 per cent, led by durable goods (–5.1 per cent), while expenditures on services increased by 0.4 per cent.
15. As for the consumption tax rate on housing, it was decided that the rate of 3 per cent would be applied if the contracts were finalised by end-September 1996. This caused a similar shift in housing investment from 1997 back to 1996.
16. Inventories at the distribution stage increased in the third and fourth quarters of 1997 as well.
17. For example, domestic auto sales declined by 4.9 per cent in 1997, while exports increased by 22.7 per cent.
18. According to the Economic Planning Agency, the effects on trade volumes of the yen's depreciation that began the second quarter of 1995 started to dominate those on prices and increased the current account surplus as from the fourth quarter of 1997.
19. The significant depreciation of the Asian countries' currencies made it costly for them to import capital goods and intermediate goods to be used in export industries. These industries suffered from a lack of export credits as well.
20. The effects of these changes on CPI are officially estimated to have been around 1.5 and 0.4 per cent (at an annual rate), respectively.
21. When income stagnates, households' propensity to consume tends to increase. However, this ratchet effect did not work in 1997, because consumers recognised that the economy was in the midst of important structural changes and lacked confidence in the future.

22. The actual decline in household net worth has been much smaller – about 5 per cent – because of the on-going accumulation of assets resulting from persistently high household saving rates.
23. This is the average figure derived by the Economic Planning Agency (1996, p. 68). It is much smaller than that believed to be operative in the United States, for example, where most studies find effects of 3 to 7 per cent for stock market wealth (Starr-McCluer, 1998) and even more for other net tangible assets (Brayton and Tinsley, 1996).
24. Firms in the construction sector were noteworthy victims, given their substantial real estate holdings.
25. Japanese banks cut off credit to over 12 000 firms last year, the highest figure for a decade.
26. For example, the People's Finance Corporation reported that it had boosted its loans by 20 per cent in the five months to April, precisely because of a lack of loan supply by private-sector institutions. These often were in the form of *marukei* loans, which are small (up to 10 million yen or around $70 000), low-interest loans without collateral or guarantees.
27. See Chapter II for a more detailed look at the balance sheet of the general government sector.
28. At the time of writing, only two or three of the largest banks are believed to still have cumulative capital gains on their equity holdings. Fukao (1998) shows that the banking sector suffered a decline in the market value of its holdings of listed equities of 43 trillion yen over the eight years to March 1997. With a rise of almost 20 trillion yen in their book values, these so-called ''hidden reserves'' dropped by some 63 trillion yen or 85 per cent. Another fall of 6.5 trillion yen was recorded for the 19 major banks in the year ended March 1998 (see Table 11 below). Since 45 per cent of these reserves are available for satisfying the BIS capital adequacy requirement of 8 per cent for banks operating internationally, the negative impact on qualifying capital, even before the 15 per cent fall in the stock price indices since 31 March 1998, amounts to over 31 trillion yen. With very little increase in other forms of capital, this has been reflected in a reduction of similar magnitude in the total amount of capital available, broadly defined (about 40 per cent). It is, therefore, hardly surprising that the ratings assigned to Japanese banks by international ratings agencies have fallen from levels which were largely in line with those assigned to major European banks in the 1980s to several notches below them already by the mid-1990s. Similarly, the stock market sub-index for bank shares has fallen even more severely than has the overall market.
29. According to the Tokyo Stock Exchange the proportion of all shares in listed companies owned by banks and investment trusts fell from 19.4 per cent in fiscal 1992 to 17.1 per cent in fiscal 1996 (Fukao, 1998). Banks hold about 10 per cent of all industrial companies' shares, while life insurance companies hold around 8½ per cent of these shares, both down about ½ percentage point since 1991. Financial institutions' ownership is much more subject to interlocking claims than is the rest of the market. Overall cross-shareholding ownership as a share of all equity holdings has fallen from around 55 per cent in 1987 to 47 per cent in 1996, according to the Daiwa Institute of Research.
30. This system of bank-centred governance may be waning because it is not as efficient as a deregulated securities-market-based system. Kang and Stulz (1997) show that, for a sample of 1 380 firms other than utilities and financial institutions, the higher the share of bank loans

in total debt, the higher were buy-and-hold returns to shareholders in the period 1986 to 1989 but the lower were such returns in the years 1990 to 1993. This could of course be due to the business cycle rather than deregulation.

31. According to the Federation of Bankers Associations of Japan collateral was sufficient to cover only 32 per cent of reported non-performing loans of the 19 major banks in September 1997. This implies the need for loan-loss provisions of the remaining 68 per cent, a target not yet reached by March 1998 when the share was 64 per cent; the difference was almost 0.9 trillion yen.

32. The reference to a convoy is apt because in the traditional Japanese system the pace of regulatory change for the whole sector (a fleet) is set according to the adaptive capacity of the weakest institution (the slowest ship in the fleet).

33. This practise, which also exists in a number of other OECD countries and is known in Japan as *amakudari* (meaning ''descent from heaven''), may have helped sustain the system of discretionary regulation. It may have had certain advantages such as an efficient use of human capital and improved information flows between the government and the private sector, but in many cases it risked creating a conflict of interest. A recent empirical study (Horiuchi and Shimizu, 1997) found that those banks which hired former public officials tended to have lower capital adequacy ratios and higher bad-loan ratios than others. Van Rixtel and Hassink (1998) show that banks appointing such officials subsequently moved to higher risk/return positions, resulting in an increase in lending to risky industries (construction, real estate and non-bank financial industries) and higher profitability over the period 1980-93. According to press reports, as of October 1996 there were 7 080 such directors in 2 483 of the 6 815 firms under some kind of government supervision. However, the numbers involved have been on the decline for some time.

34. This summer's crisis in Russia is likely to have similar although much smaller effects: the total exposure of Japan's banks to Russia is less than $1 billion.

35. According to the Bank of Japan (1998*b*), the number of internationally active banks fell from 80 in September 1997 to 45 (out of 145) in March 1998. Figures in Table 11 show that international lending by Japanese banks fell by $50 billion (11.5 per cent) in the two years to end-1997. This reduced their share of all banks' international lending from 27.9 to 20.5 per cent. Japanese banking officials estimate that overseas loans will fall by a cumulative 25 per cent in the two years 1997-98, with about half the decline in the last seven months of the period.

36. In March 1998 year-on-year growth in personal deposits at six major city banks was 6 trillion yen (8.6 per cent), while that in the overall banking system was 16 trillion yen, but only 6.3 per cent, compared to the rise in postal savings deposits, also of 16 trillion yen (7.0 per cent). The media were also rife with stories about booming sales of safes during the crisis period. A recent survey conducted by the Central Council for Savings Information showed a shift in consumer attitudes toward safety and yields in determining where to deposit their funds.

37. Ito and Sasaki (1998) show that one way banks responded to problems of low capital ratios in the years 1990-93 was to issue more subordinated debt. Another was to reduce risk assets by cutting back on lending.

38. The Bank of Japan (1998*b*) has recently demonstrated the importance of financial institutions' capital constraints in the determination of these *Tankan* results. Other surveys by the Ministry of Finance and the NLI Research Institute, for example, also show severe deterioration over the past year.

39. There are also a number of special factors, such as securitisation, write-offs and losses on sales of loans to the Co-operative Credit Purchasing Corporation. Excluding these items, the Federation of Bankers Associations estimates that major banks' loans and discounts were actually nearly stable in the first half of FY 1998 on a year-on-year basis, and that excluding discounts, loans alone were up 3.3 per cent.

40. The appropriate measure is the share of new loans, but such data are not readily available.

41. The Bank of Japan (1998*b*, p. 146) provides similar evidence. However, it is possible that there is some selection bias in these figures: to the extent that a true credit crunch has been operant some potential borrowers are being screened out due to higher risk, raising the risk-adjusted interest rate above the actual rate. But here too it is not possible to distinguish this hypothesis from the alternative based on the cyclical deterioration that occurred in 1997-98: the weakening economy worsens adverse selection problems, and lending rates would have to rise in order to prevent risk-adjusted rates from falling. It is, therefore, unclear which bias is larger.

42. Ogawa *et al.* (1998, p. 10) claim the average for major banks in Japan was 43 basis points for an unspecified recent period, compared with 100 in Switzerland, 135 in Germany, 157 in France, 239 in the United States and 268 in the United Kingdom.

43. Consistent with this hypothesis, the Bank of Japan (1998*b*, p. 147) shows that in recent years banks with higher capital adequacy ratios had lower lending margins.

44. Simultaneously, an expenditure amounting to about 0.1 trillion yen was made from the general account in order to subsidise these institutions.

45. The necessary measures will be taken in a FY 1998 supplementary budget.

46. It is unclear whether the exact output growth figures fixed for the triggering of the suspension of the various tax and spending constraints are consistent with the modest potential growth outcomes described in Chapter II.

47. The expenditures include the costs of land acquisition and compensation for affected people. Therefore, those costs should be excluded when considering the direct effects of public works on demand.

48. There are 14 such non-profit organisations included in the broadly defined central government sector in the System of National Accounts, even though a number of others (such as the Japan National Railway Settlement Corporation) are considered to be public corporations. They are engaged in a wide variety of activities related to nuclear energy, space, the environment, international co-operation, SMEs, state pensions, postal life insurance, coal mine damage and the promotion of employment. The largest of these are the Pension Welfare Service Public Corporation and the Postal Life Insurance Welfare Corporation. These corporations are owned and managed by the Ministries of Health and Welfare, and Posts and Telecommunications, respectively. One of their functions is to channel capital from the social security funds and the post office *via*, respectively, the Trust Fund Bureau (a

government financial institution) and the Postal Life Insurance Special Account to private sector financial institutions (trust banks and insurance companies, in particular) who are then responsible for ultimate placement.

49. These are Foreign Exchange Fund Securities held by the National Debt Consolidation Fund Special Account.

50. The Bank of Japan has calculated consolidated gross debt based on a broad definition of ''national government'' (see its *Economic Statistics Monthly*, Table 86). The latest figure (for May 1998) for that which is held by the public is 146 trillion yen or around 30 per cent of GDP. Adding in local level debt (from Table 18) would yield a consolidated gross debt held by the public of about 54 per cent of GDP. This figure represents the concept involving the greatest degree of consolidation.

51. Much more revenue would have been raised if land had been sold prior to the end of the ''bubble period'', but this was not done for fear that it would encourage speculation in land.

52. The Japan Federation of Economic Organisations (Keidanren) has estimated that reducing all national statutory rates by one-fifth and eliminating the top rate in the local residents' tax would cost less than 4 trillion yen per year.

53. This is the so-called ''Ku-ro-yon'' problem, which means in Japanese that the tax authority is informed of (more than) 90 per cent of income of dependent employees, while it knows only 60 and 40 per cent of that of the self-employed (excluding those engaged in agriculture) and the self-employed in agriculture, respectively. This was empirically confirmed by Honma *et al.* (1984, 1987). This difference might be justified, to some extent, by the fact that the self-employed face more variable incomes and more risks than the employed, such as with respect to income maintenance in the event of illness.

54. More exactly, ''the spousal tax deduction'' and ''the spousal special deduction'' are available, depending on the amount of income earned by the spouse. They amount to 760 000 yen in total, if fully utilised, and could discourage housewives from going to work. The pension system also distorts the labour supply decision by housewives. The current system allows housewives to receive benefits, although neither they nor their husbands have contributed anything for them. However, once they start to work and earn more than 1.3 million yen per year, the system requires them to pay contributions themselves. Therefore, being a housewife is much more advantageous than a working spouse in terms of tax advantages to the working spouse, thereby restraining female labour supply (Kimura *et al.*, 1997). The rise in female employment will benefit the financial position of the pension system.

55. The number of employees of the National Tax Administration is about 57 000 as of FY 1995, and there is said to be problem of understaffing.

56. Japan is one of seven OECD countries not to have such a system.

57. The nature of this tax is unclear: the Ministry of Finance classifies it as a direct tax, while the Economic Planning Agency treats it as an indirect tax on an SNA basis. A reason for EPA's treatment is that the payment of the tax is deductible from taxable profits the following year, and therefore the payment itself can be regarded as a kind of cost.

58. For example, 51.6 per cent of all corporations paid some tax in FY 1990 but only 35.5 per cent in FY 1995.

59. France's ''*taxe professionnelle*'' is a prime example of a similar tax whose efficiency is widely considered to be low.

60. The idea that price stability might constrain monetary policy faced with negative demand shocks because interest rates cannot be negative has been discussed by Summers (1991). Fuhrer and Madigan (1997) have recently shown that the output path of an economy faced with a variety of negative demand shocks in a zero inflation context will be below what it would be if inflation were 4 per cent by varying amounts: as much as 1 per cent if the shock is large and lasts only a few quarters (arguably Japan's case), but only 0.1 to 0.2 per cent if it is either large and permanent or small and temporary.

61. These were to a large extent offset by open market operations.

62. Repurchase operations involve two increases on the assets side of the central bank's balance sheet (government securities in custody and cash collateral against securities borrowed) and similarly two on the liabilities side (securities borrowed and current deposits). About 6 trillion yen of the increase is attributable to this double-counting.

63. The differential between rates on three-month certificates of deposit and treasury bills has been used by the Bank of Japan (1998*b*) as a measure of this flight to quality: it jumped from an average of around 15 basis points in the years 1995-97 to over 60 basis points in February 1998.

64. Such injections, for example, peaked in the first quarter when the stock outstanding averaged 10.5 trillion yen; by June, it had dwindled to a third of that amount.

65. By the end of August such injections amounted to 4 to 5 trillion yen.

66. The yield differential between corporate and government bonds has jumped by around ¼ percentage point since last winter for well-rated corporate borrowers but by more than ½ percentage point for those with lower ratings.

67. More recent historical low points were recorded by the United Kingdom in 1897 at 2.21 per cent and the United States in 1941 at 1.85 per cent.

68. This is a situation described by Keynes in his *General Theory* in which in conditions of extreme recession and deflation monetary expansion does not lead to an increase in output because interest rates are so low as to leave agents indifferent as to holding bonds or cash. Thus, monetary expansion does not lead to an increase in spending.

69. Consistent with the story, mentioned in Chapter I, that sales of personal safes have been booming as a result of households' fears over bank safety, all of the increase in cash in circulation has been recorded as being held by individuals according to the Bank of Japan. Demand for Bank of Japan notes in recent quarters has diverged by up to 6 per cent from predicted values based on an equation developed by the Bank of Japan (1998*b*) and estimated up to 1995 Q2.

70. That aggregate has been distorted by a variable shift of funds from investment trusts to deposits which boosts its size. The liquidity aggregate is unaffected by this shift.

71. For example, the Investment Trust Association reported in June that the level of foreign currency assets held by domestic mutual funds groups rose to 4.3 trillion yen at end-May, up 46.8 per cent from a year earlier. Also, the nation's five largest life insurers plan to raise their investment in foreign bonds by 1.7 trillion yen ($12 billion) this year. Press reports claimed that individuals were moving funds out of yen as well.

72. Falling rates (and their declining pension management business) have forced them to sell equities, depressing the stock market, and to move abroad, depressing the yen. Similarly, the postal life insurance system (*kampo*) was forced to reduce its guaranteed annual interest rate of its single premium annuity from 2¾ to 1¾ per cent this year, thereby requiring 5 to 10 per cent increases in premiums, the first since 1996.

73. The official definition of problem loans at the time had been the sum of loans to borrowers in bankruptcy, loans over six months overdue, loans restructured at rates below the official discount rate (0.5 per cent) and loans to customers for support.

74. However, with specific loan loss reserves of 14.0 trillion already set up, these estimates implied little need for outside assistance, since not all problem loans would ultimately become unrecoverable and underlying collateral would offset some of the final losses. However, the judgement of private ratings agencies such as IBCA is that true problem loans also comprise loans to *jusen*, loans sold to the CCPC and problem loans of banks' affiliates, which by their estimates, would have boosted total problem loans to around 81 trillion yen and unreserved losses to nearly 16 trillion yen at that point.

75. At end-September, official figures had shown 24.8 trillion in non-performing loans including those to borrowers who are receiving assistance. Consistency between the two data sets requires that 20.5 per cent of Class II loans were non-performing.

76. This became especially relevant later in 1998 when it was revealed that the Long Term Credit Bank had shifted substantial amounts of its non-performing loans to its non-consolidated affiliates, one of which, a major leasing company, collapsed in September.

77. Prior to this funding the DIC was expected to have merely 2.7 trillion yen in total premium and interest income available in the intensive reform period ending in March 2001 during which higher deposit insurance premia will be in place. But with expenses of some 2¼ trillion yen up to and including the failure of Hokkaido Takushoku Bank, there was very little left to finance any other major failures.

78. Briefly, the institution had to be at risk of failure without the capital injection (with that failure posing systemic risk) and the funding had to be used to reconstruct management with no dividends nor bonuses to executives.

79. Some banks had already reached the 45 per cent official limit on the Tier II component of capital allowed to count in the calculation of capital adequacy and, therefore, applied for preferred shares which qualify as Tier I capital.

80. Long Term Credit Bank wanted a greater allocation of subordinated loans, and Nippon Credit Bank applied for 230 billion yen in subordinated loans, in addition to the preferred shares it was accorded. These requests were rejected because these banks had already reached the official limit on the Tier II component of capital (described in the previous note) and thus could not have raised their capital adequacy ratio, even if their applications had been fully accepted.

81. These hybrid securities were attractive primarily because they qualify for Tier I status in the eyes of the regulators. Qualification requires that the securities be truly perpetual, without any redemption clause at the discretion of the lender, and that they be non-cumulative so that interest payments may be suspended if necessary. But they allow for a step-up in interest rates, and markets might look askance if any highly-rated institution tried to suspend such payments. Thus, it is unclear if they are an effective substitute for equity.

82. While these loans are in dollars and cannot therefore be directly compared with the yen rates charged (provided in Table 24), the margins over LIBOR are comparable, and in both cases they are far higher than what was charged for the injection of public capital, even though the instruments are similar.

83. For example, it is not obvious why Sumitomo Bank and Tokai Bank merited the same rates as the Bank of Tokyo Mitsubishi for essentially the same instruments but quite different credit risks.

84. Indeed, even if all overheads could be entirely eliminated, meagre spreads on bank loans and stagnant returns on banks' holdings of equities would limit their return on equity to levels still well below those achieved by many Western banks.

85. The LTCB agreed to merge with the Sumitomo Trust Bank in June 1998, but the latter was willing to implement the agreement only if the LTCB was purged of its problem loans. The authorities were willing to provide public capital, but the opposition parties insisted that the case be considered within the revised framework (the ''total plan'' – see below) which they developed over the summer but which was not legislated until October. By then, LTCB's condition had deteriorated substantially farther, and the merger was essentially called off when it became clear LTCB would have to be temporarily nationalised.

86. The problem is not limited to financial institutions either: in May 1998, for example, a major producer of coated wires for electronics parts went bankrupt because of debt guarantees which were not disclosed on its financial statements.

87. This has led to the first shareholder lawsuit against a firm's auditors in Japanese history. The government is considering a five-year limit on the length of time that the same auditors can audit a given firm.

88. The case of the Hokkaido Takushoku Bank is well known, but the Taiheiyo Bank's problem loans were revealed to be some ten times their previously disclosed level in 1996, and disclosure surprises were recorded for Hyogo Bank and Tokuyo City Bank as well. Most recently, Long Term Credit Bank admitted that its problem loans were roughly twice as large as it had claimed in its year-end financial statements issued only a couple of months earlier. Of course, the deficiency in disclosure is more widespread, as attested to by the cases of Yamaichi and Sanyo Securities.

89. The *Wall Street Journal Europe* of 16-17 January 1998 reports that margin between Long Term Credit Bank and Nationsbank and attributes it, as here, to disclosure differences.

90. For that to be effective supervisory audits will have to take place without forewarning and more frequently than in the past. The Financial Supervisory Agency has decided to increase the frequency of its inspections to once year (from every three years) for the 19 major banks and to every one to two years (from every four years) for the regional banks.

91. An 8 per cent ratio is needed to maintain international operations. The BIS measure suffers from a number of weaknesses: loans to financial institutions are considered less risky than those to non-financial corporations and those to firms based in the OECD are less risky than others, with no distinction between different OECD countries nor between blue chips and borrowers in a fragile condition. The result is still a risk of moral hazard: banks have an incentive to lend to risky borrowers within each risk category, assuming they can get a higher return for the same amount of capital required.

92. This option was taken up by seven of the major banks and 45 out of 83 surveyed banks overall. The write-up was worth an estimated 3.7 trillion yen, and, since 45 per cent of the gain counts toward Tier II capital, the impact on the average capital adequacy ratio was 0.4 percentage point.

93. The Bank of Japan (1998*b*, p. 161) shows that banks' gross unrealised losses on equities in FY 1997 reached nearly 4 trillion yen or about 0.9 percentage point on the BIS ratio.

94. The banks themselves expect a further 15 per cent fall in FY 1998.

95. In FY 1997 less than 40 per cent of these disposals were, however, made by direct write-offs, leaving about 6½ trillion yen worth of these loans on the balance sheet, even if provisioned. Cumulatively since FY 1992, precisely half of the 45 trillion in disposals have been directly written off. Note that reserving alone does not reduce the amount of non-performing loans; furthermore, some write-offs do not reduce it either, as the borrower's failure may have been unforeseen and the relevant loans, therefore, not previously included in the non-performing total.

96. The lack of property market liquidity is intimately tied up with the financial sector problems. Some unofficial estimates are that banks still hold 22 trillion yen in real estate collateral on their books. Their incentive to sell it is limited as they have written much of it down, and it is, therefore, perceived as a call option, the liquidation of which will incur the declaration of further financial losses. Nevertheless, some unloading of property-related loan packages has taken place: unofficial estimates are that a cumulative 3 trillion yen in face value was thus sold in FY 1997 at an average price of around 10 per cent of book value. Life insurance firms are also reported to have begun to sell non-performing loans in FY 1997 – also for around 10 per cent of book value – and to intend to continue to do so this year.

97. This change would seem to be wrong-headed: eliminating a tax on holding land will *reduce* the incentives to make efficient use of it and lead to more of it being held back from the market.

98. In Japan loan collateral provided by a business borrower is not generally attached exclusively to a particular loan but rather to all its loans from all its lenders.

99. Specifically, bank loans would be allowed to finance individuals' purchases of foreclosed real estate.

100. The CCPC was set up in 1993 to dispose of banks' bad loans and allow them an immediate tax deduction. In its first five years it took over 11 030 loans with a total face value of 14.92 trillion yen at a cost of 5.76 trillion yen (38.6 per cent of face value). However, it has since then managed to sell only 1.18 trillion worth of these loans – 20.5 per cent of its outlays and a microscopic 8.0 per cent of their original value.

101. The US Federal Deposit Insurance Corporation has been authorised since 1987 to take over the operations of any insolvent bank for up to three years while a buyer is being sought.

102. A supporting motion must be obtained from shareholders under Japan's Commercial Code.

103. The RCB was originally established in March 1995 as the Tokyo Kyodou Bank to assume the operations of various failed credit co-operatives. It was re-organised in September 1996 and its remit was broadened to include other failed financial institutions. It is jointly owned by the DIC (whose share is 75 per cent), the Bank of Japan (12.5 per cent) and private financial institutions (12.5 per cent).

104. This is to be financed by government-guaranteed loans from the Bank of Japan to the Deposit Insurance Corporation.

105. One concern had been whether the public funds originally approved (some 6 per cent of GDP) were sufficient to deal with the entire bad loans problem: banking crises in six other OECD countries in the past 20 years or so have incurred final net fiscal costs (initial costs were higher in several cases) varying from as little as 2 or 3 per cent of GDP (for Sweden, Norway and the United States) to as much as an expected 12 per cent in the case of Mexico and 9 to 10 per cent for Finland (see Box 1.7 in *OECD Economic Outlook 63,* June 1998). And there is every reason to suspect that the situation is more severe in Japan's case, given the size of the bubble which caused it, even if banks themselves have made great efforts to deal with the problem themselves.

106. Its FY 1999 budget request calls for a rise in inspectors at headquarters and regional bureaus to 264 and 482, respectively. Total headquarters staff is to jump 50 per cent to around 600.

107. The debate in the summer of 1998 resulting from the Long Term Credit Bank's difficulties centred on whether the unquantifiable moral hazard costs of allowing weak but solvent banks to continue operating exceed the macroeconomic and systemic costs of allowing a major bank to fail, something all other OECD countries who have experienced banking crises have also avoided. Major international banks all have a complex web of financial derivatives exposures, raising the risk that a single failure could generate a chain reaction of defaults. Japan's major banks had national (gross) derivatives positions of over 2.2 quadrillion yen (around $16 trillion) at end-March, up 6⅔ per cent over the previous year. Long Term Credit Bank's exposure was around 50 trillion yen ($360 billion) but has since been cut back to about 40 trillion yen. Net amounts – true risk exposure – are usually much smaller.

108. On a fiscal year base, the growth rate plummeted from 3.2 per cent in FY 1996 to –0.7 per cent in FY 1997, the largest drop in the post-war era.

109. The recently formed Economic Strategy Council has just recommended that the contribution rate be frozen for the time being.

110. One conclusion, recently reached by the Economic Advisory Council, an advisory body to the Prime Minister, that the economy can grow at a 2½ per cent rate per year for the next five years, if progress is made on structural reforms, is consistent with the analysis presented here. But its other conclusion – that an average growth rate of about 2 per cent can be sustained from FY 2001 to FY 2010 – is less clear (see below).

111. After edging up from 1.42 to 1.43 in 1996, the fertility rate fell to a new historical low of 1.39 births per woman in 1997, one of the lowest figures in the OECD.

112. The authors calculate this by: *i)* interpolating and extrapolating employment-to-working-age-population ratios between and beyond peaks to generate maximum potential employment; and *ii)* multiplying by the maximum working hours per person which is the sum of regular hours – assumed to be constant at its average from 1975 until the end of 1987, then to follow a negative linear trend until the end of 1993 and finally to remain constant at the new lower level – and overtime hours which are assumed to be constant at their peak level per person.

113. Seeking such effects in the data for other countries has proved exceedingly difficult. In the US case, for example, major reform efforts seem to have merely prevented any further slowdown in productivity growth.

114. As described in last year's Survey, the Action Plan for economic structural reform was adopted by the government in May 1997 and is being co-ordinated by the Ministry of International Trade and Industry. The goals of the 1 000 specific measures in the Plan are to boost new business activities in 15 key areas.

115. Official figures, now somewhat out of date, show that 39.4 per cent of output was produced in regulated sectors in FY 1995.

116. This figure includes substantial estimated dynamic gains; on a static basis alone, increases would be only about 1¾ per cent.

117. However, in 1996 only 48 per cent of the legally permitted coverage in Tokyo was actually used, a low figure but still better than the 42 per cent outcome in 1990.

118. With taxes of 61 yen per litre as of September 1997 it is fairly clear that the ex-tax price is now near world levels.

119. These IPPs can already sell power to the existing utilities.

120. Buying a taxi license costs some 10 million yen in Tokyo and 7 million yen in Osaka.

121. According to one survey, two out of three firms are using foreign currency for some domestic transactions.

122. Foreign firms are unhappy with official requests that they contribute in full measure to this fund.

123. The limits mentioned in the text are effective only as from April 2001; before that there are no ceilings.

124. Commission rates were already fully deregulated for all OTC trading in April.

125. These mainly consumer finance companies charge an average 27 per cent on their loans, while facing a cost of funds of around 2½ per cent. Surprisingly, they seem to violate without penalty the civil usury law which limits lending rates to 20 per cent on the first 100 000 yen and 18 per cent thereafter up to one million yen. But lending at these rates does not constitute a criminal violation which permits rates of up to around 40 per cent. In other countries legal maximum rates are lower: in France, for example, the ceiling on loans of up to 10 000 francs (about 250 000 yen) is 19.25 per cent; beyond that it is 12.8 per cent. In Germany, usury is defined in relation to market rates and currently limits lending rates to around 10 per cent as a rule.

126. For example, the market share of foreign brokers almost doubled in 1997 to 30 per cent. Both Morgan Stanley and Merrill Lynch moved to the top of the league of all firms in terms of Tokyo Stock Exchange trading. Foreign life insurers are also gaining market share; foreign mutual funds saw an increase of 86 per cent in their net asset value in the year to May 1998; and foreign banks recorded a 120 per cent gain in their pension fund assets under management.

127. Sumitomo Bank also sold its California holding for $546 million. Five regional banks have withdrawn entirely from foreign operations and a sixth will do so this fiscal year. That still leaves around 50 banks operating abroad, whereas the chairman of the Industrial Bank of Japan opined in February that only about 10 will be able to compete abroad. Also, regional bank overseas lending fell 34 per cent in the year to January.

128. The world leader, UBS, has about 40 times as much funds under management as the combined IBJ/Nomura.

129. *Hasan* (Bankruptcy Law) and *Tokubetsu Seisan* (Commercial Law) are liquidation procedures, while *Kaisha Kosei* (Corporate Reorganisation Law), *Wagi* (Composition Law) and *Kaisha Seiri* (Commercial Law) are for restructuring.

130. Although *Wagi* and *Kaisha Seiri* may allow speedier restructuring, the former does not ensure fulfilment of the agreed plan, and the latter effectively requires unanimity of support by all parties. In any case, despite much criticism directed at it, the Chapter 11 bankruptcy procedure in the United States seems to be shorter than the Reorganisation Law.

131. It is said that 9 per cent of the inner urban land mass in Tokyo is zoned for agriculture.

132. The Japan Harbour Transportation Authority requires shipping companies to consult with them prior to schedule changes. Costs are higher than those of other East Asian ports (excluding Hong Kong, China) for a number of other reasons: wage levels are high; stevedores work restricted hours based on the union-management agreement; and the number of containers handled per berth at Tokyo and Yokohama is lower than elsewhere. The number of ships docking at Japanese ports has fallen by 23 per cent since 1987, while enormous gains have been recorded by other Asian ports.

133. However, the October 2001 deadline by which Japan has agreed to eliminate the unfavourable tax treatment imposed on foreign spirits producers is still not satisfactory to its trading partners. The first stage of liberalisation occurred on 1 October 1997 when the duty on imported whisky dropped 44 per cent. Further cuts are to be made in 1998.

134. Canada admits US-made vehicles duty-free, but it charges a 6 per cent tariff on vehicles produced elsewhere.

135. For example, as mentioned above, in June, the Ministry of Finance abolished most financial directives and instructions of its Banking and Securities Bureaux and will henceforth just answer questions in order to build up a body of interpretative evidence in the area. Similarly, the Ministry of Agriculture, Forestry and Fisheries abandoned the 230 out of 255 directives which are not mandated by law.

136. First, 112 officials of the Ministry of Finance were punished for accepting too much free entertainment from private sector companies, especially banks; this led to the Minister's resignation. Second, a similar situation was uncovered at the Bank of Japan, where 98 officials were punished and the Governor resigned. Third, the Bank's pay structure was found to be unjustifiable, with many staff being paid more than higher-ranking Policy Board members (who have to be approved by the Diet). Fourth, 22 of 47 prefectures were shown to have misused a total of 40 billion yen ($320 million) in public funds. Fifth, bribery scandals erupted at the Japan Highway Public Corporation, the Bureau of the Mint, the Ministry of Health and Welfare and the Self-Defense Agency.

137. For example, it was just recently revealed that the Japan National Oil Corporation, which was supposed to develop oil reserves under the control of Japanese producers, has 1.42 trillion yen in loans and investments which are at risk – about five-sixths of the cumulative total – and expects to have to write off more than one trillion in assets (which would be around a third of the losses at Credit Lyonnais, for example), compared to cumulative public investment of around 1.5 trillion yen. Three oil development companies it funded are to be

shut down by 2000. Until this revelation nothing was generally known about the true state of its finances. Similar problems involving 936 billion yen in hidden bad debts (as at March 1997) have been revealed at nine government-affiliated financial institutions, and recent events point to around 180 billion yen in bad loans at the Hokkaido-Tohoku Development Finance Public Corporation. Getting outside views and providing clearer and more transparent explanations for policies are two reform principles adopted by the Ministry of Finance in March.

138. The subsidy rate is one quarter for those over 55 (one third at small firms). It will be one-sixth for those 45 to 54 (one quarter at small firms). Extending support to this younger group even at a lower rate is rather strange, as the 45 to 54 year-old group has the lowest unemployment rate of any working-age group (see Figure 6). Furthermore, it will boost the eligible population from around 11.6 million to around 27 million because those who are 45 to 54 were born in the post-war baby boom. The additional cost is officially estimated at 50 billion yen per year.

139. The National Employment Adjustment Subsidy Programme came into force in 1981. For eligible sectors – those having experienced a production downturn in excess of 5 per cent over either the previous year or two years – the government pays between one- and two-thirds of an employee's wages when (s)he is laid off temporarily or transferred to an affiliate. In July 1995 the Programme was expanded to include sectors undergoing structural adjustment, and the number of eligible industries rose to a peak of 245, before falling steadily to 88 by January 1998. Since then, the number has ballooned to 167 by 1 September, mainly producers of intermediate materials. The April 1998 stimulus package raised the ceiling on wage subsidies to 75 per cent for mid-sized firms. The proposal is to extend the minimum time a sector is eligible after initial qualification from one to two years.

140. However, at least one private-sector source has estimated that costs will be far higher, even if price inflation in the next two years is assumed to be zero. Kishida (1998) argues that the cost of long-term care will reach 8.45 trillion yen in the year 2000 and jump to around 20 trillion yen about 2025.

141. Generally, the rate is 8.5 per cent (split equally between employers and employees) on regular wages and salary and only 1 per cent on bonus payments.

142. This figure, equivalent to 2.0 per 1 000 population, remains low by OECD standards. Switzerland has over 6 per 1 000 and Italy over 5, while only 5 out of 26 other countries are below 2.

143. Of the other countries for which data are available none has a prospective rise of more than 2 percentage points, although many start from much higher levels.

144. The government polled 3 646 adults and asked their preferences among four alternatives combining feasible cuts in benefits and increases in premiums. The most popular combination was to maintain premiums and allow benefits to fall (with 44 per cent support). Another 25 per cent were willing to see premiums raised but not by as much as would be needed to maintain benefits; that latter combination drew the support of less than 14 per cent.

145. It has already been agreed that the retirement age, *i.e.* the age of pension eligibility, will be raised from 60 to 65 over a 12-year period ending in 2013 for men and in 2018 for women.

146. Currently, hours up to three-quarters of normal hours of full-time workers are exempt from the base.

147. Taxes currently cover one-third of the costs of the first-tier public pension.

148. Corporate funds are of two types. Employee funds number 1 874 with enrolment of 12.25 million people and assets of 48.69 trillion yen as of March 1998. They are managed professionally and pay part of the benefits that would otherwise be paid by the public pension system. Tax-qualified funds are managed by employers and are separate from the public system. They number 88 310 (down from a peak of 92 467 in FY 1993) and cover 10.46 million workers with assets of 19.15 trillion yen.

149. Matsui (1997) looked at the market values of pension plan assets and the projected benefit obligations for 25 Japanese firms with American Depository Receipts trading on US exchanges. She found that all were underfunded, with total underfunding of 3.8 trillion yen (over $30 billion) in FY 1996. This represents some 15.6 per cent of these firms' total equity. The situation is undoubtedly even worse for other firms without US listings. In June 1998 a panel of the Ministry of Finance decided that all firms will have to disclose their pension liabilities and the market value of the assets they hold against them as from FY 2000. But there remain no plans to impose minimum funding requirements.

150. Fully 1 222 tax-qualified funds also failed already in FY 1996 and a further 1933 in FY 1997, bringing the total to 88 310, down 4.5 per cent in the last four years.

151. The rule specified that at least 50 per cent had to be invested in ''safe assets''; no more than 30 per cent in equities, 30 per cent in foreign assets and 20 per cent in real estate.

152. It had been abolished already in April 1997 for the tax-qualified plans.

153. Matsui (1997) alleges that this led to a cosy relationship between these financial institutions and the firms whose pension assets they were managing: they provided stable share ownership in return for the right to manage these funds.

154. Even Nempuku, the Pension Service Welfare Public Corporation, has boosted the share of new funds which it is allocating to foreign managers from 15 to 52 per cent.

155. The PIO-NET system recorded 164 643 complaints in FY 1990 and 351 140 in FY 1996.

156. Currently it is possible to bring a civil action for compensation for damages, so long as no determination of liability is being sought, but the number of such actions is small.

157. This comparison excludes the Antitrust Division of the US Department of Justice, which also has substantial competition policy responsibilities.

158. Internationally comparable information is rather scarce in this domain. Available data show that in the early 1990s the new company formation rates were around 8 per cent in France and the Netherlands, 6 per cent in Norway and 5 per cent in Sweden. See EIM Small Business Research and Consultancy (1995).

159. In earlier periods of high economic growth relatively low birth rates in Japan probably did not matter much, as business expansion relied more on increases in the size of existing companies than new company creation.

160. These three decennial periods are roughly comparable in that each comprises a period of recovery/overheating as well as a period of recession.

161. The construction sector benefited from the boom related to the asset price bubble and then from the large public works spending after the bursting of the bubble.

162. Any such comparison is subject to measurement errors, particularly concerning the use of the purchasing power parity exchange rate. See Van Ark and Pilat (1993).

163. See Karatsu (1997) for many examples of such technologies.

164. The role of the government in the catching-up process was mainly to facilitate the diffusion of advanced imported technologies and encourage R&D investment through the tax system. As for the former, the government helped companies to obtain favourable terms of technology importation and sometimes restricted market access by foreign companies through formal and informal means. Research co-operatives, which were usually set up by companies interested in certain technologies and could enjoy support from the government, sometimes played important roles in disseminating advanced technologies. The main advantage for the participating firms was the resource saving and risk reduction associated with the basic research necessary to adopt the new technology. With little labour mobility, they could not have utilised the skills of engineers outside their own firms either. The importance of these policy measures should not, however, be over-emphasised because the share of industrial production using imported technologies was rather small – about 10 per cent in 1960 (Odagiri and Goto, 1996).

165. This has made R&D activities more sensitive to market requirements as well as to cyclical developments.

166. For an interesting analysis of the shift overseas of industrial clusters, see Mori (1996).

167. Much of Japan's industrial activity is densely concentrated in Japan's three metropolitan areas: Tokyo, Osaka and Nagoya (OECD, 1996*b*). These areas remain highly attractive, especially for more modern economic activities, and most factories producing computers, industrial robots, optical fibres, pharmaceuticals and optical instruments are based there. Also, many of the new high technology industries and laboratories established over the 1985-92 period are located in these large cities.

168. See Chapter III for details.

169. This situation represents golden opportunities to those with an entrepreneurial instinct. For example, in 1976 a private transport company entered the parcel delivery market which had long been a monopoly of the Post Office. Four companies followed, and by 1990 nationwide networks of next-day delivery service were formed as a result of aggressive investment in trucking terminals and on-line information systems. In FY 1992 the new service handled almost 1.2 billion parcels, triple the number handled by the Post Office.

170. See for example, Table 7, Chapter 4, of OECD (1998), and Baily (1993).

171. Employment at the 560 000 or so construction firms peaked in September 1997 at 7 million and has since declined by half a million or 7 per cent. Productivity-boosting innovations are beginning to be adopted; construction employment is being used to a diminishing extent to prop up local economic activity; and even merit-based pay is creeping into the sector.

172. These GDP share differentials represented, respectively, 21.4 and 34.2 trillion yen ($192 and $308 billion) in that year.

173. OECD data show that the capital/output ratio in the business sector is higher than in any other G7 country except the United Kingdom.

174. Corporate governance in Japan was reviewed in detail in *OECD Economic Surveys, Japan* (1996).

175. An additional factor contributing to managerial risk aversion is the change in the nature of the post of company president. Itami (1995) reports that the president's tenure among top 50 companies (excluding financial institutions and electricity companies) shortened from 9-10 years until the mid-1960s to 4-5 years in the 1980s with its standard deviation falling from 5-6 years to 2-3 years. He interprets this as indicating that the post of president has changed its nature from a position of leadership to a top institutionalised position with a fixed tenure.

176. See *OECD Economic Surveys, Japan* (1996), pp. 163-167, for various studies supporting the positive role of the main bank. A recent work (Miwa, 1998), however, casts doubt on its significance.

177. For more detail, see Ministry of International Trade and Industry (1997).

178. Comparisons with manufacturing averages are made in the absence of specific information about industries in which these companies operate. Their apparent resilience might simply reflect the effect of industry composition. For example, as a percentage of value added, R&D intensity in manufacturing in 1994 varied from 0.8 in paper products to 24.4 in office and computing equipment, and 20.3 in pharmaceuticals.

179. For detailed discussion of the Japanese labour market, see Chapter IV of *OECD Economic Surveys, Japan* (1996) which follows up the *OECD Jobs Strategy* in the specific Japanese context.

180. Management and Co-ordination Agency, *Survey of Employment Structure*, various issues.

181. The ongoing reform of the financial sector would result in improved functioning of the stock market and a substantial increase in the supply of equity capital, which in turn would raise the turnover of venture capital by facilitating initial public offers of venture company shares. For a discussion of difficulties in developing venture capital in bank-oriented financial systems, see Black and Gilson (1998).

182. It is possible under the Civil Code to form a general partnership and limit the liabilities of the partners by including a clause prohibiting borrowing. This method, however, leaves other possibilities in which the partners can be made liable for the amount exceeding the initial investment and hence makes it difficult for pension funds to invest in a general partnership venture business. The introduction of limited partnership removes this legal ambiguity.

183. According to calculations reported in the government's FY 1997 White Paper on Education, which cited the Institute for Science Information, ''National Science Indicators on Diskette, 1981-1996'', as its source, Japan is ranked second in three and third in four out of 19 areas in 1990 to 1994 with respect to the number of citations. While citation rates are a measure of the quality of scientific work, it is also a measure of contact between national systems of innovation. Japanese research has been relatively isolated from the rest of the world – in part due to language barriers and in part to the fact that the large majority of R&D has been done in the private sector.

184. See Wakasugi (1989) and Imano (1991), who provide detailed information about careers of researchers based on hearings. This practice is also regarded as a rational response to limited labour mobility.

185. Clark and Fujimoto (1992) pointed out the shorter average development times of Japanese auto makers than their US and European rivals.

186. It is necessary to note that even under the current situation academic information is obtained through various channels. According to a comparative study of Japan and the United States (Goto and Nagata, 1997), private companies pay more attention to universities as a source of information in Japan than in the United States.

187. One of the fundamental problems that make co-operation between universities and business difficult is the fact that national universities are not a legal person and do not have an independent account. Money is often paid into a special account for all national universities and cannot be used without approval by the Ministry of Education. Otherwise, complicated special treatments with restrictive conditions are required.

188. It should be noted that patenting is not the only way to ensure the appropriation of benefits from innovation: other forms of appropriation such as immediate commercialisation, keeping the information a secret, etc., have been used to a varying extent depending on the industry and country. See Goto and Nagata (1997) for a comprehensive survey of different practices by Japanese and US firms.

189. This rate reached 2.5 per cent in FY 1996, a new record. The problem is by no means restricted to the public school system; in fact, the rate is higher in private schools.

190. Survey of opinions concerning the corporate system of the 21st century, carried out by Long-Term Credit Bank Research Institute Consulting, Inc.

191. A pure holding company is one whose sole purpose is to have financial control of other companies.

192. Currently, the employees of the company concerned or of related companies cannot be appointed as an outside auditor within five years after their separation. It is often the case that these people are appointed right after this ‘‘cooling-off’’ period so that a concern is raised as to the neutrality of outside auditors.

Bibliography

Asako, Kazumi (1997), ''Public Finance: Fiscal Policies and Macroeconomy'', in Asako, Kazumi, S. Fukada and N. Yoshino (eds.), *Modern Macroeconomic Analysis: Japanese Economy in the Transformation Period,* University of Tokyo Press, Tokyo (in Japanese).

Baily, Martin (1993), ''Competition, Regulation and Efficiency'', *Brookings Papers on Economic Activity, Microeconomics,* 2.

Bank of Japan (1998*a*), ''Utilization of Financial Institutions' Self-Assessment in Enhancing Credit Risk Management'', *Quarterly Bulletin,* 6, 1, February.

Bank of Japan (1998*b*), *Annual Report 1998.*

Bernard, Henri and Stefan Gerlach (1998), ''Does the Term Structure Predict Recessions? The International Evidence'', Centre for Economic Policy Research Discussion Paper No. 1892, May.

Black, Bernard S. and Ronald J. Gilson (1998), ''Venture Capital and the Structure of Capital Markets: Banks Versus Stock Markets'', *Journal of Financial Economics,* 47, 3, March.

Brayton, Flint and Peter Tinsley (1996), ''Attitude to FRB/US – A Macroeconomic Model of the US'', Board of Governors of the Federal Reserve System, unpublished.

Clark, Kim B. and Takahiro Fujimoto (1992), ''Product Development and Competitiveness'', *Journal of the Japanese and International Economies,* No. 6, 2, June.

Economic Planning Agency (1996), *Economic Survey of Japan 1995-1996,* Tokyo.

Economic Planning Agency (1997), *Economic Survey of Japan 1996-1997,* Tokyo.

EIM Small Business Research and Consultancy (1995), *Dutch SMEs in International Perspectives: Set-ups in the Netherlands,* Zoetermeer, Netherlands, July.

Fuhrer, Jeffrey C. and Brian F. Madigan (1997), ''Monetary Policy When Interest Rates are Bounded at Zero'', *Review of Economics and Statistics,* LXXIX, 4, November.

Fukao, Mitsuhiro (1998), ''Japanese Financial Instability and Weakness in Corporate Governance Structure'', MITI Research Institute, Conference Paper 98-2-6, 1998.

Gibson, Michael S. (1997), ''More Evidence on the Link Between Bank Health and Investment in Japan'', *Journal of the Japanese and International Economies,* 11, 3, September.

Goto, Akira, and Akiya Nagata (1997), ''Technological Opportunities and Appropriating the Returns from Innovation: Comparison of Survey Results from Japan and the US'', NISTEP Report No. 48, National Institute of Science and Technology Policy, Science and Technology Agency (in Japanese).

Hayashi, Fumio (1987), ''Tests for Liquidity Constraints: A Critical Survey and Some New Observations'', in Truman F. Bewley (ed.), *Advances in Econometrics: Fifth World Congress*, Cambridge University Press.

Higuchi, Yoshio and Kazunari Shimpo (1998), ''The Recent Changes in Unemployment Rate and Job Creation and Destruction'', MITI Research Institute, Conference Paper 98-2-5, 1998.

Honma, Masaaki, Toshihiro Ihori, Naozumi Atoda and Atsuyoshi Murayama (1984), ''Reality of Varying Income Tax Burdens Among Occupations'', *Kikan Gendai Keizaigaku*, 59 (in Japanese).

Honma, Masaaki, Toshihiro Ihori, Naozumi Atoda and Masayaki Naka (1987), ''Optimal Difference in Income Tax Burdens among Occupations'', *Keizai Bunseki*, 109, Economic Research Institute, Economic Planning Agency, June (in Japanese).

Horioka, Charles Yuji (1998), ''Japan's Public Pension System: What's Wrong With It and How to Fix It'', Osaka University, Institute of Social and Economic Research, Working Paper 447, July.

Horiuchi, Akiyoshi and Katsutoshi Shimizu (1997), ''An Analysis of the Issue Who Monitors the Monitor: The Impact of Amakudari on Bank Performance'', University of Tokyo, Research Institute for the Japanese Economy, Discussion Paper Series 97-F-4.

Imano, Koichiro (1991), ''Career Path of Engineers'', in Kazuo Koike (ed.), *Human Resource Management of White-Collar Graduates*, Toyo Keizai Shinpo-sha (in Japanese).

Itami, Takayuki (1995), ''Top Management and Companies'Adaptability'', in Corporate Behaviour Study Group (ed.), *Adaptability of Japanese Firms*, Nihon Keizai Shinbun-sha (in Japanese).

Ito, Hiro and Andrew Z. Szamosszegi (1998), *A Cure for Japan's Sick Banks*, Economic Strategy Institute, Washington D.C., June.

Ito, Takatoshi and Yuri Nagataki Sasaki (1998), ''Impacts of the Basle Capital Standard on Japanese Banks' Behavior'', National Bureau of Economic Research Working Paper 6730, September.

Kang, Jun-Koo and Rene M. Stulz (1997), ''Is Bank-Centered Governance Worth It? A Cross-Sectional Analysis of the Performance of Japanese Firms During the Asset Price Deflation'', National Bureau of Economic Research Working Paper 6238, October.

Karatsu, Hajime (1997), *Japan in the Next Thirty Years: The Sun Must Rise Again*, PHP Research Institute, Tokyo (in Japanese).

Kimura, Yoko, Chizuka Hamamoto and Naomi Teramoto (1997), ''Varied Choices of Female Life Styles and Their Economic Valuation'', mimeo.

Kishida, Hiroshi (1998), ''A Market Projection for Long-Term Care in Japan'', *NLI Research*, NLI Research Institute, 116, June.

Kiyotaki, Nobuhiro and Kenneth D. West (1996), ''Business Fixed Investment and the Recent Business Cycle in Japan'', in Ben Bernanke and Julio Rotemberg (eds.), *NBER Macroeconomics Annual 1996*, MIT Press, Cambridge, Mass. and London.

Krugman, Paul (1998), ''Japan's Trap'', article on Prof. Krugman's internet home page, May.

Long-Term Credit Bank Consulting Inc. (1998), ''Opinion Surveys Concerning the Corporate System of the 21st Century'', contained in *A Report of a Study Group on the Corporate System of the 21st Century,* Ministry of International Trade and Industry, March, 1998.

Matsui, Kathy (1997), ''Demographic Tsunami'', *Look Japan,* December.

Matsuura, Hasuhiro, Katsuki Watanabe and Shuichi Uemura (1998), ''The Potential Growth Rate of the Japanese Economy: the Effects of Declining Labour Inputs due to Ageing'', Bank of Japan, Research and Statistics Department Working Paper 98-4 (in Japanese).

Mikuni, Akio (1996), ''The Changes Taking Place in the Japanese Financial System'', speech to the International Center for Monetary and Banking Studies Conference on Forces of Change in International Banking, Geneva, 7-8 November.

Ministry of International Trade and Industry (1997), ''A Survey of R&D-based Industrial Cluster in the Greater Tama Area'', mimeo (in Japanese).

Ministry of Labour (1998), *White Paper on Labour,* Tokyo.

Mishkin, Frederic S. (1998), ''International Capital Movements, Financial Volatility and Financial Instability'', National Bureau of Economic Research Working Paper 6390, January.

Miwa, Yoshiro (1998),'''Market' and Market Orientation: from Japanese Experience'', mimeo (in Japanese).

Miyagawa, Tsutomu and Hidehiko Ishihara (1997), ''Monetary Policy: Changes in Banking Behaviours and Macroeconomy'', in Kazumi Asako, Shin-ichi Fukuda and Naoyuki Yoshino (eds.), *Modern Macroeconomic Analysis: Japanese Economy in the Transformation Period,* University of Tokyo Press, Tokyo (in Japanese).

Mori, Koichiro (1996), ''A Study on Industrial Clusters: Start of Business in Asia by Japanese Electrical Appliances Industry'', Masters Dissertation, Hitotsubashi University (in Japanese).

Muellbauer, John (1995), ''The Assessment: Consumer Expenditure'', *Oxford Review of Economic Policy,* 10, 2.

Murakami, Kiyoshi (1997), *The Crisis of the Japanse Pension System,* Toyo Keizai Shinpo-sha (in Japanese).

Odagiri, Hiroyuki and Akira Goto (1996), *Technology and Industrial Development in Japan: Building Capabilities by Learning, Innovation and Public Policy,* Oxford University Press.

OECD (1996*a*), *Regional Problems and Policies in Japan,* Paris.

OECD (1996*b*), ''Networks for Rural Economic Development'', unpublished.

OECD (1997), *The OECD Report on Regulatory Reform,* Vol. II: Thematic Studies, Paris.

OECD (1998*), Science, Technology, and Industry Outlook 1998,* July.

Ogawa, Kazuo (1992), ''An Empirical Analysis of Japanese Household Behaviour'', *Financial Review,* Ministry of Finance, December (in Japanese).

Ogawa, Kazuo, Shin-ichi Kitasaka, Hiroshi Yamaoka, and Yasuharu Iwata (1996), ''Borrowing Constraints and the Role of Land Assets in Japanese Corporate Investment Decision'', *Journal of Japanese and International Economies,* 10, 2, June.

Ogawa, Alicia *et al.* (1998), ''What Credit Crunch?'', Salomon Smith Barney, Global Equity Research, Tokyo, 2 March.

Pilat, Dirk (1996), ''Labour Productivity Levels in OECD Countries'', OECD Economics Department Working Paper No. 169.

Roseveare, Deborah, Willi Leibfritz, Douglas Fore and Eckhard Wurzel (1996), ''Ageing Populations, Pension Systems and Government Budgets: Simulations for 20 OECD Countries'', OECD Economics Department Working Paper No. 168.

Small and Medium Sized Enterprise Agency (1996), *White Paper on SMEs*, Tokyo.

Starr-McCluer, M. (1998), ''Stock Market Wealth and Consumer Spending'', Board of Governors of the Federal Reserve System, unpublished paper.

Summers, Lawrence (1991), ''How Should Long-Term Monetary Policy Be Determined?'', *Journal of Money, Credit and Banking*, 23, 3, Part 2, August.

Takayama, Noriyuki, Yukinobu Kitamura and Hiroshi Yoshida (1998), ''Generational Accounting in Japan'' in Alan J. Auerbach *et al.* (eds.), *Generational Accounting Around the World*, University of Chicago Press, forthcoming.

Thompson, John K. (1998), ''Shareholder Value and the Market in Corporate Control in OECD Countries'', *OECD Financial Market Trends*, 69, February.

US Small Business Administration (1997), *The State of Small Business, A Report of the President, 1996*, US Government Printing Office, Washington D.C.

Van Ark, Bart and Dirk Pilat (1993), ''Productivity Levels in Germany, Japan and the United States: Differences and Causes'', *Brookings Papers on Economic Activity, Microeconomics*, 2.

Van Rixtel, Adrian and Walter Hassink (1998), ''Monitoring the Monitors: Amakudari and the *Ex post* Monitoring of Private Banks'', Centre for Economic Policy Research Discussion Paper No. 1785, January.

Wakasugi, Ryuhei (1989), ''R&D Organisations and Their Behaviours'' in Kenichi Imai and Ryutaro Komiya (eds.), *Japanese Firms*, University of Tokyo Press (in Japanese) (translation in English, *Business Enterprise in Japan: Views from Leading Japanese Economists*, MIT Press, Cambridge, Mass., 1994).

Wolf, Charles R. (1998), *Economic Openness: Many Facets, Many Metrics*, Rand Organisation, Santa Monica, California, forthcoming.

Yoshikawa, Hiroshi and Hidetaka Obara (1997), ''Boom and Recession and Business Investment in Heisei Era'', IMES Discussion Paper 97-J-2, Bank of Japan (in Japanese).

Annex

Chronology of main economic events

1997

October

Kyoto Kyoei Bank released its liquidation plan.

The negotiations with the United States over the harbour dispute resulted in an agreement.

November

Sanyo Securities applied to the courts for legal restructuring procedures.

Hokkaido Takushoku Bank, a major commercial bank, announced its inability to continue operating and the transfer of its operations in Hokkaido to Hokuyo Bank.

The government announced a deregulation package, listing 120 items.

Yamaichi Securities announced closure of its business.

Tokuyo City Bank announced its closure and the transfer of its operations to other regional banks.

The Diet passed the Fiscal Structural Reform Act, which aimed to reduce fiscal deficits to 3 per cent of GDP by FY 2003.

December

Global Warming Summit was held in Kyoto, agreeing country-by-country targets to reduce emission of greenhouse gases.

The Diet passed two laws concerning holding companies in the financial sector.

The Ministry of Health and Welfare released ‘‘five alternatives’’ for the future financing of the public pension system.

The Diet passed the Assisted-Care Insurance Law for long-term care effective in FY 2000.

The Administrative Reform Council released its final report, recommending reorganisation of central government ministries and agencies.

The Administrative Reform Committee released its final report, recommending further deregulation in seventeen areas.

The Diet passed the revised Deposit Insurance Law, giving the Deposit Insurance Corporation the right to cover loan losses of merging banks.

The government announced a special tax cut amounting to 2 trillion yen in CY 1998.

The Cabinet approved the government's official economic forecast of 1.9 per cent growth in FY 1998.

The first holding company since the end of the war was set up by Daiei.

The Ministry of Finance announced measures to deal with the ''credit crunch''.

The Cabinet decided the initial FY 1998 budget plan.

1998

January

The Ministry of Finance announced measures to stabilise the stock market.

The Cabinet decided the outline of the FY 1998 tax reform, which included a corporate tax cut (0.3 trillion yen) and income tax cut (2 trillion yen).

Mr. Mitsuzuka, the Minister of Finance resigned, taking responsibility for the Ministery's bribery scandal.

The Diet adopted the bill for a temporary tax cut, amounting to 2 trillion yen.

Japan's position in the film trade dispute with the United States was fully upheld by the WTO.

February

The Diet adopted the first FY 1997 supplementary budget.

The Nagano Olympic Games were held.

The Diet passed the two finance-related laws, which enabled the government to use 30 trillion yen of public money to bail out banks and protect depositors.

March

Twenty-one banks applied for public capital injections, virtually all of which were fully approved by the Financial Crisis Management Committee.

Mr. Matsushita, Governor of the Bank of Japan, resigned, taking responsibility for the Bank's bribery scandal.

The Cabinet decided a new three-year deregulation package, the Deregulation Action Programme, upon the expiry of the previous programme.

The Diet passed the bill to allow companies to re-evaluate their land holdings.

Tax reform for FY 1998 was enacted.

An Agreement with the United States over civil aviation is reached, rectifying differential treatment between the two countries.

April

New Bank of Japan Law and new Foreign Exchange Law take effect.

The Diet adopted the initial FY 1998 budget.

''Comprehensive Economic Measures'', primarily a fiscal package amounting to more than 16 trillion yen on a project cost basis, were announced.

May

The Cabinet decided the FY 1998 supplementary budget plan.

A business tie-up between Industrial Bank of Japan and Nomura Securities was announced.

Daimler Benz announced the planned acquisition of a majority of shares in Nissan Diesel.

Hanshin Bank announced a merger with Midori Bank as of April 1999.

The Cabinet decided the Decentralisation Promotion Plan.

The Diet passed the revised Fiscal Structural Reform Act, which introduced a cyclical flexibility clause and postponed the target year for consolidation to FY 2005.

June

A business tie-up between Travelers Group and Nikko Securities was announced.

The Diet passed the four bills to implement reform of the financial system (the ''Big Bang'').

The Diet passed the Basic Law for the Reorganisation of the Central Government Ministries and Agencies.

The Diet adopted the first FY 1998 supplementary budget.

The Financial Supervisory Agency was set up, taking over the functions of supervision and inspection of the financial system from the Ministry of Finance.

The Long Term Credit Bank of Japan announced a merger with the Sumitomo Trust Bank as of April 1999. In September it became likely that the merger would not ultimately take place.

EPA formally declared the peak of the business cycle was March 1997.

July

Japan decided to bring the case of the US-Canada auto pact before the WTO.

The Liberal Democratic Party lost seats in the upper house, and Mr. Hashimoto resigned as Prime Minister and president of LDP. Mr. Obuchi was elected as his replacement.

A new entry into the domestic scheduled airline business was permitted for the first time in 35 years.

August

The Cabinet approved the guidelines of the FY 1998 supplementary budget and the initial FY 1999 budget.

The Long Term Credit Bank announced substantial restructuring measures, including write-offs of 750 billion yen.

September

The Bank of Japan lowered its target rate to 0.25 per cent, leaving the official discount rate unchanged.

Japan decided to bring the case of Massachusetts' sanctions against companies doing business with or in Myanmar to the WTO.

Nippon Leasing Corporation, an affiliate of the Long Term Credit Bank, applied for the Corporate Reorganisation Law, with liabilities worth about 2.2 trillion yen, a record in the post-war era.

A business tie-up between the Tokai Bank and the Asahi Bank was announced.

The Tokyo metropolitan government announced a crisis in its financial conditions.

October

A business tie-up between the Daiichi Kangyo Bank and JP Morgan was announced.

The government's official economic forecast for FY 1998 was revised downward from +1.9 per cent to –1.8 per cent.

The Ministry of the Health and Welfare's Pension Council released its opinions about the future of the pension system.

The Economic Strategy Council under the Prime Minister released an emergency proposal recommending fiscal stimulus worth more than 10 trillion yen, a large-scale capital injection into the banking sector and the freezing of the contribution rates for the Employees Pension Scheme for the next few years.

The Diet passed eight bills to revive the banking system, including the terms under which a bridge bank could be set up and temporary nationalisation could occur. The bills also established the Financial Revitalisation Committee to oversee the process and to take over responsibility for financial regulation and planning from the Ministry of Finance as from January 2000. Separately, it passed a bill to allow the government to inject capital into banks on request even if they are solvent. The Diet also adopted the second FY 1998 budget, thereby providing government loan guarantees amounting to 43 trillion yen in total (replacing what remained from the previous 13 trillion yen) on Bank of Japan's loans to the Deposit Insurance Corporation to implement the recapitalisation of the banks.

The Diet passed the bills for the general account to take over the debt worth 24.1 and 2.8 trillion yen from JNR Settlement Corporation and National Forestry Special Account, respectively (a total of 26.9 trillion yen).

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

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